

Integrated Science

Basic 7

Complete BECE Preparation
Examination Papers & Marking Schemes

Beacon Educational Consult

2026 Edition

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Published by Beacon Educational Consult
Accra, Ghana

First Edition: 2026

ISBN: [To be assigned]

Printed in Ghana

This book has been prepared in accordance with the
Ghana Education Service Standards-Based Curriculum (2019)
and the National Council for Curriculum and Assessment (NaCCA) guidelines.

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Preface

This comprehensive Integrated Science textbook for Basic 7 has been carefully prepared to support Junior High School students in their preparation for the Basic Education Certificate Examination (BECE).

The book contains:

- Multiple choice questions with complete answer keys
- Theory questions with detailed marking schemes
- Curriculum-aligned content following the 2019 Standards-Based Curriculum
- Practice papers from BECE Mock examinations
- Diagram Mode practice papers with scientific illustrations

Each question has been verified for accuracy and aligned with the appropriate curriculum objectives. The marking schemes provide detailed explanations to help students understand the reasoning behind each answer.

We hope this book will be a valuable resource in your academic journey.

— Beacon Educational Consult

Accra, Ghana

2026

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Note: Use the running headers at the top of each page to navigate the book. Each examination paper begins with a title page showing the paper number and topic.

BEACON EDUCATIONAL SERIES

Integrated Science for Basic 7

A Comprehensive Question Bank
with Marking Schemes

Curriculum-Aligned Practice Papers
for Basic 7 (JHS 1) Students and Teachers

Beacon Educational Consult
2026 Edition

Integrated Science for Basic 7
First Edition, 2026

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This book is designed to support the Ghana Education Service (GES) curriculum for Junior High School. It is intended as a supplementary resource for teachers and students.

The questions in this book are original practice materials created for educational purposes. They are not official BECE or GES examination questions.

Published by: Beacon Educational Consult

ISBN: [To be assigned]

Printed in Ghana

PREFACE

This book has been developed to support Junior High School Basic 7 (JHS 1) students and teachers in their study of Integrated Science, as prescribed by the Ghana Education Service (GES) and the National Council for Curriculum and Assessment (NaCCA).

The book contains a comprehensive collection of practice questions organised according to the curriculum's strand and sub-strand structure. Each paper includes both multiple-choice questions (Section A) and structured theory questions (Section B), reflecting the format of BECE examinations.

Key Features

- Curriculum-Aligned Content — All questions mapped to specific content standards.
- Comprehensive Coverage — Topic-based papers covering all major strands.
- Detailed Marking Schemes — Complete answer keys with method marks and marking guidance.
- Professional Diagrams — Clear, scientifically accurate illustrations.
- Cognitive Demand — Questions spanning all levels of Bloom's taxonomy.

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ANSWER KEY AND MARKING SCHEMES

Paper 1: Agriculture

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. A simple agricultural tool is one that is
 - A. made entirely of plastic
 - B. driven by a diesel engine
 - C. operated by hand and needs no engine
 - D. used only on very large farms
2. The tool used mainly for clearing weeds and cutting down bushes on a farm is the
 - A. rake
 - B. garden trowel
 - C. watering can
 - D. cutlass
3. The farm tool used for making ridges, mounds and weeding is the
 - A. wheelbarrow
 - B. hoe
 - C. secateurs
 - D. measuring tape
4. Seedlings in a nursery are watered gently with a
 - A. watering can
 - B. spraying machine for weedicide
 - C. shovel
 - D. bucket of ice
5. The tool used for gathering dry leaves and levelling the surface of a seedbed is the
 - A. knapsack sprayer
 - B. rake
 - C. axe
 - D. cutlass
6. A tool used for digging holes for planting tree seedlings is the
 - A. rake
 - B. pruning saw
 - C. watering can
 - D. pickaxe or mattock
7. Secateurs are used on a farm mainly for
 - A. pruning or cutting small branches of plants
 - B. carrying farm produce
 - C. watering young plants
 - D. ploughing the land
8. Which of the following tools is used on an ANIMAL farm?
 - A. A garden rake
 - B. A seed drill
 - C. A feeding trough brush
 - D. A pair of secateurs
9. A wheelbarrow is used on a farm mainly for
 - A. spraying insecticide
 - B. carrying manure, soil and harvested crops
 - C. cutting grass
 - D. measuring land
10. Which of the following is used to protect the hands while working with farm tools?
 - A. A watering can
 - B. Hand gloves
 - C. A cutlass sheath
 - D. A head pan

- 11.** The handling of an agricultural tool means the way the tool is
- shared among the farmers
 - painted at the factory
 - bought from the market
 - held and used during work
- 12.** The maintenance of agricultural tools means
- keeping the tools in a locked box for ever
 - buying new tools every season
 - selling old tools to other farmers
 - keeping the tools in good working condition
- 13.** Which of the following is NOT a reason for maintaining farm tools?
- To prevent accidents during work
 - To make the tools last longer
 - To make the tools heavier to use
 - To save the money spent on new tools
- 14.** Well-maintained farm tools help the farmer to
- harvest a poorer crop
 - spend more time on the farm
 - use more fuel
 - work faster and with less effort
- 15.** A blunt cutlass is dangerous to use because it
- becomes too light to hold
 - cuts the grass too quickly
 - may slip off the stem and injure the user
 - makes no sound when used
- 16.** The main reason for washing a hoe after use is to
- add weight to the blade
 - make the hoe look new for sale
 - soften the wooden handle
 - remove soil that would hold moisture and cause rusting
- 17.** Farm tools that are poorly maintained will MOST likely
- wear out quickly and need early replacement
 - become sharper on their own
 - work better with time
 - increase the farmer's harvest
- 18.** When walking with a cutlass on a farm, the blade should be
- held in front of the face
 - swung freely while walking
 - pointed downwards and away from the body
 - carried on the shoulder with the edge up
- 19.** A cutlass should be handed to another person by
- holding the blade and offering the handle first
 - offering the sharp edge first
 - dropping it at the person's feet
 - throwing it gently to the person
- 20.** Before using any farm tool, the user should FIRST
- check that the tool is in good condition and the handle is firm
 - paint the blade of the tool
 - sharpen the handle of the tool
 - wash the tool in the stream
- 21.** Farm tools should be used only for
- digging out stones with the blade of a cutlass
 - any work at all, to save time
 - games during break
 - the work for which they are made
- 22.** While weeding with a hoe, pupils should stand
- directly behind one another
 - very close together to finish quickly
 - a safe distance apart
 - in a single line facing each other
- 23.** The correct protective clothing for working with farm tools includes
- sandals and short sleeves
 - open slippers
 - a school uniform only
 - boots and hand gloves
- 24.** A tool with a loose handle should be
- repaired or the handle refitted before use
 - thrown into the bush
 - used quickly before it falls off
 - used only by the strongest pupil

- 25.** When a tool is not in use during work on the farm, it should be
- A. placed across the farm path
 - B. kept in a safe place with the sharp edge down or covered
 - C. hung on a growing plant
 - D. left lying in the grass with the edge up
- 27.** A cutlass is kept in good cutting condition by
- A. rubbing it with wet sand only
 - B. sharpening it with a file or whetstone
 - C. leaving it in the rain to soften
 - D. heating it in a fire until it is red
- 29.** The BEST place to keep farm tools after work is
- A. in a pool of water to keep them cool
 - B. on the bare ground in the open
 - C. a dry, well-ventilated store, hung on racks
 - D. buried in the soil of the farm
- 31.** Moving parts of a knapsack sprayer are kept working smoothly by
- A. tightening them until they cannot move
 - B. washing them with hot pepper water
 - C. oiling or greasing them regularly
 - D. leaving them in the sun
- 33.** Which maintenance practice is MOST suitable for a wheelbarrow?
- A. Painting the wheel so that it cannot turn
 - B. Sharpening the tray with a file
 - C. Oiling the wheel and tightening the bolts
 - D. Soaking the whole barrow in water
- 35.** During a visit to a farm in the community, the BEST way for pupils to record what they observe about tool maintenance is to
- A. argue with the farmer about his methods
 - B. take notes and write a report afterwards
- 26.** The maintenance practice that prevents the metal parts of farm tools from rusting is
- A. burying them in wet soil
 - B. leaving them in the sun for days
 - C. soaking them in water overnight
 - D. cleaning and oiling them after use
- 28.** Wooden handles of farm tools are rubbed with oil or kept dry mainly to
- A. make them slippery to hold
 - B. make them heavier
 - C. change their colour
 - D. prevent cracking and attack by termites
- 30.** Rust on the blade of a hoe is BEST removed by
- A. painting over the rust
 - B. scraping or filing it off and then oiling the blade
 - C. heating the blade in a fire
 - D. soaking the blade in water
- 32.** A knapsack sprayer should be washed out thoroughly after use mainly because
- A. chemicals left inside will block the nozzle and corrode the parts
 - B. the chemicals will become more effective
 - C. washing makes the sprayer lighter to carry
 - D. the sprayer must be refilled with water for storage
- 34.** Tools are hung on racks or nails in the store rather than left on the floor so that they
- A. do not take up moisture from the floor and are not stepped on
 - B. will become lighter in weight
 - C. can be seen by visitors to the farm
 - D. can dry out and crack quickly

- C. remember everything without writing
- D. borrow the farmer's tools and take them home

- 36.** Before pupils handle tools on a community farm, they should
- A. race to see who finishes first
 - B. be shown the correct way to use each tool by the teacher or farmer

- C. remove their boots for comfort
D. share the tools among themselves and begin at once
- 37.** A set of written statements that tell workers how a tool should be used safely is called the
- A. harvest record of the farm
 - B. crop rotation plan
 - C. operational rules for handling the tool
 - D. price list of the tool
- 39.** A school farm club found that most tools in the store were rusty. The FIRST step they should take is to
- A. lock the store and stop farming
 - B. clean and oil the tools and repair the loose handles
 - C. throw all the tools away
 - D. buy a complete new set of tools
- 38.** A farmer in the community keeps his cutlass sharp, clean and hung in a dry store. The MOST likely result is that his cutlass will
- A. last for many years and work efficiently
 - B. rust within a few weeks
 - C. become blunt after one season
 - D. be rejected by other farmers
- 40.** The club then drew up a list of maintenance rules and put it on the store wall. This is useful mainly because it
- A. reminds every user of what to do after work with the tools
 - B. shows the price of each tool
 - C. decorates the wall of the store
 - D. stops pupils from entering the store

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 2: Animal Production

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. An animal that is tamed and kept by people in the home or on a farm is called
 - A. a game animal
 - B. a wild animal
 - C. a domestic animal
 - D. a pest
2. Which of the following is NOT a domestic animal?
 - A. The pig
 - B. The antelope
 - C. The goat
 - D. The fowl
3. A group of animals of the same kind having the same features that they pass on to their young is called
 - A. a litter
 - B. a flock
 - C. a breed
 - D. a herd
4. Sanga, N'dama and White Fulani are breeds of
 - A. goats
 - B. cattle
 - C. sheep
 - D. fowls
5. West African Dwarf is a breed commonly used in Ghana for
 - A. goats
 - B. rabbits
 - C. turkeys
 - D. cattle only
6. A farmer at Walewale says he keeps a 'flock'. The animals he is most likely talking about are
 - A. pigs
 - B. rabbits
 - C. sheep or fowls
 - D. cattle
7. An animal with a stomach divided into four chambers is described as
 - A. a monogastric animal
 - B. a scavenger
 - C. a carnivore
 - D. a ruminant
8. Which of the following animals is a ruminant?
 - A. Fowl
 - B. Sheep
 - C. Pig
 - D. Rabbit
9. The habit of bringing swallowed food back to the mouth to chew it again is called
 - A. scavenging
 - B. grazing
 - C. browsing
 - D. chewing the cud
10. Ruminants are able to feed mainly on grass because
 - A. grass contains no fibre at all
 - B. they swallow stones to grind the grass
 - C. the micro-organisms in their stomach chambers break down the grass
 - D. they have very sharp cutting teeth in the upper jaw

- 11.** Which of the following groups contains ONLY ruminants?
- Cattle, sheep and goats
 - Sheep, fowls and goats
 - Cattle, pigs and goats
 - Goats, rabbits and sheep
- 12.** A goat and a pig are given the same heap of fresh grass. The goat does better on it because the goat
- eats much faster than the pig
 - can digest the fibre in the grass with the help of its stomach micro-organisms
 - has a longer mouth than the pig
 - does not need any other feed at all
- 13.** The four chambers of the stomach of a ruminant include the
- gizzard
 - crop
 - rumen
 - caecum only
- 14.** Which of the following is a monogastric animal?
- The sheep
 - The cow
 - The pig
 - The goat
- 15.** Which of the following is NOT poultry?
- The rabbit
 - The turkey
 - The guinea fowl
 - The duck
- 16.** A pig is usually fed on kitchen waste, maize bran and other concentrates rather than on grass alone because the pig
- refuses to eat any plant material
 - has a simple stomach and cannot digest much fibre
 - is too large to walk to the grass
 - has no teeth for cutting grass
- 17.** Ruminants, monogastric animals and poultry are alike in that all of them
- have four stomach chambers
 - lay eggs
 - are kept only for their milk
 - feed, breathe, grow and produce young ones
- 18.** A pupil wrote: 'Cattle and pigs are both ruminants because both are kept on farms.' The statement is wrong because
- only wild animals are ruminants
 - pigs have four stomach chambers
 - being kept on a farm does not
- 19.** A goat and a pig are given the same heap of fresh grass. The goat does better on it because the goat
- eats much faster than the pig
 - can digest the fibre in the grass with the help of its stomach micro-organisms
 - has a longer mouth than the pig
 - does not need any other feed at all
- 20.** An animal with a simple stomach of one chamber is described as
- a cud chewer
 - a monogastric animal
 - a ruminant
 - a herbivore
- 21.** Birds kept on a farm for their meat and eggs are known as
- draught animals
 - livestock feed
 - ruminants
 - poultry
- 22.** The part of the digestive system of a fowl that grinds its food is the
- gizzard
 - rumen
 - crop only
 - udder
- 23.** A rabbit is a monogastric animal, yet it feeds mainly on leaves and grass. This shows that
- leaves contain no fibre
 - the rabbit is a ruminant
 - all monogastric animals eat only meat
 - monogastric animals can also be herbivores
- 24.** Which of the following is a difference between a fowl and a goat?
- The goat lays eggs while the fowl does not
 - The fowl has feathers and two legs while the goat has hair and four legs
 - The fowl is a ruminant while the goat is not
 - The goat has a gizzard while the fowl has a rumen

determine the type of stomach
an animal has

D. cattle are not kept on farms

- 25.** Which observation would BEST help a pupil to decide whether an animal is a ruminant?
- A. Watching whether the animal brings food back to its mouth to chew again while resting
 - B. Noting the colour of its hair
 - C. Measuring the length of its tail
 - D. Counting the number of its legs
- 27.** The use of an animal to meet the needs of the family at home is called
- A. industrial use
 - B. wild use
 - C. commercial use
 - D. domestic use
- 29.** A farmer uses his bull to pull a plough on his farm. This use of the animal is called
- A. draught (work) use
 - B. commercial sale
 - C. a breed
 - D. a by-product
- 31.** A family keeps goats and sells two of them to pay school fees. The goats have been used
- A. as draught animals
 - B. commercially
 - C. domestically only
 - D. as poultry
- 33.** Animals are also kept for social and cultural purposes in Ghana, such as
- A. providing sheep and fowls for naming ceremonies and festivals
 - B. guarding the national borders
 - C. producing electricity for the home
 - D. pulling vehicles in the city
- 35.** A useful material obtained from an animal apart from its meat is called
- A. a by-product
 - B. a carcass only
 - C. a ration
 - D. a breed
- 24.** A farmer who has very little money for feed but plenty of grass on his land should best keep
- A. broiler fowls only
 - B. rabbits only
 - C. pigs only
 - D. sheep and goats
- 26.** Poultry are kept in large numbers on a small piece of land more easily than cattle mainly because poultry
- A. produce no waste at all
 - B. are small, grow quickly and need less space
 - C. cannot move about
 - D. do not need any feed
- 28.** Which of the following is a domestic use of a fowl?
- A. Providing eggs and meat for the family
 - B. Exporting frozen chicken
 - C. Supplying a hotel with eggs every week
 - D. Selling day-old chicks in the market
- 30.** Cattle dung spread on a food crop farm is useful because it
- A. kills all insects on the farm
 - B. prevents rain from falling
 - C. makes the soil hard and dry
 - D. adds nutrients and humus to the soil
- 32.** Which of the following best shows the domestic use of a cow in a home?
- A. Renting the cow to another farmer
 - B. Supplying milk to a factory
 - C. Taking milk from the cow for the family to drink
 - D. Selling the cow at the market
- 34.** Keeping animals mainly in order to sell them or their products for money is called
- A. cultural use
 - B. domestic use
 - C. subsistence use
 - D. commercial use

- 37.** The skin of a goat is used in many Ghanaian communities to
- A. feed other animals
 - B. replace roofing sheets
 - C. make the heads of drums and leather goods
 - D. build the walls of houses
- 39.** A farmer near Kumasi keeps 500 layers and sells the eggs to shops in the town every week. The main purpose of his farm is
- A. cultural
 - B. experimental
 - C. commercial
 - D. domestic
- 36.** Which of the following is an animal by-product?
- A. Maize bran fed to the animal
 - B. Water given to the animal
 - C. Grass grown on the farm
 - D. Hide used for making leather
- 38.** Animal droppings from a poultry house are best described as
- A. a by-product that can be used as manure
 - B. a type of animal feed for cattle
 - C. a disease of poultry
 - D. a waste of no use at all
- 40.** A young farmer has a small plot behind her house and wants a quick return on the little money she has. The BEST advice is to
- A. start with a herd of cattle, because cattle are the largest animals
 - B. start with a small flock of fowls, because they grow fast and produce eggs and meat within a short time
 - C. keep one pig and wait for it to produce manure only
 - D. keep no animals until she can afford a large farm

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram, a labelled sketch or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 3: Climate Change and

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. A sustainable energy source is one that
 - A. is found only outside Ghana
 - B. is always the cheapest to buy
 - C. can be used again and again without being used up
 - D. burns with the brightest flame
2. Which of the following is a renewable source of energy?
 - A. petrol
 - B. sunlight
 - C. coal
 - D. diesel
3. The energy of moving air is called
 - A. tidal energy
 - B. wind energy
 - C. chemical energy
 - D. nuclear energy
4. Energy obtained from the heat and light of the Sun is
 - A. geothermal energy
 - B. sound energy
 - C. solar energy
 - D. fossil energy
5. Electricity produced at the Akosombo Dam is obtained from
 - A. burning petrol
 - B. burning charcoal
 - C. the wind
 - D. falling water
6. Which of the following is NOT a fossil fuel?
 - A. coal
 - B. natural gas
 - C. firewood
 - D. crude oil
7. Fossil fuels are described as non-renewable because they
 - A. cannot burn in air
 - B. take millions of years to form and are being used up quickly
 - C. are found in only one country
 - D. are made in factories every day
8. The gas produced when organic waste rots in a closed tank without air is
 - A. biogas
 - B. oxygen
 - C. nitrogen
 - D. chlorine
9. Planting trees on bare land to protect the environment is called
 - A. afforestation
 - B. urbanisation
 - C. deforestation
 - D. desertification
10. A pupil who switches off the lights and the fan before leaving the classroom is practising
 - A. energy wastage
 - B. energy production
 - C. energy conservation
 - D. energy transformation
11. Replacing filament bulbs with LED bulbs in a school helps the environment MAINLY because LED bulbs
 - A. are made of glass only

- B. are brighter than the Sun
- C. never need to be switched off
- D. use less electricity for the same amount of light

- 13.** The use of an improved cooking stove that needs less firewood helps the environment because it
- A. costs more money to buy
 - B. increases the heat lost to the air
 - C. makes the food taste better
 - D. reduces the number of trees cut down and the smoke produced
- 15.** Which of the following BEST describes a green economy?
- A. an economy based only on farming
 - B. an economy that creates wealth while protecting the environment
 - C. an economy that uses no energy at all
 - D. an economy in which only green goods are sold
- 17.** The greenhouse effect is the
- A. reflection of all sunlight back into space
 - B. growing of green plants inside a glass house
 - C. cooling of the Earth by rain clouds
 - D. trapping of heat near the Earth's surface by gases in the atmosphere
- 19.** Besides carbon dioxide, another important greenhouse gas is
- A. hydrogen
 - B. methane
 - C. neon
 - D. argon
- 21.** Green plants help to reduce the amount of carbon dioxide in the air because they
- A. take in carbon dioxide during photosynthesis
 - B. give out carbon dioxide at night only
 - C. turn carbon dioxide into soil
 - D. store carbon dioxide in their roots as salt
- 23.** Which of the following is a likely effect of global warming?
- A. a permanent end to the rainy season everywhere
- 12.** Which of these practices adds the LEAST pollution to the air?
- A. drying clothes in the Sun
 - B. running a petrol generator all day
 - C. burning rubbish in the yard
 - D. burning old tyres
- 14.** Walking or cycling to school instead of being driven helps the environment because it
- A. reduces the exhaust gases released into the air
 - B. increases the number of vehicles on the road
 - C. makes the roads wider
 - D. uses more fuel
- 16.** Sorting household waste so that bottles and tins are collected separately supports
- A. recycling
 - B. open burning
 - C. littering
 - D. land filling of everything
- 18.** Which of the following is a greenhouse gas?
- A. helium
 - B. nitrogen
 - C. oxygen
 - D. carbon dioxide
- 20.** Carbon dioxide is added to the atmosphere MAINLY by
- A. the burning of fuels such as petrol, coal and firewood
 - B. the freezing of water
 - C. the evaporation of sea water
 - D. the rusting of iron
- 22.** A steady rise in the average temperature of the Earth is called
- A. condensation
 - B. global dimming
 - C. evaporation
 - D. global warming

- B. melting of ice at the poles and a rise in sea level
- C. an increase in the number of planets
- D. a fall in the amount of carbon dioxide in the air

- 24.** Cutting down large areas of forest increases the greenhouse effect because it
- A. cools the soil too quickly
 - B. adds oxygen to the air

25. The layer of gases that surrounds the Earth and holds in some of its heat is the
- A. biosphere
 - B. atmosphere
 - C. lithosphere
 - D. hydrosphere
27. Which of the following is an effect of climate change on farming in Ghana?
- A. crops no longer need water
 - B. all soils become more fertile
 - C. rains become unreliable, so planting times are harder to predict
 - D. weeds stop growing on farms
29. A community that wants to reduce the greenhouse gases it produces should
- A. cut down trees to make more charcoal
 - B. use solar lamps and plant trees instead of burning more fuel
 - C. burn its rubbish in open heaps every evening
 - D. buy more petrol generators for every house
31. Burning farm residues in the open field after harvest is discouraged because it
- A. increases the number of earthworms
 - B. adds water to the soil
 - C. releases carbon dioxide and smoke and destroys soil organisms
 - D. makes the field cooler
33. A country is said to use clean energy when its electricity comes mainly from
- A. firewood and charcoal
 - B. old tyres and plastics
 - C. coal and crude oil
 - D. the Sun, wind and water
- C. stops the wind from blowing
D. removes the trees that would have taken in carbon dioxide
26. Climate change is BEST described as
- A. the change of weather from morning to afternoon
 - B. a long-term change in the usual pattern of weather of a place
 - C. the movement of clouds across the sky
 - D. the change of seasons within a single week
28. Flooding in coastal towns is made worse by climate change MAINLY because
- A. rivers flow backwards during the night
 - B. the Moon moves nearer to the Earth
 - C. the level of the sea rises and storms become stronger
 - D. the soil becomes lighter in colour
30. The BEST reason for using public transport, or sharing a vehicle, is that it
- A. removes the need for roads
 - B. makes journeys take longer
 - C. carries many people while releasing less exhaust gas per person
 - D. allows drivers to use more fuel
32. Which pair of actions would MOST reduce a school's contribution to climate change?
- A. switching off unused appliances and planting trees in the compound
 - B. clearing the school garden and paving all the ground
 - C. burning waste paper daily and using more generators
 - D. leaving taps running and buying more bulbs
34. A simple solar dryer used to dry pepper and fish works by
- A. burning charcoal under the trays
 - B. trapping the Sun's heat so that the warm air removes moisture

- C. adding salt to the food in a tank
- D. cooling the food with ice

- 35.** In a biogas project, the material fed into the digester is
- A. animal dung and other organic waste
 - B. plastic bags and metal
 - C. sand and gravel
 - D. broken bottles and tins
- 37.** A school wishing to light its classrooms during a power cut, without polluting the air, should install
- A. a petrol generator
 - B. a diesel generator
 - C. a charcoal stove
 - D. solar panels with a storage battery
- 39.** Which of the following would BEST show that a class project on sustainable energy has succeeded?
- A. the device uses more fuel than before
 - B. the device is the most expensive in the district
 - C. the device works and reduces the fuel or electricity the school uses
 - D. the device is kept in a locked store
- 36.** Apart from gas for cooking, a biogas plant also provides
- A. slurry that can be used as manure on the farm
 - B. pure drinking water
 - C. cement for building
 - D. petrol for vehicles
- 38.** Before building a small energy project, the FIRST step a class should take is to
- A. paint the school wall
 - B. call the whole town for a party
 - C. plan and find out which source of energy is available in the area
 - D. buy all the materials at once
- 40.** Green jobs are jobs that
- A. require workers to wear green uniforms
 - B. are done only on farms
 - C. are done only during the rainy season
 - D. help to protect the environment while providing a living

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 4: Conversion and Conservation

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. The law of conservation of energy states that energy can be
 - A. created whenever it is needed
 - B. destroyed after it has been used
 - C. stored only in dry cells
 - D. changed from one form to another but cannot be created or destroyed
2. The unit in which energy is measured is the
 - A. joule
 - B. newton
 - C. ampere
 - D. volt
3. A system into which no energy enters and from which no energy escapes is described as a
 - A. moving system
 - B. closed system
 - C. heated system
 - D. wasted system
4. In a closed system, the total amount of energy at the end of a change is
 - A. always less than at the beginning
 - B. the same as the total amount at the beginning
 - C. equal to zero
 - D. always more than at the beginning
5. A dry cell supplies 60 J of energy to a torch bulb. The bulb gives out 18 J as light. The energy given out as heat is
 - A. 60 J
 - B. 78 J
 - C. 18 J
 - D. 42 J
6. Which of the following statements about energy changes is TRUE?
 - A. Energy is only changed from one useful form into other forms
 - B. Energy disappears when a machine stops working
 - C. New energy is made inside a dry cell
 - D. Energy can be weighed on a balance
7. A pupil said that the electricity used by a fan is 'lost' when the fan is switched off. The correct statement is that the energy has been
 - A. changed into kinetic energy, sound and heat
 - B. destroyed by the motor
 - C. stored inside the blades
 - D. returned unchanged to the socket
8. A machine is said to waste energy when some of the energy supplied to it is
 - A. destroyed completely inside the machine
 - B. changed into forms that are not useful for the job
 - C. weighed and thrown away
 - D. returned to the source of the energy
9. The energy change that takes place in a dry-cell torch when it is switched on is
 - A. electrical to chemical to sound
 - B. light to electrical to chemical

- C. chemical to electrical to light and heat
- D. heat to chemical to light

- 11.** An electric pressing iron changes electrical energy mainly into
- heat energy
 - chemical energy
 - sound energy
 - light energy
- 13.** An electric fan changes electrical energy mainly into
- kinetic energy
 - chemical energy
 - light energy
 - nuclear energy
- 15.** When a mobile phone is being charged, the energy change is
- sound energy to heat energy
 - light energy to electrical energy
 - chemical energy to electrical energy
 - electrical energy to chemical energy
- 17.** Which device changes kinetic energy into electrical energy?
- A generator
 - An electric motor
 - A pressing iron
 - A dry cell
- 19.** A boy winds up a toy car and releases it. The energy change is
- heat energy to potential energy
 - sound energy to kinetic energy
 - kinetic energy to chemical energy
 - elastic potential energy to kinetic energy
- 21.** A motor is supplied with 100 J of electrical energy. It gives 70 J of kinetic energy. The energy that appears as heat and sound is
- 100 J
 - 30 J
 - 70 J
 - 170 J
- 10.** A solar panel on the roof of a clinic changes
- solar energy into electrical energy
 - sound energy into electrical energy
 - electrical energy into solar energy
 - heat energy into chemical energy
- 12.** The main energy change in a loudspeaker of a radio set is
- chemical energy to light energy
 - sound energy to electrical energy
 - heat energy to kinetic energy
 - electrical energy to sound energy
- 14.** A microphone used at a school gathering changes
- heat energy into sound energy
 - sound energy into electrical energy
 - electrical energy into sound energy
 - light energy into sound energy
- 16.** Green plants carry out an energy conversion in which
- chemical energy is changed into light energy
 - heat energy is changed into sound energy
 - light energy is changed into chemical energy in food
 - kinetic energy is changed into light energy
- 18.** In a hydroelectric station such as Akosombo, the energy of the stored water is changed in the order
- chemical to heat to electrical
 - kinetic to chemical to electrical
 - potential to kinetic to electrical
 - electrical to kinetic to potential
- 20.** A candle burning in a dark room changes chemical energy into
- electrical and nuclear energy
 - potential and elastic energy
 - light and heat energy
 - sound and kinetic energy

- 23.** In every energy conversion, some of the energy is usually changed into
- solar energy
 - nuclear energy
 - chemical energy
 - heat energy
- 25.** Which of the following energy chains is CORRECT for an electric kettle boiling water?
- kinetic energy to light energy
 - heat energy to electrical energy
 - electrical energy to heat energy
 - chemical energy to sound energy
- 27.** An LED lamp is preferred to a filament bulb for lighting a classroom because the LED lamp
- gives out more heat
 - uses a larger current
 - changes a greater part of the electrical energy into light
 - needs no switch to control it
- 29.** Of the energy supplied to it, the useful energy given out by a machine is always
- equal to the energy supplied
 - less than the energy supplied
 - equal to zero
 - more than the energy supplied
- 31.** A pupil claimed to have built a machine that works for ever without any supply of energy. The claim must be false because
- machines are always made of metal
 - no machine can give out heat
 - energy cannot be created; a machine must be supplied with energy
 - every machine must have an electric motor
- 33.** Which of the following is the BEST way of conserving electrical energy at home?
- Using filament bulbs in every room
 - Switching off lights and fans in rooms that are not in use
 - Leaving the television on all
- 22.** A lamp supplied with 50 J of electrical energy gives out 10 J of light. The fraction of the energy that is useful is
- the whole
 - four-fifths
 - one-half
 - one-fifth
- 24.** A pupil lists the energy chain of a petrol lorry as: chemical to to kinetic. The missing form is
- light energy
 - nuclear energy
 - heat energy
 - sound energy
- 26.** A filament bulb gives out far more heat than light. This means that most of the energy supplied to it is
- destroyed
 - stored
 - returned to the mains
 - wasted
- 28.** Oiling the moving parts of a machine saves energy because it reduces the energy wasted as
- light given out by the metal
 - heat and sound caused by friction
 - electrical energy in the wires
 - chemical energy in the oil
- 30.** Heat lost through the walls of an electric oven is an example of
- destroyed energy
 - created energy
 - wasted energy
 - stored energy
- 32.** Energy conservation means
- using energy carefully so that less of it is wasted
 - making new energy in a laboratory
 - destroying energy that is not needed
 - using as much energy as possible

night

D. Opening the fridge door often

- 35.** Ironing all the family's clothes at one time rather than one garment at a time conserves energy because
- A. the iron gives out no heat
 - B. the clothes need less water
 - C. the iron is heated once instead of many times
 - D. the iron works at a lower voltage
- 37.** Which of the following sources of energy will NOT run out if it is used wisely?
- A. Sunlight
 - B. Crude oil
 - C. Coal
 - D. Natural gas
- 39.** Putting a lid on a cooking pot conserves energy because it
- A. changes the heat into light
 - B. keeps the heat in the pot so that the food cooks faster
 - C. adds more heat to the food
 - D. makes the fire burn more fuel
- 34.** Drying clothes in the open air instead of using a drying machine conserves energy because it
- A. increases the cost of electricity
 - B. uses energy from the Sun, which is free and renewable
 - C. destroys the electrical energy saved
 - D. makes the clothes heavier
- 36.** A sign of energy being wasted in a school is
- A. an LED lamp used in the library
 - B. a kettle switched off after the water has boiled
 - C. a classroom with the windows opened for air
 - D. a classroom with the lights and fans left on at break time
- 38.** A reason for conserving energy in Ghana is that it
- A. increases the number of power cuts
 - B. raises the cost of electricity for everybody
 - C. makes dry cells last a shorter time
 - D. reduces the fuel burnt and so protects the environment
- 40.** A school wishes to teach the community about saving energy. The MOST suitable activity is to
- A. make posters and leaflets on energy saving and display them in the town
 - B. ask every home to stop using electricity
 - C. switch off the school water pump for a term
 - D. sell dry cells to the community

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Show all the steps in any calculation, and give the unit with every answer.

Question 7

- (a) Use a pencil and a ruler where a diagram, an energy-flow diagram or a table is required. Do not use correction fluid.

Question 8

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 5: Crop Production

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. Substances that plants take in from the soil for healthy growth are called
 - A. plant hormones
 - B. plant nutrients
 - C. plant cells
 - D. plant fibres
2. The part of the plant through which most nutrients are taken in from the soil is the
 - A. root
 - B. flower
 - C. fruit
 - D. leaf
3. Plant nutrients are taken up by the roots
 - A. as dry solid grains
 - B. only when the soil is completely dry
 - C. as a gas from the air
 - D. dissolved in soil water
4. Any substance added to the soil to supply nutrients to crops is known as
 - A. a fertilizer
 - B. a fungicide
 - C. a herbicide
 - D. a pesticide
5. Maize plants on a school farm are short, with pale yellow leaves, while those on the next plot are tall and dark green. The most likely cause is that the soil of the first plot
 - A. lacks enough plant nutrients
 - B. is in a cooler part of the school
 - C. has too many stones
 - D. contains too much air
6. Which of the following is the chief reason for adding plant nutrients to a school garden year after year?
 - A. Plants cannot take in water without fertilizer
 - B. The crops harvested each year remove nutrients from the soil
 - C. Fertilizers make the soil heavier
 - D. Nutrients make the soil change colour
7. A source of plant nutrients obtained from the remains of plants and animals is described as
 - A. mineral
 - B. artificial
 - C. organic
 - D. inorganic
8. Which of the following is an organic source of plant nutrients?
 - A. Compost
 - B. Urea
 - C. Ammonium sulphate
 - D. Muriate of potash
9. Rotted plant remains, kitchen waste and animal dung heaped together and allowed to decay form
 - A. gravel
 - B. cement
 - C. charcoal
 - D. compost
10. Dung and urine of farm animals mixed with their bedding is known as
 - A. wood ash
 - B. farmyard manure
 - C. green manure
 - D. liquid fertilizer

- 11.** A farmer ploughs a young leguminous crop into the soil while it is still fresh and green. This practice provides
- A. green manure
 - B. inorganic fertilizer
 - C. poultry litter only
 - D. farmyard manure
- 13.** Organic sources of plant nutrients improve the soil in a way that inorganic fertilizers do not, because they
- A. supply nutrients instantly to the crop
 - B. make the soil completely free of germs
 - C. add humus, which improves the structure of the soil and its ability to hold water
 - D. stop all weeds from growing
- 15.** Which of the following is an inorganic fertilizer?
- A. Compost
 - B. NPK 15-15-15
 - C. Green manure
 - D. Poultry manure
- 17.** Which of the following is supplied mainly by sulphate of ammonia and urea?
- A. Calcium
 - B. Phosphorus
 - C. Nitrogen
 - D. Potassium
- 19.** A danger of using too much inorganic fertilizer on a bed of vegetables is that it may
- A. scorch (burn) the plants and harm the soil
 - B. prevent rain from falling on the farm
 - C. turn the soil into stone
 - D. make the plants grow too slowly to harvest
- 21.** Which of the following pairs contains ONLY organic sources of plant nutrients?
- A. Compost and farmyard manure
 - B. Urea and poultry manure
 - C. Superphosphate and urea
- 12.** Which of the following organic materials is likely to be collected from a poultry house?
- A. Granules of NPK fertilizer
 - B. Bags of lime
 - C. Droppings mixed with wood shavings
 - D. Crystals of ammonium sulphate
- 14.** A source of plant nutrients that is manufactured in a factory from mineral substances is described as
- A. natural
 - B. decayed
 - C. organic
 - D. inorganic
- 16.** The letters N, P and K on a bag of fertilizer stand for
- A. nitrogen, phosphorus and potassium
 - B. nickel, phosphorus and potassium
 - C. nitrogen, phosphorus and carbon
 - D. nitrogen, potassium and calcium
- 18.** Inorganic fertilizers act faster on a crop than organic manures because they
- A. are always applied in larger quantities
 - B. are heavier than manure
 - C. contain living organisms
 - D. dissolve quickly in soil water and are taken up at once
- 20.** Which of the following statements about inorganic fertilizers is correct?
- A. They are always cheaper than organic manures
 - B. They supply nutrients in a concentrated form, so only small quantities are needed
 - C. They are made by heaping plant remains to decay
 - D. They add humus to the soil

D. NPK and green manure

- 22.** Compared with inorganic fertilizers, organic manures are
- A. quicker in their action on the crop
 - B. light, so very small quantities

- 23.** A farmer needs 2 kg of NPK fertilizer to supply the nutrients for one bed of vegetables. To supply the same nutrients with poultry manure he would need
- A. exactly 2 kg as well
 - B. less than 1 kg
 - C. no manure at all, since manure has no nutrients
 - D. a much larger quantity, measured in many kilograms
- 25.** A school has little money but keeps goats and sweeps up plenty of leaves and kitchen waste. The best source of nutrients for its garden is
- A. no nutrients at all until money is available
 - B. bags of imported NPK fertilizer
 - C. compost and goat manure made on the school compound
 - D. liquid fertilizer bought from the city
- 27.** Well-made compost is usually
- A. clear, colourless and watery
 - B. yellow, dry and powdery
 - C. white, hard and shiny
 - D. dark brown to black, crumbly and earthy-smelling
- 29.** Which of the following nutrient sources is a liquid?
- A. Farmyard manure
 - B. Liquid fertilizer mixed with water for spraying
 - C. Dry poultry litter
 - D. Muriate of potash crystals
- 31.** Inorganic fertilizer is kept in a sealed bag in a dry store mainly because it
- A. turns into compost when it is left open
 - B. is food for farm animals
 - C. absorbs moisture from the air and cakes into lumps
 - D. gives off a pleasant smell
- are enough
- C. bulky, so large quantities must be applied
 - D. always sold in sealed sacks
- 24.** One advantage of organic manure over inorganic fertilizer is that organic manure
- A. is sold in small light bags
 - B. acts within one or two days
 - C. contains only one nutrient
 - D. releases its nutrients slowly over a long period
- 26.** A farmer who uses only inorganic fertilizer year after year without adding any organic matter will most likely find that
- A. the soil becomes richer in humus each year
 - B. the soil loses its structure and holds less water
 - C. weeds disappear from the farm completely
 - D. the crop no longer needs water
- 28.** NPK fertilizer is commonly supplied as
- A. a dark crumbly heap
 - B. a wet, smelly paste
 - C. small hard granules in a sealed sack
 - D. a block of pressed leaves
- 30.** Fresh poultry droppings differ from dried poultry manure in that the fresh droppings are
- A. dry, light and odourless
 - B. hard, white crystals
 - C. clear and colourless
 - D. wet, sticky and strong-smelling
- 32.** Why is a bulky organic manure usually worked into the soil rather than left on the surface?
- A. So that it can be seen by the farmer
 - B. So that the soil can become harder
 - C. So that it decays in the soil and releases its nutrients where the roots are
 - D. So that birds can feed on it

easily

- 33.** The physical form of a nutrient source affects how it is applied. A fine powder is harder to spread by hand on a windy day than granules

- because the powder
- A. dissolves in the air
 - B. has no nutrients in it
 - C. is heavier than the granules
 - D. is light and is blown away easily
- 35.** Placing fertilizer in a circle round a plant, a short distance from the stem, is known as
- A. top dressing of the leaves
 - B. ring application
 - C. broadcasting
 - D. deep ploughing
- 37.** Liquid fertilizer that is sprayed onto the leaves of a crop is said to be applied as
- A. a foliar spray
 - B. a basal dressing
 - C. a ring application
 - D. a compost heap
- 39.** A pupil applies fertilizer to a bed in the middle of a hot, dry afternoon and the crop later shows burnt leaf edges. The best advice to the pupil is to
- A. apply twice as much fertilizer next time
 - B. stop using any fertilizer on the bed again
 - C. apply fertilizer when the soil is moist, such as after a light rain or watering
 - D. apply the fertilizer directly on the leaves instead
- 34.** Spreading fertilizer evenly over the whole surface of a bed is called
- A. broadcasting
 - B. ring application
 - C. foliar spraying
 - D. drilling
- 36.** Fertilizer is placed a short distance away from the stem of a plant rather than touching it in order to
- A. allow the wind to blow it away
 - B. avoid scorching the stem and roots while still feeding the plant
 - C. make the plant grow in a circle
 - D. keep the fertilizer away from the roots
- 38.** Compost is best applied to a vegetable bed
- A. sprinkled on the leaves of the growing crop
 - B. during land preparation, worked into the soil before planting
 - C. on the day of harvesting
 - D. only after the crop has been uprooted
- 40.** A school garden has one wheelbarrow of poultry manure and one small bag of NPK for four vegetable beds. The BEST plan is to
- A. spray both the manure and the NPK onto the leaves of the crop
 - B. mix the manure and the NPK together and heap them in one corner
 - C. pour all the NPK onto one bed and leave the manure unused
 - D. work the manure into all four beds during land preparation and side-dress the growing crop with small measured amounts of NPK

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram, a labelled sketch or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 6: Ecosystems

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. An ecosystem is best described as
 - A. a place where only animals are found
 - B. a group of animals of the same kind
 - C. a unit in which living things and their non-living surroundings interact
 - D. the soil in which plants grow
2. An ecosystem is said to be self-sustaining because
 - A. no living thing ever dies in it
 - B. it never changes from year to year
 - C. its parts supply one another with what they need to continue
 - D. it is kept going by human beings
3. Which of the following is an example of an ecosystem?
 - A. A piece of iron sheet
 - B. A pond
 - C. A bucket of sand
 - D. A single mango fruit
4. The place where a plant or an animal lives in an ecosystem is called its
 - A. nutrient
 - B. community
 - C. population
 - D. habitat
5. All the living things of different kinds living together in one place form
 - A. a community
 - B. a habitat
 - C. an organ
 - D. a population
6. Which of the following best explains why the Owabi forest can be called an ecosystem?
 - A. It is found in the Ashanti Region
 - B. It contains very many trees
 - C. Its plants, animals, soil, water and air act on one another as one unit
 - D. It is protected by the government
7. The study of ecosystems is important mainly because it
 - A. helps us to use and protect our surroundings wisely
 - B. shows us how to build better houses
 - C. helps us to kill dangerous animals
 - D. teaches us the names of all plants
8. An ecosystem found on land is described as
 - A. artificial
 - B. terrestrial
 - C. aquatic
 - D. arboreal
9. An ecosystem found in water is described as
 - A. arid
 - B. terrestrial
 - C. arboreal
 - D. aquatic
10. An ecosystem made up of the life found on trees is described as
 - A. terrestrial
 - B. arboreal
 - C. marine
 - D. aquatic

- 11.** Which of the following is an aquatic ecosystem?
- A. The Kakum forest
 - B. The Volta Lake
 - C. The Accra plains
 - D. A baobab tree
- 13.** Monkeys, birds, orchids and insects living in the branches of a tall forest tree make up
- A. a desert ecosystem
 - B. an arboreal ecosystem
 - C. an aquatic ecosystem
 - D. a soil ecosystem
- 15.** The living parts of an ecosystem are called
- A. mineral components
 - B. physical components
 - C. abiotic components
 - D. biotic components
- 17.** Green plants in an ecosystem are called producers because they
- A. produce the water of the ecosystem
 - B. break down dead bodies
 - C. are eaten by all other organisms
 - D. make their own food using sunlight
- 19.** Bacteria and fungi in an ecosystem are important as
- A. decomposers that break down dead plants and animals
 - B. producers that make food from sunlight
 - C. abiotic components of the ecosystem
 - D. animals that feed on living plants
- 21.** The non-living parts of an ecosystem are called
- A. biotic components
 - B. decomposers
 - C. abiotic components
 - D. consumers
- 23.** Which group contains ONLY abiotic components?
- A. Water, fish, soil, air
 - B. Water, air, soil, temperature
 - C. Air, water, bacteria, rock
 - D. Soil, grass, light, air
- 12.** A grassland in the Northern Region of Ghana is an example of
- A. a terrestrial ecosystem
 - B. a man-made lake
 - C. an arboreal ecosystem
 - D. an aquatic ecosystem
- 14.** Which of the following pairs are BOTH aquatic ecosystems?
- A. The sea and a grassland
 - B. A mangrove tree and a rock
 - C. A pond and a forest
 - D. A river and the sea
- 16.** Which of the following is a biotic component of a pond?
- A. Sunlight
 - B. Tilapia
 - C. Mud
 - D. Water
- 18.** Animals in an ecosystem are called consumers because they
- A. produce oxygen for other organisms
 - B. use sunlight to make food
 - C. depend on other organisms for their food
 - D. live only in water
- 20.** If all the decomposers in a forest were removed, the MOST likely result would be that
- A. more rain would fall in the forest
 - B. all the animals would become plants
 - C. the forest would grow faster
 - D. dead plants and animals would pile up and nutrients would not return to the soil
- 22.** Which of the following is an abiotic component of a forest ecosystem?
- A. Earthworm
 - B. Climbing plant
 - C. Sunlight
 - D. Mushroom
- 24.** Sunlight is an important abiotic component of an ecosystem chiefly because
- A. it keeps animals awake
 - B. it makes water salty
 - C. it helps the wind to blow

D. green plants need it to make food

- 25.** Water is important in an ecosystem because
- A. it prevents plants from growing too tall
 - B. it supplies the soil with sunlight
 - C. it is a living part of the ecosystem
 - D. all living things need it for their life processes
- 27.** Fish are suited to life in water because they
- A. have lungs and legs
 - B. have roots that take in water
 - C. have gills for breathing in water and fins for swimming
 - D. have wings and feathers
- 29.** Water lilies are suited to life in a pond because they have
- A. deep woody roots
 - B. leaves reduced to spines
 - C. broad floating leaves that stay on the surface
 - D. thick bark on the stem
- 31.** Which of the following animals is found in a terrestrial ecosystem?
- A. Grasscutter
 - B. Crab in a rock pool
 - C. Tilapia
 - D. Shrimp
- 33.** The MAIN difference between the organisms of a pond and those of a grassland is that
- A. pond organisms are all larger
 - B. each group has features suited to the conditions of its own surroundings
 - C. grassland organisms do not need water
 - D. pond organisms do not need food
- 35.** Two mango trees growing very close together struggle for light, water and soil nutrients. This is an example of
- A. competition
 - B. predation
 - C. decomposition
 - D. pollination
- 26.** A pond dries up completely during a long dry season. The fish die mainly because an abiotic component
- A. needed by the fish has been removed
 - B. has turned into a biotic component
 - C. has eaten the fish
 - D. has multiplied too fast
- 28.** Which structure helps a monkey to live in the branches of trees?
- A. A hard shell for protection
 - B. Long limbs and a gripping tail for holding on to branches
 - C. Gills for taking in air
 - D. Fins for moving about
- 30.** Grasses of the northern savanna survive the long dry season mainly because they
- A. take in food from the air
 - B. can swim to wetter places
 - C. grow best without any water
 - D. have underground parts that sprout again when the rains come
- 32.** An earthworm taken from the soil and placed in a bucket of water soon dies. This shows that
- A. an organism survives best in the ecosystem to which it is suited
 - B. earthworms are not living things
 - C. every animal can live anywhere
 - D. water is harmful to all animals
- 34.** The feeding relationship in which a hawk feeds on a chick is an example of
- A. predation
 - B. competition
 - C. decomposition
 - D. pollination
- 36.** Which of the following shows a dependence of a plant on an animal?
- A. A plant taking in water through its roots
 - B. A plant using sunlight to make

food

- C. A plant losing water from its leaves
- D. A bee carrying pollen from one flower to another

37. Green plants in an ecosystem supply

animals with

- A. wind and rain
- B. soil and rock
- C. food and oxygen
- D. light and heat

39. A factory pours waste into a river and most of the fish die. The farmers who catch the fish also suffer. This shows that

- A. fish are not part of the ecosystem
- B. rivers can clean themselves at once
- C. the parts of an ecosystem are linked, so a change in one affects the others
- D. only the fish are affected by pollution

38. Which of the following best shows how a biotic component depends on an abiotic component?

- A. A tree takes up water from the soil
- B. A bird builds a nest with twigs
- C. A fungus grows on a dead log
- D. A goat feeds on grass

40. Which of the following human activities would MOST help to protect a wetland ecosystem near a town?

- A. Catching all the fish before the rains
- B. Stopping the dumping of refuse in the wetland and replanting the plants around it
- C. Filling the wetland with sand for building
- D. Burning the plants at the edge of the water

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram, a flow chart or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 7: Electricity and E

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. Electricity produced by burning fuels such as natural gas, crude oil or coal is called
 - A. wind electricity
 - B. nuclear electricity
 - C. solar electricity
 - D. thermal electricity
2. In a thermal power station, the heat from the burning fuel is used to
 - A. cool the generator
 - B. melt the metal of the turbine
 - C. boil water and produce steam
 - D. charge large dry cells directly
3. In both thermal and nuclear power stations, the part that is turned by steam is the
 - A. pylon
 - B. transformer
 - C. turbine
 - D. chimney
4. The machine that changes the movement of the turbine into electricity is the
 - A. boiler
 - B. resistor
 - C. switch
 - D. generator
5. Nuclear electricity is produced from the heat released when
 - A. water falls from a high dam
 - B. coal is burnt in a large furnace
 - C. the nuclei of atoms of a fuel such as uranium are split
 - D. the Sun shines on a panel
6. Which of the following is the fuel used in a nuclear power station?
 - A. Uranium
 - B. Crude oil
 - C. Natural gas
 - D. Charcoal
7. The energy change that takes place in the generator of a power station is
 - A. chemical energy to heat energy
 - B. kinetic energy to electrical energy
 - C. electrical energy to kinetic energy
 - D. light energy to electrical energy
8. Which statement about thermal and nuclear power stations is correct?
 - A. Both burn fuel in a furnace to obtain heat
 - B. Both use steam to turn a turbine, but they obtain their heat in different ways
 - C. Both of them depend on falling water
 - D. Neither of them uses a generator
9. The gas given off in the largest amount when fossil fuels are burnt in a thermal power station is
 - A. nitrogen
 - B. oxygen
 - C. carbon dioxide
 - D. helium
10. Carbon dioxide released from thermal power stations is of concern because it
 - A. adds to the greenhouse gases that cause global warming
 - B. turns rain water into fresh water
 - C. makes the soil more fertile
 - D. destroys the ozone layer at once

- 11.** Smoke and soot from a thermal power station affect people living nearby mainly by
- improving the quality of their drinking water
 - causing breathing problems such as coughs and asthma
 - increasing the amount of oxygen in the air
 - making their houses stronger
- 12.** Sulphur dioxide released when some fuels are burnt may cause
- acid rain
 - a water cycle
 - an earthquake
 - a rainbow
- 13.** The MAIN environmental danger of a nuclear power station is
- the smoke from its chimney
 - the radioactive waste it produces and the risk of leakage
 - the carbon dioxide it releases
 - the noise of its boiler
- 14.** Which of the following would reduce the harmful effects of electricity generation on the environment?
- Building more chimneys for the same station
 - Pouring hot water from the station into a nearby river
 - Burning more coal instead of natural gas
 - Using more solar and hydro-electric power and fitting filters to chimneys
- 15.** Hot water released from a power station into a river may harm the river because it
- makes the river flow more quickly
 - raises the temperature of the water and kills fish and other organisms
 - removes all the salt from the water
 - adds useful nutrients to the water
- 16.** The letters LED stand for
- light energy detector
 - light-emitting diode
 - long electric drive
 - low energy device
- 17.** The component that gives out light when current passes through it in the correct direction is the
- LED
 - resistor
 - inductor
 - capacitor
- 18.** The MAIN function of a resistor in an electronic circuit is to
- reduce (limit) the current flowing in the circuit
 - change the current into heat only
 - give out light
 - store electrical charge
- 19.** The value of a small resistor is shown on its body by
- its length
 - the number of legs it has
 - its weight
 - coloured bands
- 20.** The component that stores electrical charge and releases it later is the
- switch
 - resistor
 - capacitor
 - diode
- 21.** A coil of wire that opposes a change in the current flowing through it is the
- inductor
 - LED
 - capacitor

D. resistor

23. A source of electrical energy is needed in every working circuit. In the circuit built in class, this is provided by the

- A. resistor
- B. LED
- C. switch
- D. dry cells

22. Which component allows current to pass in one direction only?

- A. A resistor
- B. A connecting wire
- C. A diode
- D. A capacitor

24. Two dry cells of 1.5 V each are joined in series. The voltage of the battery formed is

- A. 3 V

- B. 0.75 V
C. 6 V
D. 1.5 V
- 25.** A P-N junction diode is made by joining
- A. two pieces of copper wire
 - B. a resistor and a capacitor
 - C. a P-type region and an N-type region of semiconductor material
 - D. two dry cells in series
- 27.** A diode connected so that current flows through it is said to be
- A. short circuited
 - B. reverse biased
 - C. earthed
 - D. forward biased
- 29.** The longer leg of a new LED is the
- A. leg that must be cut off before use
 - B. positive leg (anode), which is joined to the positive side of the battery
 - C. negative leg, which is joined to the positive side of the battery
 - D. leg that carries no current
- 31.** When the switch in a circuit is opened, the LED goes out because
- A. the cells have run down at once
 - B. the path of the current is broken
 - C. the resistor has melted
 - D. the LED has been spoilt by the current
- 33.** Before joining an LED to a 3 V battery, it is wise to join a resistor in the circuit in order to
- A. prevent too large a current from spoiling the LED
 - B. store charge for the LED
 - C. make the LED brighter than normal
 - D. change the direction of the current
- 26.** In a P-N junction diode, the P region is the
- A. region that stores charge
 - B. positive region, which must be joined to the positive side of the battery for current to flow
 - C. region that gives off light
 - D. negative region, which must be joined to the positive side of the battery
- 28.** An LED in a circuit does not light when the cells are connected one way round, but lights when the cells are turned round. This is because the LED
- A. needs a larger resistor
 - B. stores charge before lighting
 - C. allows current to pass in one direction only
 - D. is partly spoilt
- 30.** When the switch in a circuit is closed,
- A. the circuit is broken and no current flows
 - B. the LED stores charge
 - C. the circuit is complete and current flows
 - D. the cells are destroyed
- 32.** In a circuit diagram, a long thin line beside a short thick line is the symbol for
- A. a cell
 - B. a resistor
 - C. a switch
 - D. a lamp
- 34.** A pupil built a circuit with two cells, a switch, a resistor and an LED, but the LED did not light even when the switch was closed. The FIRST thing she should check is
- A. whether the LED is warm
 - B. whether the wires are long enough
 - C. whether the room is dark enough
 - D. whether all the connections are

tight and the cells are the right
way round

- 35.** In a simple series circuit, the components are joined
- A. with the LED outside the circuit
 - B. in no particular order, since current finds its own way
 - C. side by side, so that each carries a different current
 - D. one after another, so that the same current passes through each of them
- 37.** If the resistor in an LED circuit is replaced by one of much smaller resistance, the LED will
- A. go out completely
 - B. store more charge
 - C. change the direction of the current
 - D. shine more brightly and may be damaged by the larger current
- 39.** The components of an electronic circuit are said to be interdependent because
- A. the way one component behaves affects the working of the others
 - B. each component works on its own
 - C. they are all made of the same material
 - D. they are all the same size
- 36.** When a second resistor is added in series to a circuit containing an LED, the LED becomes
- A. brighter, because the resistance is reduced
 - B. dimmer, because the voltage of the cells increases
 - C. dimmer, because the current is reduced
 - D. brighter, because the current is increased
- 38.** A diode joined into an LED circuit the wrong way round will make the LED
- A. shine more brightly
 - B. flash on and off
 - C. shine with a different colour
 - D. stay off, because the diode blocks the current
- 40.** A radio set, a mobile phone and a television all contain resistors, diodes and capacitors. This shows that
- A. all these gadgets do the same work
 - B. the same basic components are used in many different electronic gadgets
 - C. electronic gadgets need no source of energy
 - D. the components work only in radios

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram, a circuit diagram or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 8: Energy

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. Energy is best described as
 - A. the speed of a moving body
 - B. the space a body occupies
 - C. the mass of a body
 - D. the ability to do work
2. The unit in which energy is measured is the
 - A. newton
 - B. joule
 - C. kilogram
 - D. watt
3. The energy possessed by a body because of its position above the ground is
 - A. sound energy
 - B. potential energy
 - C. kinetic energy
 - D. nuclear energy
4. The energy possessed by a body because of its motion is
 - A. light energy
 - B. potential energy
 - C. chemical energy
 - D. kinetic energy
5. The energy stored in food, fuel and a dry cell is
 - A. solar energy
 - B. kinetic energy
 - C. sound energy
 - D. chemical energy
6. The energy that reaches the Earth directly from the Sun is called
 - A. solar energy
 - B. chemical energy
 - C. potential energy
 - D. nuclear energy
7. Which form of energy reaches our ears when a drum is beaten?
 - A. Light energy
 - B. Sound energy
 - C. Chemical energy
 - D. Nuclear energy
8. Which of the following is NOT a form of energy?
 - A. Heat
 - B. Light
 - C. Sound
 - D. Mass
9. The sum of the potential energy and the kinetic energy of a body is called its
 - A. nuclear energy
 - B. electrical energy
 - C. chemical energy
 - D. mechanical energy
10. A coconut hanging on a tree possesses
 - A. potential energy
 - B. sound energy
 - C. no energy at all
 - D. kinetic energy only
11. As a mango falls from a tree to the ground, its
 - A. potential energy increases while its kinetic energy decreases
 - B. potential and kinetic energy

- both increase
- C. potential energy decreases while its kinetic energy increases
- D. potential and kinetic energy both decrease

13. At the highest point of its flight, a ball thrown straight upwards has

- A. maximum potential energy and least kinetic energy
- B. equal potential and kinetic energy
- C. neither potential nor kinetic energy
- D. maximum kinetic energy and least potential energy

12. A stone thrown straight upwards has its greatest potential energy

- A. half-way up
- B. just before it hits the ground
- C. at the highest point of its flight
- D. at the moment it leaves the hand

- 15.** When a mango falls from a tree, the potential energy it loses is changed mainly into
- kinetic energy
 - nuclear energy
 - electrical energy
 - chemical energy
- 17.** An electric iron changes electrical energy into
- heat energy
 - chemical energy
 - nuclear energy
 - sound energy
- 19.** A radio set changes electrical energy mainly into
- potential energy
 - chemical energy
 - sound energy
 - nuclear energy
- 21.** Water stored in the Akosombo dam high above the turbines possesses
- chemical energy, which is changed into heat energy
 - potential energy, which is changed into electrical energy at the power station
 - sound energy, which is changed into light energy
 - nuclear energy, which is changed into kinetic energy
- 23.** The potential energy of a body is calculated using the expression
- $PE = mgh$
 - $PE = mv$
 - $PE = mg/h$
 - $PE = 1/2 mv^2$
- 25.** A bag of mass 2 kg is lifted onto a shelf 5 m above the ground. Its potential energy is ($g = 10 \text{ m/s}^2$)
- 25 J
 - 1 000 J
 - 100 J
 - 10 J
- 14.** A pendulum bob swings from side to side. At the lowest point of the swing the bob has
- more potential than kinetic energy
 - no energy at all
 - the greatest kinetic energy
 - the greatest potential energy
- 16.** A torch (flashlight) changes
- potential energy into sound energy
 - light energy into chemical energy
 - chemical energy into light energy
 - sound energy into heat energy
- 18.** A solar panel on the roof of a clinic changes
- solar energy into electrical energy
 - heat energy into sound energy
 - electrical energy into solar energy
 - chemical energy into light energy
- 20.** Which appliance changes electrical energy into kinetic energy?
- An electric fan
 - A candle
 - An electric bulb
 - A dry cell
- 22.** A burning candle changes chemical energy into
- sound and nuclear energy
 - heat and light energy
 - potential and sound energy
 - electrical and chemical energy
- 24.** The kinetic energy of a body is calculated using the expression
- $KE = 2mv$
 - $KE = mgh$
 - $KE = mv$
 - $KE = 1/2 mv^2$
- 26.** A boy of mass 40 kg runs at a speed of 3 m/s. His kinetic energy is
- 120 J
 - 180 J
 - 360 J

D. 60 J

- 27.** Two stones are held at the same height. The stone with the greater mass has

A. no potential energy

- B. the smaller potential energy
 - C. the same potential energy as the lighter stone
 - D. the greater potential energy
- 29.** The transfer of heat through a solid from the hot end to the cold end is called
- A. evaporation
 - B. conduction
 - C. radiation
 - D. convection
- 31.** The transfer of heat from the Sun to the Earth takes place by
- A. condensation
 - B. conduction
 - C. radiation
 - D. convection
- 33.** The handles of cooking pots are often made of wood or plastic because these materials are
- A. heavier than metal
 - B. good conductors of heat
 - C. poor conductors of heat
 - D. able to store heat for long
- 35.** The formation of a sharp shadow behind an object placed in sunlight shows that light
- A. is made up of sound waves
 - B. travels in a straight line
 - C. travels only in water
 - D. bends round objects
- 37.** The bouncing back of light when it strikes a smooth, shiny surface is called
- A. refraction
 - B. dispersion
 - C. absorption
 - D. reflection
- 39.** The splitting of white light into its colours as it passes through a glass prism is called
- A. dispersion
 - B. conduction
 - C. reflection
 - D. diffusion
- 28.** If the speed of a moving car is doubled while its mass stays the same, its kinetic energy becomes
- A. half as large
 - B. the same
 - C. twice as large
 - D. four times as large
- 30.** Heat is transferred through liquids and gases mainly by
- A. radiation
 - B. convection
 - C. reflection
 - D. conduction
- 32.** Which of the following is the BEST conductor of heat?
- A. Rubber
 - B. Aluminium
 - C. Plastic
 - D. Wood
- 34.** Warm air above a hot road rises. This is because warm air
- A. has no weight at all
 - B. is pushed up by sound waves
 - C. is lighter (less dense) than the cooler air around it
 - D. is heavier than the cooler air
- 36.** Three cardboard sheets, each with a small hole, are arranged so that a candle flame can be seen through them. The flame can be seen only when the three holes are
- A. covered with glass
 - B. placed at different heights
 - C. of different sizes
 - D. in a straight line
- 38.** A pencil standing in a glass of water appears bent at the surface of the water. This is due to
- A. refraction of light
 - B. absorption of light
 - C. reflection of light
 - D. dispersion of light

- 40.** A rainbow seen in the sky after a shower of rain is formed because the raindrops
- A. absorb all the colours of sunlight
 - B. reflect sound back to the Earth
 - C. split the sunlight into its different colours
 - D. give out light of their own

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Show all the steps of your working in every calculation. Take the acceleration due to gravity, g , as 10 m/s^2 .

Question 7

- (a) Use a pencil and a ruler where a diagram, a flow chart or a table is required. Do not use correction fluid.

Question 8

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 9: Farming Systems

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. A farming system is best described as
 - A. the way a farmer arranges and manages the growing of crops and the keeping of animals
 - B. the tools used to clear a piece of land
 - C. the number of workers on a farm
 - D. the type of soil found on a farm
2. Which of the following is NOT one of the farming systems studied at this level?
 - A. Cold storage farming
 - B. Organic farming
 - C. Crop rotation
 - D. Mixed farming
3. A farmer chooses a farming system mainly according to
 - A. the day of the week on which he starts
 - B. the number of children in his house
 - C. the colour of his farm tools
 - D. the land, the crops, the animals and the money available to him
4. Growing crops and keeping animals on the same farm at the same time is called
 - A. land rotation
 - B. mixed farming
 - C. mixed cropping
 - D. crop rotation
5. Growing two or more crops together on the same piece of land at the same time is called
 - A. mixed cropping
 - B. mixed farming
 - C. crop rotation
 - D. monocropping
6. Growing only one type of crop on a piece of land season after season is called
 - A. mixed farming
 - B. organic farming
 - C. mixed cropping
 - D. monocropping (sole cropping)
7. In land rotation, the farmer
 - A. uses only manure and no chemicals
 - B. keeps animals and crops on the same land
 - C. grows different crops on the same land each year
 - D. cultivates one piece of land for a few years, then leaves it and moves to another piece
8. Land rotation is also known as
 - A. crop rotation
 - B. shifting cultivation
 - C. plantation farming
 - D. mixed farming
9. Land that has been left uncultivated so that it can regain its fertility is said to be under
 - A. rotation of crops
 - B. fallow

- C. mulch
- D. irrigation

11. Land rotation is practised mostly in areas where

- A. land is scarce and the population is large
- B. the soil is already very rich
- C. there are many tractors and fertilizers
- D. land is plentiful and the population is small

10. During the fallow period the fertility of the land is restored mainly by

- A. the fertilizer spread before the land is left
- B. the heat of the Sun on the bare soil
- C. the ploughing done by the farmer
- D. the natural vegetation that grows and decays on it

- 13.** Land rotation is becoming less common in many parts of Ghana today mainly because
- the government has banned all farming
 - the soil has become too rich for it
 - the growing population has reduced the amount of land available
 - farmers no longer know how to practise it
- 15.** The MAIN reason for including a legume such as groundnut in a crop rotation is that the legume
- takes all the pests away
 - adds nitrogen to the soil
 - dries the soil quickly
 - gives the largest harvest
- 17.** Crop rotation helps to control pests and diseases because
- the pests of one crop do not find their food plant in the following season
 - the crops are sprayed with more chemicals
 - the pests are frightened by the new crops
 - the land is burnt before every planting
- 19.** Which of the following is the MAIN difference between crop rotation and land rotation?
- In land rotation the crops are changed each season
 - In crop rotation the same land is used and the crops are changed; in land rotation the land itself is changed
 - In crop rotation only one crop is grown
 - Crop rotation can be practised only with animals
- 21.** Which of the following is an example of mixed cropping?
- Maize, cassava and cowpea growing together in the same field
 - Maize this year and cassava on the same field next year
 - Maize grown alone on a large
- 12.** Which of the following is a disadvantage of land rotation?
- It gives the land no time to rest
 - It needs a large area of land and the burning of bush destroys soil organisms
 - It cannot be practised without a tractor
 - It makes the farmer grow only one crop
- 14.** In crop rotation, the farmer
- grows different crops on the same piece of land in a planned order, season after season
 - keeps animals instead of growing crops
 - moves to a new piece of land each year
 - grows the same crop on the land every year
- 16.** The nitrogen added to the soil by a groundnut crop comes from
- the leaves falling on the soil
 - bacteria living in nodules on its roots
 - the water the plant takes in
 - the sunlight trapped by the plant
- 18.** In a good crop rotation plan, a deep-rooted crop is followed by a shallow-rooted one so that
- the soil becomes harder
 - only one type of nutrient is used up
 - the plants draw nutrients from different depths of the soil
 - the farmer can harvest at the same time
- 20.** A farmer at Wenchi grows maize, then beans, then cassava and then rests the land on the same plot year after year. This farmer is practising
- crop rotation
 - monocropping
 - mixed cropping
 - land rotation

field

- D. Maize grown on one field and goats kept on another

22. One advantage of mixed cropping is that

- A. harvesting is always done by machine
- B. the crops never compete for

23. Mixed cropping helps to protect the soil because
- A. the different crops cover the soil and reduce erosion
 - B. the soil is left bare between the plants
 - C. the farmer ploughs the soil more often
 - D. the crops are burnt after harvesting
25. Mixed cropping is difficult to carry out with machines mainly because
- A. the crops are planted too deep
 - B. the farmer plants only in the dry season
 - C. machines cannot move on a farm
 - D. the crops are of different kinds and ripen at different times
27. Which of the following describes a farm practising mixed farming?
- A. A farm where only cattle are kept
 - B. A farm where the land is left to rest for five years
 - C. A farm at Ejura where maize is grown and cattle are also kept
 - D. A farm where only maize and cassava are grown together
29. In mixed farming, the crops are useful to the animals mainly because they
- A. provide medicine for the animals
 - B. supply water to the animals
 - C. keep the animals warm at night
 - D. supply feed such as crop residues and grain
31. A disadvantage of mixed farming is that it
- A. gives the farmer only one source of income
 - B. needs more capital, more labour and more skill than a single enterprise
 - C. cannot be practised in Ghana
 - D. always destroys the soil
- anything
- C. the farmer still has something to harvest if one crop fails
- D. only one type of nutrient is removed from the soil
24. A disadvantage of mixed cropping is that
- A. the soil cannot hold water at all
 - B. no crop can be grown with another
 - C. the crops may compete with one another for light, water and nutrients
 - D. the farmer can harvest only once in ten years
26. A farmer plants cowpea between his rows of maize. Apart from extra food, the cowpea also helps by
- A. covering the ground and adding nitrogen to the soil
 - B. making the maize grow taller than usual
 - C. preventing rain from reaching the soil
 - D. driving away all the birds
28. In mixed farming, the animals are useful to the crops mainly because they
- A. shade the young crops from the Sun
 - B. supply manure that improves the fertility of the soil
 - C. carry the seeds to the field
 - D. eat the pests of the crops at night
30. One financial advantage of mixed farming is that the farmer
- A. is paid by the government each month
 - B. earns income from both crops and animals during the year
 - C. does not need to buy any tools
 - D. pays no money for labour
32. Mixed farming is said to make good use of a farm's resources because
- A. the farmer buys everything from the market
 - B. nothing is ever harvested from the farm
 - C. the waste of one enterprise is

used by the other

- D. the animals are kept far away from the crops

- 33.** Organic farming is a system in which the farmer
- A. keeps animals in cages all year round
 - B. grows only one crop on a large farm
 - C. grows crops and keeps animals without using artificial fertilizers and chemical pesticides
 - D. uses only artificial fertilizers
- 35.** An organic farmer controls insect pests mainly by
- A. leaving the pests to multiply freely
 - B. burning the whole farm every month
 - C. using natural methods such as neem extract, hand picking and natural enemies
 - D. spraying strong chemical insecticides
- 37.** A disadvantage of organic farming is that
- A. yields are often lower and more labour is needed
 - B. the food produced cannot be eaten
 - C. it cannot be practised without machines
 - D. it destroys the structure of the soil
- 39.** A farmer with a small piece of land near Kumasi, little money and a family to feed all year would be BEST advised to practise
- A. leaving the land to fallow for five years
 - B. land rotation on several large plots
 - C. mixed cropping with manure from his own animals
 - D. monocropping of one crop on the whole land
- 34.** Which of the following would an organic farmer use to feed his crops?
- A. NPK fertilizer
 - B. Sulphate of ammonia
 - C. Weedicide
 - D. Compost and animal manure
- 36.** One advantage of organic farming is that
- A. it works only on very poor soils
 - B. it needs very little labour
 - C. the yields are always the highest possible
 - D. the food produced is free from chemical residues and the soil organisms are protected
- 38.** Organic produce often sells at a higher price than other produce because
- A. the government fixes a higher price for it
 - B. it can be stored for ten years
 - C. many buyers are willing to pay more for food grown without chemicals
 - D. it is always larger in size
- 40.** Which statement about the farming systems is correct?
- A. Organic farming cannot be combined with crop rotation
 - B. Only one system may be used on a farm
 - C. Mixed farming cannot include any crops
 - D. More than one system can be used on the same farm at the same time

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram, a flow chart or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 10: Force and Motion

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. A force is best described as
 - A. the speed of a moving body
 - B. the mass of a body at rest
 - C. a push or a pull acting on a body
 - D. the energy stored in a body
2. The unit in which force is measured is the
 - A. newton
 - B. kilogram
 - C. watt
 - D. joule
3. Two forces acting on a body are said to be balanced when they are
 - A. applied at different times
 - B. equal in size and act in the same direction
 - C. unequal and act in opposite directions
 - D. equal in size and act in opposite directions
4. When the forces acting on a body at rest are balanced, the body will
 - A. move with increasing speed
 - B. begin to move slowly
 - C. turn round and round
 - D. remain at rest
5. A body begins to move only when the forces acting on it are
 - A. unbalanced
 - B. very small
 - C. equal and opposite
 - D. balanced
6. Two pupils pull a box in opposite directions with forces of 30 N and 50 N. The box will move
 - A. in neither direction
 - B. in the direction of the 50 N force
 - C. at right angles to both forces
 - D. in the direction of the 30 N force
7. The tendency of a body to resist a change in its state of rest or of motion is called
 - A. gravity
 - B. friction
 - C. inertia
 - D. pressure
8. Of the following bodies, the one with the greatest inertia is a
 - A. bicycle
 - B. football
 - C. wheelbarrow
 - D. loaded lorry
9. Newton's first law of motion states that a body continues in its state of rest or of uniform motion in a straight line unless
 - A. an unbalanced force acts on it
 - B. it is made of metal
 - C. it is placed on a smooth surface
 - D. its mass is increased
10. Newton's first law of motion is also known as the law of
 - A. conservation of energy
 - B. inertia
 - C. gravity
 - D. friction
11. A metal ball rolling on a smooth floor finally stops. The force that stops it is

- A. the push that started it
- B. the inertia of the ball
- C. the weight of the ball
- D. friction between the ball and the floor

- 12.** According to Newton's first law, a ball rolling on a perfectly smooth surface with no opposing force would
- A. move in a circle
 - B. keep moving in a straight line for ever

- C. rise into the air
D. stop after a short distance
- 13.** A body moving with uniform velocity has
- A. a force acting on it in the direction of motion only
 - B. a large unbalanced force acting on it
 - C. no unbalanced force acting on it
 - D. no mass
- 15.** The quantity that measures the inertia of a body is its
- A. weight
 - B. mass
 - C. volume
 - D. speed
- 17.** Seat belts are worn in vehicles mainly to
- A. hold the passenger back when the vehicle stops suddenly
 - B. make the seat comfortable
 - C. reduce the speed of the vehicle
 - D. keep the passenger warm
- 19.** Knocking the handle of a hoe against a hard surface tightens the blade. This is because the
- A. handle pushes the blade upwards
 - B. blade continues to move downwards when the handle stops
 - C. blade becomes heavier on hitting the surface
 - D. blade loses its inertia
- 21.** Dust is removed from a mat by beating it because, when the mat is struck, the
- A. mat attracts the dust strongly
 - B. dust is burnt by the stick
 - C. dust gains weight and falls
 - D. mat moves away while the dust particles stay at rest and are left behind
- 23.** Which of the following materials will be attracted by a magnet?
- C. rise into the air
 - D. stop after a short distance
- 14.** Which of the following is the BEST example of inertia of rest?
- A. A ball that bounces off a wall
 - B. A lorry that is difficult to stop when moving
 - C. A heavy stone that is difficult to start moving
 - D. A magnet that attracts a pin
- 16.** Passengers in a trotro jerk forward when the driver brakes suddenly because their bodies
- A. tend to continue moving forward
 - B. lose their inertia
 - C. are pushed forward by the brakes
 - D. become heavier at that moment
- 18.** When a vehicle at rest starts off suddenly, the passengers are thrown backwards because the
- A. seats push the passengers backwards
 - B. engine pulls the passengers backwards
 - C. weight of the passengers increases
 - D. lower part of the body moves with the vehicle while the upper part stays at rest
- 20.** A tablecloth can be pulled quickly from under a glass without the glass falling because the glass
- A. tends to stay at rest
 - B. has no mass
 - C. is firmly stuck to the cloth
 - D. is pushed upwards by the cloth
- 22.** A magnet is a material that can
- A. attract all materials
 - B. attract iron and certain other materials
 - C. produce electricity by itself
 - D. attract all metals

- A. A copper wire
- B. A wooden pencil
- C. A plastic ruler
- D. An iron nail

25. The poles of a magnet are the parts where the magnetic force is

- A. weakest
- B. completely absent
- C. strongest
- D. the same as at the middle

24. A magnet that keeps its magnetism for a long time is described as a

- A. soft magnet
- B. temporary magnet
- C. permanent magnet
- D. weak magnet

- 27.** The surest test that a bar of metal is a magnet is that it
- is made of steel
 - repels one end of a known magnet
 - attracts an iron nail
 - is shiny and hard
- 29.** Which of the following is a use of a magnet in everyday life?
- Separating iron pieces from other waste at a scrap yard
 - Cutting metal sheets
 - Boiling water in a kettle
 - Measuring the mass of a body
- 31.** Which of the following is NOT a simple machine?
- A pulley
 - An inclined plane
 - A wheel and axle
 - A dry cell
- 33.** A lever in which the fulcrum lies between the effort and the load is a
- second-class lever
 - third-class lever
 - fourth-class lever
 - first-class lever
- 35.** In a pair of sugar tongs, the effort is applied between the fulcrum and the load. The tongs are therefore a
- second-class lever
 - wheel and axle
 - third-class lever
 - first-class lever
- 37.** The efficiency of a machine is the ratio of
- work input to work output, expressed as a percentage
 - load to effort, expressed in newtons
 - effort to distance, expressed in joules
 - work output to work input, expressed as a percentage
- 39.** No simple machine has an efficiency of 100% because some of the work input is used in
- increasing the mass of the machine
- 26.** When the north pole of one magnet is brought near the north pole of another magnet, the two magnets
- repel each other
 - attract each other
 - both lose their magnetism at once
 - stick together firmly
- 28.** A bar magnet hung freely by a thread finally comes to rest with its ends pointing
- in no fixed direction
 - east and west
 - north and south
 - upwards and downwards
- 30.** A simple machine is a device that
- creates energy for the user
 - makes work easier to do
 - works without any force
 - destroys the force applied to it
- 32.** The fixed point about which a lever turns is called the
- effort
 - load
 - fulcrum
 - pivot arm
- 34.** A wheelbarrow is an example of a
- second-class lever
 - first-class lever
 - pulley
 - third-class lever
- 36.** The work done by the effort on a machine is called the
- work input
 - work output
 - efficiency
 - mechanical advantage
- 38.** A machine is supplied with 200 J of energy and does 150 J of useful work. Its efficiency is
- 133%
 - 75%
 - 35%
 - 50%

- B. overcoming friction between the moving parts
- C. creating new energy in the machine
- D. destroying the load

- 40.** The efficiency of a machine can be improved by
- A. adding more load to the machine
 - B. removing the fulcrum
 - C. oiling the moving parts to reduce friction
 - D. making the parts rougher

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Show all the steps in any calculation, and give the unit with every answer.

Question 7

- (a) Use a pencil and a ruler where a diagram or a table is required. Do not use correction fluid.

Question 8

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 11: Human Body D

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. Any substance taken into the body to provide energy, growth and health is called
 - A. waste
 - B. enzyme
 - C. food
 - D. mucus
2. Which of the following is NOT a food nutrient?
 - A. Vitamin
 - B. Sand
 - C. Carbohydrate
 - D. Protein
3. The nutrient that supplies the body with most of its energy is
 - A. protein
 - B. mineral salt
 - C. carbohydrate
 - D. vitamin
4. Foods such as beans, fish, meat and eggs are valuable in the diet mainly because they supply
 - A. energy only
 - B. protein for growth and repair of the body
 - C. roughage only
 - D. water only
5. A person who does not eat enough food over a long time will
 - A. store more energy in the body
 - B. grow taller than others
 - C. become stronger and more active
 - D. lose weight, become weak and fall ill easily
6. Roughage (dietary fibre) is important in the diet because it
 - A. helps the movement of food through the alimentary canal and prevents constipation
 - B. builds strong bones and teeth
 - C. is digested into amino acids
 - D. supplies most of the body's energy
7. Two pupils of the same age are compared. One is active, alert and of normal weight; the other is thin, tired and often ill. The BEST conclusion is that the second pupil
 - A. takes too much water
 - B. does too little homework
 - C. may not be getting enough of the right food
 - D. was born to be weak
8. The long tube through which food passes from the mouth to the anus is called
 - A. the alimentary canal
 - B. the windpipe
 - C. the blood vessel
 - D. the spinal cord
9. The correct order of the parts of the alimentary canal is
 - A. mouth, oesophagus, stomach, small intestine, large intestine, anus
 - B. mouth, oesophagus, small intestine, stomach, large

intestine, anus

- C. mouth, windpipe, stomach, small intestine, large intestine, anus
- D. mouth, stomach, oesophagus, large intestine, small intestine, anus

11. Digestion begins in the

- A. large intestine
- B. small intestine
- C. mouth
- D. stomach

10. The tube that carries food from the mouth to the stomach is the

- A. duodenum
- B. trachea
- C. rectum
- D. oesophagus

- 13.** The main work of the large intestine is to
- A. produce saliva
 - B. absorb amino acids
 - C. digest protein
 - D. absorb water from the undigested material
- 15.** Food is pushed along the alimentary canal by
- A. the air we breathe in
 - B. the force of gravity alone
 - C. the beating of the heart
 - D. the squeezing movement of the muscles of its wall
- 17.** Saliva is produced in the
- A. small intestine
 - B. stomach
 - C. liver
 - D. mouth
- 19.** The breaking down of food into smaller pieces by the teeth and the tongue, without any chemical change, is called
- A. physical (mechanical) digestion
 - B. absorption
 - C. assimilation
 - D. chemical digestion
- 21.** The digestion of protein, as in a piece of meat, begins in the
- A. rectum
 - B. large intestine
 - C. mouth
 - D. stomach
- 23.** Bile, which is produced by the liver and helps in the digestion of palm oil, acts by
- A. absorbing the oil into the blood
 - B. changing the oil into starch
 - C. breaking the oil into tiny droplets so that enzymes can act on it
 - D. turning the oil into protein
- 12.** The part of the alimentary canal where most digested food is absorbed is the
- A. stomach
 - B. small intestine
 - C. large intestine
 - D. oesophagus
- 14.** Undigested material is passed out of the body through the
- A. lungs
 - B. anus
 - C. kidney
 - D. skin
- 16.** The teeth help in digestion by
- A. cutting and grinding the food into smaller pieces
 - B. adding enzymes to the food
 - C. turning starch into sugar
 - D. absorbing nutrients from the food
- 18.** The substance in saliva that acts on boiled yam or bread is an enzyme that begins the digestion of
- A. protein
 - B. fat
 - C. starch
 - D. vitamin
- 20.** Boiled rice chewed for a long time begins to taste sweet. This is because
- A. the enzyme in saliva has changed some of the starch into sugar
 - B. the rice has started to go bad
 - C. sugar has been added by the tongue
 - D. the teeth produce sugar
- 22.** The acid produced in the stomach helps digestion by
- A. absorbing the digested food
 - B. making the food sweeter
 - C. killing germs in the food and helping the enzymes to work
 - D. changing fats into sugar
- 24.** An orange eaten at breakfast supplies vitamin C. In the alimentary canal this vitamin is
- A. digested into amino acids
 - B. absorbed without being digested

- C. changed into starch
- D. passed out unchanged in the faeces

- 26.** Proteins are digested into
- A. simple sugars
 - B. amino acids
 - C. fatty acids and glycerol
 - D. minerals

- 25.** The end product of the digestion of starch is
- A. glucose
 - B. amino acids
 - C. fatty acids
 - D. glycerol only

- 27.** Oils and fats are digested into
- glucose
 - amino acids
 - fatty acids and glycerol
 - vitamins
- 29.** A meal of kenkey and fried fish will produce which end products of digestion?
- Glucose and amino acids
 - Fatty acids only
 - Glucose only
 - Amino acids only
- 31.** Which of the following statements about digestion is correct?
- All food is absorbed without any change
 - Large food substances are broken down into smaller ones that the body can absorb
 - Small food substances are joined into larger ones
 - Food is burnt into ash inside the stomach
- 33.** The inner wall of the small intestine is folded and covered with millions of tiny finger-like structures. This arrangement
- makes the intestine shorter
 - increases the surface for absorbing digested food
 - helps to chew the food
 - produces the teeth of the body
- 35.** Glucose carried to the cells of the body is used mainly to
- build the walls of cells
 - make the hair grow dark
 - form the enamel of teeth
 - release energy during respiration
- 37.** Iodine solution added to boiled yam turns
- blue-black
 - purple
 - colourless
 - brick-red
- 28.** Which of the following pairs correctly matches a food with the end product of its digestion?
- Palm oil - glucose
 - Boiled yam - fatty acids
 - Egg - amino acids
 - Bread - amino acids
- 30.** The end products of digestion are described as 'simple' because they are
- sweet in taste
 - small enough to pass through the wall of the small intestine into the blood
 - produced only in the mouth
 - always liquid in form
- 32.** The passing of digested food through the wall of the small intestine into the blood is called
- ingestion
 - egestion
 - digestion
 - absorption
- 34.** Digested food is carried from the small intestine to all parts of the body by
- the air passages
 - the blood
 - the nerves
 - the bones
- 36.** A patient whose small intestine has been damaged will suffer mainly because
- food cannot be chewed properly
 - water cannot be drunk at all
 - digested food cannot be properly absorbed into the blood
 - saliva can no longer be produced

- 39.** Sodium hydroxide followed by copper sulphate solution added to egg white gives a
- A. purple colour
 - B. blue-black colour
 - C. green colour
 - D. brick-red colour
- 38.** Benedict's solution added to a glucose solution and heated turns
- A. purple
 - B. milky white
 - C. blue-black
 - D. brick-red
- 40.** A piece of paper rubbed with a food leaves a mark through which light passes. The food contains
- A. protein
 - B. vitamin C
 - C. starch
 - D. oil or fat

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram, a flow chart or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 12: Human Body

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. The class of food that supplies the body with most of its energy is
 - A. carbohydrates
 - B. proteins
 - C. mineral salts
 - D. vitamins
2. Proteins are described as body-building foods because they are used for
 - A. cooling the blood
 - B. growth and the repair of worn-out tissues
 - C. supplying heat only
 - D. protecting the body against all diseases
3. Which of the following foods is the RICHEST source of protein?
 - A. Yam
 - B. Cassava
 - C. Rice
 - D. Beans
4. Fats and oils are important in the diet because they
 - A. supply energy and help to store it in the body
 - B. are digested in the mouth
 - C. supply water to the cells
 - D. cannot be stored in the body
5. Vitamins and mineral salts are needed in the diet mainly to
 - A. build the bones only
 - B. replace the blood lost daily
 - C. protect the body and keep it working properly
 - D. supply all the energy for work
6. Which nutrient is needed for the formation of strong bones and teeth?
 - A. Sodium chloride
 - B. Calcium
 - C. Starch
 - D. Iodine
7. Iron is needed in the diet for the formation of
 - A. bile in the liver
 - B. sweat in the skin
 - C. haemoglobin in red blood cells
 - D. enamel on the teeth
8. A diet that contains all the classes of food in the correct proportions is described as a
 - A. balanced diet
 - B. liquid diet
 - C. protein diet
 - D. staple diet
9. A deficiency disease is a disease caused by
 - A. the lack of a particular nutrient in the diet
 - B. eating too much of one food
 - C. a virus entering the body
 - D. drinking cold water
10. Kwashiorkor in young children is caused by the lack of
 - A. fibre in the diet
 - B. protein in the diet
 - C. carbohydrate in the diet
 - D. water in the diet
11. Night blindness is caused by a lack of
 - A. iron
 - B. vitamin C
 - C. vitamin D
 - D. vitamin A

- 13.** Rickets in children is caused by a lack of
- vitamin C and iron
 - iodine and sodium
 - protein and water
 - vitamin D and calcium
- 15.** Goitre, a swelling in the neck, is caused by a lack of
- iodine
 - vitamin D
 - calcium
 - protein
- 17.** A child with kwashiorkor usually shows
- a large swelling in front of the neck
 - bowed legs and a soft skull
 - a swollen belly and feet, with thin arms and reddish hair
 - bleeding gums and loose teeth
- 19.** A person who cannot see well in dim light is suffering from
- night blindness
 - goitre
 - rickets
 - scurvy
- 21.** The BEST way to prevent scurvy in a family is to
- boil all fruits before eating them
 - eat more fried yam
 - drink more sweetened drinks
 - include fresh fruits such as oranges and guava in the daily meals
- 23.** Deficiency diseases in a community are BEST prevented by
- educating families to eat balanced diets from locally available foods
 - giving injections to every child each week
 - boiling all drinking water twice
 - wearing nose masks at meal times
- 25.** Which of the following diseases is caused by a virus?
- COVID-19
 - Ringworm
 - Cholera
- 12.** Scurvy, in which the gums bleed, is caused by a lack of
- vitamin K
 - vitamin C
 - vitamin A
 - calcium
- 14.** Anaemia, in which a person is always weak and pale, is caused by a lack of
- vitamin C
 - fat
 - iron
 - iodine
- 16.** A child who eats only kenkey and sugar every day, with no fish, beans or eggs, is MOST likely to suffer from
- goitre
 - night blindness
 - scurvy
 - kwashiorkor
- 18.** Bowed legs and soft, weak bones in a young child are symptoms of
- scurvy
 - rickets
 - anaemia
 - night blindness
- 20.** Rickets is BEST prevented by eating foods rich in vitamin D and by
- bathing the child in warm water
 - giving the child plenty of sugar
 - allowing the child to play in the early morning sunshine
 - keeping the child indoors at all times
- 22.** Which of these local foods would BEST help to prevent anaemia?
- Sugar-cane juice
 - Kontomire and liver
 - Plain white rice
 - Fried plantain only
- 24.** A virus is BEST described as a
- worm that lives in the intestines
 - large single-celled animal found in water
 - type of bacterium that lives in the soil
 - very small organism that can multiply only inside the cells of a living host

D. Malaria

- 27.** COVID-19, Ebola and H1N1 are alike in that all three are
- spread only by mosquitoes
 - deficiency diseases of childhood
 - caused by viruses and spread from person to person
 - caused by bacteria in food
- 29.** Ebola is spread MAINLY through
- direct contact with the body fluids of an infected person
 - the bite of a housefly
 - breathing the air of a cold room
 - sharing a classroom desk
- 31.** Air travel helped COVID-19 to spread round the world quickly because
- aeroplanes produce the virus in their engines
 - the virus is carried in aviation fuel
 - the virus grows faster in an aeroplane
 - infected people moved from country to country before showing symptoms
- 33.** A person suspected of having a viral disease such as COVID-19 should FIRST
- visit many friends for advice
 - isolate himself and report to a health facility
 - buy antibiotics from a drug store
 - continue to attend school as usual
- 35.** Wearing a nose mask in a crowded place helps to control a viral disease because the mask
- kills the virus in the whole room
 - traps the droplets breathed out and keeps out those from other people
 - cools the air that is breathed in
 - supplies oxygen to the wearer
- 37.** The MOST important role of an individual in controlling a viral disease is to
- share medicines with neighbours
 - hide any illness from the health workers
 - observe the safety measures and report any symptoms early
- 26.** Viral diseases cannot be cured by
- vaccination before infection
 - antibiotics
 - the body's own defence system
 - rest and good feeding
- 28.** COVID-19 is spread MAINLY through
- drinking water from a tap
 - droplets released when an infected person coughs, sneezes or talks
 - eating food that lacks protein
 - the bite of an infected mosquito
- 30.** A disease that spreads quickly from community to community and from country to country, affecting a very large number of people, is declared a
- deficiency disease
 - chronic disease
 - pandemic
 - hereditary disease
- 32.** Common symptoms of COVID-19 include
- a swelling in front of the neck
 - bleeding gums and loose teeth
 - fever, dry cough, tiredness and loss of taste or smell
 - bowled legs and soft bones
- 34.** Washing the hands with soap under running water helps to prevent COVID-19 because it
- removes and destroys virus particles picked up on the hands
 - makes the hands too cold for the virus
 - kills mosquitoes on the skin
 - adds vitamins to the skin
- 36.** Besides illness and death, a serious effect of a pandemic on a country is that
- the population of mosquitoes falls
 - the rainfall of the country increases
 - the soil of the country loses its fertility
 - schools and businesses close and the economy suffers

- D. wait for the government to act alone

- 39.** Which of the following is a role of government in managing a pandemic?
- A. Preventing health workers from giving information
 - B. Closing all health facilities to save money
 - C. Providing vaccines, testing centres and treatment facilities
 - D. Asking every family to make its own medicine
- 38.** A useful role for community members during an outbreak is to
- A. drive infected persons out of the community
 - B. hold large gatherings to discuss the disease
 - C. stop all health workers from entering the area
 - D. educate one another and support those in isolation with food and care
- 40.** A class wishes to educate the community about a viral disease. The MOST suitable material to design and produce is a
- A. list of the names of sick people in the town
 - B. chart of the prices of medicines
 - C. map of the roads in the district
 - D. poster showing the symptoms and the ways of preventing the disease

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 13: Life Cycle Ho

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. The series of changes an organism passes through from the egg to the adult is called
 - A. a life cycle
 - B. a food chain
 - C. a life span
 - D. a water cycle
2. The change in form that an insect undergoes as it grows from the egg to the adult is known as
 - A. fertilisation
 - B. photosynthesis
 - C. metamorphosis
 - D. respiration
3. An insect whose life cycle has the four stages egg, larva, pupa and adult shows
 - A. incomplete metamorphosis
 - B. continuous growth
 - C. no metamorphosis
 - D. complete metamorphosis
4. The life cycle of an organism is described as a cycle because
 - A. the organism returns to the egg stage as it grows old
 - B. it takes exactly one year to finish
 - C. the adult produces eggs and the same stages begin again
 - D. the organism grows round in shape
5. Which of the following animals has a life cycle most like that of the housefly?
 - A. The mosquito
 - B. The goat
 - C. The lizard
 - D. The fowl
6. How many stages are there in the life cycle of the housefly?
 - A. Two
 - B. Three
 - C. Four
 - D. Five
7. The correct order of the stages in the life cycle of the housefly is
 - A. larva, egg, pupa, adult
 - B. egg, larva, pupa, adult
 - C. egg, larva, adult, pupa
 - D. egg, pupa, larva, adult
8. The first stage in the life cycle of the housefly is the
 - A. larva
 - B. adult
 - C. pupa
 - D. egg
9. The young housefly that hatches from the egg is called
 - A. a nymph
 - B. a larva
 - C. an adult
 - D. a pupa
10. The larva of the housefly is commonly known as
 - A. a caterpillar
 - B. a maggot
 - C. a tadpole
 - D. a grub-fly

- 11.** The stage of the housefly that does not feed and appears to be at rest is the
- A. larva
 - B. egg
 - C. pupa
 - D. adult
- 13.** The stage of the life cycle in which the housefly lays eggs is the
- A. larva stage
 - B. egg stage
 - C. pupa stage
 - D. adult stage
- 15.** The eggs of the housefly are usually laid
- A. on dry stones in the sun
 - B. inside clean drinking water
 - C. on the leaves of growing crops only
 - D. in rotting rubbish, manure and other decaying matter
- 17.** During the pupa stage, the housefly
- A. feeds heavily on rubbish
 - B. changes inside a hard case into the adult form
 - C. flies about looking for food
 - D. lays many eggs
- 19.** The complete life cycle of the housefly may take only about one to three weeks. This is important because
- A. each housefly lives for many years
 - B. only a few eggs are laid at a time
 - C. the housefly cannot spread diseases quickly
 - D. very large numbers of houseflies can be produced in a short time
- 21.** Houseflies are mostly found around
- A. refuse dumps, faeces and uncovered food
 - B. clean, dry and well-swept rooms
 - C. the open sea
 - D. deep wells of clear water
- 12.** The adult housefly comes out of the
- A. soil only
 - B. pupa
 - C. egg
 - D. larva
- 14.** A pupil arranged cut-outs of the stages of the housefly in this order: pupa, egg, adult, larva. The arrangement is wrong because
- A. the larva must come before the egg
 - B. the pupa and the adult are the same stage
 - C. the egg comes first and the pupa comes after the larva
 - D. the adult comes first in every life cycle
- 16.** Which of the following best describes the larva of the housefly?
- A. A tiny white oval object laid in a group
 - B. A winged insect with six legs
 - C. A hard, dark, motionless case
 - D. A white, legless, worm-like creature
- 18.** Which stage of the housefly feeds most and grows fastest?
- A. The larva
 - B. The pupa
 - C. The egg
 - D. The adult
- 20.** The warm, moist conditions of a refuse dump affect the life cycle of the housefly by
- A. making the stages develop faster
 - B. making the adult unable to fly
 - C. stopping the pupa from forming
 - D. killing the eggs at once
- 22.** Which of the following is NOT a feeding place of the housefly?
- A. An open pan of cooked rice
 - B. A dead animal by the roadside
 - C. A sealed tin of milk
 - D. A heap of animal manure

- 23.** The mouth parts of the housefly are suited for
- A. tearing flesh from bones
 - B. sucking up liquid and softened food
 - C. biting and chewing hard grains
 - D. piercing the skin and sucking blood
- 25.** A housefly carries germs mainly on its
- A. eyes only
 - B. eggs only
 - C. legs, body hairs and mouth parts
 - D. wings only
- 27.** A gutter behind a chop bar is blocked with food waste and water. The best reason why many houseflies are seen there is that the gutter
- A. has clean water for the flies to drink
 - B. provides food and a breeding place for the flies
 - C. keeps away the enemies of the flies
 - D. is warmer than the air around
- 29.** Which of the following diseases can be spread by the housefly?
- A. Dysentery
 - B. Measles
 - C. Ringworm
 - D. Malaria
- 31.** Apart from spreading disease, houseflies are a problem because they
- A. bite people at night and suck blood
 - B. destroy clothes in the wardrobe
 - C. are a nuisance in the home and environment
 - D. eat the roots of crops
- 33.** Which of the following is the strongest evidence that houseflies spread disease in a community?
- A. Houseflies make a buzzing sound while flying
 - B. Houseflies fly faster in the afternoon
 - C. Houseflies are seen in every town
- 24.** A housefly feeding on solid sugar first
- A. swallows it whole
 - B. bites it into small pieces
 - C. warms it with its wings
 - D. pours saliva on it to dissolve it
- 26.** Houseflies move between a refuse dump and the food on a table. This habit is dangerous because
- A. the flies lay eggs that people can see
 - B. the flies make the food cold
 - C. the flies eat all the food
 - D. germs picked up from the refuse are left on the food
- 28.** An insect that lives close to human beings and causes harm and discomfort is described as
- A. a predator
 - B. a menace
 - C. a decomposer
 - D. a pollinator
- 30.** Food that has been visited by houseflies may cause
- A. sickle cell disease
 - B. food poisoning
 - C. night blindness
 - D. asthma
- 32.** A child in a home where food is left uncovered keeps having diarrhoea. The most likely explanation is that
- A. houseflies have bitten the child
 - B. the food was cooked for too long
 - C. the child drinks too much boiled water
 - D. houseflies have carried germs from faeces or refuse onto the food

- D. Communities with poor refuse disposal and many flies record more cases of diarrhoea

35. The best single way to stop houseflies from breeding in a compound is to

- A. spray perfume in the rooms each morning
- B. sweep the compound once a month
- C. collect and dispose of refuse properly
- D. keep the windows open all day

34. Which disease of the ones listed is spread by the mosquito and NOT by the housefly?

- A. Malaria
- B. Dysentery
- C. Typhoid fever
- D. Cholera

- 37.** Which of the following practices will NOT help to reduce the number of houseflies in a community?
- A. Throwing kitchen waste behind the house
 - B. Using a pit latrine with a tight cover
 - C. Clearing choked gutters regularly
 - D. Covering refuse containers with lids
- 39.** A Basic 7 class wants to teach their community how to reduce the housefly menace. The MOST effective plan would be to
- A. ask each pupil to kill ten flies at home
 - B. hold a clean-up of the refuse dump and gutters and give a talk on covering food and refuse
 - C. buy insecticide and spray the classroom only
 - D. write the information in their exercise books
- 36.** Covering food with a net cover helps to control the housefly menace because it
- A. keeps the flies away from the food
 - B. prevents the eggs from hatching
 - C. stops the food from getting cold
 - D. kills the flies that come near
- 38.** Faeces left in the open is a serious health problem mainly because
- A. it attracts rain to the area
 - B. it makes the soil too rich for crops
 - C. houseflies breed in it and carry the germs to food
 - D. it dries up quickly in the sun
- 40.** A household covers its food carefully but still throws its kitchen waste in an open heap beside the kitchen. The weakness of this arrangement is that
- A. the flies still breed in the heap and will reach the food whenever it is opened
 - B. covering food is of no use at all
 - C. the heap will stop the flies from laying eggs
 - D. the covered food will go bad faster

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram, a flow chart or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 14: Living Cells

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. The smallest unit of a living organism that can carry out all the activities of life is
 - A. the system
 - B. the cell
 - C. the organ
 - D. the tissue
2. The instrument used in the laboratory to see cells clearly is
 - A. the microscope
 - B. the barometer
 - C. the stethoscope
 - D. the thermometer
3. A thin piece of onion was placed on a glass slide and viewed with a microscope. The many small boxes seen were
 - A. drops of water
 - B. crystals of salt
 - C. grains of sugar
 - D. cells
4. Which of the following statements about living things is correct?
 - A. Only animals are made up of cells
 - B. Both plants and animals are made up of cells
 - C. Only plants are made up of cells
 - D. Neither plants nor animals are made up of cells
5. A group of similar cells working together to perform the same work forms
 - A. an organism
 - B. a tissue
 - C. an organelle
 - D. an organ
6. The small structures found inside a cell, each doing a special work, are called
 - A. systems
 - B. tissues
 - C. organs
 - D. organelles
7. Which of the following parts is found in an animal cell?
 - A. Cell membrane
 - B. Cell wall
 - C. Chloroplast
 - D. Large central vacuole
8. The part of an animal cell that forms its outer boundary is the
 - A. cell wall
 - B. nucleus
 - C. cell membrane
 - D. cytoplasm
9. The jelly-like material in which the organelles of a cell float is the
 - A. nucleus
 - B. cell sap
 - C. chloroplast
 - D. cytoplasm
10. The largest organelle in a typical animal cell is usually the
 - A. ribosome
 - B. mitochondrion
 - C. nucleus
 - D. vacuole

- 11.** Which of the following is NOT found in an animal cell?
- A. Mitochondrion
 - B. Cell wall
 - C. Cytoplasm
 - D. Nucleus
- 13.** An animal cell has an irregular shape MAINLY because
- A. it has no cell wall to hold it in a fixed shape
 - B. it has no cell membrane
 - C. it has no nucleus
 - D. its cytoplasm is too thick to move
- 15.** The part of the cell that controls all the activities of the cell is the
- A. nucleus
 - B. cytoplasm
 - C. ribosome
 - D. cell membrane
- 17.** The organelle in which energy is released from food in a cell is the
- A. mitochondrion
 - B. cell membrane
 - C. vacuole
 - D. nucleus
- 19.** Proteins are built up in a cell at the
- A. chloroplasts
 - B. cell membrane
 - C. vacuoles
 - D. ribosomes
- 21.** If the nucleus is removed from a living cell, the cell will MOST likely
- A. produce more energy than before
 - B. change into a plant cell
 - C. die, because its activities are no longer controlled
 - D. continue to grow faster than before
- 23.** The cell wall of a plant cell is made MAINLY of
- A. cellulose
 - B. starch
 - C. protein
 - D. fat
- 12.** The sausage-shaped organelles found in the cytoplasm of an animal cell are the
- A. nuclei
 - B. chloroplasts
 - C. mitochondria
 - D. cell walls
- 14.** A cell was taken from the cheek lining of a goat. The cell would contain
- A. chloroplasts, a cell wall and a nucleus
 - B. a cell wall, cytoplasm and a large vacuole
 - C. a nucleus, cytoplasm and a cell membrane
 - D. a cell wall, chloroplasts and a large vacuole
- 16.** The cell membrane is important because it
- A. stores the water of the cell
 - B. gives the cell its green colour
 - C. controls the substances that enter and leave the cell
 - D. makes food for the cell
- 18.** Most of the chemical reactions of a cell take place in the
- A. cell membrane
 - B. cytoplasm
 - C. nucleus
 - D. cell wall
- 20.** A muscle cell has very many mitochondria. This is because the muscle cell
- A. has no nucleus of its own
 - B. needs to store a large amount of water
 - C. needs a large amount of energy
 - D. has to make its own food
- 22.** The outermost part of a plant cell is the
- A. cell membrane
 - B. nucleus
 - C. cytoplasm
 - D. cell wall

- 25.** The green colouring matter found inside the chloroplast is
- haemoglobin
 - cell sap
 - cytoplasm
 - chlorophyll
- 27.** In a mature plant cell the nucleus is usually found
- outside the cell wall
 - exactly in the centre of the cell
 - inside the large vacuole
 - pushed to one side, near the cell membrane
- 29.** The chief function of the cell wall is to
- release energy from food
 - control what enters and leaves the cell
 - give the cell a fixed shape and support
 - make food for the plant
- 31.** The large vacuole of a plant cell is important because it
- makes the proteins of the cell
 - stores cell sap and keeps the cell firm
 - controls the activities of the cell
 - releases energy from food
- 33.** The cell membrane of a plant cell lies
- just inside the cell wall
 - just outside the cell wall
 - inside the large vacuole
 - inside the nucleus
- 35.** Which of the following parts is found in a plant cell but NOT in an animal cell?
- Nucleus
 - Cell membrane
 - Cytoplasm
 - Chloroplast
- 24.** The green organelles found in the cells of a leaf are the
- mitochondria
 - ribosomes
 - nuclei
 - chloroplasts
- 26.** A large space filled with cell sap in the middle of a mature plant cell is the
- nucleus
 - cell wall
 - vacuole
 - chloroplast
- 28.** Which part of a plant cell would NOT be found in a cell taken from the root of the plant?
- Vacuole
 - Chloroplast
 - Nucleus
 - Cell wall
- 30.** Food is made in the plant cell in the
- chloroplasts
 - nucleus
 - mitochondria
 - vacuole
- 32.** A plant that has not been watered for several days becomes limp MAINLY because
- the vacuoles of its cells have lost water
 - the cell walls have been destroyed
 - the chloroplasts have lost their colour
 - the nuclei of the cells have died
- 34.** A plant cell placed in the dark for a long time would be unable to
- keep its shape
 - make its own food
 - release energy from stored food
 - take in water
- 36.** Which parts are found in BOTH plant and animal cells?
- Chloroplast, vacuole and cell wall
 - Nucleus, cytoplasm and cell membrane
 - Cell wall, nucleus and

- chloroplast
D. Cell wall, cytoplasm and chloroplast

38. Plant cells usually have a regular box-like shape while animal cells do not. The reason is that the plant cell has

- A. a firm cell wall outside the cell membrane
- B. no cytoplasm in its cell
- C. a larger nucleus than the animal cell
- D. more mitochondria than the animal cell

37. A pupil looked at a cell under a microscope and saw a cell wall, chloroplasts and a large central vacuole. The cell was

- A. a red blood cell
- B. a plant cell
- C. a cheek cell
- D. an animal cell

- 39.** A pupil was asked to model a plant cell and an animal cell with local materials. The BEST way to show the difference between them would be to
- A. use a larger nucleus in the plant cell model
 - B. colour both models green
 - C. leave out the cytoplasm in the animal cell model
 - D. put a firm box round the model of the plant cell only

- 40.** Which observation would BEST convince a class that both plants and animals are made up of cells?
- A. Feeling the surface of a leaf and the skin of the arm
 - B. Looking at a chart of a plant cell only
 - C. Viewing slides of leaf tissue and cheek tissue and seeing similar small units in both
 - D. Reading in a textbook that living things are made of cells

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram, a labelled sketch or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 15: Material

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. Anything that has mass and occupies space is called
 - A. a force
 - B. a shadow
 - C. matter
 - D. energy
2. The three states in which materials are commonly found are
 - A. wood, metal and glass
 - B. hard, soft and rough
 - C. natural, artificial and waste
 - D. solid, liquid and gas
3. A pupil described a material as hard, rough and grey, and said that it kept its own shape. The material is MOST likely
 - A. palm oil
 - B. a stone
 - C. cooking gas
 - D. kerosene
4. Which of the following properties would a pupil record by LOOKING at a material only?
 - A. Its smell
 - B. Its colour
 - C. Its temperature
 - D. Its mass
5. Which group contains only solids?
 - A. Chalk, sand, iron nail
 - B. Sand, petrol, oxygen
 - C. Chalk, water, iron nail
 - D. Iron nail, air, palm oil
6. Which of the following is a liquid at room temperature?
 - A. Oxygen
 - B. Charcoal
 - C. Groundnut oil
 - D. Common salt
7. A material that has no colour, cannot be held in the hand and spreads to fill the whole room is
 - A. a liquid
 - B. a gas
 - C. a solid
 - D. a mixture of solids
8. A material that has a fixed shape and a fixed volume is
 - A. a solid
 - B. a gas
 - C. a vapour
 - D. a liquid
9. A liquid has
 - A. both a fixed shape and a fixed volume
 - B. a fixed shape but no fixed volume
 - C. neither a fixed volume nor a fixed mass
 - D. a fixed volume but no fixed shape
10. Water poured from a bucket into a bottle takes the shape of the bottle. This shows that a liquid
 - A. has no mass of its own
 - B. has no shape of its own
 - C. has no volume of its own
 - D. changes into a gas in the bottle
11. In which state of matter are the particles packed closely together in a

regular pattern?

- A. Vapour
- B. Gas
- C. Liquid
- D. Solid

12. The particles of a gas are

- A. far apart and move about freely
- B. joined in long chains that cannot move
- C. close together but able to slide

- over one another
D. close together and fixed in position
- 13.** A cylinder of cooking gas feels heavy because
- A. a gas has no mass but the air presses on it
 - B. a gas has mass even though it cannot be seen
 - C. the gas has a fixed shape inside the cylinder
 - D. the gas has turned into a solid inside
- 15.** The liquid that is most important for the survival of humans is
- A. kerosene
 - B. petrol
 - C. water
 - D. palm oil
- 17.** Blood is an important liquid in the body because it
- A. carries food and oxygen to all parts of the body
 - B. stores all the fat of the body
 - C. removes bones that are damaged
 - D. keeps the bones stiff and strong
- 19.** Which of the following shows the need to preserve liquids in a community?
- A. Pouring left-over drinking water on the ground
 - B. Leaving the tap running while brushing the teeth
 - C. Covering the water in a storage drum and using it carefully
 - D. Washing the compound with pipe-borne water every morning
- 21.** Which practice would BEST protect the liquid stored in a water tank at a school?
- A. Keeping the tank covered and cleaning it regularly
 - B. Siting the tank beside the refuse dump
 - C. Allowing pupils to dip their cups into the tank
- 14.** Which property is shared by solids, liquids and gases?
- A. They can all be compressed easily
 - B. They all have mass and occupy space
 - C. They all have a fixed shape
 - D. They can all be poured from one container into another
- 16.** Which of the following is a use of water in the human body?
- A. It hardens the bones of the body
 - B. It changes food into oxygen
 - C. It helps in the digestion and transport of food
 - D. It makes the body produce light
- 18.** A farmer at Techiman uses a liquid obtained from the oil palm fruit for cooking. The liquid is
- A. palm oil
 - B. vinegar
 - C. kerosene
 - D. petrol
- 20.** A tanker of petrol is important to a town MAINLY because petrol is used
- A. for drinking during the dry season
 - B. as a fuel in vehicles and machines
 - C. for watering crops on a farm
 - D. for washing cooking utensils

- D. Leaving the lid open so that air can enter
- 23.** A solid obtained from plants and used for making tables and chairs is
- A. clay
 - B. cement
 - C. iron
 - D. wood
- 25.** Which of the following solids is used by a mason to build a wall?
- A. Palm fronds
 - B. Rubber tyres
 - C. Cement blocks
 - D. Cotton cloth
- 22.** Which solid material is used MAINLY for roofing buildings in many Ghanaian towns?
- A. Glass
 - B. Cotton wool
 - C. Rubber
 - D. Aluminium sheet
- 24.** Common salt is an important solid in the home because it
- A. adds taste to food and supplies minerals the body needs
 - B. makes food cook without fire
 - C. keeps the kitchen cool
 - D. turns food into a liquid

- 27.** A Basic 7 class was asked to explain why useful solids in the environment should be preserved. The BEST reason is that
- they are found only in the forest zone
 - they are limited in supply and are needed by humans and other living things
 - they are too heavy to be carried away
 - they lose their colour when they are used
- 29.** The chemical symbol of sodium is
- So
 - Sn
 - S
 - Na
- 31.** Which of the following gives the correct symbol of calcium?
- Cl
 - C
 - Cm
 - Ca
- 33.** The elements in the periodic table are arranged in order of increasing
- atomic number
 - the date on which the element was discovered
 - mass of one litre of the element
 - the first letter of the name of the element
- 35.** Which of the following is a property of MOST metals?
- They break easily into powder
 - They are poor conductors of heat
 - They are always liquids at room temperature
 - They conduct heat and electricity
- 37.** Helium, neon and argon are grouped together in the periodic table because they
- burn easily in air
 - are all liquids at room temperature
 - are all metals
 - are unreactive gases
- 26.** Iron is widely used for making farm tools such as cutlasses MAINLY because it is
- shiny and colourful
 - cheap and smooth
 - hard and strong
 - light and soft
- 28.** A substance that cannot be broken down into simpler substances by a chemical method is
- an element
 - a mixture
 - an alloy
 - a solution
- 30.** The element whose chemical symbol is K is
- potassium
 - carbon
 - calcium
 - krypton
- 32.** The first element in the periodic table is
- oxygen
 - helium
 - hydrogen
 - lithium
- 34.** An element has atomic number 8. The element is
- sulphur
 - carbon
 - oxygen
 - neon
- 36.** Which of the following elements is a non-metal?
- Aluminium
 - Sulphur
 - Potassium
 - Magnesium
- 38.** Elements that are found in the same group of the periodic table
- have exactly the same mass
 - have similar chemical properties
 - are all metals
 - have the same atomic number

- 39.** Sodium, magnesium and aluminium are all found in Period 3 of the periodic table. This means that they
- A. belong to the same group
 - B. have the same number of protons
 - C. are in the same horizontal row of the table
 - D. are all noble gases
- 40.** A pupil found that a shiny element conducted electricity, could be beaten into sheets and did not break when it was bent. The element is BEST described as
- A. a mixture
 - B. a non-metal
 - C. a noble gas
 - D. a metal

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram, a labelled sketch or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 16: Science and

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. A career in science is BEST described as
 - A. work that needs no training at all
 - B. work done only by teachers
 - C. any work done inside a building
 - D. work in which a person uses scientific knowledge and skills to earn a living
2. A scientist who studies living things such as plants and animals is a
 - A. meteorologist
 - B. geologist
 - C. biologist
 - D. physicist
3. The scientist who studies substances, how they are made up and how they react is a
 - A. biologist
 - B. astronomer
 - C. surveyor
 - D. chemist
4. A person who designs and supervises the building of roads, bridges and dams is a
 - A. veterinary officer
 - B. civil engineer
 - C. nurse
 - D. pharmacist
5. A scientist who treats sick animals on a farm is a
 - A. veterinary doctor
 - B. agronomist
 - C. pharmacist
 - D. radiographer
6. The health worker who prepares and dispenses medicines in a hospital is the
 - A. physiotherapist
 - B. pharmacist
 - C. midwife
 - D. laboratory technician
7. A person who studies the weather and prepares the daily forecast is a
 - A. food scientist
 - B. meteorologist
 - C. zoologist
 - D. geologist
8. A scientist who works to improve the processing and preservation of foods such as cassava and maize is a
 - A. civil engineer
 - B. astronomer
 - C. surveyor
 - D. food scientist
9. Which of the following careers is MOST directly concerned with the study of rocks and minerals?
 - A. Food science
 - B. Computer programming
 - C. Nursing
 - D. Geology
10. Professor Francis Allotey, a Ghanaian scientist of international repute, worked chiefly in the field of
 - A. surgery and nursing
 - B. music and drama
 - C. mathematics and physics
 - D. law and government

- 11.** Professor Ibok Oduro of KNUST is known for her work in
- food science and the use of local food crops
 - the mining of gold
 - the design of aeroplane engines
 - the printing of newspapers
- 13.** Professor Jophus Anamuah-Mensah is best known in Ghana as a
- science educator
 - military officer
 - professional footballer
 - textile trader
- 15.** Studying the lives of Ghanaian scientists is useful to pupils chiefly because it
- helps them to pass examinations without studying
 - shows them that Ghanaians can succeed in science and gives them models to follow
 - shows that only professors can study science
 - proves that science began in Ghana
- 17.** Which institution in Ghana is BEST known for training engineers and scientists?
- The Ghana Standards Authority library
 - The National Theatre
 - The Ghana Police Training School
 - Kwame Nkrumah University of Science and Technology
- 19.** Penicillin has improved human life chiefly because it
- cures bacterial infections that once killed many people
 - supplies vitamins to the body
 - prevents all accidents on the roads
 - removes waste from the environment
- 21.** Charles Darwin is remembered for his explanation of
- how electricity flows in a wire
 - how living things change over time by natural selection
 - how the heart pumps blood
- 12.** Professor Ewurama Addy, a Ghanaian woman scientist, is known for her work in
- forecasting the weather
 - repairing motor vehicles
 - building bridges
 - biochemistry and the biological sciences
- 14.** Professors Theophilus Ossei-Anto and Christian Anthony-Krueger have both contributed to
- the building of the Akosombo Dam
 - the printing of currency notes
 - the teaching and learning of science in Ghana
 - the export of cocoa beans
- 16.** A science educator is a person who
- sells science textbooks
 - repairs laboratory apparatus
 - studies and improves the way science is taught and learnt
 - builds laboratories for a living
- 18.** Alexander Fleming is remembered for the discovery of
- the electric bulb
 - the law of gravity
 - penicillin, the first antibiotic
 - the structure of the atom
- 20.** Albert Einstein is famous for his work in
- the classification of plants
 - physics, especially the theory of relativity
 - the discovery of vaccines
 - the invention of the telephone

D. how rain is formed

23. Louis Pasteur's work on germs led directly to

- A. the discovery of gold in rocks
- B. the pasteurisation of milk and the use of vaccines
- C. the building of the first computer
- D. the invention of the aeroplane

22. Stephen Hawking was a scientist noted for his study of

- A. the design of motor cars
- B. the growth of crops
- C. the universe, including black holes
- D. the treatment of malaria

24. The MAIN lesson pupils can learn from the lives of great scientists is that

- A. a scientist must be born outside Africa

25. A pupil preparing a presentation on a named scientist should include
- A. the scientist's field of work, the discovery made and how it has helped people
 - B. only the scientist's date of birth
 - C. a list of the scientist's relatives
 - D. the scientist's favourite food
27. The use of mobile phones for money transfer in Ghana is an example of the impact of science and technology on
- A. the movement of the planets
 - B. the growth of trees in the forest
 - C. trade and daily life in the community
 - D. the rainfall of the country
29. The use of solar panels in schools without a reliable supply of electricity shows that science and technology can
- A. remove the need for teachers
 - B. make the Sun shine more brightly
 - C. provide practical solutions to local problems
 - D. increase the number of power cuts
31. Which of the following shows the impact of science in the school?
- A. The sweeping of the compound each morning
 - B. The wearing of school uniform
 - C. The use of computers and projectors in teaching
 - D. The singing of the national anthem
33. Innovation in science means
- B. discoveries are made only by accident
 - C. science needs no experiments at all
 - D. careful observation, hard work and curiosity lead to discovery
26. Technology is BEST described as
- A. the study of living things only
 - B. the use of scientific knowledge to make useful things and solve problems
 - C. the collection of old machines
 - D. the writing of science examinations
28. A refrigerator improves life in the home chiefly because it
- A. cooks food without fire
 - B. removes salt from food
 - C. makes food richer in protein
 - D. keeps food cold so that it does not spoil quickly
30. The use of tractors, improved seeds and irrigation on farms has MAINLY led to
- A. higher yields and more food for the population
 - B. a fall in the quality of all crops
 - C. the end of rainfall in farming areas
 - D. the loss of all farming jobs
32. A NEGATIVE effect of some modern technology on the community is
- A. easier travel from town to town
 - B. pollution of the air and water by factories and vehicles
 - C. faster communication between people
 - D. better treatment of diseases

- A. memorising scientific facts
- B. copying a classmate's work
- C. repeating exactly what others have done
- D. putting a new or improved idea into practical use

35. Training more medical and laboratory scientists in Ghana would MOST directly

- A. improve the diagnosis and treatment of diseases in the country
- B. lower the quality of local foods
- C. reduce the number of schools
- D. increase the rainfall of the country

34. Ghana needs to train more agricultural scientists MAINLY in order to

- A. stop the export of cocoa
- B. reduce the number of farms in the country
- C. increase food production and reduce the importation of food
- D. increase the price of fertiliser

- 37.** A pupil who wishes to become an engineer should pay particular attention to
- A. science and mathematics
 - B. handwriting only
 - C. games and sports only
 - D. memorising long poems
- 39.** A school science club wishing to learn about careers in science would benefit MOST from
- A. reading only the names of scientists
 - B. collecting money from pupils
 - C. holding a football competition each week
 - D. inviting scientists and visiting laboratories, farms and factories
- 36.** A reason for Ghana to focus on careers in renewable energy is that they would help the country to
- A. use up its crude oil more quickly
 - B. produce cleaner electricity and depend less on imported fuel
 - C. stop the use of all machines
 - D. increase the smoke in the cities
- 38.** The BEST way for a Basic 7 pupil to prepare for a career in science is to
- A. take part in practical work, read widely and ask questions
 - B. wait until senior high school before studying science
 - C. avoid difficult topics
 - D. copy answers from friends
- 40.** The MAIN purpose of building a portfolio on the impact of science and technology is to
- A. gather and organise evidence of what has been found out, for presentation
 - B. keep the findings away from other pupils
 - C. replace the class textbook
 - D. avoid doing any research

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 17: Solar Sys

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. The solar system is made up of
 - A. the Earth and the Moon only
 - B. all the stars in the sky
 - C. the clouds and the air above the Earth
 - D. the Sun and all the bodies that move round it
2. At the centre of the solar system is
 - A. the Sun
 - B. Mercury
 - C. the Earth
 - D. the Moon
3. The Sun is best described as
 - A. a moon
 - B. a star
 - C. a cloud of water
 - D. a planet
4. The Sun is important to the solar system because it
 - A. moves round the Earth each day
 - B. supplies the planets with water
 - C. reflects light from the planets
 - D. supplies light and heat and holds the planets in their paths
5. A planet differs from a star in that a planet
 - A. gives out its own light and heat
 - B. is always nearer to the Earth than the Moon
 - C. is always larger than a star
 - D. does not give out its own light but shines by reflecting sunlight
6. The force that keeps the planets moving round the Sun instead of flying away is
 - A. the heat of the Sun
 - B. the magnetism of the Earth
 - C. the gravitational pull of the Sun
 - D. the push of the wind in space
7. Which of the following is NOT a member of the solar system?
 - A. The Moon
 - B. The Milky Way
 - C. Mars
 - D. The Sun
8. The planets nearest to the Sun are called
 - A. the outer planets
 - B. the stars
 - C. the satellites
 - D. the inner planets
9. How many inner planets are there in the solar system?
 - A. Six
 - B. Four
 - C. Two
 - D. Eight
10. The inner planets, in order from the Sun, are
 - A. Venus, Mercury, Mars, Earth
 - B. Mercury, Venus, Earth, Mars
 - C. Mercury, Earth, Venus, Mars
 - D. Earth, Mars, Mercury, Venus
11. The inner planets are also called the rocky planets because they
 - A. are covered with ice
 - B. are made mainly of gas
 - C. have solid, rocky surfaces
 - D. are made of burning rock
12. The planet nearest to the Sun is
 - A. Earth
 - B. Mars
 - C. Venus
 - D. Mercury

- 13.** The fourth planet from the Sun is
- Earth
 - Venus
 - Mercury
 - Mars
- 15.** Mars is often called the red planet because
- it is very close to the Sun
 - it is the hottest planet
 - the rocks and dust on its surface are reddish in colour
 - it gives out red light of its own
- 17.** A very large group of stars, dust and gas held together by gravity is called
- a planet
 - a solar system
 - a satellite
 - a galaxy
- 19.** On a clear, dark night away from town lights, the Milky Way appears as
- a faint band of light stretching across the sky
 - a single very bright star
 - a line of moving lights
 - a round red object
- 21.** Arrange the following from the smallest to the largest: galaxy, planet, solar system.
- Planet, galaxy, solar system
 - Planet, solar system, galaxy
 - Galaxy, solar system, planet
 - Solar system, planet, galaxy
- 23.** The shape of the path along which a planet moves round the Sun is
- elliptical (oval)
 - a perfect circle
 - a square
 - a straight line
- 25.** The turning of a planet on its own axis is called
- rotation
 - radiation
 - revolution
 - reflection
- 14.** Which inner planet is known to support life?
- Earth
 - Mars
 - Venus
 - Mercury
- 16.** Which of the following is a property shared by all the inner planets?
- They all have many moons
 - They all have rings around them
 - They are all small and have solid surfaces
 - They are all covered with liquid water
- 18.** The galaxy in which our solar system is found is called
- the Milky Way
 - the Great Bear
 - the Orion Belt
 - the Southern Cross
- 20.** Which of the following statements about the Milky Way is correct?
- The Milky Way is the path followed by the Earth
 - The Milky Way is found inside the solar system
 - The solar system is only a very small part of the Milky Way
 - The Milky Way is another name for the Sun
- 22.** Street lights make it difficult to see the Milky Way from a town in Ghana because
- tall buildings block the light of the stars
 - the Milky Way moves away from towns
 - the bright lights of the town drown the faint light from the stars
 - the Milky Way can be seen only in the rainy season
- 24.** The path along which a planet travels round the Sun is called its
- axis
 - orbit
 - equator
 - satellite

- 27.** The rotation of the Earth on its axis causes
- the seasons of the year
 - the tides of the sea
 - day and night
 - the phases of the Moon
- 29.** One complete revolution of the Earth round the Sun takes about
- 24 hours
 - 365 days
 - 88 days
 - 30 days
- 31.** Mercury has a very hot day side and a very cold night side mainly because it
- is far from the Sun
 - turns very fast on its axis
 - is covered with white ice
 - has almost no atmosphere to hold in the heat
- 33.** Venus is the hottest planet in the solar system because
- it is the planet nearest to the Sun
 - its thick atmosphere traps the heat from the Sun
 - it gives out heat of its own
 - it is made of burning coal
- 35.** Which statement about the sizes of Venus and the Earth is correct?
- Venus is nearly the same size as the Earth
 - Venus is the largest planet in the solar system
 - Venus is many times larger than the Earth
 - Venus is smaller than the Moon
- 26.** The movement of a planet round the Sun is called
- orbit tilting
 - rotation
 - vibration
 - revolution
- 28.** A planet that is far from the Sun takes a longer time to go round the Sun than one that is near. This is mainly because the far planet
- has a much longer path to travel
 - stops from time to time on the way
 - is pulled backwards by the stars
 - is heavier than the near planet
- 30.** Which of the following is a property of Mercury?
- It is the smallest of the inner planets and has almost no atmosphere
 - It has a thick atmosphere of oxygen
 - It is covered with large oceans of water
 - It is the largest planet in the solar system
- 32.** Mercury takes about 88 Earth days to go round the Sun once. This means that Mercury
- does not move round the Sun at all
 - has a much shorter year than the Earth
 - has a much longer year than the Earth
 - has a year of the same length as that of the Earth
- 34.** Venus is often called the morning star or the evening star because it
- appears only once each year
 - is the brightest star in the sky
 - can be seen shining brightly near sunrise or sunset
 - gives out its own light at dawn
- 36.** Mercury and Venus are alike in that both
- have surfaces covered with water
 - have many moons

- C. are larger than the Earth
- D. move round the Sun in paths inside that of the Earth

- 38.** In building a model of the solar system from balls of different sizes, the largest ball should stand for
- A. Venus
 - B. the Earth
 - C. the Sun
 - D. Mercury

- 37.** A model of the solar system is useful mainly because it
- A. helps us to see the arrangement and movement of bodies that are too far away to visit
 - B. can be used to measure the heat of the Sun
 - C. shows the true sizes and distances of the planets
 - D. makes the planets move faster

- 39.** A pupil at Koforidua made a model in which the four inner planets were

placed in a straight line beside the Sun. The main fault in this model is that

- A. the planets should be bigger than the Sun
- B. the Sun should be placed at the edge of the model
- C. the planets move round the Sun along separate paths and are not in a line
- D. the planets should all be the same size

40. Which material would be the MOST suitable for showing the path of a planet in a model of the solar system?

- A. A square piece of cardboard
- B. A loop of wire bent into an oval shape
- C. A straight stick
- D. A short piece of rope tied in a knot

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram, a flow chart or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 18: Understanding the

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. A landform is BEST described as
 - A. a map drawn by a surveyor
 - B. a kind of soil used for farming
 - C. a natural feature of the surface of the Earth, such as a mountain or a plain
 - D. a building put up by people
2. A wide stretch of flat, low land is called a
 - A. plateau
 - B. valley
 - C. mountain
 - D. plain
3. A piece of high land with a broad, flat top is a
 - A. plateau
 - B. valley
 - C. delta
 - D. plain
4. The low land lying between two hills or mountains is a
 - A. plateau
 - B. plain
 - C. peak
 - D. valley
5. The highest mountain in Ghana is
 - A. Mount Afadja (Afadjato)
 - B. Mount Kenya
 - C. Mount Kilimanjaro
 - D. Mount Cameroon
6. The Accra Plains and the Afram Plains are examples of the landform called the
 - A. plateau
 - B. plain
 - C. valley
 - D. mountain
7. Which group of animals would one expect to find MOST commonly on a grassy plain?
 - A. bats, baboons and rock hyraxes
 - B. grasscutters, antelopes and cattle
 - C. crabs, oysters and starfish
 - D. tilapia, catfish and mudfish
8. Mangrove plants and crabs are found MAINLY in
 - A. the roof of a cave
 - B. the middle of a desert
 - C. dry rocky hill tops
 - D. muddy ground along the coast and river mouths
9. The special features that help a living thing to survive where it lives are called
 - A. habitats
 - B. associations
 - C. adaptations
 - D. competitions
10. The thick fur of an animal living on a cold mountain helps it to
 - A. keep its body warm
 - B. store water in its body
 - C. find its food in the dark
 - D. run faster on the rocks
11. Mountain goats and rock hyraxes can move about on steep rocky slopes because they have
 - A. webbed feet for swimming

- B. strong legs and hooves or pads that grip the rock
- C. wide flat feet for walking on mud
- D. long tails for hanging from trees

- 12.** The long legs of the egret that feeds in a marshy plain help it to
- A. dig deep holes in dry soil
 - B. climb tall trees quickly
 - C. wade through shallow water

- without wetting its body
D. fly faster than other birds
- 13.** An antelope living on an open plain escapes from its enemies MAINLY by
A. swimming across the plain
B. burrowing into the rock
C. changing into another animal
D. running very fast on its long slender legs
- 15.** The sandy colour of many animals of the dry northern plains is useful because it
A. makes them attractive to hunters
B. helps them to dig faster
C. helps them to blend with their surroundings and escape notice
D. keeps insects away from them
- 17.** Plants growing in a dry, open plain often have long roots in order to
A. catch insects for food
B. hold leaves above the clouds
C. produce more flowers in the dry season
D. reach water deep in the soil
- 19.** The thick fleshy stem of a cactus-like plant is useful for
A. killing insects
B. storing water
C. attracting rainfall
D. storing oxygen
- 21.** Trees of the thick forest in a wet valley are usually
A. short, with needle-like leaves
B. tall, with broad leaves that catch the sunlight
C. leafless throughout the year
D. grown only in sandy soil
- 23.** Some plants on rocky hillsides grow close to the ground. This helps them to
A. lose more water quickly
B. resist being blown away by strong winds
C. receive less sunlight
- 14.** A fish living in a fast-flowing stream in a valley is helped to stay in place by its
A. streamlined body and strong fins
B. hard shell and eight legs
C. hollow bones and feathers
D. thick fur and long ears
- 16.** Burrowing into the ground during the hottest part of the day helps a small animal of the plains to
A. grow a longer tail
B. find more sunlight
C. increase the loss of water from its body
D. avoid the heat and reduce the loss of water from its body
- 18.** The small, needle-like leaves of some plants in dry places help them to
A. lose more water by transpiration
B. lose less water by transpiration
C. trap more rain water
D. store more seeds
- 20.** Mangrove plants growing in soft muddy soil are supported by
A. a single straight tap root only
B. woody stems that lie flat in the mud
C. floating leaves
D. stilt (prop) roots that spread out above the mud
- 22.** Grasses survive the yearly bush fires of the savanna plains MAINLY because they
A. grow taller than the flames
B. are too wet to burn at all
C. have underground parts that sprout again after the fire
D. are protected by their flowers

- D. escape from grazing animals completely

25. An association in which BOTH living things benefit is called

- A. predation
- B. parasitism
- C. commensalism
- D. mutualism

- 24.** A plant that grows on a plateau with shallow, stony soil is likely to have
- A. roots that spread out widely near the surface
 - B. no roots at all
 - C. roots that float in water
 - D. one very deep root only

- 27.** An association in which one living thing benefits while the other is neither helped nor harmed is
- mutualism
 - parasitism
 - competition
 - commensalism
- 29.** A tapeworm living in the intestine of a goat shows
- mutualism
 - competition
 - commensalism
 - parasitism
- 31.** Which of the following is a parasite of a flowering plant?
- an earthworm in the soil below the tree
 - a bee resting on a flower
 - mistletoe growing on the branch of a tree
 - a bird building a nest in the branches
- 33.** A lion hunting and killing an antelope is BEST described as
- predation
 - mutualism
 - commensalism
 - parasitism
- 35.** Two goats feeding on the same small patch of grass are said to be in
- parasitism
 - competition
 - mutualism
 - commensalism
- 37.** Birds flying from one region to another when the dry season comes are showing
- migration
 - camouflage
 - germination
 - parasitism
- 26.** In parasitism, the parasite
- is harmed while the host benefits
 - and the host are both harmed
 - benefits while the host is harmed
 - and the host both benefit equally
- 28.** A bee visiting a flower for nectar and carrying pollen to the next flower is an example of
- predation
 - mutualism
 - parasitism
 - commensalism
- 30.** A cattle egret that feeds on the insects disturbed by a grazing cow, without affecting the cow, shows
- mutualism
 - commensalism
 - parasitism
 - predation
- 32.** Root nodule bacteria living in the roots of a groundnut plant supply the plant with nitrogen and receive food from it. This is
- commensalism
 - a food chain
 - mutualism
 - parasitism
- 34.** The place where a plant or an animal naturally lives is its
- habitat
 - population
 - adaptation
 - association
- 36.** A chameleon changing its colour to match the branch it rests on is an example of
- camouflage
 - pollination
 - hibernation
 - migration
- 38.** A pupil studying the animals of a nearby valley should FIRST
- copy the names of animals from a friend
 - write the report before the visit
 - visit the valley and observe and

- record the animals seen there
- D. collect money from the class

- 39.** Which of the following is the BEST record to keep in an album of the plants and animals of a landform?
- A. a labelled picture or drawing of each one, with the place and date it was seen
 - B. the price of each animal in the

market

- C. a list of names only, without any detail
- D. the names of the pupils in the class only

40. If the trees of a valley are all cut down, the animals living there will MOST likely

- A. change into plants
- B. grow larger and multiply quickly
- C. lose their shelter and food and move away or die
- D. remain exactly as they were

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 19: Waste Management

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. Waste is BEST described as
 - A. any material that is sold in the market
 - B. any material that is buried in the soil
 - C. any material that is no longer wanted by the person who produced it
 - D. only the rubbish swept from a room
2. Peels, leftover food and dry leaves are examples of
 - A. metallic waste
 - B. electronic waste
 - C. organic (biodegradable) waste
 - D. chemical waste
3. Water sachets, plastic bottles and rubber bags are examples of
 - A. organic waste
 - B. metallic waste
 - C. glass waste
 - D. plastic waste
4. Empty milk tins and old roofing sheets are examples of
 - A. liquid waste
 - B. paper waste
 - C. plastic waste
 - D. metallic waste
5. Used needles, soiled bandages and expired drugs from a clinic are classified as
 - A. paper waste
 - B. domestic waste
 - C. medical waste
 - D. agricultural waste
6. Dirty water from a bathroom and a kitchen is an example of
 - A. gaseous waste
 - B. toxic metallic waste
 - C. solid waste
 - D. liquid waste
7. Smoke from a lorry and fumes from a factory chimney are examples of
 - A. gaseous waste
 - B. liquid waste
 - C. solid waste
 - D. organic waste
8. Spoilt mobile phones, old radios and broken television sets are known as
 - A. medical waste
 - B. electronic waste
 - C. agricultural waste
 - D. paper waste
9. Biodegradable waste is waste that
 - A. cannot be broken down at all
 - B. can be burnt in an open fire
 - C. can be broken down by micro-organisms into simple substances
 - D. can be sold to a recycling factory
10. Which of the following wastes is NON-biodegradable?
 - A. A sheet of old newspaper
 - B. A plastic water sachet
 - C. A banana peel
 - D. Maize husks
11. The organisms chiefly responsible for

breaking down biodegradable waste
are

- A. birds and bats
- B. bacteria and fungi
- C. fish and frogs
- D. grasshoppers and bees

12. A plastic bottle buried in the soil is still whole after twenty years because plastic

- A. absorbs water from the soil
- B. is a living material
- C. is heavier than soil
- D. is not broken down by

micro-organisms

- 13.** Waste that rots and gives off a bad smell is MOST likely to be
- A. food remains and other organic waste
 - B. broken glass
 - C. aluminium cans
 - D. pieces of rubber tyre
- 15.** Choked gutters filled with plastic waste in a town lead directly to
- A. an increase in soil fertility
 - B. flooding when it rains
 - C. a fall in the price of plastics
 - D. cleaner drinking water
- 17.** The three Rs of good waste management are
- A. remove, rinse and rest
 - B. reduce, reuse and recycle
 - C. read, record and report
 - D. repair, rebuild and resell
- 19.** Using an empty tin as a container for seedlings is an example of
- A. recycling waste
 - B. reducing waste
 - C. dumping waste
 - D. reusing waste
- 21.** Separating waste into different containers at the point where it is produced is called
- A. waste recycling
 - B. waste burning
 - C. waste collection
 - D. waste segregation (sorting)
- 23.** Household refuse kept in a covered container is better than refuse left in an open pan because the cover
- A. keeps out flies and prevents bad smells from spreading
 - B. makes the refuse rot faster
 - C. turns the refuse into compost at once
 - D. adds weight to the refuse
- 14.** Heaps of rotting refuse near homes are dangerous mainly because they
- A. attract rain to the area
 - B. lower the temperature of the air
 - C. make the soil too rich for crops
 - D. breed houseflies, mosquitoes and rats that spread disease
- 16.** Burning rubber and plastic waste in the open is discouraged because it
- A. cools the surrounding air
 - B. destroys the waste completely with no effect
 - C. releases poisonous gases and smoke into the air
 - D. makes the plastics biodegradable
- 18.** Carrying a basket to the market instead of collecting several plastic bags is an example of
- A. recycling waste
 - B. burying waste
 - C. reducing waste
 - D. burning waste
- 20.** Collecting plastic bottles and sending them to a factory where they are melted and made into new articles is
- A. recycling
 - B. incineration
 - C. composting
 - D. reusing
- 22.** The MAIN advantage of sorting waste at home is that it
- A. makes it easier to compost, reuse or recycle each kind of waste
 - B. removes the need for dustbins
 - C. makes the waste disappear
 - D. increases the total amount of waste
- 24.** Treating waste in a properly designed incinerator is better than open burning because the incinerator
- A. uses no fuel at all
 - B. burns the waste at a high temperature and controls the gases released
 - C. turns all the waste into drinking water

D. produces no ash

- 25.** Which of the following is the BEST method of disposing of broken glass at home?
- Wrapping it securely and placing it in a bin for hard waste
 - Putting it loose into a sack of food waste
 - Burying it in the vegetable bed
 - Throwing it into the gutter
- 27.** The materials MOST suitable for making compost are
- old batteries and spoilt phones
 - grass cuttings, peels, dry leaves and animal droppings
 - tins and broken glass
 - water sachets and plastic bottles
- 29.** A compost heap is turned over from time to time mainly to
- allow air to reach all parts of the heap
 - remove the manure formed
 - make the heap look tidy
 - stop the micro-organisms from working
- 31.** Compost added to garden soil is useful because it
- kills all the organisms in the soil
 - turns the soil into sand
 - supplies plant nutrients and improves the structure of the soil
 - makes the soil hold no water
- 33.** Dumping refuse into a stream used by a village is dangerous because it
- pollutes the water and can cause diseases such as cholera
 - increases the number of fish
 - purifies the water naturally
 - makes the stream flow faster
- 35.** Mosquitoes that spread malaria breed MOST easily in
- water collected in discarded tins and old tyres
 - dry heaps of sand
 - burning rubbish
 - swept and clean compounds
- 26.** Composting is the process by which
- waste water is boiled to make it pure
 - plastic waste is melted into new bottles
 - organic waste is allowed to decay to form manure
 - metal waste is beaten into new tools
- 28.** Water is sprinkled on a compost heap in order to
- wash away the bad smell
 - cool the heap and stop decay
 - keep it moist so that the micro-organisms can act
 - make the heap heavier
- 30.** The warmth felt inside a compost heap is evidence that
- the Sun has heated the whole heap
 - micro-organisms are active and respiration is taking place
 - water is freezing inside the heap
 - the waste is burning with a flame
- 32.** The use of a compost pit to treat kitchen waste shows that waste can be
- changed into plastic
 - stored for ever without change
 - made to disappear completely
 - turned into a useful resource
- 34.** A common disease spread by houseflies that breed on uncovered refuse is
- sickle-cell disease
 - typhoid fever
 - night blindness
 - ringworm
- 36.** Before carrying out a clean-up project in the community, pupils should FIRST
- research and plan the work and obtain permission from the authorities
 - begin by burning all the waste they find

- C. clean only their own homes
- D. collect money from passers-by

37. The MOST suitable protective items for pupils taking part in a clean-up

exercise are

- A. sunglasses and wrist watches
- B. hand gloves, nose masks and boots
- C. school bags and books
- D. sandals and short sleeves

39. The BEST way for a school to keep its compound permanently clean is to

- A. burn all waste at the back of the classrooms
- B. ask pupils to carry their waste home every day
- C. provide labelled bins and agree on a routine for emptying and sorting them
- D. sweep only when visitors are expected

38. At the end of a waste management project, the pupils should

- A. keep the findings a secret
- B. forget about the project at once
- C. repeat exactly the same work the next day
- D. write and present a report on what they did and found

40. A community that practises good waste management will MOST likely experience

- A. more flooding in the rainy season
- B. an increase in malaria cases
- C. a greater number of refuse heaps
- D. fewer diseases and a cleaner environment

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 20: Water Cycle

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. Which of the following is a natural source of water?
 - A. A water tanker
 - B. A sachet of water
 - C. A tap
 - D. A river
2. The largest natural store of water on the earth is
 - A. the sea
 - B. the stream
 - C. the well
 - D. the lagoon
3. Water that collects under the ground and is reached by digging a well is called
 - A. rain water
 - B. sea water
 - C. ground water
 - D. pipe water
4. Lake Bosomtwe, the Volta Lake and the Pra River are all examples of
 - A. underground streams
 - B. sources of sea water
 - C. man-made reservoirs only
 - D. surface water bodies
5. Which of the following best explains why the water in the sea cannot be drunk straight away?
 - A. It contains a large amount of dissolved salt
 - B. It never evaporates
 - C. It is always too cold
 - D. It contains no dissolved substances
6. The change of water from the liquid state into water vapour is called
 - A. precipitation
 - B. evaporation
 - C. condensation
 - D. melting
7. The main source of the energy that drives the water cycle is
 - A. the moon
 - B. the wind
 - C. the soil
 - D. the sun
8. Clothes spread on a line in the open dry quickly because the water in them
 - A. condenses on the cloth
 - B. freezes in the air
 - C. evaporates into the air
 - D. sinks into the ground
9. The loss of water vapour from the leaves of plants to the atmosphere is called
 - A. respiration
 - B. transpiration
 - C. condensation
 - D. germination
10. Which of the following conditions will make water in an open pan evaporate fastest?
 - A. A shaded, humid evening
 - B. A cool, still night
 - C. A hot, windy afternoon
 - D. A cold, rainy morning
11. Two equal volumes of water were poured into a wide basin and a

narrow bottle and left in the sun. The water in the basin disappeared faster because

- A. it had a larger surface exposed to the air
- B. it was heavier than the water in the bottle
- C. it was colder than the water in the bottle
- D. it contained more dissolved air

12. During evaporation from the sea, the salt in the sea water

- A. is left behind in the sea
- B. falls back as salty rain
- C. rises with the water vapour

- D. changes into water vapour
- 14.** The change of water vapour into tiny drops of liquid water is called
- A. evaporation
 - B. transpiration
 - C. precipitation
 - D. condensation
- 15.** A farmer at Techiman covered the soil around his young plants with dry grass. This practice mainly reduces
- A. condensation in the air
 - B. transpiration from the leaves
 - C. precipitation over the farm
 - D. evaporation of water from the soil
- 16.** Clouds are made up of
- A. pure oxygen gas
 - B. tiny drops of water and ice crystals
 - C. smoke from factories
 - D. dry dust from the desert
- 17.** Water vapour rising into the atmosphere condenses mainly because the air above
- A. becomes warmer with height
 - B. moves faster with height
 - C. becomes cooler with height
 - D. contains no dust
- 18.** Dew that is seen on grass early in the morning is formed by
- A. condensation of water vapour on the cool grass
 - B. light rain that falls at dawn
 - C. evaporation of water from the soil
 - D. transpiration by the grass at night
- 19.** Drops of water form on the outside of a cold bottle of mineral water left on a table because
- A. the glass melts a little
 - B. water leaks out through the glass
 - C. the water inside evaporates through the glass
 - D. water vapour in the air condenses on the cold surface
- 20.** In the water cycle, condensation takes place
- A. on the surface of the sea
 - B. high in the atmosphere where the air is cool
 - C. inside the roots of plants
 - D. deep inside the soil
- 21.** Water that falls from clouds to the earth as rain, hail or snow is known as
- A. transpiration
 - B. precipitation
 - C. evaporation
- 22.** Which process is the direct opposite of evaporation?
- A. Condensation
 - B. Precipitation
 - C. Infiltration
 - D. Transpiration

D. condensation

23. The commonest form of precipitation in Ghana is

- A. hail
- B. rain
- C. sleet
- D. snow

25. The movement of rain water downwards into the soil is called

- A. evaporation
- B. transpiration
- C. infiltration
- D. precipitation

22. Rain falls from a cloud when the water drops in the cloud

- A. join together and become too heavy to stay in the air
- B. become warmer than the air around them
- C. change back into water vapour
- D. are pushed up by the wind

24. Rain water that flows over the surface of the land into streams and rivers is called

- A. condensation
- B. transpiration
- C. run-off
- D. evaporation

26. After a heavy downpour at Kumasi, most of the rain water that fell on the tarred roads

- A. soaked into the tar
- B. was taken up by the roads as

- ground water
- C. evaporated immediately
- D. ran off quickly into gutters and streams
- 27.** Flooding is common in towns where many open spaces have been covered with concrete, mainly because
- less water soaks into the ground, so run-off increases
 - plants take up less water from the air
 - more water evaporates from the concrete
 - concrete causes heavier rainfall
- 28.** The correct order of the stages of the water cycle is
- collection, precipitation, condensation, evaporation
 - evaporation, condensation, precipitation, collection
 - condensation, evaporation, collection, precipitation
 - precipitation, evaporation, condensation, collection
- 29.** The water cycle is described as a repeated pattern of change in nature because
- it happens only in the rainy season
 - it takes place once in every year
 - new water is made each time it rains
 - the same stages follow one another again and again without ending
- 30.** The total amount of water on the earth
- decreases each time the sun shines
 - remains about the same, as the water is used over and over again
 - increases each time it rains
 - is made fresh by plants each year
- 31.** Which of the following returns water from the land back to the sea, completing the cycle?
- Rivers and streams
 - Wind
 - Dew
 - Clouds
- 32.** A pupil drew a flow chart of the water cycle but ended it at precipitation. The chart is incomplete because it fails to show that
- clouds are made of water drops
 - rain water is always clean
 - the water returns to the sources from which it evaporates again
 - plants also lose water
- 33.** The water that a pupil drinks today may once have been part of a river that flowed many years ago. This is possible because
- fresh water is created by rainfall
 - water does not change state

- C. water is used again and again in the water cycle
 - D. rivers never dry up
- 35.** As water vapour condenses in the atmosphere, energy is released. This released energy helps to
- A. form salt in the air
 - B. increase run-off on land
 - C. warm the surrounding environment
 - D. cool the sea water
- 37.** Water moving through the cycle is described as a carrier of nutrients because it
- A. stores nutrients only in clouds
 - B. destroys the nutrients it meets
 - C. turns mineral salts into gas
 - D. dissolves mineral salts and carries them to soils, rivers and living things
- 34.** The water cycle is important to farmers mainly because it
- A. removes weeds from the farm
 - B. provides rain for the growth of crops
 - C. makes the soil harder
 - D. provides salt for the soil
- 36.** The water cycle raises the water table of an area by
- A. allowing rain water to soak into the ground and collect below the surface
 - B. keeping the soil surface dry
 - C. carrying water away in rivers
 - D. increasing evaporation from the sea

39. A town has cut down the trees on the hills around it and the rains have become less reliable. The most likely reason is that

- A. the trees used to block the clouds
- B. the soil now holds too much water
- C. trees prevent water from evaporating from rivers
- D. with fewer trees there is less transpiration, so less water vapour enters the air

38. Which of the following best explains how the water cycle helps to provide clean water for a community?

- A. Water is boiled in the clouds before it falls
- B. Water that evaporates leaves its dissolved salts and dirt behind and returns as fairly pure rain
- C. Clouds filter out the water and keep the clean part
- D. Rain water collects salt from the air as it falls

40. A community wants to reduce the flooding that follows every heavy rain while also raising its water table. The BEST action to take is to

- A. pump the flood water into the nearest stream each time
- B. build higher walls round the houses
- C. plant trees and grasses and keep open ground so that rain water soaks in
- D. pave all the open ground with concrete

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram, a flow chart or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

PART 2

Diagram Practice Papers

This section contains 22 practical papers focused on diagram interpretation and analysis. Each paper includes diagrams with questions that test students' ability to study, interpret, and explain scientific diagrams.

Diagram Paper 1: Agricultural Tools and

Time Allowed: ** 1 hour 30 minutes

INSTRUCTIONS TO CANDIDATES

1. Answer ALL the questions in this paper.
2. Every question is based on a diagram. Study each diagram carefully before you answer the questions that follow it.
3. Write your answers in the spaces provided. The marks for each question are printed beside it.
4. Where a question asks for a stated number of points, give that number. Any extra answers will not be marked.
5. You may refer back to any diagram at any time.

Basic and Simple Agricultural Tools

CUTTING TOOLS				
CUTLASS	HOE	AXE	SICKLE	SHEARS
				
				
				
clearing	cutting	felling	reaping	pruning
bush	weeds	trees	rice	sheep
MACHETE	SCYTHE	PRUNING	SECATEUR	CHISEL
				
				
sawing	large	thick	fine	shaping
branches	areas of	branches	pruning	wood
	grass			

Fig. 1: Cutting tools

1. Every cutting tool in Fig. 1 is a WEDGE. Explain what this means, and use it to explain why a cutlass with a thin blade cuts through a stem more easily than one with a thick blade. [4 marks]
2. Name any THREE cutting tools shown in the diagram. [3 marks]
3. For each job, name the most suitable cutting tool and give a reason for your choice: (a) clearing thick bush before planting; (b) reaping a field of rice by hand; (c) trimming a hedge into shape. [3 marks]

15.2 15.2 Importance of Proper Handling and Maintenance of Agricultural To

Importance**DIGGING AND BORING TOOLS**

SPADE	FORK	DIBBER	AUGER
			
digging and turning soil	breaking up clods, lifting roots	making a hole for a seed or seedling	boring deep holes for posts

WHICH TOOL FOR WHICH JOB

turning a whole bed over [SPADE]	lifting root crops without cutting [FORK]
planting a single maize seed	deep hole for a fence post

Fig. 2: Digging and boring tools

4. Explain why a GARDEN FORK is better than a SPADE for lifting root crops such as cassava or yam, and explain why a spade is better than a fork for turning a bed over. [4 marks]
5. State the use of a DIBBER and the use of an AUGER. **[3 marks]**
6. A farmer wants to plant maize on a plot where the soil is hard and dry. Suggest which digging tool he should use to open the holes, and explain why a DIBBER would not do. [3 marks]

15.3 15.3 Rules for Handling Agricultural Tools Safely

4. Explain why a GARDEN FORK is better than a SPADE for lifting root crops such as cassava or yam, and explain why a spade is better than a fork for turning a bed over. [4 marks]
5. State the use of a DIBBER and the use of an AUGER. [3 marks]
6. A farmer wants to plant maize on a plot where the soil is hard and dry. Suggest which digging tool he should use to open the holes, and explain why a DIBBER would not do. [3 marks]

Rules for Handling Agricultural Tools Safely**SIMPLE FARM TOOLS AND MACHINERY**

WHEELBARROW	KNAPSACK SPRAYER	PLOUGH	HARROW
			
SEED DRILL	TRACTOR	COMBINE HARVESTER	SHELLER
			

TOOL = handheld, worked by muscle
 MACHINE = has moving parts, driven by an engine, an animal or a motor

*Fig. 3: Simple

7. The diagram separates TOOLS from MACHINES. Explain the difference, and say which of the following are tools and which are machines, giving a reason each time: a hoe, a knapsack sprayer, a tractor-drawn plough, a combine harvester. [4 marks]
8. Give TWO reasons why a farmer might use a tractor-drawn plough rather than a hand hoe. [3 marks]
9. Suggest which item from Fig. 3 a farmer should use for each of the following, and give a reason each time: (a) moving a heap of compost to the far end of the field; (b) applying weedkiller to a maize field; (c) harvesting and threshing a large area of rice. [3 marks]

15.4 15.4 Maintaining Agricultural Tools in the Community

7. The diagram separates TOOLS from MACHINES. Explain the difference, and say which of the following are tools and which are machines, giving a reason each time: a hoe, a knapsack sprayer, a tractor-drawn plough, a combine harvester. [4 marks]
8. Give TWO reasons why a farmer might use a tractor-drawn plough rather than a hand hoe. [3 marks]
9. Suggest which item from Fig. 3 a farmer should use for each of the following, and give a reason each time: (a) moving a heap of compost to the far end of the field; (b) applying weedkiller to a maize field; (c) harvesting and threshing a large area of rice. [3 marks]

Maintaining Agricultural Tools in the Community

CARE AND MAINTENANCE					
WHY TOOLS FAIL			THE ROUTINE		
RUST	left wet		1	CLEAN	after use
BLUNT	edge		2	DRY	thoroughly
	rolled		3	SHARPEN	the edge
LOOSE	wood		4	OIL	the metal
HANDLE	shrank		5	STORE	dry, hung
SPLIT	levered			up off the floor	
SHAFT	too hard				
SAFETY - AND WHY IT IS NOT OPTIONAL					
cut AWAY	wear		never		
from the	boots and		leave a		
body	gloves		tool in		
			the grass		

Fig. 4: Care and maintenance

10. Explain why iron tools rust, and explain why BOTH drying and oiling are need
11. State THREE things that should be done to a hand tool after use. **[3 mark]
12. The head of a cutlass is loose and is likely to come off. Describe what should be done.
- ## SECTION B - PRACTICAL TASK

DOES A SHARP BLADE REALLY CUT BETTER?					
THE TEST RIG			THE RESULTS		
handle			BLADE	1st	2nd
				3rd	MEAN
		blade	sharp	3.0	3.4
				3.2	3.2 N
[N] === stem			blunt	11.5	12.0
<--pull				11.6	11.7 N
[block]					
=====			the sharp blade		
the stem to be cut			needed about a		
			QUARTER of the		
			force		

WHY: a sharp edge is very thin, so the push is spread over a tiny area and the PRESSURE is enormous. A blunt edge spreads the same push wide, so the pressure is lower and the stem is crushed rather than cut.

SAFETY: cut away from the body, keep the other hand

13. Use the results to explain why the SHARP blade needed so much less force, re
 14. Explain why each cut was repeated THREE times and a mean taken, and suggest
 15. Suggest TWO safety precautions the pupils must take, and explain what accide

 ## ANSWER KEY AND MARKING SCHEME
 15.1 15.1 Basic and Simple Agricultural Tools
 - **Q1** (4 marks) A wedge is two sloping faces meeting at a thin edge, and a cu
 - **Q2** (3 marks) Any three of: cutlass, hoe, axe, sickle, shears, machete saw,
 - **Q3** (3 marks) (a) A cutlass (or machete). It has a long heavy blade that wi
 **15.2 15.2 Importance of Proper Handling and Maintenance of Agricultural Tools

*Fig. 5: Does a sha

- 10.** Explain why iron tools rust, and explain why BOTH drying and oiling are needed rather than just one of them. [4 marks]
- 11.** State THREE things that should be done to a hand tool after use. [3 marks]
- 12.** The head of a cutlass is loose and is likely to come off. Describe what should be done about it, and explain why working with it in that state is dangerous. [3 marks]
- 13.** Use the results to explain why the SHARP blade needed so much less force, referring to PRESSURE. [4 marks]
- 14.** Explain why each cut was repeated THREE times and a mean taken, and suggest ONE other thing the pupils must keep the same for the comparison to be fair. [3 marks]
- 15.** Suggest TWO safety precautions the pupils must take, and explain what accident each one prevents. [3 marks]

Diagram Paper 2: Animal Production

Time Allowed: ** 1 hour 30 minutes

INSTRUCTIONS TO CANDIDATES

1. Answer ALL the questions in this paper.
2. Every question is based on a diagram. Study each diagram carefully before you answer the questions that follow it.
3. Write your answers in the spaces provided. The marks for each question are printed beside it.
4. Where a question asks for a stated number of points, give that number. Any extra answers will not be marked.
5. You may refer back to any diagram at any time.

Types of Domestic Animals in Ghana

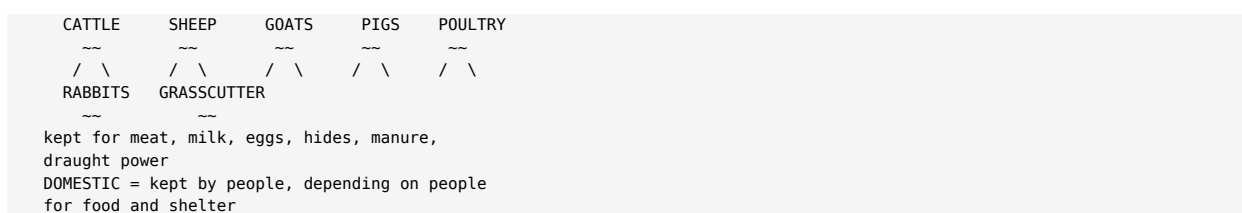


Fig. 1: Domestic animals kept in Ghana

1. The diagram calls all these animals DOMESTIC. Explain what makes an animal domestic.
2. From the diagram, state TWO ways in which POULTRY differ from the other animals shown.
3. From the diagram, name TWO animals kept mainly for MILK and TWO kept mainly for MEAT.

1. The diagram calls all these animals DOMESTIC. Explain what makes an animal domestic, and use your answer to explain why a grasscutter kept in a cage is domestic while one living wild in the bush is not. [4 marks]
2. From the diagram, state TWO ways in which POULTRY differ from the other animals shown. [3 marks]
3. From the diagram, name TWO animals kept mainly for MILK and TWO kept mainly for MEAT. [2 marks]

Classifying Domestic Animals: Ruminants, Monogastrics and Poultry

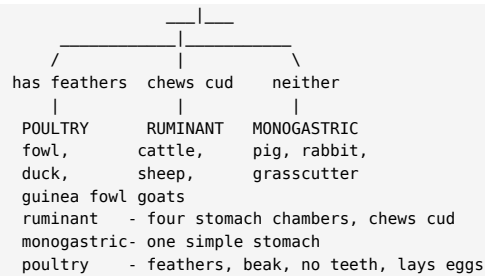


Fig. 2: Sorting domestic animals by their features

4. Use the key to explain what 'chews the cud' means, and explain why doing it p

5. A farmer has just bought an animal. It has hooves, but it does not chew the c

6.3 6.3 Differences and Similarities Among Ruminants, Monogastrics and Pou

4. Use the key to explain what 'chews the cud' means, and explain why doing it puts an animal in the ruminant group. [3 marks]
5. A farmer has just bought an animal. It has hooves, but it does not chew the cud. Use the key to say which group it belongs to, and name one animal it could be. [3 marks]

Differences and Similarities Among Ruminants, Monogastrics and Poultry

RUMINANT - four chambers food brought back up to chew the cud <-- [rumen] [reticulum] [omasum] [abomasum]	MONOGASTRIC - one [one simple stomach] pigs, rabbits, poultry, grasscutter
cattle, sheep, goats	
a ruminant can digest grass and leaves	
a monogastric cannot	

Fig. 3: The two kinds of stomach compared

6. Use the diagrams to explain why a ruminant such as a cow can live on grass, w

7. A very young calf lives mainly on its mother's milk and cannot digest grass w

6.4 6.4 Domestic and Commercial Uses of Animals

6. Use the diagrams to explain why a ruminant such as a cow can live on grass, while a pig cannot. [4 marks]
7. A very young calf lives mainly on its mother's milk and cannot digest grass well. Suggest why, using the diagram. [3 marks]

Domestic and Commercial Uses of Animals

DOMESTIC ANIMALS					
MEAT	MILK	EGGS	DRAUGHT POWER	MANURE	HIDES & SKINS
meat					
milk					
eggs					
draught power					
manure					
hides & skins					

food - protein for family and sale

food - fresh, soured, or cheese

food - from poultry, sold for cash

work - ploughing and carting

the soil - dung feeds crops

clothing - leather for shoes, bags,

drums

8. A smallholder farmer has no tractor and cannot afford to hire one. State which use of animals from the diagram matters most to him, and explain why. [4 marks]

9. A pupil says the most important use of animals is MEAT. Explain why this is too narrow a way of looking at it for a smallholder farmer. [3 marks]

6.5 6.5 By-Products of Animals and Their Importance

*Fig. 4: What w

8. A smallholder farmer has no tractor and cannot afford to hire one. State which use of animals from the diagram matters most to him, and explain why. [4 marks]
9. A pupil says the most important use of animals is MEAT. Explain why this is too narrow a way of looking at it for a smallholder farmer. [3 marks]

By-Products of Animals and Their Importance

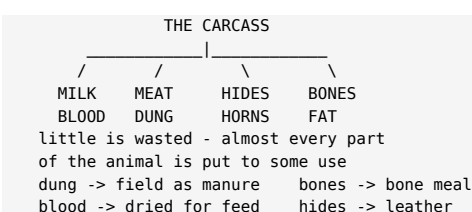


Fig. 5: By-products from one cow

10. The diagram lists DUNG among the by-products. Explain why, on a mixed farm, dung is a by-product undervalues it. [4 marks]

11. Give TWO things that hides and skins are used for, and explain what has to happen to a hide before it can be used. [3 marks]

SECTION B - PRACTICAL TASK

A	B	C	D			
COW	SHEEP	PIG	FOWL			
FEATURE			A	B	C	D
has feathers			no	no	no	YES
has hooves		YES	YES	YES	YES	no
chews the cud		YES	YES	YES	no	no
has horns		YES	no	no	no	no
a feature that only ONE animal has is the one that identifies it						

Fig. 6: Using a key to identify four animals

12. Use the table to identify animals A, B, C and D, giving the reason in each case. [4 marks]

13. Explain why 'has hooves' is a poor feature for identifying these animals, while 'has horns' is a good one. [4 marks]

14. A pupil adds a fifth animal, a GOAT, to the table. State what you would expect in its column, and explain what problem this would cause. [3 marks]

15. State what a DICHOTOMOUS KEY is, and explain why each step in it offers only one choice. [3 marks]

ANSWER KEY AND MARKING SCHEME

6.1 6.1 Types of Domestic Animals in Ghana

- **Q1** (4 marks) An animal is domestic when people keep it, feed it, house it

- **Q2** (3 marks) Any two, each stated: poultry have feathers, while the others have horns

- **Q3** (2 marks) Milk: cattle, goats (sheep are also correct). Meat: cattle, sheep

10. The diagram lists DUNG among the by-products. Explain why, on a mixed farm, calling dung a by-product undervalues it. [4 marks]
11. Give TWO things that hides and skins are used for, and explain what has to happen to a hide before it can be used. [3 marks]
12. Use the table to identify animals A, B, C and D, giving the reason in each case. [4 marks]
13. Explain why 'has hooves' is a poor feature for identifying these animals, while 'has horns' is a good one. [4 marks]
14. A pupil adds a fifth animal, a GOAT, to the table. State what you would expect in its column, and explain what problem this would cause. [3 marks]

15.

Diagram Paper 3: Climate Change

Time Allowed: ** 1 hour 30 minutes

INSTRUCTIONS TO CANDIDATES

1. Answer ALL the questions in this paper.
2. Every question is based on a diagram. Study each diagram carefully before you answer the questions that follow it.
3. Write your answers in the spaces provided. The marks for each question are printed beside it.
4. Where a question asks for a stated number of points, give that number. Any extra answers will not be marked.
5. You may refer back to any diagram at any time.

What is Sustainable Energy?

WEATHER AND CLIMATE ARE NOT THE SAME	
WEATHER	CLIMATE
+-----+	+-----+
the state of the air	the average pattern
right now	over many years
Mon Tue Wed Thu	/
sun rain cloud storm	/
* ~ 0 %	/
	/
changes hour to hour -	1990 2020
forecast only a few	
days ahead	measured over 30 years
	or more
+-----+	+-----+

the test: one hot day is WEATHER.
Thirty years of hot years is CLIMATE.
a single flood proves nothing - the RECORD of many
years does

Fig. 1: Weather and climate are not the same

1. Explain the difference between WEATHER and CLIMATE. A pupil says that because it rained heavily in Accra last Tuesday, the climate must be changing. Explain why this does not follow. [4 marks]
2. State what is meant by WEATHER and by CLIMATE. **[3 marks]**
3. A farmer wants to know what to plant this season. Explain why knowing the CLIMATE of his area is more useful to him than knowing tomorrow's WEATHER. [3 marks]

19.2 19.2 Greenhouse Effects on the Environment

1. Explain the difference between WEATHER and CLIMATE. A pupil says that because it rained heavily in Accra last Tuesday, the climate must be changing. Explain why this does not follow. [4 marks]
2. State what is meant by WEATHER and by CLIMATE. [3 marks]
3. A farmer wants to know what to plant this season. Explain why knowing the CLIMATE of his area is more useful to him than knowing tomorrow's WEATHER. [3 marks]

Greenhouse

HOW WE KNOW IT IS CHANGING

```

+-----+ +-----+ +-----+ +-----+
| TEMPERATURE | MELTING | SEA LEVEL | CHANGING |
| RECORDS      | ICE       | RISING   | SEASONS  |
| thermo-      | glaciers | measured | rains come |
| meters for   | retreating, | by tide  | late,    |
| over a      | Arctic sea | gauges and | harmattan |
| century     | shrinking | satellite | shorter  |
+-----+ +-----+ +-----+ +-----+

```

WHAT THIS LOOKS LIKE IN GHANA

```

+-----+ +-----+ +-----+ +-----+
| the rains | the | FLOODS and | the sea |
| come LATER | HARMATTAN | DROUGHTS | is eating |
| than they | is shorter | are both | the coast |
| did      | | commoner | at Keta, |
|          | |         | Ada     |
+-----+ +-----+ +-----+ +-----+
cocoa and maize yields are less predictable

```

Fig. 2: How we know it is changing

4. Explain why scientists need several DIFFERENT kinds of evidence before they will say the climate is changing, and describe TWO of those kinds of evidence. [4 marks]
5. State THREE pieces of evidence that the climate is changing. **[3 marks]**
6. The rains now arrive in your district about three weeks later than they did thirty years ago. Suggest TWO ways a farmer could check whether this is really happening rather than just remembering it differently. [3 marks]

19.3 Sustainable Energy Choices and Scientific Ideas

4. Explain why scientists need several DIFFERENT kinds of evidence before they will say the climate is changing, and describe TWO of those kinds of evidence. [4 marks]
5. State THREE pieces of evidence that the climate is changing. [3 marks]
6. The rains now arrive in your district about three weeks later than they did thirty years ago. Suggest TWO ways a farmer could check whether this is really happening rather than just remembering it differently. [3 marks]

Sustainable Energy Choices and Scientific Ideas

THE GREENHOUSE EFFECT

```

* sun
| \ short-wave radiation from the sun
| \ passes straight through
v v v v
===== the earth's surface
^ ^ ^ ^
| / long-wave (infrared) heat tries to go back out
|/ greenhouse gases absorb it and send some back down
+-----+ +-----+ +-----+ +-----+
| CARBON DIOXIDE | burning coal, oil and gas; |
|                | cutting and burning forest |
| METHANE        | rice paddies, cattle, rotting |
|                | waste in dumps |
| NITROUS OXIDE  | artificial fertiliser, |
|                | burning crop residue |
| WATER VAPOUR   | a feed-back: warmer air holds |
|                | more of it |
+-----+ +-----+ +-----+ +-----+

```

*Fig. 3: The gr

7. Explain how the greenhouse effect works, and explain why it is wrong to say that the greenhouse effect is itself a bad thing. [4 marks]

8. Name TWO human activities that add carbon dioxide to the air, and explain for each how it does so. [3 marks]

9. Name THREE greenhouse gases. [3 marks]

19.4 19.4 Designing Local Energy Solutions

7. Explain how the greenhouse effect works, and explain why it is wrong to say that the greenhouse effect is itself a bad thing. [4 marks]
8. Name TWO human activities that add carbon dioxide to the air, and explain for each how it does so. [3 marks]
9. Name THREE greenhouse gases. [3 marks]

Designing Local Energy Solutions

THE TWO THINGS WE CAN DO		ADAPTATION - live with what is here	
MITIGATION - stop making it worse			
plant trees,	a tree	plant	so a bad
stop burning	takes	DROUGHT-	season
the forest	CO2 out	RESISTANT	is not a
		varieties	famine
use less	solar		
electricity,	and wind	harvest	store it
make it	put out	RAINWATER	against
without coal	no CO2		dry spell
walk, cycle,	one bus	plant when	not by
use a bus	replaces	the rains	the old
	30 cars	actually come	calendar
grow food	food	build DRAINS	and keep
nearer home	flown	and raise	the
	in has	flood defences	coastal

Fig. 4: The two things we can do

10. Explain the difference between MITIGATION and ADAPTATION, and give ONE example.

11. Explain why a farmer should plant several DIFFERENT crops rather than only one.

12. State TWO effects of climate change on Ghana. [3 marks]

SECTION B - PRACTICAL TASK

THE APPARATUS		THE RESULTS	
* sun			
two identical jars in the sun, one covered with a clear plastic sheet			
		TIME	JAR A JAR B
		open	covered
		0 min	28 C 28 C
		5 min	31 C 34 C
		10 min	33 C 38 C
		15 min	34 C 41 C
		20 min	34 C 43 C
		in shade	falls falls
		fast	slowly
the covered jar ran 9 C hotter, and cooled more slowly			
WHY: sunlight passes through clear plastic and warms the air inside. The warm air tries to radiate heat back out,			

13. From the results, explain why the covered jar got hotter than the open one,
 14. State THREE things the pupils must keep the same for this comparison to be f
 15. The covered jar also COOLED more slowly when the jars were moved into the sh

 ## ANSWER KEY AND MARKING SCHEME
 19.1 19.1 What is Sustainable Energy?
 - **Q1** (4 marks) Weather is the state of the atmosphere at a particular place
 - **Q2** (3 marks) Weather is the condition of the atmosphere at a particular pl
 - **Q3** (3 marks) The weather forecast tells him whether to harvest or spray to
 19.2 19.2 Greenhouse Effects on the Environment

*Fig. 5: Can you tr

- 10.** Explain the difference between MITIGATION and ADAPTATION, and give ONE example of each that a farming family in northern Ghana could actually carry out. [4 marks]
- 11.** Explain why a farmer should plant several DIFFERENT crops rather than only one, as a way of adapting to a changing climate. [3 marks]
- 12.** State TWO effects of climate change on Ghana. [3 marks]
- 13.** From the results, explain why the covered jar got hotter than the open one, and explain carefully how this models the greenhouse effect - and ONE important way in which the model is NOT like the real thing. [4 marks]
- 14.** State THREE things the pupils must keep the same for this comparison to be fair, and explain why each matters. [3 marks]
- 15.** The covered jar also COOLED more slowly when the jars were moved into the shade. Explain why, and suggest ONE improvement that would make the experiment a better test. [3 marks]

Diagram Paper 4: Conservation of

Time Allowed: ** 1 hour 30 minutes

INSTRUCTIONS TO CANDIDATES

1. Answer ALL the questions in this paper.
2. Every question is based on a diagram. Study each diagram carefully before you answer the questions that follow it.
3. Write your answers in the spaces provided. The marks for each question are printed beside it.
4. Where a question asks for a stated number of points, give that number. Any extra answers will not be marked.
5. You may refer back to any diagram at any time.

The Law of Conservation of Energy

ENERGY CHANGING FROM ONE FORM TO ANOTHER

+-----+ +-----+			+-----+ +-----+		
A BALL ROLLING DOWN A		A PERSON LIFTING A LOAD			
SLOPE					
potential kinetic heat		chemical kinetic potent.			
[at top]->[rolling]->[+]		[in food]->[muscle]->[up]			
sound					
+-----+ +-----+			+-----+ +-----+		
A BATTERY LIGHTING A BULB		THE SUN WARMING A ROOF			
chemical electric light		nuclear light heat			
[in cell]->[in wire]->[+]		[in sun]->[travel]->[roof]			
heat		+ heat			
+-----+ +-----+			+-----+ +-----+		

the arrow is not a loss - it is the change of form
energy is never USED UP: it changes form, and at every
change some ends up as heat, which spreads out and can
never be gathered back

Fig. 1: Energy changing from one form to another

1. A ball rolls down a slope. Describe the energy changes that take place, and explain why the ball eventually comes to a stop even though energy cannot be destroyed. [4 marks]
2. Name any THREE forms of energy. **[3 marks]**
3. For each of the following, state the energy change that takes place: (a) a torch being switched on; (b) a piece of charcoal burning; (c) a bowstring being pulled back and then released. [3 marks]

13.2 13.2 Energy Conversion

1. A ball rolls down a slope. Describe the energy changes that take place, and explain why the ball eventually comes to a stop even though energy cannot be destroyed. [4 marks]
2. Name any THREE forms of energy. [3 marks]
3. For each of the following, state the energy change that takes place: (a) a torch being switched on; (b) a piece of charcoal burning; (c) a bowstring being pulled back and then released. [3 marks]

A FLOW CHART FOR A HYDRO-ELECTRIC DAM

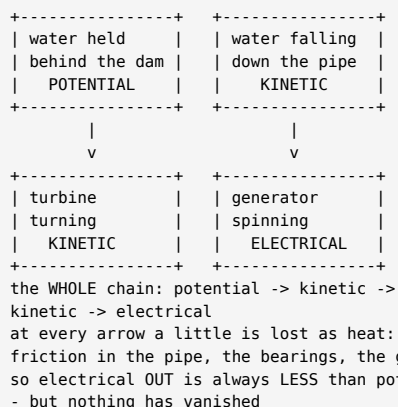


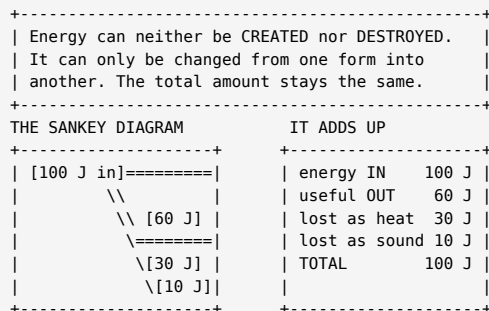
Fig. 2: A flow chart for a hydro-electric dam

4. Describe the energy flow chart for a hydro-electric dam, naming the form of energy at each stage. [4 marks]
5. Explain why the ELECTRICAL energy produced by a hydro-electric dam is always less than the POTENTIAL energy stored in the water behind it, and say what has happened to the difference. [3 marks]
6. State what is meant by the EFFICIENCY of an energy conversion. [3 marks]

13.3 13.3 Everyday Examples of Energy Conversion

Everyday Examples of Energy Conversion

THE LAW OF CONSERVATION OF ENERGY



the width of each band is the amount of energy
nothing has disappeared - the 40 J of heat and
sound is still there, spread out into the air

Fig. 3: The law of conservation of energy

7. State the law of conservation of energy. A pupil says that if energy cannot be destroyed, a car ought to run for ever on one tank of fuel. Explain why the pupil is wrong. [4 marks]
8. A machine takes in 100 J of energy and delivers 60 J of useful energy, 30 J as heat and 10 J as sound. Show that this agrees with the law of conservation of energy. [4 marks]

13.4 13.4 Conserving Energy for Future Use

energy, and calculate the efficiency. [3 marks]

9. State what a SANKEY diagram shows, and what the width of each band represents. [3 marks]

Conserving Energy for Future Use

WHERE OUR ENERGY COMES FROM	
RENEWABLE - replaced as fast as we use it	NON-RENEWABLE - a fixed stock that runs down
+-----+	+-----+
SUN	CRUDE OIL
solar panels,	petrol, diesel,
drying crops	kerosene
+-----+	+-----+
WIND	NATURAL GAS
wind pumps,	cooking, power
wind turbines	stations
+-----+	+-----+
FLOWING WATER	COAL
hydro-electric dams	power stations,
	industry
+-----+	+-----+
BIOMASS	URANIUM
firewood, charcoal,	nuclear power
biogas	stations
+-----+	+-----+

Fig. 4: Where our energy comes from

10. Explain the difference between a RENEWABLE and a NON-RENEWABLE source of energy.

11. A village in the Northern Region needs energy for COOKING and for LIGHTING.

12. Name THREE non-renewable sources of energy, and state one use of each. **[

SECTION B - PRACTICAL TASK

WHERE DOES THE BOUNCE GO? THE METHOD	THE RESULTS
+-----+	+-----+
_	DROP 1st 2nd 3rd
0 <-- drop	100cm 62cm 38 23
	80cm 50cm 31 19
	60cm 37cm 23 14
0 <-- 1st	
	each bounce reaches
	about 62% of the
0 <-- 2nd	height before it
o <-- 3rd	
=====	
hard floor	
+-----+	+-----+

AT THE TOP: all POTENTIAL. ON THE WAY DOWN: KINETIC.
AT THE FLOOR: the ball squashes - kinetic becomes

Fig. 5: Where does the bounce go?

13. Describe the energy changes as the ball falls and bounces, and explain why.

14. From the results, state what fraction of its height the ball reaches after each bounce.

15. State TWO difficulties in measuring the bounce height accurately, and say how you can overcome them.

ANSWER KEY AND MARKING SCHEME

13.1 13.1 The Law of Conservation of Energy

- **Q1** (4 marks) At the top of the slope the ball has gravitational POTENTIAL

- **Q2** (3 marks) Any three of: gravitational potential, kinetic, heat, light,

- **Q3** (3 marks) (a) Chemical energy stored in the cells becomes electrical energy

13.2 13.2 Energy Conversion

10. Explain the difference between a RENEWABLE and a NON-RENEWABLE source of energy, and explain why calling firewood renewable is true only under one condition. [4 marks]

11. A village in the Northern Region needs energy for COOKING and for LIGHTING. Suggest ONE renewable source for each, and explain why each

suits that village. [3 marks]

- 12.** Name THREE non-renewable sources of energy, and state one use of each. [3 marks]
- 13.** Describe the energy changes as the ball falls and bounces, and explain why each bounce is lower than the one before it even though energy is conserved. [4 marks]
- 14.** From the results, state what fraction of its height the ball reaches after each bounce, and use it to calculate the height of the SECOND bounce if the ball is dropped from 40 cm. [3 marks]
- 15.** State TWO difficulties in measuring the bounce height accurately, and say how each one could be reduced. [3 marks]

Diagram Paper 5: Crop Production

Time Allowed: ** 1 hour 30 minutes

INSTRUCTIONS TO CANDIDATES

1. Answer ALL the questions in this paper.
2. Every question is based on a diagram. Study each diagram carefully before you answer the questions that follow it.
3. Write your answers in the spaces provided. The marks for each question are printed beside it.
4. Where a question asks for a stated number of points, give that number. Any extra answers will not be marked.
5. You may refer back to any diagram at any time.

What are Plant Nutrients?

NITROGEN, N --> LEAVES AND SHOOTS
 leaves turn pale yellow, starting with
 the OLDEST leaves; plant is stunted
 PHOSPHORUS, P --> ROOTS, FLOWERS, SEEDS
 roots stay small; few flowers and
 seeds; leaves may go purple
 POTASSIUM, K --> STEM STRENGTH, FRUIT
 leaf edges scorch and curl brown;
 stems weak, lodging common
 the symptom shows in the part the nutrient builds

Fig. 1: What nitrogen, phosphorus and potassium do

1. A farmer's maize has pale yellow leaves, but only the OLDEST leaves are yellow
2. From the diagram, name TWO plant nutrients and state one thing each does for

5.2 5.2 Organic and Inorganic Nutrient Sources

1. A farmer's maize has pale yellow leaves, but only the OLDEST leaves are yellow while the youngest ones at the top are still green. Use the diagram to name the nutrient that is missing, and explain why the symptom appears in the oldest leaves first. [4 marks]
2. From the diagram, name TWO plant nutrients and state one thing each does for the crop. [2 marks]

Organic and Inorganic Nutrient Sources

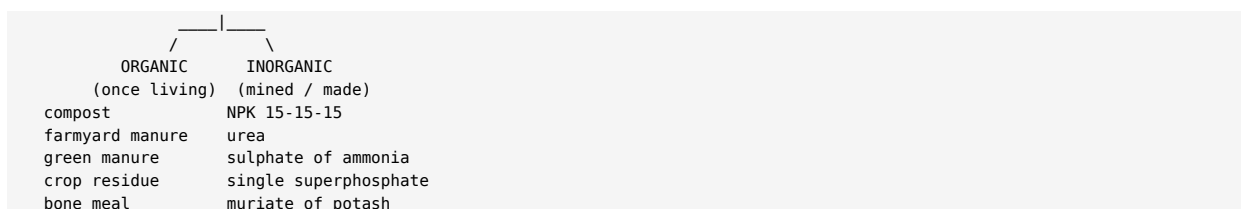


Fig. 2: How plant nutrient sources are classified

3. Urea is made in a factory from ammonia gas. Use the diagram to say whether ur

4. At the end of harvest a farmer digs the old maize stalks and leaves back into

5.3 5.3 Differences Between Organic and Inorganic Fertilizers

3. Urea is made in a factory from ammonia gas. Use the diagram to say whether urea is an ORGANIC or an INORGANIC nutrient source, and explain why. [3 marks]
4. At the end of harvest a farmer digs the old maize stalks and leaves back into the soil instead of burning them. Classify this material using the diagram, and explain why it will not feed the next crop straight away. [3 marks]

Differences Between Organic and Inorganic Fertilizers

	ORGANIC	INORGANIC
where it comes from	dead things	rock / factory
how fast it feeds	slow	fast
nutrients per kg	low	high
effect on soil	improves it	none
how long it lasts	seasons	one season
risk of leaching	little	high
cost and handling	cheap/heavy	costs/easy
a good farmer uses	BOTH	

Fig. 3: Organic against inorganic fertiliser

5. The diagram shows that organic fertiliser improves the soil while inorganic d

6. Heavy rain falls the day after fertiliser is applied to two fields, one given

5.4 5.4 Physical Characteristics of Plant Nutrients

5. The diagram shows that organic fertiliser improves the soil while inorganic does not. Explain why a farmer who can afford both would still choose to apply inorganic fertiliser as well. [4 marks]
6. Heavy rain falls the day after fertiliser is applied to two fields, one given compost and one given NPK. Use the diagram to state which field loses more of its fertiliser, and explain why. [3 marks]

Physical Characteristics of Plant Nutrients

A. COMPOST	B. NPK GRANULES
dark brown to black	grey, white or pinkish
crumbs of all shapes	every granule the same
smells of damp earth	little or no smell
bits of leaf and stem visible	nothing recognisable
irregular = decayed	uniform = manufactured
plant and animal matter	in a factory

*Fig. 4: Telling

7. You are given a fertiliser sample with no label. Describe how you would tell from looking at it whether it is compost or NPK, and explain WHY the two look so different. [4 marks]

8. A pupil says that compost must be better than NPK because it is natural. Use the diagram to explain why 'natural' is not the point. [3 marks]

5.5 5.5 How Plant Nutrients Are Applied in the Field

7. You are given a fertiliser sample with no label. Describe how you would tell from looking at it whether it is compost or NPK, and explain WHY the two look so different. [4 marks]
8. A pupil says that compost must be better than NPK because it is natural. Use the diagram to explain why 'natural' is not the point. [3 marks]

How Plant Nutrients Are Applied in the Field

A. BROADCASTING	scattered over the whole field before planting
B. SIDE-DRESSING	band placed beside the row while the crop grows
C. RING APPLICATION	ring around each plant, not on the stem
D. FOLIAR SPRAY	sprayed onto the leaves as a liquid feed

Fig. 5: Four ways of putting fertiliser on a crop

9. A farmer wants to feed a maize crop that is already knee-high and growing fast. State which method from the diagram is best, and explain why broadcasting would be a poorer choice. [4 marks]

10. In ring application, the diagram shows the fertiliser placed in a ring AROUND the plant rather than against it. Explain why putting it directly against the stem would harm the plant. [3 marks]

5.6 5.6 Practical Application in the School Garden

9. A farmer wants to feed a maize crop that is already knee-high and growing fast. State which method from the diagram is best, and explain why broadcasting would be a poorer choice. [4 marks]
10. In ring application, the diagram shows the fertiliser placed in a ring AROUND the plant rather than against it. Explain why putting it directly against the stem would harm the plant. [3 marks]

Practical Application in the School Garden

20 m		
Bed 1	: Bed 2	: Bed 3
maize	: tomato	: groundnut
+ NPK	: + compost	: (no fertiliser)
..... path		
Bed 4	: Bed 5	: Bed 6
pepper	: okra	: NURSERY
+ NPK	: + compost	:
12 m		
spacing: maize 75 cm between plants, 90 cm between rows		

Fig. 6: Plan of the school garden plot

11. The plan leaves a path between every bed. Explain why the paths are needed,

12. The plan gives NPK to Bed 1 and compost to Bed 2 rather than giving NPK to e

SECTION B - PRACTICAL TASK

no fertiliser	compost	NPK 15-15-15
=====	=====	=====

six weeks after planting

PLOT	TREATMENT	MEAN HEIGHT AT 6 WEEKS
------	-----------	------------------------

A	nothing added	42 cm
---	---------------	-------

B	compost	68 cm
---	---------	-------

C	NPK 15-15-15	86 cm
---	--------------	-------

PLOT A is the CONTROL

kept the same: maize variety, soil, watering, spacing, plot size

Fig. 7: Does compost feed maize as well as NPK?

13. From the results, state which treatment produced the tallest maize, and calc

14. Explain why Plot A, which is given no fertiliser at all, is needed in this e

15. State TWO things the experiment keeps the same in all three plots, and expla

ANSWER KEY AND MARKING SCHEME

**5.1 5.1 What are Plant Nutrients?*

- **Q1** (4 marks) Nitrogen. The diagram links nitrogen with the leaves and shoo

- **Q2** (2 marks) Any two, each with a correct job. Nitrogen - builds the leave

**5.2 5.2 Organic and Inorganic Nutrient Sources*

- **Q3** (3 marks) Inorganic. The diagram sorts nutrient sources by where they c

11. The plan leaves a path between every bed. Explain why the paths are needed, describing the harm that walking on the beds would do. [4 marks]
12. The plan gives NPK to Bed 1 and compost to Bed 2 rather than giving NPK to every bed. Suggest TWO reasons why a plan might do this. [3 marks]
13. From the results, state which treatment produced the tallest maize, and calculate how much taller it was than the maize given no fertiliser at all. [4 marks]
14. Explain why Plot A, which is given no fertiliser at all, is needed in this experiment. [3 marks]
15. State TWO things the experiment keeps the same in all three plots, and explain why each one matters. [3 marks]

Diagram Paper 6: Digestion

Time Allowed: ** 1 hour 30 minutes

INSTRUCTIONS TO CANDIDATES

1. Answer ALL the questions in this paper.
2. Every question is based on a diagram. Study each diagram carefully before you answer the questions that follow it.
3. Write your answers in the spaces provided. The marks for each question are printed beside it.
4. Where a question asks for a stated number of points, give that number. Any extra answers will not be marked.
5. You may refer back to any diagram at any time.

What is Food? Nutrients and Their Importance

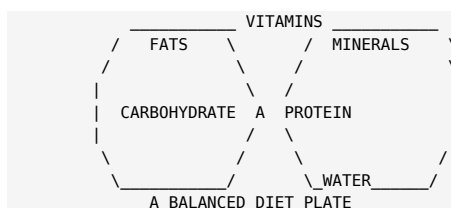


Fig. 1: A balanced diet

1. Use the diagram to explain why a person who eats only maize and fish every day is not getting a balanced diet, even though they never feel hungry. [4 marks]
2. From the diagram, name TWO classes of food and give one food source of each.

7.2 7.2 Why Humans Need to Eat

1. Use the diagram to explain why a person who eats only maize and fish every day is not getting a balanced diet, even though they never feel hungry. [4 marks]
2. From the diagram, name TWO classes of food and give one food source of each. [2 marks]

Why Humans Need to Eat

BALANCED - the beam stays level	
[ENERGY IN] the food eaten	[ENERGY OUT] work, growth, keeping warm
IN MORE THAN OUT	-> stored as fat, gain weight
OUT MORE THAN IN	-> stored fat used, lose weight
IN EQUALS OUT	-> mass steady, stay healthy

3. A pupil says that if you want to lose weight you should simply stop eating fat, because fat is what makes people fat. Use the diagram to explain why this is only half right. [3 marks]

*Fig. 2: Energy in

The Digestive System: Parts of the Alimentary Canal

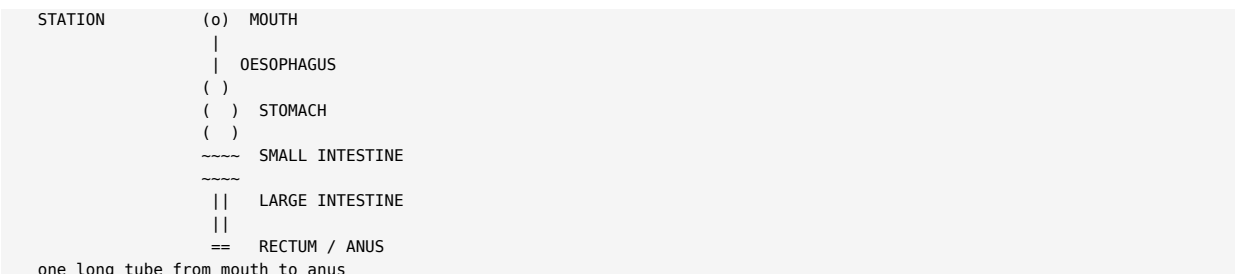


Fig. 3: The alimentary canal

4. Use the diagram to explain why the alimentary canal has to be a LONG tube, and state what would happen to a person if a large part of the small intestine had to be removed by surgery. [4 marks]

5. The diagram shows that the oesophagus is part of the alimentary canal, but that no digestion takes place there. Explain how it can be part of the canal and yet do no digesting. [3 marks]

7.4 7.4 What Happens to Food at Each Stage of Digestion

- Use the diagram to explain why the alimentary canal has to be a LONG tube, and state what would happen to a person if a large part of the small intestine had to be removed by surgery. [4 marks]
- The diagram shows that the oesophagus is part of the alimentary canal, but that no digestion takes place there. Explain how it can be part of the canal and yet do no digesting. [3 marks]

What Happens to Food at Each Stage of Digestion

STATION	MECHANICAL	CHEMICAL
MOUTH	teeth chew and grind	saliva acts on starch
OESOPHAGUS	muscles squeeze food down	nothing - only a passage
STOMACH	wall churns it to a liquid	acid and enzymes on protein
SMALL INTESTINE	nothing much	enzymes finish starch, protein, fats
LARGE INTESTINE	nothing	nothing - water taken back

Fig. 4: What happens at each station

6. Use the diagram to explain the difference between MECHANICAL and CHEMICAL digestion, giving one example of each. [3 marks]

7. Explain why chewing food thoroughly helps CHEMICAL digestion later on, even though almost no chemical digestion is finished in the mouth. [4 marks]

7.5 7.5 End Products of Digestion: Starch, Protein and Fats

- Use the diagram to explain the difference between MECHANICAL and CHEMICAL digestion, giving one example of each. [3 marks]
- Explain why chewing food thoroughly helps CHEMICAL digestion later on, even though almost no chemical digestion is finished in the mouth. [4 marks]

End Product

BIG, INSOLUBLE		SMALL, SOLUBLE
STARCH	----->	GLUCOSE (a simple sugar)
PROTEIN	----->	AMINO ACIDS
FATS	----->	FATTY ACIDS
& OILS		+ GLYCEROL
digested by enzymes, so the products can pass through the gut wall into the blood		

Fig. 5: End products of digestion

8. Use the diagram to explain why a patient who cannot digest food properly can

9. Explain, using the diagram, why the body cannot use starch directly, even tho

7.6 7.6 Absorption of Digested Food in the Small Intestine

8. Use the diagram to explain why a patient who cannot digest food properly can be kept alive by giving glucose solution straight into a vein. [3 marks]
9. Explain, using the diagram, why the body cannot use starch directly, even though starch is made entirely of glucose units joined together. [4 marks]

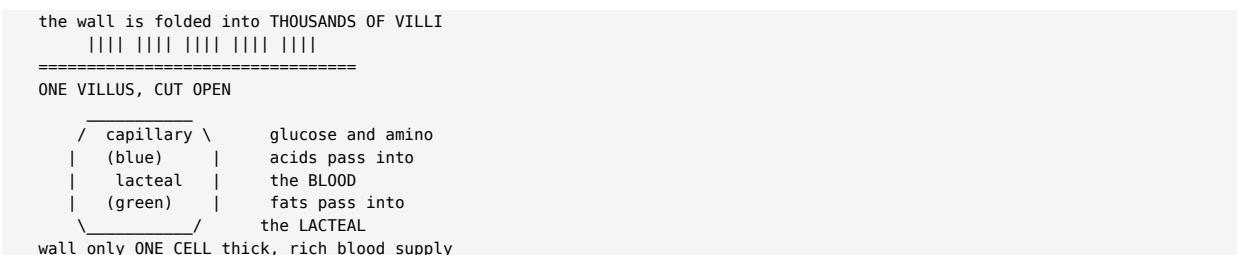
Absorption of Digested Food in the Small Intestine

Fig. 6: Villi in the small intestine

10. Explain, using the diagram, how the villi make absorption FAST. Refer to TWO

11. From the diagram, state where the digested FATS go after absorption, and exp

7.7 7.7 Food Tests: Starch, Glucose, Protein, Fats and Oils

10. Explain, using the diagram, how the villi make absorption FAST. Refer to TWO features shown. [4 marks]
11. From the diagram, state where the digested FATS go after absorption, and explain why they do not travel the same way as glucose. [3 marks]

Food Tests: Starch, Glucose, Protein, Fats and Oils

TESTING FOR	REAGENT	START	POSITIVE
STARCH	iodine	brown	blue-black
GLUCOSE	Benedict's, then HEAT	blue	brick-red green = a little
PROTEIN	Biuret	blue	purple
FATS & OILS	ethanol, then water	clear	cloudy white emulsion
a CONTROL of pure water is run alongside			

*Fig. 7: Food tests

12. Describe how you would test a sample of food for STARCH. State the reagent you would use, what you would do, and the colour that shows a positive result. [3 marks]

13. A pupil tests a food sample with Benedict's solution, heats it, and gets a GREEN colour instead of the brick-red shown in the diagram. Explain what this result shows. [3 marks]

SECTION B - PRACTICAL TASK

A	B	C	D
			
a few drops of iodine added to each			
SAMPLE	COLOUR AFTER IODINE	STARCH?	
A	stayed brown	NO	
B	went blue-black	YES	
C	stayed brown	NO	
D	went blue-black	YES	

tube A is the control

Fig. 8: Testing four unknown food samples

14. From the diagram, state which samples contain starch, and describe what you would do to check that this result is reliable. [4 marks]

15. The pupil writes in the conclusion that samples A and C contain no food nutrients at all. Explain why the results do not support this. [3 marks]

ANSWER KEY AND MARKING SCHEME

7.1 7.1 What is Food? Nutrients and Their Importance

- **Q1** (4 marks) Maize supplies carbohydrate and fish supplies protein, so two

- **Q2** (2 marks) Any two, each with a correct source: carbohydrate - bread, rice

7.2 7.2 Why Humans Need to Eat

- **Q3** (3 marks) The diagram shows that body mass depends on the BALANCE between energy intake and energy expenditure

7.3 7.3 The Digestive System: Parts of the Alimentary Canal

12. Describe how you would test a sample of food for STARCH. State the reagent you would use, what you would do, and the colour that shows a positive result. [3 marks]
13. A pupil tests a food sample with Benedict's solution, heats it, and gets a GREEN colour instead of the brick-red shown in the diagram. Explain what this result shows. [3 marks]
14. From the diagram, state which samples contain starch, and describe what you would do to check that this result is reliable. [4 marks]
15. The pupil writes in the conclusion that samples A and C contain no food nutrients at all. Explain why the results do not support this. [3 marks]

Diagram Paper 7: Earth Science - The

Time Allowed: ** 1 hour 30 minutes

INSTRUCTIONS TO CANDIDATES

1. Answer ALL the questions in this paper.
2. Every question is based on a diagram. Study each diagram carefully before you answer the questions that follow it.
3. Write your answers in the spaces provided. The marks for each question are printed beside it.
4. Where a question asks for a stated number of points, give that number. Any extra answers will not be marked.
5. You may refer back to any diagram at any time.

Natural Sources of Water

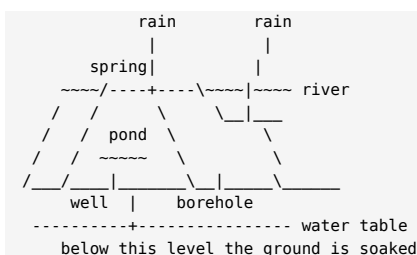


Fig. 1: Where our water comes from

1. The diagram shows both a well and a borehole going down below the water table
2. From the diagram, name THREE sources of water and say for each whether it is
3. Name TWO sources of water shown in the diagram. **[2 marks]**

3.2 3.2 Stages of the Water Cycle: Evaporation, Condensation, Precipitation

1. The diagram shows both a well and a borehole going down below the water table. Explain why both have to reach below it, and explain what would happen to the well during a long drought. [4 marks]
2. From the diagram, name THREE sources of water and say for each whether it is SURFACE water or water from UNDERGROUND. [3 marks]
3. Name TWO sources of water shown in the diagram. [2 marks]

Stages of the Water Cycle: Evaporation, Condensation, Precipitation and Transpiration

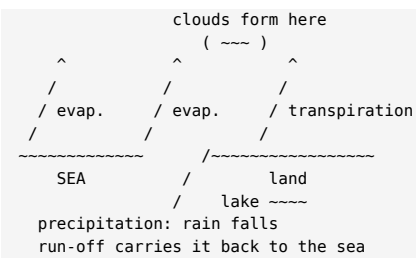


Fig. 2: The water cycle

4. Explain why the SUN is essential to the water cycle, and describe what would happen to the supply of rain if the sun stopped heating the sea. [4 marks]
5. The diagram shows TRANSPIRATION as well as evaporation. In a thick forest, which of the two puts more water into the air? Explain why. [3 marks]

3.3 3.3 Why the Water Cycle is a Repeated Pattern in Nature

4. Explain why the SUN is essential to the water cycle, and describe what would happen to the supply of rain if the sun stopped heating the sea. [4 marks]
5. The diagram shows TRANSPIRATION as well as evaporation. In a thick forest, which of the two puts more water into the air? Explain why. [3 marks]

Why the Water Cycle is a Repeated Pattern in Nature

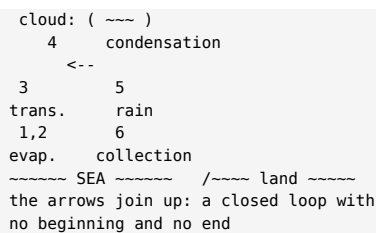


Fig. 3: The same water, going round and round

6. The diagram shows the water cycle as a closed loop. Explain what is meant by saying it is a CLOSED system, and use that idea to explain why the water falling as rain today is the same water that has been falling on the earth for millions of years. [4 marks]
7. A pupil says that because the stages are numbered 1 to 6, stage 1 must be the beginning of the cycle. Use the diagram to explain why this is wrong. [3 marks]

3.4 3.4 Importance of the Water Cycle to the Environment

6. The diagram shows the water cycle as a closed loop. Explain what is meant by saying it is a CLOSED system, and use that idea to explain why the water falling as rain today is the same water that has been falling on the earth for millions of years. [4 marks]
7. A pupil says that because the stages are numbered 1 to 6, stage 1 must be the beginning of the cycle. Use the diagram to explain why this is wrong. [3 marks]

Importance of the Water Cycle to the Environment

and dirt behind - the vapour
that rises is pure
CONDENSATION gathers the vapour into clouds
and carries it inland
PRECIPITATION brings fresh water down onto
the land, filling streams,
lakes and the water table
COLLECTION gathers water where living
things can reach it
TRANSPIRATION returns water to the air and
cools the air around plants

Fig. 4: What each stage does for us

8. The diagram says that evaporation DISTILLS the water. Explain what this means,

9. State TWO things the water cycle does for the environment. **[3 marks]**

3.5 3.5 The Water Cycle and Weather Patterns

8. The diagram says that evaporation DISTILLS the water. Explain what this means, and use it to explain why rain is fresh even though most of the water that evaporates comes from the salty sea. [4 marks]

9. State TWO things the water cycle does for the environment. [3 marks]

The Water Cycle and Weather Patterns

SEA coast inland
~~~~~  
^ ^  
/ / evap. / clouds drift inland,  
/ / / shedding rain as they go  
(~~) (~~) (~~) (~)  
||| || | |  
heavy less less least  
the further the air travels from the sea,  
the less moisture it still carries

\*Fig. 5: Why the sea side gets more rain\*

10. Use the diagram to explain why a town near the coast gets more rain than a town

11. Explain why clearing a large area of forest would reduce the rainfall in the

## SECTION B - PRACTICAL TASK

| DISH A       | DISH B         | DISH C         | DISH D            |
|--------------|----------------|----------------|-------------------|
| wide,<br>sun | wide,<br>shade | narrow,<br>sun | wide,<br>sun, fan |
| ~~~~~        | ~~~~~          | ~~~            | ~~~~~             |
|              |                |                |                   |
| +-----+      | +-----+        | +---+          | +-----+           |

vapour rising from each dish

| DISH | CONDITION      | WATER LEFT AFTER 3 DAYS |
|------|----------------|-------------------------|
| A    | wide, sun      | 35 ml                   |
| B    | wide, shade    | 78 ml                   |
| C    | narrow, sun    | 62 ml                   |
| D    | wide, sun, fan | 18 ml                   |

each dish started with 100 ml

\*Fig. 6: What makes water evaporate faster?\*

12. List the four dishes in order, from the one where water evaporated FASTEST to

13. Explain, using the idea of particles, why dish D lost water faster than dish

14. State TWO things that must be kept the same in this experiment, and explain

15. A pupil wants to test whether putting a LID on a dish slows evaporation. Describe

---

## ANSWER KEY AND MARKING SCHEME

\*\*3.1 3.1 Natural Sources of Water\*\*

- \*\*Q1\*\* (4 marks) The water table is the level below which the pores and cracks

- \*\*Q2\*\* (3 marks) Any three, each correctly classified. Stream - surface. River

- \*\*Q3\*\* (2 marks) Any two of: rain, stream, river, pond, spring, well, borehole

10. Use the diagram to explain why a town near the coast gets more rain than a town far inland. [4 marks]

11. Explain why clearing a large area of forest would reduce the rainfall in the region downwind of it. [3 marks]

**12. 13. 14**

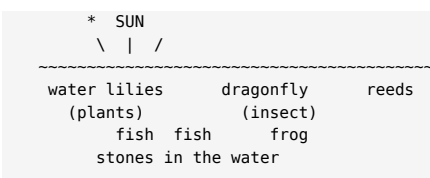
## Diagram Paper 8: Ecosystem

Time Allowed: \*\* 1 hour 30 minutes

### INSTRUCTIONS TO CANDIDATES

1. Answer ALL the questions in this paper.
2. Every question is based on a diagram. Study each diagram carefully before you answer the questions that follow it.
3. Write your answers in the spaces provided. The marks for each question are printed beside it.
4. Where a question asks for a stated number of points, give that number. Any extra answers will not be marked.
5. You may refer back to any diagram at any time.

### What is an Ecosystem?



\*Fig. 1: A pond ecosystem\*

1. Explain why the pond in the diagram is described as an ECOSYSTEM rather than
2. Name TWO non-living (abiotic) things shown in the diagram, and state why the

#### 9.2 9.2 Types of Ecosystems: Terrestrial, Aquatic and Arboreal

1. Explain why the pond in the diagram is described as an ECOSYSTEM rather than simply as a HABITAT. [4 marks]
2. Name TWO non-living (abiotic) things shown in the diagram, and state why the living things depend on each. [3 marks]

### Types of Ecosystems: Terrestrial, Aquatic and Arboreal

| TERRESTRIAL | AQUATIC      | ARBOREAL |
|-------------|--------------|----------|
| grass       | fish         | birds    |
| shrubs      | water plants | monkeys  |
| grazing     | insects      | lizards  |
| animals     | frogs        | insects  |

3. Choose TWO of the three ecosystems. For EACH, name one organism shown and explain its role in the ecosystem.  
4. A pupil says a frog could live happily in ANY of the three ecosystems. Explain why you agree or disagree.

#### 9.3 9.3 Components of an Ecosystem: Biotic and Abiotic Factors

\*Fig. 2: Three kind

3. Choose TWO of the three ecosystems. For EACH, name one organism shown and explain ONE way it is suited to living there. [4 marks]
4. A pupil says a frog could live happily in ANY of the three ecosystems. Explain why the frog would struggle in the DRIEST of them. [3 marks]

## Components of an Ecosystem: Biotic and Abiotic Factors

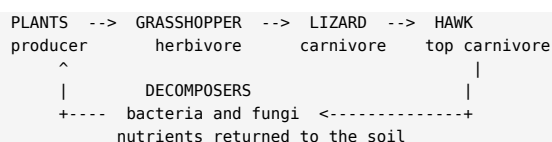
1 sunlight      2 water      3 water lily      4 stones  
5 fish      6 dragonfly      7 temperature      8 frog  
sort these eight into LIVING and NON-LIVING

\*Fig. 3: Eight things in a pond, numbered\*

5. Copy and complete the table by sorting the eight numbered items into LIVING ( )  
6. Choose ONE abiotic factor from the diagram. Explain what would happen to the  
7. A pupil argues that sunlight is NOT part of the pond ecosystem because it comes from the sun.  
#### 9.4 9.4 How Components of an Ecosystem Interact

5. Copy and complete the table by sorting the eight numbered items into LIVING (biotic) and NON-LIVING (abiotic). [3 marks]
6. Choose ONE abiotic factor from the diagram. Explain what would happen to the living things in the pond if that factor were removed or greatly reduced. [3 marks]
7. A pupil argues that sunlight is NOT part of the pond ecosystem because it comes from the sky, outside the pond. Explain why the pupil is wrong. [4 marks]

## How Components of an Ecosystem Interact



\*Fig. 4: How the parts of an ecosystem interact\*

8. The diagram has arrows between the organisms. State what an arrow means in a

9. All the LIZARDS in the ecosystem are removed. PREDICT what happens to the num

10. Explain the role of the DECOMPOSERS shown in the diagram, and what would hap

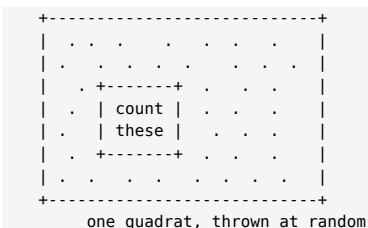
#### 9.5 9.5 Interdependence of Organisms in an Ecosystem

- 8.** The diagram has arrows between the organisms. State what an arrow means in a food chain, and explain why the direction of the arrow matters. [3 marks]
- 9.** All the LIZARDS in the ecosystem are removed. PREDICT what happens to the number of grasshoppers, and then to the number of plants. Explain the steps. [4 marks]
- 10.** Explain the role of the DECOMPOSERS shown in the diagram, and what would happen in the ecosystem without them. [3 marks]



\*Fig. 5: A food web\*

11. Write out TWO different food chains from the web, each with at least four links.
  12. Explain why a food WEB, where an animal feeds on more than one thing, makes the ecosystem more STABLE than a single food chain.
  13. A disease kills every GRASSHOPPER in the web. PREDICT TWO effects on the other organisms.
- ## SECTION B - PRACTICAL TASK



\*Fig. 6: Using a quadrat to sample plants\*

14. Describe how you would use the quadrat in the diagram to estimate how many plants are in the whole field, and explain why you would throw it more than once.
15. Explain why it is better to count the plants in several small squares than to try to count every plant in the whole field.

## ANSWER KEY AND MARKING SCHEME

\*\*9.1 9.1 What is an Ecosystem?\*

- \*\*Q1\*\* (4 marks) A habitat is only the place where an organism lives. An ecosystem is a community of organisms and their physical environment.
- \*\*Q2\*\* (3 marks) Any two with a reason: SUNLIGHT - the plants need it to make food; WATER - plants need it to grow.
- \*\*Q3\*\* (4 marks) Any two, organism plus adaptation: AQUATIC - fish have gills; TERRESTRIAL - birds have wings.
- \*\*Q4\*\* (3 marks) A frog breathes partly through its skin, which must stay moist.

11. Write out TWO different food chains from the web, each with at least four links. Remember what the arrow means. [3 marks]
12. Explain why a food WEB, where an animal feeds on more than one thing, makes the ecosystem more STABLE than a single food chain. [4 marks]
13. A disease kills every GRASSHOPPER in the web. PREDICT TWO effects on the other organisms. [3 marks]
14. Describe how you would use the quadrat in the diagram to estimate how many plants are in the whole field, and explain why you would throw it more than once. [4 marks]
15. Explain why it is better to count the plants in several small squares than to try to count every plant in the whole field. [2 marks]

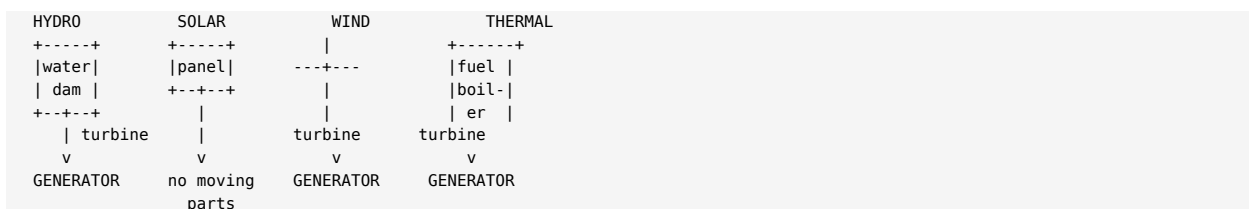
# Diagram Paper 9: Electricity and Ele

Time Allowed: \*\* 1 hour 30 minutes

## INSTRUCTIONS TO CANDIDATES

1. Answer ALL the questions in this paper.
2. Every question is based on a diagram. Study each diagram carefully before you answer the questions that follow it.
3. Write your answers in the spaces provided. The marks for each question are printed beside it.
4. Where a question asks for a stated number of points, give that number. Any extra answers will not be marked.
5. You may refer back to any diagram at any time.

## Forms of Electricity Generation



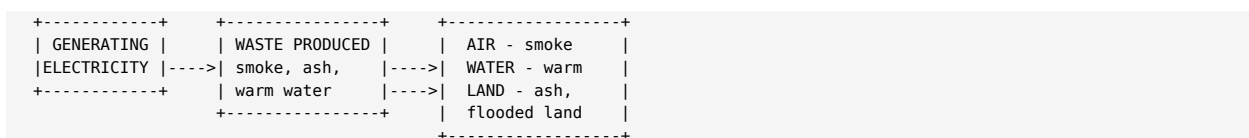
\*Fig. 1: Four ways of generating electricity\*

1. Choose TWO methods of generation shown in the diagram. For EACH one state the energy change that takes place, naming the form of energy at the start and at the end.
2. A town needs electricity every day of the year, including through a long dry season. Use the diagram to explain why a HYDRO dam alone would be a poor choice, and which method would be more reliable.

#### 12.2 12.2 Impact on the Environment

1. Choose TWO methods of generation shown in the diagram. For EACH one state the energy change that takes place, naming the form of energy at the start and at the end. [4 marks]
2. A town needs electricity every day of the year, including through a long dry season. Use the diagram to explain why a HYDRO dam alone would be a poor choice, and which method would be more reliable. [3 marks]

## Impact on the Environment





\*Fig. 2: How ge

3. Use the diagram to explain how generating electricity can harm the water in a river BELOW a dam. [3 marks]

4. A community objects to a THERMAL plant being built near their town. SUGGEST TWO environmental reasons for their objection, and ONE way the harm could be reduced. [3 marks]

#### 12.3 12.3 Basic Electronic Components

3. Use the diagram to explain how generating electricity can harm the water in a river BELOW a dam. [3 marks]
4. A community objects to a THERMAL plant being built near their town. SUGGEST TWO environmental reasons for their objection, and ONE way the harm could be reduced. [3 marks]

## Basic Electronic Components

|        |                 |      |         |        |          |
|--------|-----------------|------|---------|--------|----------|
| cell   | battery         | lamp | switch  | switch | resistor |
| -  -   | -  -            | (X)  | o-- o-- | o---o  | [===]    |
|        | -  -            |      | OPEN    | CLOSED |          |
| diode  | connecting wire |      | ammeter |        |          |
| --> -- |                 |      | ( A )   |        |          |

\*Fig. 3: Circuit symbols for common components\*

5. Name the component represented by each of these symbols: the rectangle; the triangle with a bar; the circle with an A in it. [3 marks]

6. The diagram shows a switch in two positions. Explain the difference between them in terms of what happens to the CURRENT. [3 marks]

7. Explain why a resistor is needed in a circuit that contains an LED, and what would happen without one. [3 marks]

#### 12.4 12.4 Assembling a Simple Circuit

5. Name the component represented by each of these symbols: the rectangle; the triangle with a bar; the circle with an A in it. [3 marks]
6. The diagram shows a switch in two positions. Explain the difference between them in terms of what happens to the CURRENT. [3 marks]
7. Explain why a resistor is needed in a circuit that contains an LED, and what would happen without one. [3 marks]

## Assembling a Simple Circuit

```

CIRCUIT 1 - SERIES
+---[ bulb ]---[ bulb ]---+
|                               |
+---[ switch ]---||-----+
      one loop only
CIRCUIT 2 - PARALLEL
+---[ bulb ]-----+
|                   |
|       [ bulb ]   [ bulb ]
|                   |
+---[ switch ]-||-+-----+
      each bulb has its own branch

```

\*Fig. 4: A series circuit and a parallel circuit\*

8. In CIRCUIT 1 (series) one of the two bulbs is unscrewed. PREDICT what happens to the other bulb, and explain why. [4 marks]

9. In CIRCUIT 2 (parallel) one of the bulbs is unscrewed. PREDICT what happens to the other bulbs, and explain why. [3 marks]

10. Explain why the rooms of a house are wired in PARALLEL and not in series. Give TWO reasons. [4 marks]

#### 12.5 12.5 The P-N Junction Diode

8. In CIRCUIT 1 (series) one of the two bulbs is unscrewed. PREDICT what happens to the other bulb, and explain why. [4 marks]
9. In CIRCUIT 2 (parallel) one of the bulbs is unscrewed. PREDICT what happens to the other bulbs, and explain why. [3 marks]
10. Explain why the rooms of a house are wired in PARALLEL and not in series. Give TWO reasons. [4 marks]

```

A - FORWARD BIAS          B - REVERSE BIAS
+--[sw]-->|--( bulb )----+  +--[sw]--<|--( bulb )----+
|                               |                               |
+-----||-----+          +-----||-----+
      bulb LIGHTS             bulb does NOT light
      (diode arrow points the way the current goes)

```

\*Fig. 5: The same diode connected two ways round\*

11. State in which circuit, A or B, the bulb lights. Explain, using the diagram,

12. State what is meant by FORWARD BIAS. **\*\*[3 marks]\*\***

#### 12.6 12.6 Functions and Interdependence

**11.** State in which circuit, A or B, the bulb lights. Explain, using the diagram, why it lights in one and not the other. [4 marks]

**12.** State what is meant by FORWARD BIAS. [3 marks]

## Functions and Interdependence

```

SOURCE  -> CONDUCTOR  -> CONTROL  -> LOAD
battery  copper        switch      bulb
or gen.  wires          iron        radio
remove ANY one part and the circuit stops working

```

\*Fig. 6: What every circuit needs\*

13. Name the four parts shown, and state what each one does. **\*\*[3 marks]\*\***

14. Use the diagram to explain why a bulb would not light if the conductor were

## SECTION B - PRACTICAL TASK

```

A PUPIL'S CIRCUIT - THE LAMP DOES NOT LIGHT
+--[sw closed]--( bulb )--+
|                               |
+---||----- X -----+
      break in the wire

```

\*Fig. 7: A circuit that does not work\*

15. The switch is closed, yet the lamp does not light. SUGGEST TWO possible faults

---

## ANSWER KEY AND MARKING SCHEME

**\*\*12.1 12.1 Forms of Electricity Generation\*\***

- **\*\*Q1\*\*** (4 marks) HYDRO: the potential energy of water held behind the dam changes

- **\*\*Q2\*\*** (3 marks) A hydro dam depends on water held behind it. In a long dry season

**\*\*12.2 12.2 Impact on the Environment\*\***

- **\*\*Q3\*\*** (3 marks) The dam floods the land behind it, which changes the river's

- **\*\*Q4\*\*** (3 marks) Any two reasons: the smoke from burning the fuel pollutes the

**\*\*12.3 12.3 Basic Electronic Components\*\***

**13.** Name the four parts shown, and state what each one does. [3 marks]

**14.** Use the diagram to explain why a bulb would not light if the conductor were broken at ANY single point. [3 marks]

**15.** The switch is closed, yet the lamp does not light. SUGGEST TWO possible faults, and describe how you would test for each one. [4 marks]

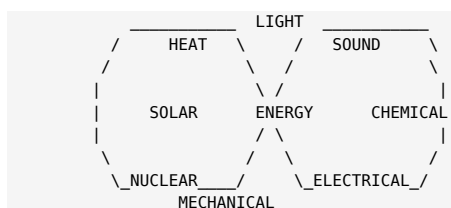
## Diagram Paper 10: Energy

Time Allowed: \*\* 1 hour 30 minutes

### INSTRUCTIONS TO CANDIDATES

1. Answer ALL the questions in this paper.
2. Every question is based on a diagram. Study each diagram carefully before you answer the questions that follow it.
3. Write your answers in the spaces provided. The marks for each question are printed beside it.
4. Where a question asks for a stated number of points, give that number. Any extra answers will not be marked.
5. You may refer back to any diagram at any time.

### What is Energy?



\*Fig. 1: The forms of energy\*

1. State what energy is, and use the diagram to explain why a stone held high above the ground has energy even though nothing is happening to it. [4 marks]
2. From the diagram, name TWO forms of energy and give one everyday example of each. [2 marks]

#### 11.2 11.2 Forms of Energy

1. State what energy is, and use the diagram to explain why a stone held high above the ground has energy even though nothing is happening to it. [4 marks]
2. From the diagram, name TWO forms of energy and give one everyday example of each. [2 marks]

### Forms of Energy

| FORM       | WHERE YOU MEET IT                   |
|------------|-------------------------------------|
| CHEMICAL   | -> food, fuel, a battery            |
| HEAT       | -> a fire, a hot iron, the sun      |
| LIGHT      | -> a lit bulb, the sun, a torch     |
| SOUND      | -> a drum, a radio, a school bell   |
| ELECTRICAL | -> current flowing in a fan's wires |
| MECHANICAL | -> a moving car, a falling stone    |
| NUCLEAR    | -> released in a nuclear reactor    |
| SOLAR      | -> sunlight on a solar panel        |

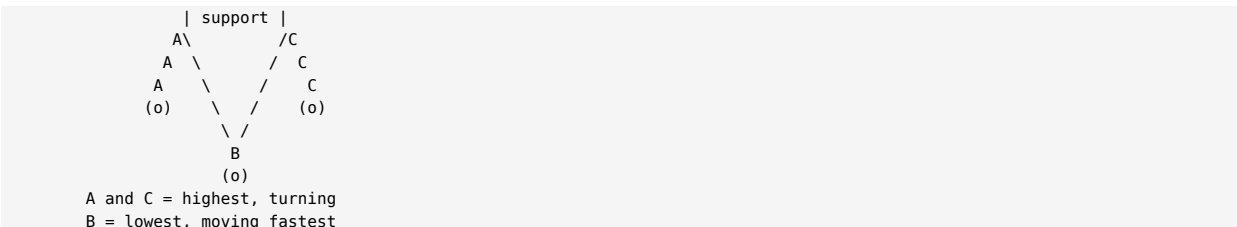
3. A pupil says a battery stores electrical energy. Use the diagram to correct this, and explain why the correction matters. [3 marks]

#### 11.3 11.3 Potential and Kinetic Energy

\*Fig. 2: Each form

3. A pupil says a battery stores electrical energy. Use the diagram to correct this, and explain why the correction matters. [3 marks]

## Potential and Kinetic Energy



\*Fig. 3: A swinging pendulum\*

4. Explain, using the diagram, why the bob is moving fastest at B. \*\*[4 marks]

5. In a real pendulum the swings gradually die away. State what has happened to

#### 11.4 11.4 Calculating PE and KE

4. Explain, using the diagram, why the bob is moving fastest at B. [4 marks]

5. In a real pendulum the swings gradually die away. State what has happened to the energy, and explain why this does not contradict the idea that energy is conserved. [3 marks]

## Calculating PE and KE

| h (m) | speed(m/s) | PE(J) | KE(J) | PE+KE(J) |
|-------|------------|-------|-------|----------|
| 20    | 0          | 400   | 0     | 400      |
| 15    | 10         | 300   | 100   | 400      |
| 10    | 14.1       | 200   | 200   | 400      |
| 5     | 17.3       | 100   | 300   | 400      |
| 0     | 20         | 0     | 400   | 400      |

A 2 kg stone dropped from 20 m,  $g = 10 \text{ m/s}^2$

\*Fig. 4: A 2 kg stone dropped from 20 m\*

6. State what the last column of the table shows, and explain what this tells us

7. A pupil says: 'the stone has 400 J of potential energy at the top and 400 J of

#### 11.5 11.5 Daily Applications of Energy

6. State what the last column of the table shows, and explain what this tells us about energy. [4 marks]

7. A pupil says: 'the stone has 400 J of potential energy at the top and 400 J of kinetic energy at the bottom, so it finishes with 800 J and energy has been created.' Explain what the pupil has got wrong. [3 marks]

## Daily Applications of Energy

|                |            |    |              |
|----------------|------------|----|--------------|
| Electric iron  | electrical | -> | heat         |
| Filament bulb  | electrical | -> | light + heat |
| Radio          | electrical | -> | sound        |
| Electric fan   | electrical | -> | mechanical   |
| Charcoal stove | chemical   | -> | heat         |
| Torch          | chemical   | -> | light        |
| Solar panel    | solar      | -> | electrical   |

\*Fig. 5: Energy conversion in everyday appliances\*

8. From the diagram, name the energy change that takes place in a torch, and explain why a torch with a flat battery gives no light. [3 marks]

9. The diagram shows that a filament bulb gives out light AND heat. A pupil says the heat is wasted. Explain why it is wasted, and suggest ONE thing that could be done about it. [3 marks]

#### 11.6 11.6 Heat Transfer

8. From the diagram, name the energy change that takes place in a torch, and explain why a torch with a flat battery gives no light. [3 marks]

9. The diagram shows that a filament bulb gives out light AND heat. A pupil says the heat is wasted. Explain why it is wasted, and suggest ONE thing that could be done about it. [3 marks]

## Heat Transfer

1. CONDUCTION - through a solid  
(o)(o)(o)(o)(o)(o)(o)(o) particles vibrate,  
hot end =====> cool end solid stays put
2. CONVECTION - through a liquid or gas  
^ hot water rises, cool water sinks  
/ \ a convection current
3. RADIATION - no material needed  
(sun) -----> -----> -----> you  
infrared waves across empty space

\*Fig. 6: The three ways heat travels\*

10. State which of the three ways of transferring heat does NOT need any material to travel through, and use the diagram to justify your answer. [3 marks]

11. Explain why the heated water in the convection panel RISES, using the idea of particles. [4 marks]

#### 11.7 11.7 Characteristics of Light

10. State which of the three ways of transferring heat does NOT need any material to travel through, and use the diagram to justify your answer. [3 marks]

11. Explain why the heated water in the convection panel RISES, using the idea of particles. [4 marks]

## Characteristics of Light

1. REFLECTION  
/ | reflected ray  
/ |  
/ | normal  
---/---|----- mirror  
angle i = angle r
2. REFRACTION  
air\  
\  
glass |\  
| \ air
3. LIGHT TRAVELS IN STRAIGHT LINES  
(lamp) ---> [object] -----> |screen  
shadow behind it

\*Fig. 7: What light

12. State the law of reflection, and use the diagram to say what the dashed line  
 13. Use the diagram to explain why a stick standing in water looks bent, and sta  
 ## SECTION B - PRACTICAL TASK

| HEATED BLOCK             |              |      |      |
|--------------------------|--------------|------|------|
| [=====]                  |              |      |      |
| pin                      | /            |      | \    |
|                          | /            |      | \    |
| copper                   | aluminium    | iron | wood |
| TIME FOR PIN TO DROP (s) |              |      |      |
| copper                   | 48           | 1st  |      |
| aluminium                | 95           | 2nd  |      |
| iron                     | 210          | 3rd  |      |
| wood                     | did not drop | 4th  |      |

\*Fig. 8: Investigating conduction\*

14. State, with a reason, which rod you would expect the pin to fall from FIRST,  
 15. Explain why this experiment uses a pin stuck on with WAX rather than just fe  
 ---

## ANSWER KEY AND MARKING SCHEME

\*\*11.1 11.1 What is Energy?\*\*

- \*\*Q1\*\* (4 marks) Energy is the ability to do work. The raised stone has stored  
 - \*\*Q2\*\* (2 marks) Any two, each with a correct example: chemical - energy store

\*\*11.2 11.2 Forms of Energy\*\*

- \*\*Q3\*\* (3 marks) A battery stores CHEMICAL energy, not electrical energy. The

\*\*11.3 11.3 Potential and Kinetic Energy\*\*

12. State the law of reflection, and use the diagram to say what the dashed line is for. [3 marks]
13. Use the diagram to explain why a stick standing in water looks bent, and state what this effect is called. [4 marks]
14. State, with a reason, which rod you would expect the pin to fall from FIRST, and explain what the order tells you about the rods. [4 marks]
15. Explain why this experiment uses a pin stuck on with WAX rather than just feeling which rod gets hot first. [3 marks]

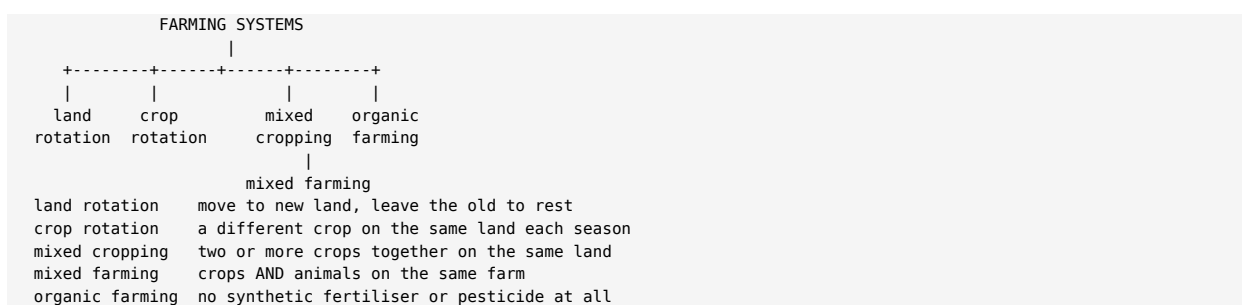
## Diagram Paper 11: Farming Systems

Time Allowed: \*\* 1 hour 30 minutes

### INSTRUCTIONS TO CANDIDATES

1. Answer ALL the questions in this paper.
2. Every question is based on a diagram. Study each diagram carefully before you answer the questions that follow it.
3. Write your answers in the spaces provided. The marks for each question are printed beside it.
4. Where a question asks for a stated number of points, give that number. Any extra answers will not be marked.
5. You may refer back to any diagram at any time.

### Types of Farming Systems in Ghana



\*Fig. 1: Farming systems found in Ghana\*

1. A farmer in the Northern Region has a large area of land and only a small family to work it. A farmer just outside Kumasi has a small plot and a large family. Suggest a SUITABLE farming system for each, and explain why each suits that farmer. [4 marks]
2. Name any THREE farming systems shown in the diagram. \*\*[3 marks]\*\*

#### 10.2 10.2 Land Rotation Farming

1. A farmer in the Northern Region has a large area of land and only a small family to work it. A farmer just outside Kumasi has a small plot and a large family. Suggest a SUITABLE farming system for each, and explain why each suits that farmer. [4 marks]
2. Name any THREE farming systems shown in the diagram. [3 marks]

### Land Rotation Farming

```

+-----+ +-----+ +-----+ +-----+ +-----+
| PLOT A | | PLOT B | | PLOT C | | PLOT D | | PLOT E | | | | | | | | | | |
|  ||  | |  ||  | |  ||  | |  ||  | |  ||  |
| maize | | maize | | maize | | ww  | | ww  |
|  ||  | |  ||  | |  ||  | | weeds| | weeds|
+-----+ +-----+ +-----+ +-----+ +-----+
    -->      -->      -->      -->
the farmer farms one plot, moves on, and returns
years later when the soil has recovered

```

\*Fig. 2: Land rotation\*

3. Explain why leaving a plot FALLOW restores the fertility of the soil, and explain why the system stops working when a village grows and the spare land runs out. [4 marks]

#### 10.3 10.3 Crop Rotation Farming

3. Explain why leaving a plot FALLOW restores the fertility of the soil, and explain why the system stops working when a village grows and the spare land runs out. [4 marks]

### Crop Rotation Farming

```

SEASON 1          SEASON 2          SEASON 3
+-----+        +-----+        +-----+
| MAIZE |  ---> | GROUNDNUT |  ---> | CASSAVA | | | | |
|  ||  |        | oo  |        |  ||  |
|  ||  |        | legume |        |  ()  |
+-----+        +-----+        +-----+
takes much      a legume -      a root crop,
nitrogen out    puts nitrogen   feeding from
of the soil     BACK into it    deeper down
different crops take different nutrients, so no
one nutrient is drained year after year

```

\*Fig. 3: Crop rotation\*

4. Explain why growing GROUNDNUTS between two maize crops helps the second maize

5. State TWO advantages of crop rotation. \*\*[3 marks]\*\*

#### 10.4 10.4 Mixed Cropping

4. Explain why growing GROUNDNUTS between two maize crops helps the second maize crop, referring to what each crop does to the soil. [4 marks]

5. State TWO advantages of crop rotation. [3 marks]

### Mixed Cropping

```

MAIZE WITH BEANS          WHY FARMERS DO IT
+-----+                +-----+
|  ||  |  ||  |  ||  |  | beans put nitrogen in |
| oo  | oo  | oo  |  | -> the maize uses it |
|  ||  |  ||  |  ||  | | different root depths |
| oo  | oo  | oo  |  | -> no competition  |
|  ||  |  ||  |  ||  | | the beans cover ground|
|  ||  |  ||  |  ||  | | -> less weed, keeps |
| beans fill the |  | its water |
| gaps between  |  | two harvests not one |
| the maize rows |  | -> food all season  |
+-----+        | if one crop fails, |
                  | the other may do  |
                  +-----+

```



\*Fig. 4: Mixed

6. The diagram shows maize grown with beans. Explain TWO ways in which the beans help the maize, and suggest ONE way in which the two crops might harm each other. [4 marks]

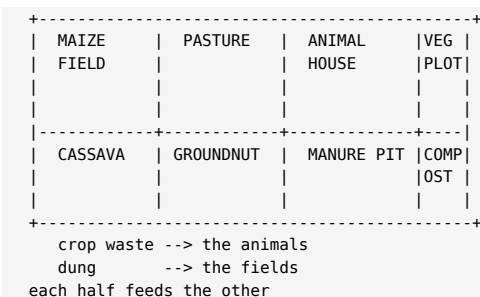
7. State TWO reasons why a farmer practises mixed cropping. [3 marks]

#### 10.5 10.5 Mixed Farming

6. The diagram shows maize grown with beans. Explain TWO ways in which the beans help the maize, and suggest ONE way in which the two crops might harm each other. [4 marks]

7. State TWO reasons why a farmer practises mixed cropping. [3 marks]

## Mixed Farming



\*Fig. 5: Mixed farming\*

8. On a mixed farm the animals depend on the crops and the crops depend on the animals. Give TWO examples of each. [4 marks]

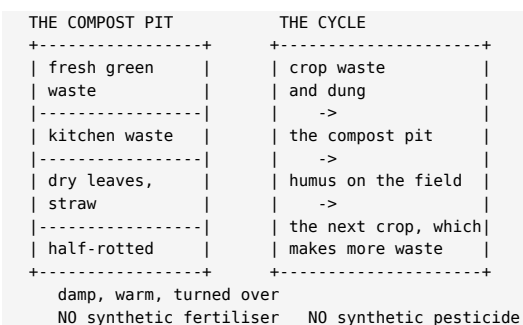
9. Explain what is meant by a MIXED FARM. [3 marks]

#### 10.6 10.6 Organic Farming

8. On a mixed farm the animals depend on the crops and the crops depend on the animals. Give TWO examples of each. [4 marks]

9. Explain what is meant by a MIXED FARM. [3 marks]

## Organic Farming



\*Fig. 6: Organic farming\*

10. Explain why organic farming depends so heavily on the compost pit, and explain why yields are often LOWER in the first years after a farmer converts to it. [4 marks]

11. The diagram shows different layers in the compost pit. Explain why the heap needs BOTH fresh green waste and dry material, and why it has to be turned over. [3 marks]

#### 10.7 10.7 Merits and Demerits of Each Farming System

10. Explain why organic farming depends so heavily on the compost pit, and explain why yields are often LOWER in the first years after a farmer converts to it. [4 marks]

11. The diagram shows different layers in the compost pit. Explain why the heap needs BOTH fresh green waste and dry material, and why it has to be turned over. [3 marks]

## Merits and

| SYSTEM    | MERITS            | DEMERITS             |
|-----------|-------------------|----------------------|
| land rot. | soil recovers     | needs lots of land   |
| crop rot. | keeps fertility   | needs planning       |
| mixed     | two harvests,     | crops compete,       |
| cropping  | less risk         | harder to weed       |
| mixed     | dung and crop     | more skill, labour   |
| farming   | waste exchanged   | and capital          |
| organic   | builds the soil   | bulky, slow,         |
|           | for the long term | lower yield at first |

\*Fig. 7: Weighing the systems up\*

12. A farmer has a small plot on the edge of a large town, plenty of family labour

## SECTION B - PRACTICAL TASK

| PLOT A - MAIZE EVERY SEASON |            |            |            |         | PLOT B - MAIZE, BEANS, MAIZE |    |    |    |    |
|-----------------------------|------------|------------|------------|---------|------------------------------|----|----|----|----|
| +-----+                     |            |            |            |         | +-----+                      |    |    |    |    |
| Y1                          |            |            |            |         | Y1                           |    |    |    |    |
| Y2                          |            |            |            |         | Y2                           | oo | oo | oo | oo |
| Y3                          | !          | !          | !          | !       | Y3                           |    |    |    |    |
| +-----+                     |            |            |            |         | +-----+                      |    |    |    |    |
| PLOT                        | YEAR 1     | YEAR 2     | YEAR 3     | MEAN    |                              |    |    |    |    |
| A                           | maize 2400 | maize 1850 | maize 1300 | 1850 kg |                              |    |    |    |    |
|                             | kg         | kg         | kg         |         |                              |    |    |    |    |
| B                           | maize 2380 | beans (not | maize 2260 | 2320 kg |                              |    |    |    |    |
|                             | kg         | weighed)   | kg         |         |                              |    |    |    |    |

\*Fig. 8: Does rotation keep the maize yield up?\*

13. Use the results to explain why plot B held its yield while plot A did not, r

14. Suggest TWO things the pupils must keep the same for this comparison to be f

15. State what is meant by the YIELD of a crop. \*\*[2 marks]\*\*

---

## ANSWER KEY AND MARKING SCHEME

\*\*10.1 10.1 Types of Farming Systems in Ghana\*\*

- \*\*Q1\*\* (4 marks) For the Northern Region farmer: land rotation. It asks little

- \*\*Q2\*\* (3 marks) Any three of: land rotation, crop rotation, mixed cropping, m

\*\*10.2 10.2 Land Rotation Farming\*\*

- \*\*Q3\*\* (4 marks) While a plot lies fallow, wild plants grow on it and die back

12. A farmer has a small plot on the edge of a large town, plenty of family labour, and very little money to buy chemicals. Which farming system would you advise, and why? [3 marks]
13. Use the results to explain why plot B held its yield while plot A did not, referring to what each crop did to the soil. [3 marks]
14. Suggest TWO things the pupils must keep the same for this comparison to be fair, and explain why each matters. [3 marks]
15. State what is meant by the YIELD of a crop. [2 marks]

## Diagram Paper 12: Force and Motion

Time Allowed: \*\* 1 hour 30 minutes

### INSTRUCTIONS TO CANDIDATES

1. Answer ALL the questions in this paper.
2. Every question is based on a diagram. Study each diagram carefully before you answer the questions that follow it.
3. Write your answers in the spaces provided. The marks for each question are printed beside it.
4. Where a question asks for a stated number of points, give that number. Any extra answers will not be marked.
5. You may refer back to any diagram at any time.

### What is Force?



\*Fig. 1: Types of force\*

1. For each of the following, say whether the force is a CONTACT force or a NON-
2. State what is meant by a FORCE, and name any TWO forces. \*\*[3 marks]\*\*

#### 14.2 14.2 Balanced and Unbalanced Forces

1. For each of the following, say whether the force is a CONTACT force or a NON-CONTACT force, and give a reason in each case: (a) the force that makes a compass needle swing; (b) the force that stops your foot slipping on a tiled floor; (c) the force that pulls a coconut down from a tree; (d) the force of a bulldozer blade on a mound of earth. [4 marks]
2. State what is meant by a FORCE, and name any TWO forces. [3 marks]

### Balanced and Unbalanced Forces

| START or STOP | CHANGE THE SPEED | CHANGE THE DIRECTION | CHANGE THE SHAPE |
|---------------|------------------|----------------------|------------------|
| [ ] -->       | [ ] -->          | /                    | [ ]              |
| at rest       | speeds up        | (o)                  | squashed         |
| until         | while            | by the               | by the           |
| pushed        | pushed           | push                 | squeeze          |

a bat hitting a ball stops it, turns it and squashes it at the same time

\*Fig. 2: What a force can do\*

3. A goalkeeper catches a fast ball. Use the diagram to state THREE different effects the force of his hands has on the ball. [3 marks]

### 14.3 14.3 Inertia

3. A goalkeeper catches a fast ball. Use the diagram to state THREE different effects the force of his hands has on the ball. [3 marks]

## Inertia

| FRICITION - THE FORCE THAT FIGHTS MOTION | IT CUTS BOTH WAYS             |
|------------------------------------------|-------------------------------|
| HOW IT ARISES                            |                               |
| [ ]                                      | USEFUL: walk without slipping |
| pull --> <-- fr.                         | USEFUL: tyres grip            |
| =====                                    | USEFUL: knots hold            |
|                                          | USEFUL: brakes stop           |
| friction acts the                        |                               |
| OPPOSITE way to                          | WASTED: wears soles           |
| the motion                               | WASTED: wastes                |
| even a smooth                            | energy as                     |
| surface is rough                         | heat                          |
| under a microscope                       | WASTED: machines              |
|                                          | run hot                       |

rougher -> more heavier load -> more  
oil, wheels, air cushion -> much less

\*Fig. 3: Friction\*

4. Friction is said both to HELP us and to WASTE our effort. Give ONE everyday example of each, and explain in each case what the friction is actually doing. [4 marks]

5. State TWO ways of REDUCING friction and TWO ways of INCREASING it. [3 marks]

### 14.4 14.4 Newton's First Law of Motion

4. Friction is said both to HELP us and to WASTE our effort. Give ONE everyday example of each, and explain in each case what the friction is actually doing. [4 marks]
5. State TWO ways of REDUCING friction and TWO ways of INCREASING it. [3 marks]

## Newton's First Law of Motion

| BALANCED AND UNBALANCED FORCES |                         |
|--------------------------------|-------------------------|
| BALANCED - NO CHANGE           | UNBALANCED - IT CHANGES |
| 3 N <-- [ ] -->                | 5 N <-- [ ] -->         |
| 3 N                            | 2 N                     |
| equal and opposite             | one is bigger: there    |
| -> they cancel                 | is a RESULTANT          |
| at rest -> stays               | -> it accelerates in    |
| moving -> same                 | the direction of        |
| speed, straight line           | the bigger force        |

the test is not 'are there forces?' but  
'do they cancel?'

\*Fig. 4: Balance

6. A book lies on a table. Gravity pulls it down with a force of 6 N and the table pushes up on it with a force of 6 N. Explain why the book does not move even though two forces act on it, and describe what would happen if the table were taken away. [4 marks]

7. Two teams pull in a tug of war. Team A pulls with 800 N and team B pulls with 800 N, and the rope does not move. State the resultant force on the rope, and explain what changes when team A begins to win. [3 marks]

#### 14.5 14.5 Applications of Newton's First Law

6. A book lies on a table. Gravity pulls it down with a force of 6 N and the table pushes up on it with a force of 6 N. Explain why the book does not move even though two forces act on it, and describe what would happen if the table were taken away. [4 marks]
7. Two teams pull in a tug of war. Team A pulls with 800 N and team B pulls with 800 N, and the rope does not move. State the resultant force on the rope, and explain what changes when team A begins to win. [3 marks]

## Applications of Newton's First Law

### CIRCULAR MOTION

.....

o = hand, at the centre  
the stone is at the rim

string

(o) <- stone

if the string broke it

would fly off along

this straight line

pull IN towards the centre

the speed may be steady but the VELOCITY is not:

velocity includes direction, and that changes

\*Fig. 5: Circular motion\*

8. A stone is tied to a string and whirled round in a circle at a steady speed.

#### 14.6 14.6 Magnets: Types and Characteristics

8. A stone is tied to a string and whirled round in a circle at a steady speed. Explain why the stone is ACCELERATING even though its speed never changes, and state what supplies the force causing it. [4 marks]

## Magnets: Types and Characteristics

### CENTRIPETAL - REAL

-----+

o

(o)

--|--

v

IN (real)

string tension,

gravity, grip of

the tyres

-----+

### CENTRIFUGAL - APPARENT

-----+

o

(o)

/|

v

OUT (apparent)

not a real force:

your body trying

to go straight on

-----+

in a turning bus you feel thrown OUT, but nothing pushes you out - the seat pushes you IN

9. A passenger standing in a bus that turns sharply to the left feels herself thrown over to the right. Explain what is really happening to her, and explain why the sensation is misleading. [4 marks]

10. State the DIFFERENCE between a centripetal force and the centrifugal force. [3 marks]

#### 14.7 14.7 Repulsion, Attraction and Orientation

\*Fig. 6: Centripetal

9. A passenger standing in a bus that turns sharply to the left feels herself thrown over to the right. Explain what is really happening to her, and explain why the sensation is misleading. [4 marks]
10. State the DIFFERENCE between a centripetal force and the centrifugal force. [3 marks]

## Repulsion, Attraction and Orientation

### PUTTING CIRCULAR MOTION TO WORK

| SPIN DRYER        | CENTRIFUGE         | CREAM SEPARATOR   | BANKED ROAD         |
|-------------------|--------------------|-------------------|---------------------|
| (*)               | (*)                | (*)               | /o \                |
| '                 | '                  | '                 | /----\              |
| water             | dense              | heavier           | the bank            |
| flies             | part               | milk out,         | tilts the           |
| out               | forced             | cream             | normal              |
|                   | to bottom          | stays in          | force in            |
| STONE ON A STRING | SATELLITE IN ORBIT | A BUCKET OVERHEAD | CORNER ON A BICYCLE |

\*Fig. 7: Putting circular motion to work\*

11. Explain how a SPIN DRYER gets water out of wet clothes, using the idea of circular motion. [3 marks]

#### 14.8 14.8 Uses of Magnets

11. Explain how a SPIN DRYER gets water out of wet clothes, using the idea of circular motion. [3 marks]

## Uses of Magnets

### WHY A SATELLITE DOES NOT FALL DOWN

(satellite) .....-> moving this way

```

      | \
      |  \
      |   \ v gravity pulls IN
      |_____|
      | EARTH |
      |_____|
  
```

it IS falling - all the time. But it is also moving sideways so fast that as it falls the ground curves away beneath it. It falls and misses, forever.

\*Fig. 8: Why a satellite does not fall down\*

12. Explain why a satellite in orbit does not fall to the earth, even though gravity is pulling it down.

## SECTION B - PRACTICAL TASK

### MEASURING FRICTION WITH A SPRING BALANCE

| HOW IT IS DONE     | THE READING     |       |       |
|--------------------|-----------------|-------|-------|
| +-----+            | +-----+         |       |       |
|                    | 6 -             |       |       |
| pull [ ] [N]       | 4 -             |       |       |
| --> block          | 2 - <- <-       |       |       |
| =====              | 0 - pointer     |       |       |
| <-- friction       |                 |       |       |
|                    | read at EYE     |       |       |
| pull gently so the | LEVEL           |       |       |
| block slides at a  |                 |       |       |
| STEADY speed       | at steady speed |       |       |
|                    | pull = friction |       |       |
| +-----+            | +-----+         |       |       |
| SURFACE            | no load         | +1 kg | +2 kg |
| bare bench         | 2.0 N           | 3.1 N | 4.3 N |
| sandpaper          | 4.6 N           | 6.8 N | 9.0 N |
| oiled bench        | 1.1 N           | 1.8 N | 2.5 N |
| on rollers         | 0.3 N           | 0.5 N | 0.7 N |

\*Fig. 9: Measur

13. From the results table, state TWO conclusions the pupils can draw, quoting figures from the table to support each. [3 marks]
14. Explain why the block has to be pulled at a STEADY speed for this method to give the friction, and explain what the reading would show if the block were accelerating instead. [3 marks]
15. State TWO things the pupils must keep the same when they compare the four surfaces, and explain why each matters. [3 marks]

---  
**## ANSWER KEY AND MARKING SCHEME**

**\*\*14.1 14.1 What is Force?\*\***

- **\*\*Q1\*\*** (4 marks) (a) Non-contact. The magnet and the compass needle never touch.
- **\*\*Q2\*\*** (3 marks) A force is a push or a pull acting on an object; it can change the motion of an object.

**\*\*14.2 14.2 Balanced and Unbalanced Forces\*\***

- **\*\*Q3\*\*** (3 marks) 1. It STOPS the ball - the ball was moving and the force from the hand stopped it.

- 12.** Explain why a satellite in orbit does not fall to the earth, even though gravity is pulling on it the whole time. [3 marks]
- 13.** From the results table, state TWO conclusions the pupils can draw, quoting figures from the table to support each. [3 marks]
- 14.** Explain why the block has to be pulled at a STEADY speed for this method to give the friction, and explain what the reading would show if the block were accelerating instead. [3 marks]
- 15.** State TWO things the pupils must keep the same when they compare the four surfaces, and explain why each matters. [3 marks]

# Diagram Paper 13: Human Health Part 1 - I

Time Allowed: \*\* 1 hour 30 minutes

## INSTRUCTIONS TO CANDIDATES

1. Answer ALL the questions in this paper.
2. Every question is based on a diagram. Study each diagram carefully before you answer the questions that follow it.
3. Write your answers in the spaces provided. The marks for each question are printed beside it.
4. Where a question asks for a stated number of points, give that number. Any extra answers will not be marked.
5. You may refer back to any diagram at any time.

## Food Nutrients and the Human Body

| THE FOOD NUTRIENTS  |                                              |                                                      |
|---------------------|----------------------------------------------|------------------------------------------------------|
| NUTRIENT            | WHAT IT DOES                                 | GOOD SOURCES IN GHANA                                |
| CARBO-<br>  HYDRATE | the body's main fuel<br>  releases energy    | rice, maize, yam,<br>  cassava, plantain, bread      |
| PROTEIN             | growth and repair:<br>  muscle, skin, blood  | fish, meat, eggs, beans,<br>  groundnuts, soya, milk |
| FAT                 | stores energy,<br>  keeps you warm           | palm oil, groundnut oil,<br>  shea butter, coconut   |
| VITAMINS            | tiny amounts, to<br>  keep the body working  | fruit, dark green<br>  leaves, palm oil, liver       |
| MINERALS            | teeth, bone, blood,<br>  nerves              | salt, milk, fish bones,<br>  green leaves, liver     |
| WATER               | the solvent of the<br>  body: carries, cools | water, soup, fruit, any<br>  liquid you drink        |

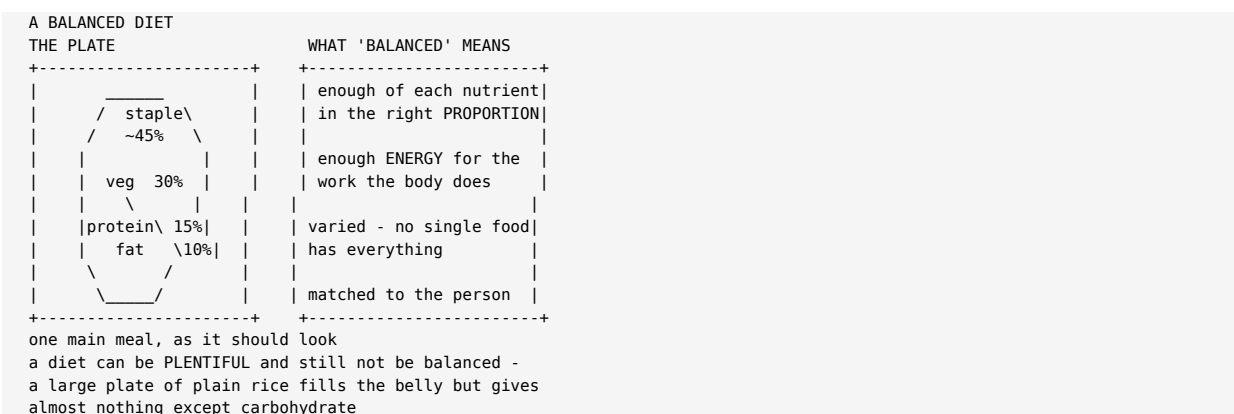
\*Fig. 1: The food nutrients\*

1. Explain why a child who eats a large plate of plain rice every day can still be
  2. Name THREE food nutrients. \*\*[3 marks]\*\*
  3. A pupil eats gari and beans for lunch. State which nutrient each part mainly
  4. State the FUNCTION of each of the following in the body: carbohydrate; protei
- #### 17.2 17.2 Deficiency Diseases: Causes, Symptoms, Effects and Prevention

1. Explain why a child who eats a large plate of plain rice every day can still be MALNOURISHED, referring to what the rice does and does not supply. [4 marks]
2. Name THREE food nutrients. [3 marks]
3. A pupil eats gari and beans for lunch. State which nutrient each part mainly supplies, and explain why the two together make a better meal than gari alone. [3 marks]
4. State the FUNCTION of each of the following in the body: carbohydrate; protein; roughage. [3 marks]



## Deficien



\*Fig. 2: A balanced diet\*

5. Explain what is meant by a BALANCED DIET, and explain why the diet that suits  
6. Use the plate in Fig. 2 to describe how a school lunch should be made up, and  
7. State TWO reasons why a diet of only ONE food is not a balanced diet. \*\*[3  
#### 17.3 17.3 Night Blindness, Rickets, Scurvy, Kwashiorkor and Anaemia

5. Explain what is meant by a BALANCED DIET, and explain why the diet that suits a labourer working on a farm does not suit an office worker of the same age and the same weight. [4 marks]
6. Use the plate in Fig. 2 to describe how a school lunch should be made up, and explain why the staple should be the largest part. [3 marks]
7. State TWO reasons why a diet of only ONE food is not a balanced diet. [3 marks]

## Night Blindness, Rickets, Scurvy, Kwashiorkor and Anaemia

| WHEN SOMETHING IS MISSING |                 |                                      |                                       |
|---------------------------|-----------------|--------------------------------------|---------------------------------------|
| MISSING                   | THE DISEASE     | WHAT YOU SEE                         | EAT MORE OF                           |
| protein                   | KWASHIORKOR     | swollen belly,<br>thin hair          | fish, beans,<br>eggs, ground-<br>nuts |
| energy                    | MARASMUS        | wasted: skin<br>and bone only        | more food of<br>every kind            |
| vitamin A                 | night blindness | cannot see in<br>dim light           | palm oil,<br>liver, leaves            |
| vitamin B1                | BERIBERI        | weak, numb<br>limbs, swollen<br>feet | unrefined<br>rice, whole<br>grains    |
| vitamin C                 | SCURVY          | bleeding gums,<br>slow healing       | oranges,<br>mango, tomato             |

8. Explain why KWASHIORKOR can develop in a child who is getting enough to eat but  
 9. One child is pale, tires easily and is breathless after only slight exertion.  
 10. Explain why a child who eats plenty of WHITE, well-milled rice may develop beriberi.  
 11. State what is meant by a DEFICIENCY disease, and name TWO. \*\*[3 marks]\*\*  
**## SECTION B - PRACTICAL TASK**

\*Fig. 3: When some

| TESTING FOODS FOR NUTRIENTS<br>HOW EACH TEST WORKS    | THE RESULTS                                            |
|-------------------------------------------------------|--------------------------------------------------------|
| STARCH add IODINE                                     | FOOD IOD BEN BIU                                       |
| POSITIVE: blue-black                                  | rice BBl -- --                                         |
| NEGATIVE: orange                                      | egg white -- -- viol                                   |
| GLUCOSE add BENEDICT'S<br>and HEAT in a water<br>bath | orange -- red --<br>juice<br>ground- -- -- viol<br>nut |
| POSITIVE: brick-red                                   | yam BBl -- --                                          |
| NEGATIVE: stays blue                                  | milk -- orange<br>viol                                 |
| PROTEIN add BIURET                                    |                                                        |
| POSITIVE: violet                                      |                                                        |
| NEGATIVE: stays blue                                  |                                                        |

always run a CONTROL with plain water: it shows the reagents start off negative  
 iodine stains; Biuret burns the skin; Benedict's must be

\*Fig. 4: Testing foods for nutrients\*

12. From the results, explain why MILK gave a positive result with both Benedict  
 13. Explain why a CONTROL containing plain water must be set up alongside the fo  
 14. A pupil tests a food with Benedict's solution and sees NO change. Describe w  
 15. State TWO safety precautions needed for these tests, and explain the hazard

**## ANSWER KEY AND MARKING SCHEME****\*\*17.1 17.1 Food Nutrients and the Human Body\*\***

- \*\*Q1\*\* (4 marks) Plain rice is almost entirely carbohydrate, so it supplies en  
 - \*\*Q2\*\* (3 marks) Any three of: carbohydrate, protein, fat, vitamins, minerals,  
 - \*\*Q3\*\* (3 marks) Gari is made from cassava and is mainly carbohydrate - it sup  
 - \*\*Q4\*\* (3 marks) Carbohydrate: the body's main source of energy, released by r

8. Explain why KWASHIORKOR can develop in a child who is getting enough to eat but eat by volume, and explain why it is most often seen soon after a child has been weaned. [4 marks]
9. One child is pale, tires easily and is breathless after only slight exertion. A second child cannot see properly in dim light. Name the deficiency in each case, and for each suggest TWO foods that would correct it. [4 marks]
10. Explain why a child who eats plenty of WHITE, well-milled rice may develop beriberi, and what should be done about it. [3 marks]
11. State what is meant by a DEFICIENCY disease, and name TWO. [3 marks]
12. From the results, explain why MILK gave a positive result with both Benedict's solution and Biuret reagent, while COOKED RICE gave a positive result with iodine only. Say what each result tells you about the food. [4 marks]
13. Explain why a CONTROL containing plain water must be set up alongside the food tests, and say what the control should show. [3 marks]
14. A pupil tests a food with Benedict's solution and sees NO change. Describe what she must do before concluding that the food contains no glucose, and explain why. [3 marks]

**15.**

# Diagram Paper 14: Human Health Part 2 - V

Time Allowed: \*\* 1 hour 30 minutes

## INSTRUCTIONS TO CANDIDATES

1. Answer ALL the questions in this paper.
2. Every question is based on a diagram. Study each diagram carefully before you answer the questions that follow it.
3. Write your answers in the spaces provided. The marks for each question are printed beside it.
4. Where a question asks for a stated number of points, give that number. Any extra answers will not be marked.
5. You may refer back to any diagram at any time.

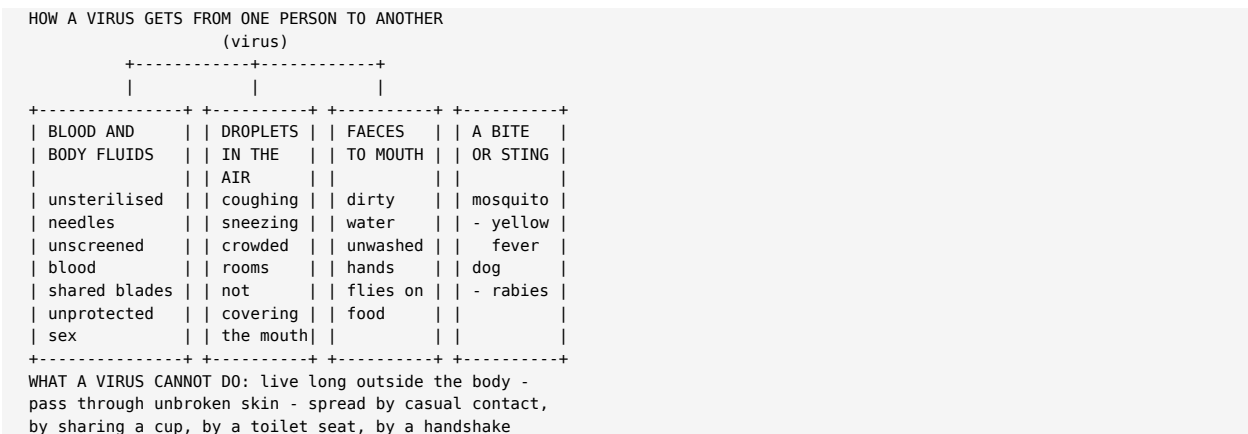
## Nature of Viral Diseases

| VIRAL DISEASES |                               |                                            |                         |
|----------------|-------------------------------|--------------------------------------------|-------------------------|
| DISEASE        | WHAT IT ATTACKS               | THE SIGN                                   | HOW IT SPREADS          |
| HIV/AIDS       | the immune system itself      | no early sign; weight loss, repeat infect. | blood and body fluids   |
| HEPATITIS B    | the liver                     | yellow eyes, dark urine                    | blood and body fluids   |
| MEASLES        | the airways and the skin      | fever, then red blotchy rash               | droplets in the air     |
| POLIO          | the nerves that drive muscles | sudden weakness or paralysis               | faeces to mouth         |
| RABIES         | the brain and nerves          | fear of water, confusion                   | bite of an infected dog |
| CHICKEN-POX    | the skin                      | itchy blisters                             | droplets, and           |

\*Fig. 1: Viral diseases\*

1. Explain why HIV is more serious than most viral infections even though the virus itself does not usually kill the person directly. [4 marks]
2. Name any THREE viral diseases shown in the diagram. \*\*[3 marks]\*\*
3. One child has a fever followed by a red blotchy rash that began behind the ears. A second child has yellow eyes and dark urine. Name the likely disease in each case, and say which part of the body each one attacks. [3 marks]

1. Explain why HIV is more serious than most viral infections even though the virus itself does not usually kill the person directly. [4 marks]
2. Name any THREE viral diseases shown in the diagram. [3 marks]
3. One child has a fever followed by a red blotchy rash that began behind the ears. A second child has yellow eyes and dark urine. Name the likely disease in each case, and say which part of the body each one attacks. [3 marks]

**COVID-19**

\*Fig. 2: How a virus gets from one person to another\*

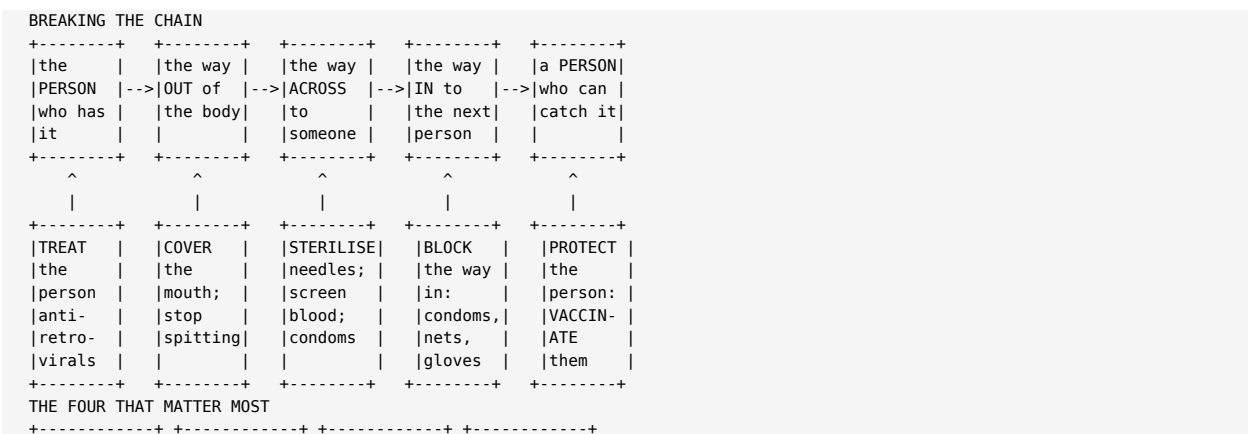
4. Explain why HIV CANNOT be spread by sharing a cup, by a handshake or by a mosquito bite, when it IS spread by sharing a needle. Refer to what the virus needs in order to infect someone. [4 marks]

5. State THREE ways in which viruses can be spread. \*\*[3 marks]\*\*

6. A village has an outbreak of a disease spread by the FAECES-TO-MOUTH route. Suggest THREE things the village should do, and explain how each one breaks the chain. [3 marks]

#### 17.6 Ebola and H1N1 Diseases

4. Explain why HIV CANNOT be spread by sharing a cup, by a handshake or by a mosquito bite, when it IS spread by sharing a needle. Refer to what the virus needs in order to infect someone. [4 marks]
5. State THREE ways in which viruses can be spread. [3 marks]
6. A village has an outbreak of a disease spread by the FAECES-TO-MOUTH route. Suggest THREE things the village should do, and explain how each one breaks the chain. [3 marks]

**Ebola and H1N1 Diseases**

7. Explain how VACCINATION prevents a viral disease, and explain why it works for measles but there is still no vaccine that prevents HIV infection. [4 marks]
8. A health worker is about to take blood from a patient. State THREE precautions she should take, and explain what each one prevents. [3 marks]
9. State what is meant by SCREENING blood, and state TWO diseases it prevents. [3 marks]
- #### 17.7 17.7 Role of Individuals, Communities and Government

\*Fig. 3: Breaking t

7. Explain how VACCINATION prevents a viral disease, and explain why it works for measles but there is still no vaccine that prevents HIV infection. [4 marks]
8. A health worker is about to take blood from a patient. State THREE precautions she should take, and explain what each one prevents. [3 marks]
9. State what is meant by SCREENING blood, and state TWO diseases it prevents. [3 marks]

## Role of Individuals, Communities and Government

### WHAT THE ILLNESS DOES TO A FAMILY

#### THE FACTS

```

+-----+
| HIV is a virus, not a |
| punishment or a curse |
|
| it cannot be caught by |
| touching, sharing food, |
| or caring for someone |
|
| with treatment a person |
| can live a long, |
| working life |
|
| treated properly the |
| virus becomes unde- |
| tectable and cannot be |
| passed on |
|
| a person with HIV can |

```

#### WHAT STIGMA DOES

```

+-----+
| the person hides the |
| illness and is not |
| tested |
|
| they delay treatment |
| until it is far |
| advanced |
|
| others go untested too, |
| afraid of the same |
| treatment |
|
| so the virus spreads |
| further, in silence |
|
| the person loses work, |
| friends and family for |

```

\*Fig. 4: What the illness does to a family\*

10. Explain why STIGMA makes the spread of HIV worse rather than better, and explain why it works for measles but there is still no vaccine that prevents HIV infection. [4 marks]
11. A man in a village is found to have HIV and his family want him to move out. State THREE facts about living with HIV that are true today and that many people believe. [3 marks]
12. State THREE facts about living with HIV that are true today and that many people believe. [3 marks]
- ## SECTION B - PRACTICAL TASK

### HOW FAST DOES AN INFECTION SPREAD?

no real virus is used - only water, an indicator, and one hidden 'infected' cup

#### THE METHOD

```

+-----+
| o o o o o |
| / \ / \ / \ / \ / \ |
| [ ] [ ] [ ] [ ] [ ] |
| --> --> --> --> |
| swap swap swap swap |
| P1 P2 P3 P4 P5 |
+-----+

```

#### THE RESULT

```

+-----+
| - - - - - |
| | | | | | | | | | | |
| |P| |P| |P| |P| |P| |
| |1| |2| |3| |4| |5| |
| | | | | | |
| pink blue pink pink blue |
+-----+

```

| ROUND | SWAPS | NUMBER 'INFECTED' | OUT OF |
|-------|-------|-------------------|--------|
| start | 0     | 1                 | 5      |
| 1     | 5     | 1                 | 5      |
| 2     | 10    | 3                 | 5      |
| 3     | 15    | 5                 | 5      |

every exchange is a risk

\*Fig. 5: How fa

13. From the results, explain why the number 'infected' rose so quickly. Explain  
 14. Explain why NO REAL VIRUS was used in this practical, and explain why the 'i  
 15. State TWO safety precautions for this practical, and suggest ONE change to t  
 ---  
 ## ANSWER KEY AND MARKING SCHEME  
 \*\*17.4 17.4 Nature of Viral Diseases\*\*  
 - \*\*Q1\*\* (4 marks) HIV attacks the immune system itself - the very cells that co  
 - \*\*Q2\*\* (3 marks) Any three of: HIV/AIDS, hepatitis B, measles, polio, rabies,  
 - \*\*Q3\*\* (3 marks) The first child has MEASLES. The rash begins behind the ears  
 \*\*17.5 17.5 COVID-19: Causes, Symptoms, Transmission and Prevention\*\*

- 10.** Explain why STIGMA makes the spread of HIV worse rather than better, and explain what difference knowing the facts in Fig. 4 makes to how a family should treat a relative who has the virus. [4 marks]
- 11.** A man in a village is found to have HIV and his family want him to move out and eat separately. Explain why this is unnecessary, and suggest what the family should do instead. [3 marks]
- 12.** State THREE facts about living with HIV that are true today and that many people do not know. [3 marks]
- 13.** From the results, explain why the number 'infected' rose so quickly. Explain in detail what this shows about why you cannot tell by looking whether someone has a virus. [4 marks]
- 14.** Explain why NO REAL VIRUS was used in this practical, and explain why the 'infected' cup could not be identified until the indicator was added. [3 marks]
- 15.** State TWO safety precautions for this practical, and suggest ONE change to the method that would slow the spread, explaining why it would work. [3 marks]

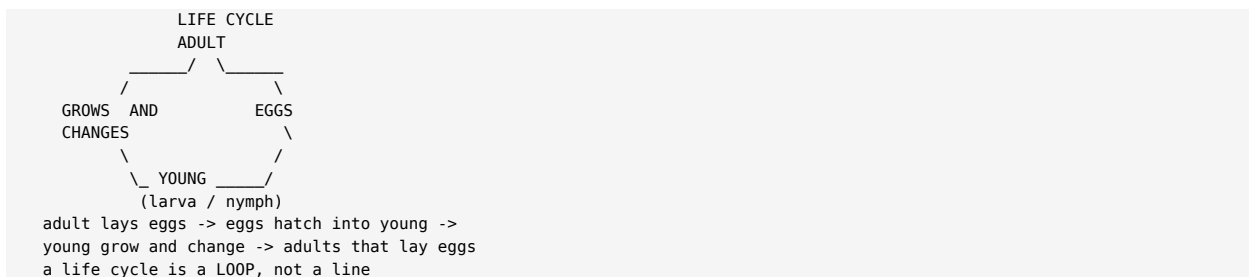
# Diagram Paper 15: Life Cycle of Organisms

Time Allowed: \*\* 1 hour 30 minutes

## INSTRUCTIONS TO CANDIDATES

1. Answer ALL the questions in this paper.
2. Every question is based on a diagram. Study each diagram carefully before you answer the questions that follow it.
3. Write your answers in the spaces provided. The marks for each question are printed beside it.
4. Where a question asks for a stated number of points, give that number. Any extra answers will not be marked.
5. You may refer back to any diagram at any time.

## What is a Life Cycle?



\*Fig. 1: A life cycle is a loop, not a line\*

1. The diagram shows a life cycle as a wheel with arrows going round it. Explain
2. Name the four stages shown in the life cycle wheel. \*\*[2 marks]\*\*

#### 4.2 4.2 Stages of the Life Cycle of the Housefly: Egg → Larva → Pupa → Adu

1. The diagram shows a life cycle as a wheel with arrows going round it. Explain why a life cycle is drawn as a CYCLE rather than as a straight line, and explain what would happen to the species if the loop were broken at any one point. [4 marks]
2. Name the four stages shown in the life cycle wheel. [2 marks]

## Stages of the Life Cycle of the Housefly: Egg → Larva → Pupa → Adult

| EGG                                                                            | LARVA                           | PUPA                                 | ADULT                                 |
|--------------------------------------------------------------------------------|---------------------------------|--------------------------------------|---------------------------------------|
| [8-24 h]                                                                       | [3-5 days]                      | [3-6 days]                           | [2-4 weeks]                           |
| up to 150 eggs laid on rotting matter                                          | feeds and grows, moulting twice | inside a hard brown case; no feeding | lays eggs, and the cycle begins again |
| COMPLETE METAMORPHOSIS - four stages, and the young do NOT look like the adult |                                 |                                      |                                       |



\*Fig. 2: The Li

3. The diagram shows the housefly larva moulting as it grows. Explain why a maggot has to moult, and explain what makes the housefly's development a COMPLETE metamorphosis. [4 marks]

4. Use the times given on the diagram to explain why one uncovered heap of rubbish can suddenly produce a very large number of flies within about two weeks. [3 marks]

#### 4.3 4.3 Feeding Habits of the Housefly

3. The diagram shows the housefly larva moulting as it grows. Explain why a maggot has to moult, and explain what makes the housefly's development a COMPLETE metamorphosis. [4 marks]
4. Use the times given on the diagram to explain why one uncovered heap of rubbish can suddenly produce a very large number of flies within about two weeks. [3 marks]

### Feeding Habits of the Housefly

| THE MOUTH                                                                                | HOW IT GETS FOOD                                         |
|------------------------------------------------------------------------------------------|----------------------------------------------------------|
| (o) ---- compound eyes                                                                   | 1. lands on faeces, rubbish or a corpse                  |
| [proboscis]                                                                              | 2. spits saliva onto the food and soaks it into a liquid |
|                                                                                          | 3. sponges it up                                         |
| (_) sponging pad                                                                         | 4. some filth stays on its legs and body                 |
| no teeth, no jaws                                                                        |                                                          |
| - cannot bite or chew                                                                    |                                                          |
| because it cannot bite, the fly must LIQUEFY its food first - and that carries the germs |                                                          |

\*Fig. 3: How the housefly feeds\*

5. Explain the connection between the housefly's MOUTHPARTS and the way it spreads disease. Use the diagram. [4 marks]

6. The diagram shows the fly spitting saliva onto its food. Explain why it has to do this. [3 marks]

#### 4.4 4.4 How the Housefly Affects Human Health

5. Explain the connection between the housefly's MOUTHPARTS and the way it spreads disease. Use the diagram. [4 marks]
6. The diagram shows the fly spitting saliva onto its food. Explain why it has to do this. [3 marks]

### How the Housefly Affects Human Health

| 1. FILTH                                           | 2. FLY LANDS                                  | 3. FLY LANDS AGAIN            | 4. PERSON                            |
|----------------------------------------------------|-----------------------------------------------|-------------------------------|--------------------------------------|
|                                                    |                                               |                               |                                      |
| faeces, rubbish, a corpse                          | germs stick to the hairs on its legs and body | on cooked food left uncovered | eats the food and swallows the germs |
| the fly does not MAKE the germs - it CARRIES them  |                                               |                               |                                      |
| it cannot bite, so nothing is injected             |                                               |                               |                                      |
| break the chain at ANY step to prevent the illness |                                               |                               |                                      |

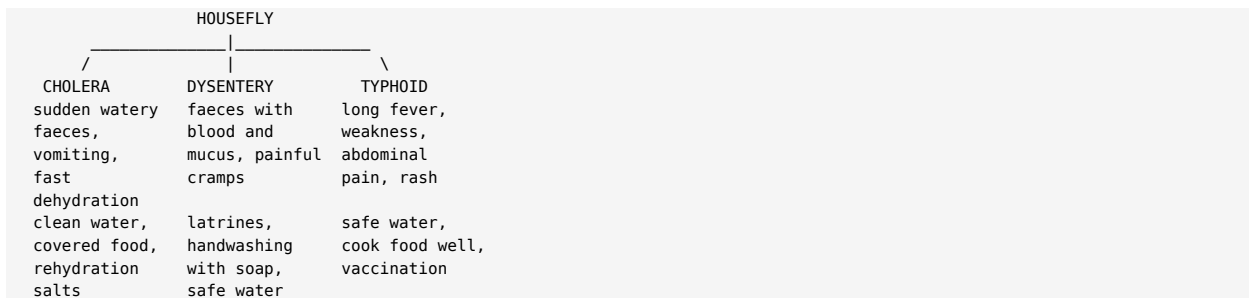
\*Fig. 4: How germs reach a person\*

7. Use the diagram to describe how germs get from faeces into a person's body, and state ONE point at which you would break the chain. [4 marks]

8. A pupil says 'the housefly GIVES you cholera'. Explain why this is not accurate, and why the difference matters. [3 marks]

#### 4.5 4.5 Diseases Spread by the Housefly

7. Use the diagram to describe how germs get from faeces into a person's body, and state ONE point at which you would break the chain. [4 marks]
8. A pupil says 'the housefly GIVES you cholera'. Explain why this is not accurate, and why the difference matters. [3 marks]



\*Fig. 5: Diseases the housefly spreads\*

9. From the diagram, name the disease whose prevention includes VACCINATION, and 10. State TWO symptoms of dysentery, and state ONE way of preventing it. \*\*[3

### 4.6 4.6 How to Reduce the Effects of Housefly Activities

9. From the diagram, name the disease whose prevention includes VACCINATION, and explain why vaccination on its own will not protect against the other two. [3 marks]

10. State TWO symptoms of dysentery, and state ONE way of preventing it. [3 marks]

## How to Reduce the Effects of Housefly Activities

|                                                                                                  |                                                                    |                                                         |
|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|---------------------------------------------------------|
| SCREENED<br>WINDOWS<br>stops it<br>getting in                                                    | COVERED<br>BIN<br>stops it<br>breeding                             | PIT LATRINE<br>stops it reaching<br>human faeces        |
| SPRAYING<br>kills the<br>adults<br>already<br>present                                            | COVERING<br>FOOD<br>stops it<br>walking<br>germs onto<br>your food | CLEARING BUSH<br>removes where<br>it rests and<br>hides |
| killing adults only deals with today's flies<br>stopping BREEDING deals with the next generation |                                                                    |                                                         |

\*Fig. 6: Breaking the chain\*

11. The diagram shows six ways of controlling houseflies. Explain why spraying t 12. State which control method stops flies getting INTO the house, and explain h

## SECTION B - PRACTICAL TASK

| JAR A                                     | JAR B                                    | JAR C                            | JAR D                     |
|-------------------------------------------|------------------------------------------|----------------------------------|---------------------------|
| rotting<br>fruit,<br>OPEN                 | rotting<br>fruit,<br>COVERED<br>with net | fresh<br>cooked<br>rice,<br>OPEN | dry soil<br>only,<br>OPEN |
| four jars left in the shade for five days |                                          |                                  |                           |
| JAR                                       | CONTENTS                                 | MAGGOTS                          |                           |
| A                                         | rotting fruit, open                      | many                             |                           |
| B                                         | rotting fruit, covered                   | none                             |                           |
| C                                         | fresh cooked rice, open                  | none                             |                           |
| D                                         | dry soil, open                           | none                             |                           |

\*Fig. 7: Where do houseflies breed?\*

13. From the results, state what the experiment shows about where houseflies bre

14. Before doing the experiment, a pupil predicted that maggots would appear in

15. State TWO things that must be kept the same for all four jars, and explain w

---

## ANSWER KEY AND MARKING SCHEME

\*\*4.1 4.1 What is a Life Cycle?\*

- \*\*Q1\*\* (4 marks) It is a cycle because it comes back to where it started: the

- \*\*Q2\*\* (2 marks) Adult, eggs, young (larva or nymph), and growing and changing

\*\*4.2 4.2 Stages of the Life Cycle of the Housefly: Egg → Larva → Pupa → Adult\*

- \*\*Q3\*\* (4 marks) A maggot has to moult because it is wrapped in an outer skin,

11. The diagram shows six ways of controlling houseflies. Explain why spraying to kill the adult flies works less well in the long run than removing the places where they breed. [4 marks]

**12. 13.**

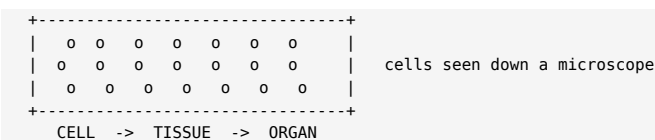
# Diagram Paper 16: Living Cells

Time Allowed: \*\* 1 hour 30 minutes

## INSTRUCTIONS TO CANDIDATES

1. Answer ALL the questions in this paper.
2. Every question is based on a diagram. Study each diagram carefully before you answer the questions that follow it.
3. Write your answers in the spaces provided. The marks for each question are printed beside it.
4. Where a question asks for a stated number of points, give that number. Any extra answers will not be marked.
5. You may refer back to any diagram at any time.

## What is a Cell? Introduction to Living Cells

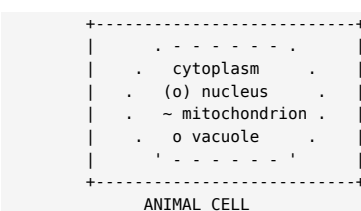


\*Fig. 1: Cells seen down a microscope, and how they build up\*

1. Explain why a single muscle CELL on its own cannot do the work of a muscle TISSUE. Use the diagram to support your answer. [3 marks]
2. An amoeba is a living thing made of ONE cell. State, with a reason from the diagram, whether it is a TISSUE or an ORGANISM. [3 marks]

1. Explain why a single muscle CELL on its own cannot do the work of a muscle TISSUE. Use the diagram to support your answer. [3 marks]
2. An amoeba is a living thing made of ONE cell. State, with a reason from the diagram, whether it is a TISSUE or an ORGANISM. [3 marks]

## Structure and Functions of the Animal Cell



\*Fig. 2: An ani

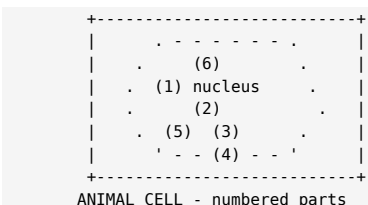
3. Explain how the function of the CELL MEMBRANE differs from the function of the CYTOPLASM. [3 marks]

4. A poison stops the mitochondria from working. PREDICT what would happen to the cell, and explain why. [4 marks]

#### 2.3 2.3 Organelles of the Animal Cell and Their Functions

3. Explain how the function of the CELL MEMBRANE differs from the function of the CYTOPLASM. [3 marks]
4. A poison stops the mitochondria from working. PREDICT what would happen to the cell, and explain why. [4 marks]

## Organelles of the Animal Cell and Their Functions



\*Fig. 3: An animal cell with its parts numbered\*

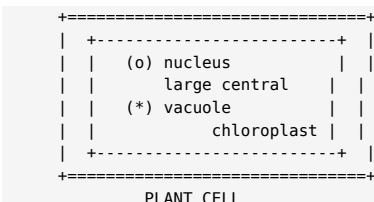
5. Name the parts numbered 2 and 5, and state ONE function of each. \*\*[3 marks]

6. Explain why part 1, the nucleus, is often described as the CONTROL CENTRE of the cell. [3 marks]

#### 2.4 2.4 Structure and Functions of the Plant Cell

5. Name the parts numbered 2 and 5, and state ONE function of each. [3 marks]
6. Explain why part 1, the nucleus, is often described as the CONTROL CENTRE of the cell. [3 marks]

## Structure and Functions of the Plant Cell



\*Fig. 4: A plant cell, with its parts labelled\*

7. Explain why a plant cell has a CELL WALL but an animal cell does not. Relate your answer to how each organism supports itself. [4 marks]

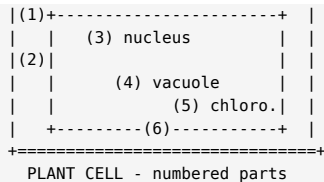
8. State the function of the LARGE CENTRAL VACUOLE, and explain why an animal cell does not need one as large. [3 marks]

#### 2.5 2.5 Organelles of the Plant Cell and Their Functions

7. Explain why a plant cell has a CELL WALL but an animal cell does not. Relate your answer to how each organism supports itself. [4 marks]
8. State the function of the LARGE CENTRAL VACUOLE, and explain why an animal cell does not need one as large. [3 marks]

## Organelles of the Plant Cell and Their Functions

+=====



\*Fig. 5: A plant cell with its parts numbered\*

9. Name the parts numbered 1, 4 and 5. **\*\*[3 marks]\*\***

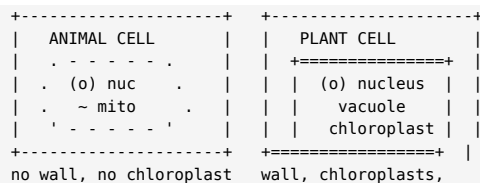
10. A plant kept in a dark cupboard makes FEWER chloroplasts than the same plant

#### 2.6 2.6 Differences and Similarities Between Plant and Animal Cells

**9.** Name the parts numbered 1, 4 and 5. [3 marks]

**10.** A plant kept in a dark cupboard makes FEWER chloroplasts than the same plant kept in the light. SUGGEST why, and explain what this shows about the function of the chloroplast. [3 marks]

## Differences and Similarities Between Plant and Animal Cells



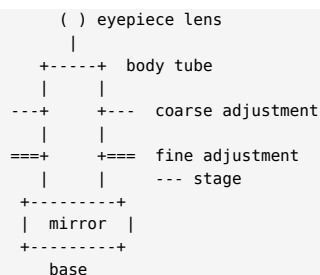
\*Fig. 6: An animal cell and a plant cell compared\*

11. State THREE differences between the animal cell and the plant cell shown in

12. Explain why a plant cell needs BOTH a cell wall AND a cell membrane, whereas

13. A pupil sees a cell down a microscope and says it must be a PLANT cell because

## SECTION B - PRACTICAL TASK



\*Fig. 7: A light microscope\*

14. A pupil looks down the microscope and sees only darkness. SUGGEST TWO things

15. Explain why the specimen on the slide must be thin enough for light to pass

16. A pupil turns the coarse adjustment knob while looking down the eyepiece, and

---

## ANSWER KEY AND MARKING SCHEME

\*\*2.1 2.1 What is a Cell? Introduction to Living Cells\*\*

- \*\*Q1\*\* (3 marks) A single cell can only contract with the force of one cell. A

- \*\*Q2\*\* (3 marks) It is an organism, not a tissue. The diagram shows that a tis

\*\*2.2 2.2 Structure and Functions of the Animal Cell\*\*

- \*\*Q3\*\* (3 marks) The cell membrane is the outer boundary: it holds the cell to

**11.** State THREE differences between the animal cell and the plant cell shown in the diagram. [3 marks]

**12.** Explain why a plant cell needs BOTH a cell wall AND a cell membrane, whereas an animal cell needs only a membrane. [4 marks]

**13.** A pupil sees a cell down a microscope and says it must be a PLANT cell because it has a nucleus. Explain why this reasoning is wrong, and say what the pupil should look for instead. [3 marks]

**14. 15.**

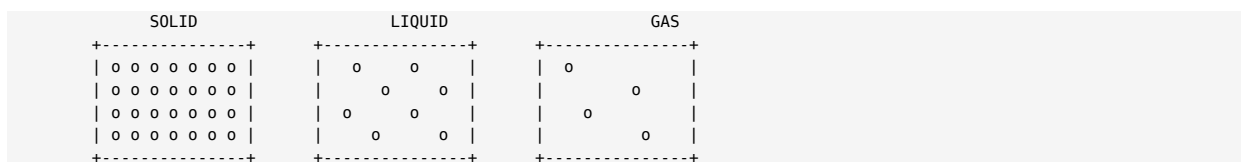
# Diagram Paper 17: Material

Time Allowed: \*\* 1 hour 30 minutes

## INSTRUCTIONS TO CANDIDATES

1. Answer ALL the questions in this paper.
2. Every question is based on a diagram. Study each diagram carefully before you answer the questions that follow it.
3. Write your answers in the spaces provided. The marks for each question are printed beside it.
4. Where a question asks for a stated number of points, give that number. Any extra answers will not be marked.
5. You may refer back to any diagram at any time.

## States of Matter: Solids, Liquids and Gases



\*Fig. 1: The particles in a solid, a liquid and a gas\*

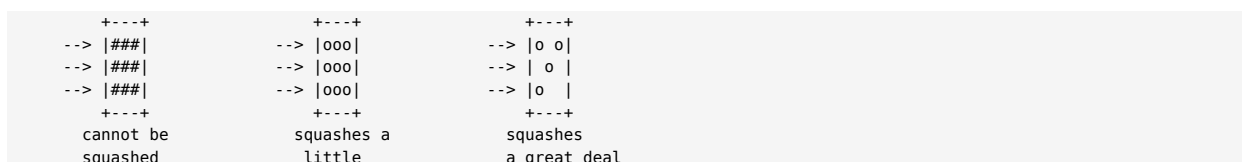
1. Use the diagram to explain why a solid has BOTH a fixed shape and a fixed vol
2. A pupil says that the particles in the gas must be BIGGER than the particles
3. Describe how the ARRANGEMENT of the particles and their MOVEMENT change as th

### 1.2 1.2 Properties and Differences Among Solids, Liquids and Gases

1. Use the diagram to explain why a solid has BOTH a fixed shape and a fixed volume, but a gas has NEITHER. [4 marks]
2. A pupil says that the particles in the gas must be BIGGER than the particles in the solid, because they are drawn further apart. Use the diagram to explain why this is wrong. [3 marks]
3. Describe how the ARRANGEMENT of the particles and their MOVEMENT change as the substance is heated from the solid state to the gas state. [2 marks]

## Properties and Differences Among Solids, Liquids and Gases





\*Fig. 2: Three syringes, with the plunger pushed in\*

4. Explain, in terms of the particles, why syringe R squashes easily but syringe

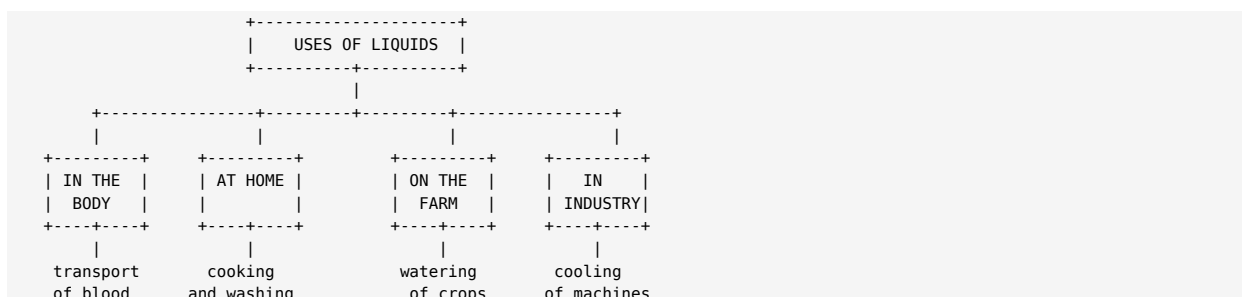
5. The plunger of syringe R is now pulled OUT instead of pushed in. PREDICT what

#### 1.3 1.3 Importance of Liquids to Human Life

4. Explain, in terms of the particles, why syringe R squashes easily but syringe P cannot be squashed at all. [3 marks]

5. The plunger of syringe R is now pulled OUT instead of pushed in. PREDICT what happens to the particles, and explain why it happens. [3 marks]

## Importance of Liquids to Human Life



\*Fig. 3: A concept map of the uses of liquids\*

6. Choose TWO uses of liquids from the diagram. For EACH one, name the property

7. Explain why water is used for washing but a solid such as a stone is not. Ref

#### 1.4 1.4 Importance of Solids to Human Life

6. Choose TWO uses of liquids from the diagram. For EACH one, name the property of liquids that makes them suitable for that use, and explain why the property matters. [4 marks]

7. Explain why water is used for washing but a solid such as a stone is not. Refer to the properties shown in the diagram. [2 marks]

## Importance of Solids to Human Life



8. Choose TWO solids from the diagram. For EACH one, explain how one PROPERTY of  
 9. Give ONE use of solids from the diagram that depends on the solid being HARD,  
 ### 1.5 1.5 The Periodic Table: Metals, Non-Metals and Noble Gases

\*Fig. 4: A concept

- 8.** Choose TWO solids from the diagram. For EACH one, explain how one PROPERTY of the solid makes it suitable for the use shown. [4 marks]
- 9.** Give ONE use of solids from the diagram that depends on the solid being HARD, and explain why hardness matters for that use. [2 marks]

## The Periodic Table: Metals, Non-Metals and Noble Gases

|                      |                                                 |   |  |                  |                    |                    |
|----------------------|-------------------------------------------------|---|--|------------------|--------------------|--------------------|
| GROUP:               | 1                                               | 2 |  | 16               | 17                 | 18                 |
| PERIOD 1             | (left-hand side and the middle<br>of the table) |   |  |                  | He<br>(noble)      |                    |
| PERIOD 2             |                                                 |   |  |                  |                    |                    |
| PERIOD 3             |                                                 |   |  |                  | F<br>non-<br>metal | Ne<br>noble<br>gas |
| PERIOD 4             |                                                 |   |  |                  |                    |                    |
| <----- METALS -----> |                                                 |   |  | <- NON-METALS -> |                    |                    |

\*Fig. 5: The main regions of the periodic table\*

10. A new element is discovered and placed in Group 18 at the bottom of the tabl  
 11. Fluorine (F) and lithium (Li) are both near the TOP of the table, yet fluori  
 ### 1.6 1.6 Arrangement of Elements in the Periodic Table

- 10.** A new element is discovered and placed in Group 18 at the bottom of the table. PREDICT whether it would be a metal or a non-metal, and give the reason from the diagram. [3 marks]
- 11.** Fluorine (F) and lithium (Li) are both near the TOP of the table, yet fluorine is a non-metal and lithium is a metal. Use the diagram to explain why. [3 marks]

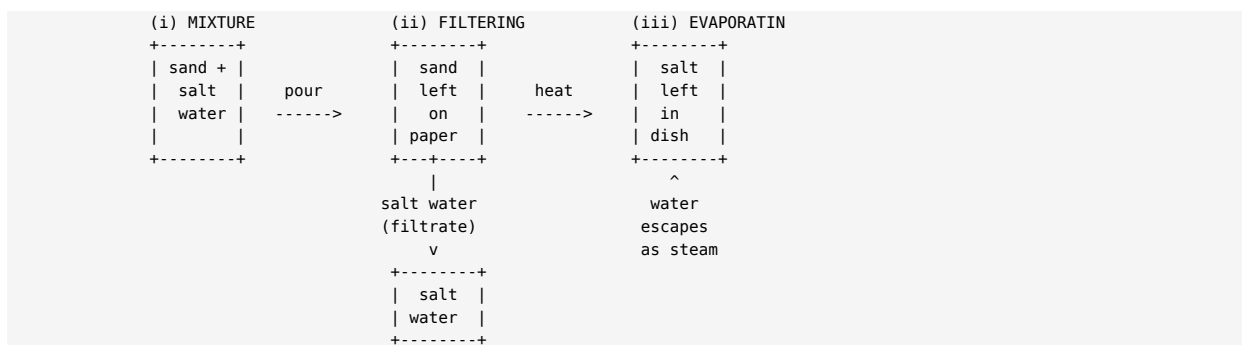
## Arrangement of Elements in the Periodic Table

|          |                                                                      |         |       |          |          |
|----------|----------------------------------------------------------------------|---------|-------|----------|----------|
|          | GROUP 1                                                              | GROUP 2 | ..... | GROUP 17 | GROUP 18 |
| PERIOD 1 | H                                                                    |         |       |          | He       |
| PERIOD 2 | Li                                                                   | Be      |       | F        | Ne       |
| PERIOD 3 | Na                                                                   | Mg      |       | Cl       | Ar       |
| PERIOD 4 | K                                                                    | Ca      |       | Br       | Kr       |
|          | v                                                                    | v       |       | v        | v        |
|          | elements in a COLUMN are a GROUP;<br>elements in a ROW are a PERIOD. |         |       |          |          |

\*Fig. 6: Period

12. From the diagram, write down the period number and the group number of magnesium (Mg) and of chlorine (Cl). [2 marks]
13. Sodium (Na) and potassium (K) are both in Group 1. Use what the diagram shows about groups to suggest TWO things the two elements would have in common, and give a reason for each. [3 marks]
14. Explain the difference between a period and a group, and say which of the two tells you more about how an element reacts. [2 marks]

## ## SECTION B - PRACTICAL TASK



\*Fig. 7: Apparatus for separating a mixture of sand and salt water\*

15. Name the process shown at stage (ii). \*\*[1 mark]\*\*
16. State what is left on the filter paper and what passes through it. \*\*[2 marks]
17. Explain, in terms of PARTICLE SIZE, why the sand is held back by the paper but the salt passes through with the water. [3 marks]
18. Describe how you would change the apparatus so that the water is collected instead of being allowed to escape, and name the process it would then be. [2 marks]
19. A pupil heats the salt water very strongly at stage (iii) and the salt spits out of the dish and is lost. Suggest ONE change to the METHOD that would prevent this, and explain why it works. [2 marks]

## ## ANSWER KEY AND MARKING SCHEME

\*\*1.1 1.1 States of Matter: Solids, Liquids and Gases\*\*

- \*\*Q1\*\* (4 marks) In a solid the particles are packed closely together in a regular arrangement.
- \*\*Q2\*\* (3 marks) The particles are exactly the same size in all three states - solid, liquid and gas.

12. From the diagram, write down the period number and the group number of magnesium (Mg) and of chlorine (Cl). [2 marks]
13. Sodium (Na) and potassium (K) are both in Group 1. Use what the diagram shows about groups to suggest TWO things the two elements would have in common, and give a reason for each. [3 marks]
14. Explain the difference between a period and a group, and say which of the two tells you more about how an element reacts. [2 marks]
15. Name the process shown at stage (ii). [1 marks]
16. State what is left on the filter paper and what passes through it. [2 marks]
17. Explain, in terms of PARTICLE SIZE, why the sand is held back by the paper but the salt passes through with the water. [3 marks]
18. Describe how you would change the apparatus so that the water is collected instead of being allowed to escape, and name the process it would then be. [2 marks]
19. A pupil heats the salt water very strongly at stage (iii) and the salt spits out of the dish and is lost. Suggest ONE change to the METHOD that would prevent this, and explain why it works. [2 marks]

## Diagram Paper 18: Science and Industry

Time Allowed: \*\* 1 hour 30 minutes

### INSTRUCTIONS TO CANDIDATES

1. Answer ALL the questions in this paper.
2. Every question is based on a diagram. Study each diagram carefully before you answer the questions that follow it.
3. Write your answers in the spaces provided. The marks for each question are printed beside it.
4. Where a question asks for a stated number of points, give that number. Any extra answers will not be marked.
5. You may refer back to any diagram at any time.

### Careers in Science and How They Improve Human Life

| FROM RAW MATERIAL TO FINISHED PRODUCT |                                   |                                       |                    |
|---------------------------------------|-----------------------------------|---------------------------------------|--------------------|
| RAW MATERIAL                          | THE PROCESS                       | THE PRODUCT                           | WHERE IN GHANA     |
| cocoa pods                            | ferment, dry, roast, grind, press | cocoa butter, cocoa powder, chocolate | Tema, Takoradi     |
| crude oil                             | refine: separate by boiling point | petrol, diesel, LPG                   | Tema refinery      |
| gold ore                              | crush, dissolve, recover          | gold bars                             | Obuasi, Tarkwa     |
| bauxite                               | refine to alumina, smelt          | aluminium                             | Ada Foah, Tema     |
| cassava                               | peel, grate, press, dry           | gari, starch, flour, ethanol          | across the country |
| shea nuts                             | crush, roast,                     | shea butter                           | Northern,          |

\*Fig. 1: From raw material to finished product\*

1. Explain what is meant by an INDUSTRY, and explain why the cocoa that Ghana exports as raw beans earns less than the cocoa that is finally sold as chocolate. [4 marks]
2. Choose ONE raw material from Fig. 1 that is found in your own region. Describe how it is processed into a product, and suggest TWO difficulties a factory using it would face. [4 marks]
3. Name THREE raw materials shown in the diagram. [3 marks]
4. State what is meant by a RAW MATERIAL, and give TWO examples. [3 marks]

#### 18.2 18.2 Ghanaian Scientists and Their Contributions

1. Explain what is meant by an INDUSTRY, and explain why the cocoa that Ghana exports as raw beans earns less than the cocoa that is finally sold as chocolate. [4 marks]
2. Choose ONE raw material from Fig. 1 that is found in your own region. Describe how it is processed into a product, and suggest TWO difficulties a factory using it would face. [4 marks]
3. Name THREE raw materials shown in the diagram. [3 marks]
4. State what is meant by a RAW MATERIAL, and give TWO examples. [3 marks]

| THE PEOPLE BEHIND THE SCIENCE                        |                                                                                 | SCIENTISTS FROM ELSEWHERE |                                             |
|------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------|---------------------------------------------|
| GHANAIAN SCIENTISTS                                  |                                                                                 |                           |                                             |
| Francis Allotey                                      | mathematical physics<br>soft X-ray spectroscopy;<br>the Allotey Formalism       | Isaac Newton              | motion, gravity, light                      |
| Marian Ewurama Addy                                  | biochemistry<br>tested herbal medicines<br>to find what really<br>works in them | Charles Darwin            | evolution by natural<br>selection           |
| Ibok Oduro                                           | food science<br>post-harvest loss,<br>fortified sweet potato                    | Marie Curie               | radioactivity; found<br>radium and polonium |
|                                                      |                                                                                 | Louis Pasteur             | germ theory,<br>pasteurisation,<br>vaccines |
| WHY WE LEARN THEIR NAMES: not to memorise dates, but |                                                                                 |                           |                                             |

\*Fig. 2: The people behind the science\*

5. Choose ONE Ghanaian scientist from Fig. 2 and ONE scientist from elsewhere. F

6. Name TWO Ghanaian scientists and state the field each worked in. \*\*[3 marks

7. A pupil says that remembering the NAMES of scientists is not real science. Gi

8. State ONE contribution of any scientist named in Fig. 2. \*\*[3 marks]\*\*

#### 18.4 18.4 Science, Technology and Innovation in Ghana

- Choose ONE Ghanaian scientist from Fig. 2 and ONE scientist from elsewhere. For each, state what they worked on, and explain why their work mattered to the country they lived in. [4 marks]
- Name TWO Ghanaian scientists and state the field each worked in. [3 marks]
- A pupil says that remembering the NAMES of scientists is not real science. Give TWO reasons why learning what they did is still worth doing. [3 marks]
- State ONE contribution of any scientist named in Fig. 2. [3 marks]

## Science, Technology and Innovation in Ghana

| WHERE A SCIENCE SUBJECT CAN TAKE YOU                                                                                            |                                                                                                                |                                                                                                                       |                          |
|---------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|--------------------------|
| BIOLOGY                                                                                                                         | CHEMISTRY                                                                                                      | PHYSICS                                                                                                               |                          |
| doctor, nurse,<br>midwife<br>pharmacist<br>lab technician<br>veterinary<br>officer<br>agricultural<br>officer<br>soil scientist | industrial<br>chemist<br>water quality<br>officer<br>pharmaceutical<br>maker<br>food<br>technologist<br>brewer | engineer -<br>civil,<br>electrical,<br>mechanical<br>electrician<br>geologist,<br>surveyor<br>pilot,<br>meteorologist |                          |
| WHAT EVERY ONE OF THEM NEEDS                                                                                                    |                                                                                                                |                                                                                                                       |                          |
| OBSERVE<br>carefully<br>write what                                                                                              | MEASURE<br>accurately<br>repeat it,                                                                            | KEEP RECORDS<br>a result                                                                                              | WORK<br>SAFELY<br>in the |

9. A pupil wants to become an AGRICULTURAL OFFICER. State which science subject  
 10. State TWO careers that need PHYSICS, and for each say what the person does.  
 11. Explain why an employer in ANY of these careers would value a pupil who keeps careful records.

## SECTION B - PRACTICAL TASK

#### WHICH LOCAL CROP GIVES THE MOST STARCH?

the same method a starch factory uses, done on a small scale

| THE METHOD            | THE RESULT                               |
|-----------------------|------------------------------------------|
| 1 PEEL and CUT 100 g  | CROP 1st 2nd MEAN<br>cassava 24g 26g 25g |
| 2 GRATE it finely     | yam 17g 19g 18g                          |
| 3 SIEVE with water    | plantain 14g 14g 14g                     |
| through a cloth       | cocoyam 12g 13g 13g                      |
| 4 let the milky water | sweet 11g 10g 11g                        |
| STAND 20 minutes,     | potato                                   |
| pour off the water,   | cassava gives more than                  |
| dry the sediment      | twice as much as sweet                   |
|                       | potato                                   |

WHY INDUSTRY CARES: a starch factory chooses its raw material on exactly this question - how much product comes out of each kilogram it buys. That decides the price it can

\*Fig. 4: Which local crop gives the most starch?\*

12. From the results, state which crop gives the MOST starch and which the LEAST  
 13. Explain why each crop was tested TWICE and a mean taken, and suggest ONE other thing the pupils must keep the same.  
 14. In this method the starch settles to the bottom and the water above is poured off. Explain why the starch settles and the water can be poured away, and explain what would happen if the mixture were shaken instead of left to stand.  
 15. Suggest TWO improvements to the method that would give a more accurate yield, and explain what each one fixes.

#### ## ANSWER KEY AND MARKING SCHEME

\*\*18.1 18.1 Careers in Science and How They Improve Human Life\*\*

- \*\*Q1\*\* (4 marks) An industry is an organised process that takes a raw material  
 - \*\*Q2\*\* (4 marks) Accept any raw material from the figure, marked on the quality  
 - \*\*Q3\*\* (3 marks) Any three of: cocoa pods, crude oil, gold ore, bauxite, cassava  
 - \*\*Q4\*\* (3 marks) A raw material is a natural substance, taken from the ground,

\*Fig. 3: Where a sc

9. A pupil wants to become an AGRICULTURAL OFFICER. State which science subject she must study, describe TWO things she would do in the job, and explain why the skills listed in Fig. 3 matter in that work. [4 marks]
10. State TWO careers that need PHYSICS, and for each say what the person does. [3 marks]
11. Explain why an employer in ANY of these careers would value a pupil who keeps careful records. [3 marks]
12. From the results, state which crop gives the MOST starch and which the LEAST, and explain why a starch factory would care both about this figure and about whether the comparison was FAIR. [4 marks]
13. Explain why each crop was tested TWICE and a mean taken, and suggest ONE other thing the pupils must keep the same. [3 marks]
14. In this method the starch settles to the bottom and the water above is poured off. Explain why the starch settles and the water can be poured away, and explain what would happen if the mixture were shaken instead of left to stand. [3 marks]
15. Suggest TWO improvements to the method that would give a more accurate yield, and explain what each one fixes. [3 marks]

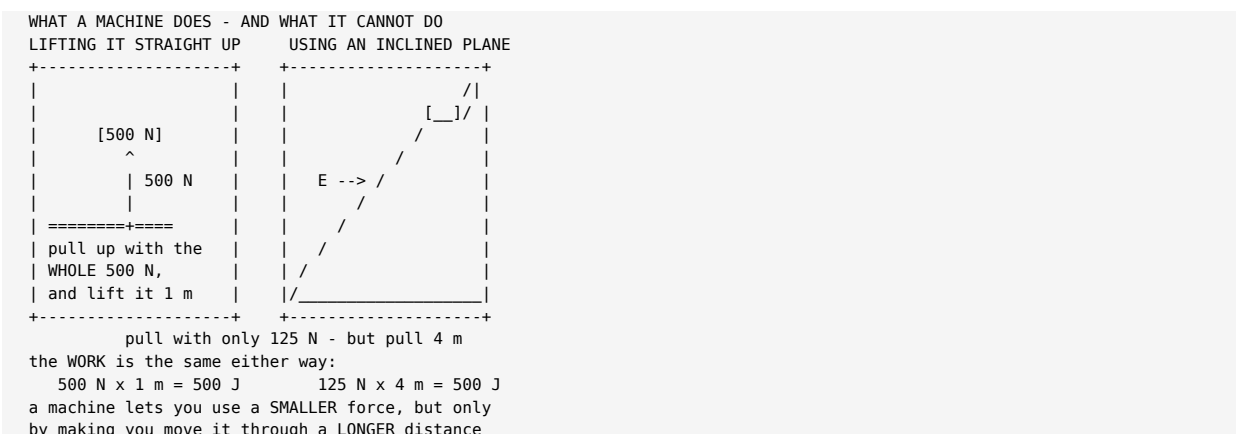
## Diagram Paper 19: Simple Machines

Time Allowed: \*\* 1 hour 30 minutes

### INSTRUCTIONS TO CANDIDATES

1. Answer ALL the questions in this paper.
2. Every question is based on a diagram. Study each diagram carefully before you answer the questions that follow it.
3. Write your answers in the spaces provided. The marks for each question are printed beside it.
4. Where a question asks for a stated number of points, give that number. Any extra answers will not be marked.
5. You may refer back to any diagram at any time.

### Simple Machines and Their Types



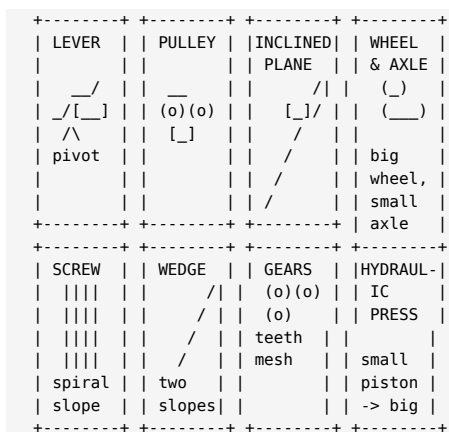
\*Fig. 1: What a machine does - and what it cannot do\*

1. The diagram shows a 500 N load lifted 1 m straight up, and the same load pushed 4 m up a slope with a force of 125 N. Explain why the slope is said to MAKE THE WORK EASIER when the work done is exactly the same in both cases. [4 marks]
2. State what is meant by a MACHINE, and state the ONE thing no machine can do. [3 marks]

#### 14.10 14.10 Types and Functions of Levers

1. The diagram shows a 500 N load lifted 1 m straight up, and the same load pushed 4 m up a slope with a force of 125 N. Explain why the slope is said to MAKE THE WORK EASIER when the work done is exactly the same in both cases. [4 marks]
2. State what is meant by a MACHINE, and state the ONE thing no machine can do. [3 marks]

### Types and Functions of Levers



\*Fig. 2: The simple machines\*

3. For each of the following, name the simple machine it is an example of, and say whether it mainly makes the force BIGGER or changes the DIRECTION of the force: (a) a bottle opener; (b) a rope over a wheel at the top of a flag pole; (c) an axe blade; (d) a ramp used to load a truck. [4 marks]

4. Name any THREE simple machines shown in the diagram. \*\*[3 marks]\*\*

5. Explain why a LONG spanner undoes a tight nut more easily than a short one. [3 marks]

6. A screw is described as a spiral inclined plane. Explain what this means, and explain why a screw with CLOSELY spaced threads is easier to drive in than one with widely spaced threads. [3 marks]

### 14.11 14.11 Work Input, Work Output and Efficiency

3. For each of the following, name the simple machine it is an example of, and say whether it mainly makes the force BIGGER or changes the DIRECTION of the force: (a) a bottle opener; (b) a rope over a wheel at the top of a flag pole; (c) an axe blade; (d) a ramp used to load a truck. [4 marks]
4. Name any THREE simple machines shown in the diagram. [3 marks]
5. Explain why a LONG spanner undoes a tight nut more easily than a short one. [3 marks]
6. A screw is described as a spiral inclined plane. Explain what this means, and explain why a screw with CLOSELY spaced threads is easier to drive in than one with widely spaced threads. [3 marks]

## Work Input, Work Output and Efficiency

| MEASURING HOW GOOD A MACHINE IS |                                                      |
|---------------------------------|------------------------------------------------------|
| THE THREE QUANTITIES            | WHY IT IS NEVER 100%                                 |
| MECHANICAL ADVANTAGE            | work in = useful work out + work lost                |
| load                            |                                                      |
| effort                          | useful work out   lost work                          |
| VELOCITY RATIO                  | work out   as   heat                                 |
| dist effort moves               | some effort always overcomes friction in the machine |
| dist load moves                 |                                                      |
| EFFICIENCY                      |                                                      |
| M.A.                            |                                                      |
| ----- x 100%                    |                                                      |
| V.R.                            |                                                      |



\*Fig. 3: Measur

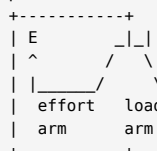
7. Explain why the MECHANICAL ADVANTAGE of a real machine is always less than it  
 8. A machine has a velocity ratio of 5. An effort of 50 N lifts a load of 200 N.  
 9. Suggest TWO ways of improving the EFFICIENCY of a pulley system, and explain  
 10. State what is meant by the VELOCITY RATIO of a machine. \*\*[3 marks]\*\*

## ## SECTION B - PRACTICAL TASK

## THE LEVER: WHERE SHOULD THE PIVOT GO?

## SET-UP 1

pivot near LOAD



## SET-UP 2

pivot in MIDDLE



## SET-UP 3

pivot near EFFORT



| SET-UP | EFFORT ARM cm | LOAD ARM cm | V.R. | EFFORT NEEDED N | M.A. | EFFICIENCY |
|--------|---------------|-------------|------|-----------------|------|------------|
| 1      | 80            | 20          | 4.0  | 2.6             | 3.8  | 95%        |
| 2      | 50            | 50          | 1.0  | 10.4            | 0.96 | 96%        |
| 3      | 20            | 80          | 0.25 | 42.0            | 0.24 | 95%        |

the load is a 10 N weight in every set-up

\*Fig. 4: The lever - where should the pivot go?\*

11. From the results, explain the relationship between the EFFORT ARM, the LOAD  
 12. Using the results for SET-UP 1, calculate the velocity ratio, the mechanical  
 13. A pupil wants to use a lever to lift a heavy stone with the SMALLEST possible  
 14. State what is meant by the EFFORT ARM of a lever. \*\*[2 marks]\*\*  
 15. State what is meant by the LOAD of a machine. \*\*[2 marks]\*\*

## ## ANSWER KEY AND MARKING SCHEME

\*\*14.9 14.9 Simple Machines and Their Types\*\*

- \*\*Q1\*\* (4 marks) The work is identical: lifting straight up is  $500 \text{ N} \times 1 \text{ m} = 500 \text{ J}$   
 - \*\*Q2\*\* (3 marks) A machine is a device that makes work easier by changing the

7. Explain why the MECHANICAL ADVANTAGE of a real machine is always less than its VELOCITY RATIO, and explain what has become of the difference. [4 marks]
8. A machine has a velocity ratio of 5. An effort of 50 N lifts a load of 200 N. Calculate the mechanical advantage and the efficiency, and explain what has happened to the missing percentage. [4 marks]
9. Suggest TWO ways of improving the EFFICIENCY of a pulley system, and explain how each one helps. [3 marks]
10. State what is meant by the VELOCITY RATIO of a machine. [3 marks]
11. From the results, explain the relationship between the EFFORT ARM, the LOAD ARM and the EFFORT NEEDED. Refer to all three set-ups. [4 marks]
12. Using the results for SET-UP 1, calculate the velocity ratio, the mechanical advantage and the efficiency. Comment on what the efficiency figure tells you. [4 marks]
13. A pupil wants to use a lever to lift a heavy stone with the SMALLEST possible effort. Describe what he should do with the pivot, and explain what disadvantage this brings. [4 marks]
14. State what is meant by the EFFORT ARM of a lever. [2 marks]

**15.**

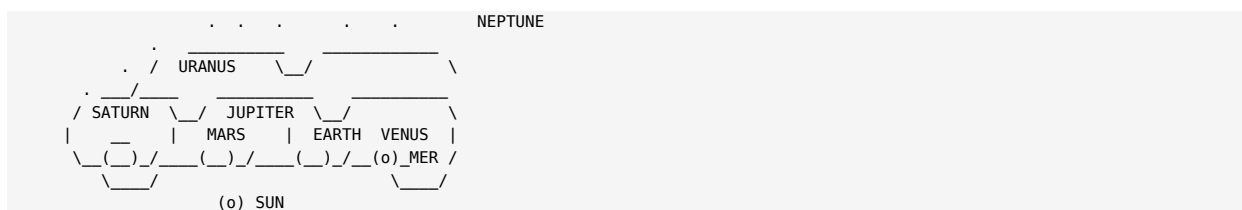
## Diagram Paper 20: The Solar System

Time Allowed: \*\* 1 hour 30 minutes

### INSTRUCTIONS TO CANDIDATES

1. Answer ALL the questions in this paper.
2. Every question is based on a diagram. Study each diagram carefully before you answer the questions that follow it.
3. Write your answers in the spaces provided. The marks for each question are printed beside it.
4. Where a question asks for a stated number of points, give that number. Any extra answers will not be marked.
5. You may refer back to any diagram at any time.

### What is the Solar System?



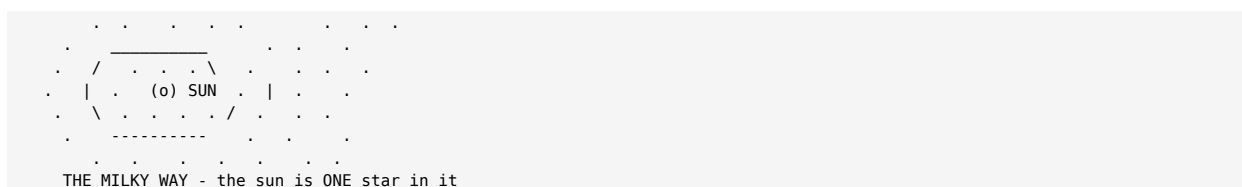
\*Fig. 1: The solar system\*

1. State what is at the centre of the solar system, and use the diagram to explain why the planets do not fly off into space. [4 marks]
2. Use the diagram to state ONE way the four inner planets differ from the four outer planets, and explain the difference. [3 marks]

#### 8.2 8.2 The Galaxy and the Milky Way

1. State what is at the centre of the solar system, and use the diagram to explain why the planets do not fly off into space. [4 marks]
2. Use the diagram to state ONE way the four inner planets differ from the four outer planets, and explain the difference. [3 marks]

### The Galaxy and the Milky Way

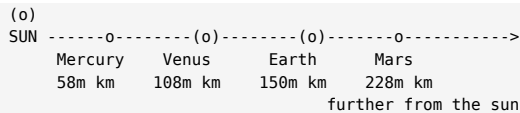


3. State what the Milky Way is, and mark or describe where the sun is in it. \*
4. A pupil says that the Milky Way and the solar system are the same thing seen from different places. Explain why this is wrong.
- #### 8.3 8.3 The Inner Planets: Mercury, Venus, Earth and Mars

\*Fig. 2: The Milky

3. State what the Milky Way is, and mark or describe where the sun is in it. [3 marks]
4. A pupil says that the Milky Way and the solar system are the same thing seen from different places. Explain why this is wrong. [3 marks]

### The Inner Planets: Mercury, Venus, Earth and Mars



\*Fig. 3: The four inner planets\*

5. From the diagram, list the four inner planets in order of distance from the sun.
6. Use the diagram to explain why a YEAR on Mercury is much shorter than a year on the earth.
- #### 8.4 8.4 Properties of the Inner Planets

5. From the diagram, list the four inner planets in order of distance from the sun, and state which one is nearest. [3 marks]
6. Use the diagram to explain why a YEAR on Mercury is much shorter than a year on the earth. [4 marks]

### Properties of the Inner Planets

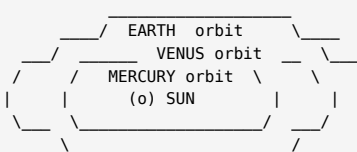
| Planet  | Diameter  | One day    | One year   | Moons |
|---------|-----------|------------|------------|-------|
| Mercury | 4 879 km  | 59 e-days  | 88 e-days  | 0     |
| Venus   | 12 104 km | 243 e-days | 225 e-days | 0     |
| Earth   | 12 756 km | 24 hours   | 365 days   | 1     |
| Mars    | 6 792 km  | 24 h 37 m  | 687 e-days | 2     |

\*Fig. 4: The inner planets compared\*

7. Use the table to describe how the length of a YEAR changes as you move outwards from Mercury to Mars, and explain why.
8. The table shows that a DAY on Venus is longer than a YEAR on Venus. Explain how this can be true.
- #### 8.5 8.5 Movement of Mercury and Venus Around the Sun

7. Use the table to describe how the length of a YEAR changes as you move outwards from Mercury to Mars, and explain why. [4 marks]
8. The table shows that a DAY on Venus is longer than a YEAR on Venus. Explain how this can be true. [3 marks]

### Movement of Mercury and Venus Around the Sun



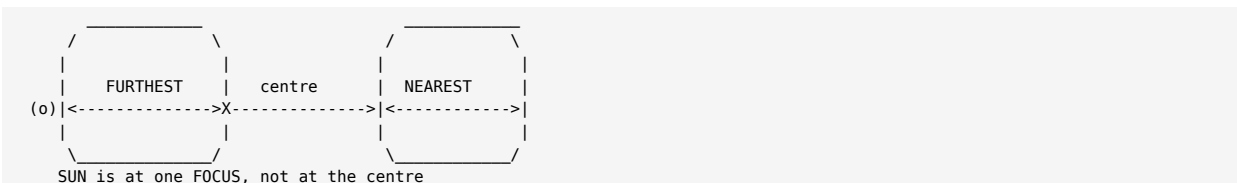
\*Fig. 5: The or

9. Use the diagram to explain why Mercury and Venus are always seen in the sky close to the sun, and are never seen in the middle of the night. [4 marks]  
 10. From the diagram, work out which of Mercury and Venus takes LESS time to go once round the sun, and explain how you can tell. [3 marks]

#### 8.6 8.6 Elliptical Paths of the Planets

9. Use the diagram to explain why Mercury and Venus are always seen in the sky close to the sun, and are never seen in the middle of the night. [4 marks]
10. From the diagram, work out which of Mercury and Venus takes LESS time to go once round the sun, and explain how you can tell. [3 marks]

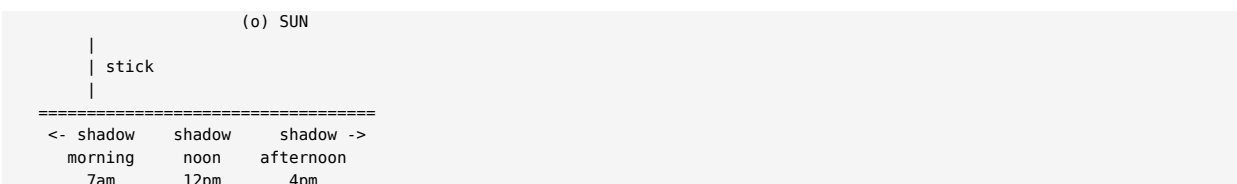
## Elliptical Paths of the Planets



\*Fig. 6: An elliptical orbit\*

11. State what is meant by an ELLIPSE, and state where the sun is in the orbit shown. [3 marks]  
 12. Explain why the statement 'the earth is always exactly the same distance from the sun' is wrong, using the diagram. [3 marks]

## SECTION B - PRACTICAL TASK



\*Fig. 7: Watching a shadow through the day\*

13. Describe how you would use a stick and the sun to show that the EARTH ROTATES, and explain how your observations show it. [4 marks]  
 14. Explain why the shadow is SHORTEST at midday. [3 marks]  
 15. State ONE source of error in this experiment, and say how you would reduce it. [3 marks]

## ANSWER KEY AND MARKING SCHEME

\*\*8.1 8.1 What is the Solar System?\*

- \*\*Q1\*\* (4 marks) The sun is at the centre. The sun's gravity pulls on every planet.  
 - \*\*Q2\*\* (3 marks) Any one with an explanation. The inner four are small and rocky.  
 \*\*8.2 8.2 The Galaxy and the Milky Way\*\*  
 - \*\*Q3\*\* (3 marks) The Milky Way is a galaxy - an enormous collection of many stars.

11. State what is meant by an ELLIPSE, and state where the sun is in the orbit shown. [3 marks]
12. Explain why the statement 'the earth is always exactly the same distance from the sun' is wrong, using the diagram. [3 marks]
13. Describe how you would use a stick and the sun to show that the EARTH ROTATES, and explain how your observations show it. [4 marks]
14. Explain why the shadow is SHORTEST at midday. [3 marks]
15. State ONE source of error in this experiment, and say how you would reduce it. [3 marks]

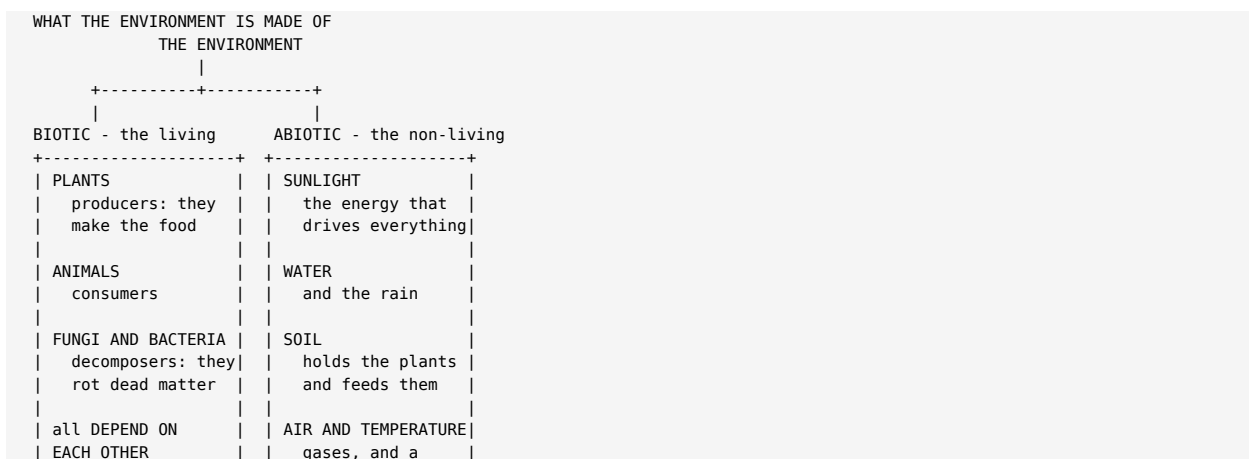
# Diagram Paper 21: Understanding the Environment

Time Allowed: \*\* 1 hour 30 minutes

## INSTRUCTIONS TO CANDIDATES

1. Answer ALL the questions in this paper.
2. Every question is based on a diagram. Study each diagram carefully before you answer the questions that follow it.
3. Write your answers in the spaces provided. The marks for each question are printed beside it.
4. Where a question asks for a stated number of points, give that number. Any extra answers will not be marked.
5. You may refer back to any diagram at any time.

## Landforms in Ghana: Plateau, Plain, Mountain and Valley



\*Fig. 1: What the environment is made of\*

1. Explain the difference between the BIOTIC and ABIOTIC parts of the environment.
2. Name THREE living (biotic) and THREE non-living (abiotic) parts of the environment.
3. A dam is built across a river and the land below it no longer floods each year. Explain TWO effects this would have on the living things in that area.

1. Explain the difference between the BIOTIC and ABIOTIC parts of the environment, and explain why a change in a NON-LIVING part can destroy the living part. [4 marks]
2. Name THREE living (biotic) and THREE non-living (abiotic) parts of the environment. [3 marks]
3. A dam is built across a river and the land below it no longer floods each year. Explain TWO effects this would have on the living things in that area. [3 marks]

## Plants and Animals

| WHAT PEOPLE DO THAT HARMS IT                   |                                                |                                               |
|------------------------------------------------|------------------------------------------------|-----------------------------------------------|
| DEFORESTATION                                  | BUSH BURNING                                   | OVERFISHING                                   |
| cutting for firewood, charcoal, farming        | firing the bush to clear land or drive game    | taking fish faster than they breed, fine nets |
| the trees that held the soil and made the rain | kills the soil life, drives off the animals    | the stock collapses                           |
| MINING                                         | POLLUTION                                      | OVERGRAZING                                   |
| galamsey: river banks, mercury, cyanide        | waste, sewage and chemicals into water and air | too many animals on too little land           |

\*Fig. 2: What people do that harms it\*

4. Explain why galamsey does far more damage than simply removing the gold. Refer to what it does to the soil, to the water, and to the people downstream. [4 marks]

5. Name THREE human activities that harm the environment. \*\*[3 marks]\*\*

#### 20.3 20.3 How Animals Adapt to Different Landforms

4. Explain why galamsey does far more damage than simply removing the gold. Refer to what it does to the soil, to the water, and to the people downstream. [4 marks]

5. Name THREE human activities that harm the environment. [3 marks]

## How Animals Adapt to Different Landforms

| HOW THE DAMAGE TRAVELS                  |                                                                                                                                                 |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| EROSION - THE SOIL LEAVES               | AND THEN                                                                                                                                        |
| rain running straight off a bare slope  | soil washes away<br>land grows less bush<br>people clear MORE<br>[ ]->[ ]->[ ]->[ ]<br>which erodes in turn<br>a circle that tightens on itself |
| topsoil lost: the best layer goes first |                                                                                                                                                 |
| THE OTHER DAMAGE                        |                                                                                                                                                 |
| LOSS OF BIODIVERSITY                    | DRYING UP cut the<br>FLOODING no roots<br>SILTED RIVERS                                                                                         |

\*Fig. 3: How the damage travels\*

6. Explain how soil erosion can set going a CIRCLE that makes itself worse, so that poor farmers end up damaging more land. [4 marks]

7. Explain why cutting down a forest on a hillside makes flooding worse in the valley below. [3 marks]

8. State THREE effects of damaging the environment. \*\*[3 marks]\*\*

#### 20.4 20.4 How Plants Survive in Different Landforms

6. Explain how soil erosion can set going a CIRCLE that makes itself worse, so that poor farmers end up damaging more land. [4 marks]

7. Explain why cutting down a forest on a hillside makes flooding worse in the valley below. [3 marks]

8. State THREE effects of damaging the environment. [3 marks]

## How Plant

## PUTTING IT RIGHT

|             |             |             |            |
|-------------|-------------|-------------|------------|
| PLANT       | MAKE        | LEAVE THE   | PROTECT    |
| TREES       | RIDGES ON   | LAND FALLOW | THE        |
| roots hold  | THE CONTOUR | give the    | WATER'S    |
| the soil,   | slope, not  | soil a      | EDGE       |
| canopy      | down it     | season to   | keep a     |
| breaks the  |             | recover     | strip of   |
| rain        |             |             | vegetation |
| ROTATE AND  | MAKE        | USE A       | PROTECT    |
| MIX CROPS   | COMPOST     | BETTER      | WHOLE      |
| keeps the   | returns the | STOVE       | AREAS      |
| ground      | goodness    | less wood,  | forest     |
| covered     | instead of  | so fewer    | reserves,  |
| and fertile | burning     | trees cut   | parks,     |
|             |             |             | sacred     |
|             |             |             | groves     |

\*Fig. 4: Putting it right\*

9. A village's hillside farms are losing their soil. Suggest TWO practical things

10. Explain why keeping a strip of vegetation along a river bank protects both t

### 20.5 20.5 Associations Among Plants and Animals

9. A village's hillside farms are losing their soil. Suggest TWO practical things the farmers could do, and explain in each case exactly HOW it would stop the soil being lost. [4 marks]

10. Explain why keeping a strip of vegetation along a river bank protects both the RIVER and the FARMER. [3 marks]

## Associations Among Plants and Animals

## USING IT WITHOUT USING IT UP

## TAKING ONLY THE INTEREST

## THE THREE QUESTIONS

|                       |                           |
|-----------------------|---------------------------|
| [THE INTEREST]        | 1 Is it being replaced as |
| what it grows         | fast as it is taken?      |
| each year             | a tree for every tree     |
| ^                     | cut; no fine nets         |
|                       |                           |
| [THE CAPITAL]         | 2 Does today's use leave  |
| the forest, the       | tomorrow's choice open?   |
| stock, the soil       | if it closes a door for   |
|                       | our children, it is not   |
|                       | sustainable               |
| take the interest     |                           |
| and the capital stays | 3 Who pays, and who       |
| eat the capital and   | benefits?                 |
| next year there is    | if one profits and the    |
| nothing to take       | village carries the       |
|                       | cost, it will not last    |

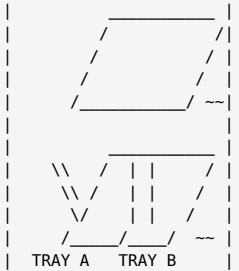
\*Fig. 5: Using it without using it up\*

11. Explain what is meant by using the environment SUSTAINABLY, using the idea o

12. State what is meant by the SUSTAINABLE use of the environment, and give ONE

## SECTION B - PRACTICAL TASK



| THE APPARATUS                                                                                                                                                                                                            |  | THE RESULTS                                                                                    |         |       |        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|------------------------------------------------------------------------------------------------|---------|-------|--------|
|                                                                                                                                         |  | TRAY                                                                                           | RUN-OFF | SOIL  | SOAKED |
|                                                                                                                                                                                                                          |  |                                                                                                |         | in it | IN     |
|                                                                                                                                                                                                                          |  | A bare                                                                                         | 390 ml  | 31 g  | 110    |
|                                                                                                                                                                                                                          |  | B grass                                                                                        | 180 ml  | 4 g   | 320    |
|                                                                                                                                                                                                                          |  | the grassy tray lost less than a SEVENTH of the soil and held almost three times as much water |         |       |        |
| <p>WHY: the leaves break the force of the drops; the roots hold the particles; the stems slow the water so it soaks in. Run-off is what carries the soil away - and it carries the best material first, the topsoil.</p> |  |                                                                                                |         |       |        |

\*Fig. 6: Does plant cover stop the soil washing away?\*

13. From the results, explain why the grassy tray lost so much less soil and held

14. State THREE things the pupils must keep the same for this comparison to be f

15. Suggest ONE change to the method that would test something further, and say

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## ANSWER KEY AND MARKING SCHEME

\*\*20.1 20.1 Landforms in Ghana: Plateau, Plain, Mountain and Valley\*\*

- \*\*Q1\*\* (4 marks) The biotic part is everything that is alive - plants, animals

- \*\*Q2\*\* (3 marks) Biotic, any three: plants, animals, fungi, bacteria. Abiotic,

- \*\*Q3\*\* (3 marks) Any two, each explained. The floodplain plants that depended

\*\*20.2 20.2 Plants and Animals Found in Different Landforms\*\*

11. Explain what is meant by using the environment SUSTAINABLY, using the idea of capital and interest from Fig. 5. [4 marks]
12. State what is meant by the SUSTAINABLE use of the environment, and give ONE example. [3 marks]
13. From the results, explain why the grassy tray lost so much less soil and held so much more water, referring to what the grass actually does. [3 marks]
14. State THREE things the pupils must keep the same for this comparison to be fair, and explain why each matters. [3 marks]
15. Suggest ONE change to the method that would test something further, and say what you would expect to find and why. [3 marks]

## Diagram Paper 22: Waste Management

Time Allowed: \*\* 1 hour 30 minutes

### INSTRUCTIONS TO CANDIDATES

1. Answer ALL the questions in this paper.
2. Every question is based on a diagram. Study each diagram carefully before you answer the questions that follow it.
3. Write your answers in the spaces provided. The marks for each question are printed beside it.
4. Where a question asks for a stated number of points, give that number. Any extra answers will not be marked.
5. You may refer back to any diagram at any time.

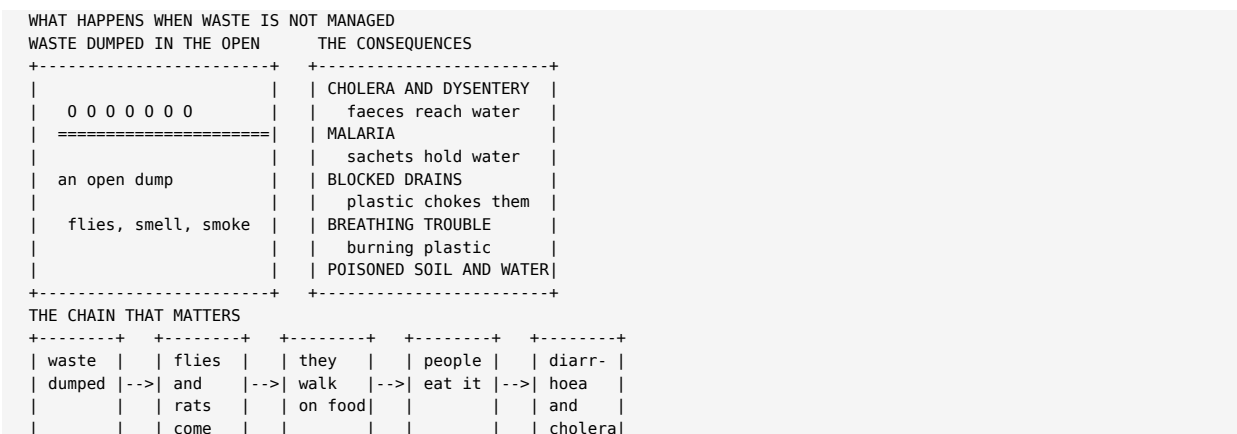
### Types of Waste in the Environment

| WHERE OUR WASTE COMES FROM                                                                                                                                                            |                       |                   |             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-------------------|-------------|
| HOUSEHOLD                                                                                                                                                                             | MARKET                | SCHOOL AND OFFICE | FARM        |
| food scraps                                                                                                                                                                           | vegetable             | paper, books      | crop        |
| plastic bags                                                                                                                                                                          | trimmings             | plastic           | residues    |
| old clothes                                                                                                                                                                           | rotten fruit          | bottles           | animal dung |
| grey water                                                                                                                                                                            | polythene             | leftovers         | fertiliser  |
|                                                                                                                                                                                       |                       |                   | sacks       |
| HOSPITAL                                                                                                                                                                              | INDUSTRY AND WORKSHOP | ELECTRONIC        | THE STREET  |
| used needles                                                                                                                                                                          | metal offcuts         | dead phones       | litter      |
| dressings                                                                                                                                                                             | waste oil             | batteries         | sachet      |
| drug packs                                                                                                                                                                            | containers            | old computers     | water bags  |
| medicines                                                                                                                                                                             |                       | burnt-out         | drain       |
|                                                                                                                                                                                       |                       | appliances        | clearance   |
| <p>waste is any material the owner no longer wants - but one person's waste is often another person's raw material<br/>sort it by what it IS MADE OF, not by where it was dropped</p> |                       |                   |             |

\*Fig. 1: Where our waste comes from\*

1. Explain why it is better to sort waste by the MATERIAL it is made of rather than by the place it was found, and give an example of two very different wastes produced in the same place. [4 marks]
2. Name THREE sources of waste. \*\*[3 marks]\*\*
3. For each of the following, say where it comes from and what it is mainly made of: (a) an empty pesticide container; (b) a sachet water bag; (c) a heap of vegetable trimmings. [3 marks]

1. Explain why it is better to sort waste by the MATERIAL it is made of rather than by the place it was found, and give an example of two very different wastes produced in the same place. [4 marks]
2. Name THREE sources of waste. [3 marks]
3. For each of the following, say where it comes from and what it is mainly made of: (a) an empty pesticide container; (b) a sachet water bag; (c) a heap of vegetable trimmings. [3 marks]



\*Fig. 2: What happens when waste is not managed\*

4. Explain how a heap of household rubbish that nobody touches can give a child

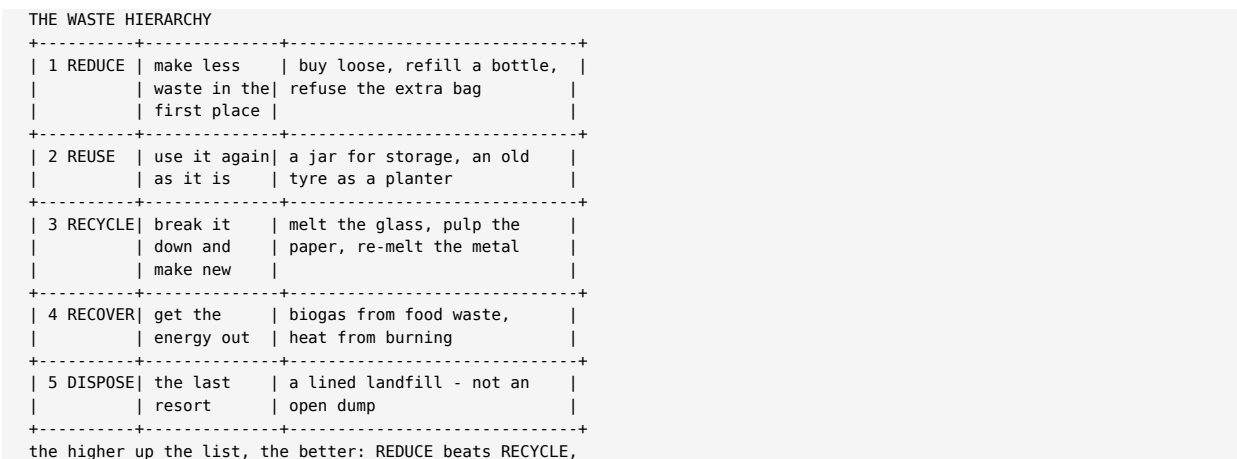
5. State THREE effects of poor waste management on health. \*\*[3 marks]\*\*

6. A town's drains are choked with plastic bags and the streets flood every time

#### 16.3 16.3 The Science Behind Waste Management

4. Explain how a heap of household rubbish that nobody touches can give a child cholera. [4 marks]
5. State THREE effects of poor waste management on health. [3 marks]
6. A town's drains are choked with plastic bags and the streets flood every time it rains. Suggest TWO things the town should do, and explain how each helps. [3 marks]

## The Science Behind Waste Management



\*Fig. 3: The waste

7. The diagram puts REDUCE above RECYCLE. Explain why reducing waste is better than recycling it, even when the recycling works perfectly. [4 marks]

8. For each of the following, say whether it is an example of REDUCE, REUSE or RECYCLE, and give a reason: (a) taking a cloth bag with you when you shop; (b) melting old glass to make new bottles; (c) using an old tyre as a planter. [3 marks]

9. State what is meant by RECYCLING, and name TWO materials that are commonly recycled. [3 marks]

7. The diagram puts REDUCE above RECYCLE. Explain why reducing waste is better than recycling it, even when the recycling works perfectly. [4 marks]
8. For each of the following, say whether it is an example of REDUCE, REUSE or RECYCLE, and give a reason: (a) taking a cloth bag with you when you shop; (b) melting old glass to make new bottles; (c) using an old tyre as a planter. [3 marks]
9. State what is meant by RECYCLING, and name TWO materials that are commonly recycled. [3 marks]

## Community Waste Management Projects

### THE WASTES THAT NEED SPECIAL CARE

| WASTE            | WHY IT IS DANGEROUS                                                             | WHAT MUST BE DONE                                                                        |
|------------------|---------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| PLASTIC          | does not rot: it breaks into smaller pieces. Chokes drains, is eaten by animals | reduce sachet use; collect and recycle; never burn it, never let it reach a watercourse  |
| MEDICAL / SHARPS | needles can carry HIV and hepatitis B into the blood of anyone who handles them | into a puncture-proof sharps box, then incinerated or buried deep. Never an ordinary bin |
| E-WASTE          | lead, mercury and cadmium leak into soil and water                              | to a collection point for recovery of the metals. Never burn, never dump                 |
| CHEMICAL         | empty pesticide                                                                 | triple-rinse, puncture,                                                                  |

\*Fig. 4: The wastes that need special care\*

10. Explain why a USED NEEDLE needs completely different handling from a FOOD SCRAP.

11. Explain why an EMPTY PESTICIDE CONTAINER must never be reused to store drink.

12. State why E-WASTE is dangerous, and name TWO items that count as e-waste.

### ## SECTION B - PRACTICAL TASK

#### WHAT IS ACTUALLY IN THE SCHOOL'S WASTE?

one day's waste from Class 2, sorted and weighed

| One day's waste from class 2, sorted and weighed |  |  |                      |  |  |  |
|--------------------------------------------------|--|--|----------------------|--|--|--|
| SORTING IT                                       |  |  | THE RESULT           |  |  |  |
| +-----+                                          |  |  | +-----+              |  |  |  |
|                                                  |  |  | MATERIAL MASS % DO   |  |  |  |
| -        -        -                              |  |  | organic 4.2 51% comp |  |  |  |
|                                                  |  |  | plastic 2.1 25% redu |  |  |  |
|                                                  |  |  | paper 1.6 19% recy   |  |  |  |
|                                                  |  |  | glass 0.25 3% recy   |  |  |  |
|                                                  |  |  | metal 0.15 2% recy   |  |  |  |
| ORG       PLA       PAP                          |  |  | TOTAL 8.3 100%       |  |  |  |
|                                                  |  |  |                      |  |  |  |
| sort by MATERIAL, not                            |  |  |                      |  |  |  |
| by where it was found                            |  |  |                      |  |  |  |
| gloves and tongs -                               |  |  |                      |  |  |  |
| never bare hands                                 |  |  |                      |  |  |  |

#### THE THREE QUESTIONS THIS RAISES

- Why is 51% being burnt when it could be composted?
- Why is a quarter plastic, mostly sachet water bags?

\*Fig. 5: What i

13. From the results, explain what the audit shows about the school's waste, and  
 14. The school decides to compost the organic fraction. Describe what should be  
 15. A quarter of the waste is sachet water bags. Suggest TWO things the school c

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## ## ANSWER KEY AND MARKING SCHEME

## \*\*16.1 16.1 Types of Waste in the Environment\*\*

- \*\*Q1\*\* (4 marks) Sorting by where the waste was found tells you who dropped it
- \*\*Q2\*\* (3 marks) Any three of: household, market, school or office, farm, hosp
- \*\*Q3\*\* (3 marks) (a) From a farm; made of plastic, with chemical residue still

## \*\*16.2 16.2 Good Waste Management Practices\*\*

- 10.** Explain why a USED NEEDLE needs completely different handling from a FOOD SCRAP, and describe exactly what should be done with each. [4 marks]
- 11.** Explain why an EMPTY PESTICIDE CONTAINER must never be reused to store drinking water, and describe what should be done with it. [3 marks]
- 12.** State why E-WASTE is dangerous, and name TWO items that count as e-waste. [3 marks]
- 13.** From the results, explain what the audit shows about the school's waste, and explain why MEASURING the waste changes what can be done about it. [4 marks]
- 14.** The school decides to compost the organic fraction. Describe what should be done with it, and explain TWO conditions the heap needs. [3 marks]
- 15.** A quarter of the waste is sachet water bags. Suggest TWO things the school could do to reduce this, and explain the difficulty with each. [3 marks]

# **ANSWER KEY**

## **and Marking Schemes**

This section contains correct answers and detailed marking schemes for the BECE Mock examinations.

# Agricultural Tools

## ANSWER KEY

### SECTION A: OBJECTIVE TEST

Total Marks: 40

| Q  | Ans | Curriculum Area                               | Cognitive Level |
|----|-----|-----------------------------------------------|-----------------|
| 1  | C   | 15.1 Simple Agricultural Tools and Their Uses | Remember        |
| 2  | D   | 15.1 Simple Agricultural Tools and Their Uses | Remember        |
| 3  | B   | 15.1 Simple Agricultural Tools and Their Uses | Remember        |
| 4  | A   | 15.1 Simple Agricultural Tools and Their Uses | Understand      |
| 5  | B   | 15.1 Simple Agricultural Tools and Their Uses | Remember        |
| 6  | D   | 15.1 Simple Agricultural Tools and Their Uses | Understand      |
| 7  | A   | 15.1 Simple Agricultural Tools and Their Uses | Remember        |
| 8  | C   | 15.1 Simple Agricultural Tools and Their Uses | Analyse         |
| 9  | B   | 15.1 Simple Agricultural Tools and Their Uses | Remember        |
| 10 | B   | 15.1 Simple Agricultural Tools and Their Uses | Remember        |
| 11 | D   | 15.2 The Meaning and Importance of Handlin... | Remember        |
| 12 | D   | 15.2 The Meaning and Importance of Handlin... | Remember        |
| 13 | C   | 15.2 The Meaning and Importance of Handlin... | Analyse         |
| 14 | D   | 15.2 The Meaning and Importance of Handlin... | Understand      |
| 15 | C   | 15.2 The Meaning and Importance of Handlin... | Understand      |
| 16 | D   | 15.2 The Meaning and Importance of Handlin... | Understand      |
| 17 | A   | 15.2 The Meaning and Importance of Handlin... | Analyse         |
| 18 | C   | 15.3 Basic Rules for Handling Tools Safely    | Understand      |
| 19 | A   | 15.3 Basic Rules for Handling Tools Safely    | Understand      |
| 20 | A   | 15.3 Basic Rules for Handling Tools Safely    | Apply           |
| 21 | D   | 15.3 Basic Rules for Handling Tools Safely    | Understand      |
| 22 | C   | 15.3 Basic Rules for Handling Tools Safely    | Apply           |
| 23 | D   | 15.3 Basic Rules for Handling Tools Safely    | Remember        |
| 24 | A   | 15.3 Basic Rules for Handling Tools Safely    | Apply           |
| 25 | B   | 15.3 Basic Rules for Handling Tools Safely    | Apply           |
| 26 | D   | 15.4 Maintenance Practices for Particular ... | Understand      |
| 27 | B   | 15.4 Maintenance Practices for Particular ... | Remember        |
| 28 | D   | 15.4 Maintenance Practices for Particular ... | Understand      |
| 29 | C   | 15.4 Maintenance Practices for Particular ... | Understand      |
| 30 | B   | 15.4 Maintenance Practices for Particular ... | Apply           |
| 31 | C   | 15.4 Maintenance Practices for Particular ... | Remember        |
| 32 | A   | 15.4 Maintenance Practices for Particular ... | Analyse         |
| 33 | C   | 15.4 Maintenance Practices for Particular ... | Apply           |
| 34 | A   | 15.4 Maintenance Practices for Particular ... | Understand      |
| 35 | B   | 15.5 Applying Tool Handling and Maintenanc... | Apply           |
| 36 | B   | 15.5 Applying Tool Handling and Maintenanc... | Understand      |

| Q  | Ans      | Curriculum Area                               | Cognitive Level |
|----|----------|-----------------------------------------------|-----------------|
| 37 | <b>C</b> | 15.5 Applying Tool Handling and Maintenanc... | Remember        |
| 38 | <b>A</b> | 15.5 Applying Tool Handling and Maintenanc... | Analyse         |
| 39 | <b>B</b> | 15.5 Applying Tool Handling and Maintenanc... | Evaluate        |
| 40 | <b>A</b> | 15.5 Applying Tool Handling and Maintenanc... | Evaluate        |

**Answer Distribution:**

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

**Cognitive Level Distribution:**

Analyse: 5 (12%) | Apply: 7 (17%) | Evaluate: 2 (5%) | Remember: 13 (32%) | Understand: 13 (32%)



## SECTION B: THEORY — MARKING SCHEME

### Question 1 [Total: 12 marks]

**(a)** [4 marks]

- A - **cutlass (machete)** (1 mark).
- B - **hoe** (1 mark).
- C - **rake** (1 mark).
- D - **watering can** (1 mark).
- \*Marking note: Accept 'matchet' for A.\*

**(b)** [4 marks]

- A - **clearing weeds and bushes (accept harvesting)** (1 mark).
- B - **weeding, making ridges or mounds** (1 mark).
- C - **gathering leaves and trash, levelling a seedbed** (1 mark).
- D - **watering seedlings and nursery beds** (1 mark).
- \*Marking note: 'Farming' is too general and earns no mark.\*

**(c)** [2 marks]

- Any TWO of: **feeding or drinking trough**; **broom, brush or scraper for cleaning pens**; **shovel or spade**; **bucket**; **wheelbarrow**; **rope or halter** (1 mark each).
- \*Marking note: A tool already shown in Fig. 1 earns no mark.\*

**(d)** [2 marks]

- **They are operated by hand and need no engine or fuel** (1 mark).
- **They are cheap, easy to use and easy to repair and maintain** (1 mark).
- \*Marking note: 'They are small' alone earns no mark.\*

### Question 2 [Total: 10 marks]

**(a)** [3 marks]

- P - **the blade is rusty (not cleaned and oiled)** (1 mark).
- Q - **the handle is split or cracked** (1 mark).
- R - **the blade is left lying in water** (1 mark).
- \*Marking note: 'The cutlass is old' earns no mark.\*

**(b)** [3 marks]

- P - **wash off the soil, dry the blade and oil it after use** (1 mark).
- Q - **keep the handle dry and oiled; refit or replace it when it splits** (1 mark).
- R - **store the tool in a dry store, hung on a rack, not on wet ground** (1 mark).
- \*Marking note: The practice must match the fault it is written against.\*

**(c) [2 marks]**

- Any TWO of: **\*\*the tools are hung on racks or nails off the floor\*\***; **\*\*the tools are clean and free of rust\*\***; **\*\*oil is kept ready for oiling the tools\*\***; **\*\*the store is dry and under cover\*\*** (1 mark each).

- **\*Marking note: 'The store is nice' earns no mark.\***

**(d) [2 marks]**

- Any TWO of: **\*\*the tool is spoilt early and money must be spent replacing it\*\***; **\*\*work on the farm becomes slow and tiring with a blunt, rusty tool\*\***; **\*\*it is dangerous and can cause injury\*\*** (1 mark each).

- **\*Marking note: Two answers that both say 'it will spoil' count once.\***

**Question 3 [Total: 10 marks]****(a) [2 marks]**

- (i) **\*\*The way a tool is held, carried and used during work\*\*** (1 mark).  
 - (ii) **\*\*The care given to a tool to keep it in good working condition\*\*** (1 mark).

- **\*Marking note: Examples alone, with no explanation, earn no mark.\***

**(b) [4 marks]**

- Any FOUR of: **\*\*check the tool and the handle before use\*\***; **\*\*carry a cutlass with the blade down and away from the body\*\***; **\*\*give a cutlass to another person handle first\*\***; **\*\*work a safe distance apart\*\***; **\*\*use a tool only for its proper work\*\***; **\*\*wear boots and hand gloves\*\***; **\*\*keep idle tools together with the edges down or covered\*\*** (1 mark each).

- **\*Marking note: 'Be careful with the cutlass' is too vague for a mark.\***

**(c) [2 marks]**

- Any TWO of: **\*\*the edge is chipped, bent or blunted\*\***; **\*\*the blade may break\*\***; **\*\*the blade may fly off the handle and injure someone\*\*** (1 mark each).

- **\*Marking note: The rule broken is that a tool must be used only for its proper work.\***

**(d) [2 marks]**

- Any TWO of: **\*\*they prevent injury to the user and to others\*\***; **\*\*they protect the tools from damage\*\***; **\*\*work is in fact finished faster and better with safe, sound tools\*\*** (1 mark each).

- **\*Marking note: 'Because the teacher said so' earns no mark.\***

**Question 4 [Total: 10 marks]****(a) [4 marks]**

- Cutlass - **\*\*sharpen with a file or whetstone; clean and oil the blade\*\*** (1 mark).

- Hoe - **\*\*clean, dry and oil the blade; keep the wooden handle dry and oiled\*\*** (1

mark).

- Knapsack sprayer - \*\*wash out the tank and nozzle after use and oil the moving parts\*\* (1 mark).

- Wheelbarrow - \*\*oil the wheel and axle and tighten the bolts\*\* (1 mark).

- \*Marking note: A practice written against the wrong tool earns no mark.\*

**(b)** [3 marks]

- \*\*Scrape and wash off the soil, then dry the blade\*\* (1 mark).

- \*\*Rub the blade with oil and check or wipe the handle\*\* (1 mark).

- \*\*Hang it in a dry store, off the floor\*\* (1 mark).

- \*Marking note: The three steps must be in a sensible order.\*

**(c)** [3 marks]

- \*\*Rust eats away the metal, so the tool becomes blunt, weak and finally breaks\*\* (1 mark).

- \*\*Condition 1: moisture, from water, rain, dew or damp soil\*\* (1 mark).

- \*\*Condition 2: air (oxygen) in contact with the bare metal, especially under soil left on it\*\* (1 mark).

- \*Marking note: 'Dirty tools' alone is not a condition; the moisture must be named.\*

## Question 5 [Total: 10 marks]

**(a)** [3 marks]

- Any THREE of: \*\*the tools used and the work each does\*\*; \*\*how the farmer holds, carries and uses them, and the protective clothing worn\*\*; \*\*what is done to the tools after work (cleaning, sharpening, oiling, repair)\*\*; \*\*where and how the tools are stored\*\*; \*\*the condition of the tools\*\* (1 mark each).

- \*Marking note: 'Everything about the farm' earns no mark.\*

**(b)** [3 marks]

- Any THREE of: \*\*greet the farmer and ask permission\*\*; \*\*stay together and follow instructions\*\*; \*\*do not touch tools or animals without being told\*\*; \*\*ask questions politely and do not criticise the farmer\*\*; \*\*do not trample the crops or frighten the birds\*\*; \*\*thank the farmer before leaving\*\* (1 mark each).

- \*Marking note: Accept any clear rule of courtesy or safety.\*

**(c)** [4 marks]

- (i) Any TWO of: \*\*oil the wheelbarrow wheel and tighten the bolts\*\*; \*\*refit or replace the loose hoe handles\*\*; \*\*wash out the sprayer after every use\*\* (1 mark each).

- (ii) \*\*A correct reason beside each of the two recommendations given\*\* (1 mark each).

- \*Marking note: The reason must match the recommendation it is written

against.\*

## Question 6 [Total: 8 marks]

(a) [4 marks]

- **Four clear rules, each stating an action** (1 mark each), for example: **sign for and return each tool to its rack**; **clean and dry tools before returning them**; **oil the metal parts**; **report damaged tools at once**; **use each tool only for its proper work**; **keep the store dry and locked**.

- **Marking note:** A rule with no action, such as 'tools are important', earns no mark.\*

(b) [2 marks]

- **Two tools named with a correct use for each** (1 mark each), for example **hoe - to dig and loosen the soil**; **rake - to level the bed**.

- **Marking note:** A tool named without its use earns no mark.\*

(c) [2 marks]

- **Well-kept tools last longer, so money is not spent replacing them each season** (1 mark).

- **Work is done faster and with fewer accidents, so less is spent on hired labour and on treatment** (1 mark).

- **Marking note:** A general statement about saving money, with no link to the tools, earns no mark.\*

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### ## CURRICULUM ASSESSMENT BLUEPRINT

**Strand:** 4: Forces and Energy | **Sub-strand:** 5: Agricultural Tools |  
**Class:** Basic 7 (JHS 1)

**Content Standard:** **B7/JHS1.4.5.1** - Demonstrate knowledge and skills in handling and maintenance of basic and simple agricultural tools.

> S

# Animal Product

## ANSWER KEY

### SECTION A: OBJECTIVE TEST

Total Marks: 40

| Q  | Ans | Curriculum Area                               | Cognitive Level |
|----|-----|-----------------------------------------------|-----------------|
| 1  | C   | 6.1 Domestic Animals and their Breeds         | Remember        |
| 2  | B   | 6.1 Domestic Animals and their Breeds         | Understand      |
| 3  | C   | 6.1 Domestic Animals and their Breeds         | Remember        |
| 4  | B   | 6.1 Domestic Animals and their Breeds         | Remember        |
| 5  | A   | 6.1 Domestic Animals and their Breeds         | Remember        |
| 6  | C   | 6.1 Domestic Animals and their Breeds         | Apply           |
| 7  | D   | 6.2 Ruminants                                 | Remember        |
| 8  | B   | 6.2 Ruminants                                 | Remember        |
| 9  | D   | 6.2 Ruminants                                 | Remember        |
| 10 | C   | 6.2 Ruminants                                 | Understand      |
| 11 | A   | 6.2 Ruminants                                 | Understand      |
| 12 | B   | 6.2 Ruminants                                 | Analyse         |
| 13 | C   | 6.2 Ruminants                                 | Remember        |
| 14 | B   | 6.3 Monogastric Animals and Poultry           | Remember        |
| 15 | C   | 6.3 Monogastric Animals and Poultry           | Remember        |
| 16 | D   | 6.3 Monogastric Animals and Poultry           | Remember        |
| 17 | A   | 6.3 Monogastric Animals and Poultry           | Understand      |
| 18 | A   | 6.3 Monogastric Animals and Poultry           | Remember        |
| 19 | B   | 6.3 Monogastric Animals and Poultry           | Understand      |
| 20 | D   | 6.3 Monogastric Animals and Poultry           | Analyse         |
| 21 | D   | 6.4 Comparing Ruminants, Monogastrics and ... | Understand      |
| 22 | B   | 6.4 Comparing Ruminants, Monogastrics and ... | Understand      |
| 23 | C   | 6.4 Comparing Ruminants, Monogastrics and ... | Evaluate        |
| 24 | D   | 6.4 Comparing Ruminants, Monogastrics and ... | Apply           |
| 25 | A   | 6.4 Comparing Ruminants, Monogastrics and ... | Analyse         |
| 26 | B   | 6.4 Comparing Ruminants, Monogastrics and ... | Understand      |
| 27 | D   | 6.5 Domestic Uses of Animals                  | Remember        |
| 28 | A   | 6.5 Domestic Uses of Animals                  | Understand      |
| 29 | A   | 6.5 Domestic Uses of Animals                  | Remember        |
| 30 | D   | 6.5 Domestic Uses of Animals                  | Understand      |
| 31 | B   | 6.5 Domestic Uses of Animals                  | Apply           |
| 32 | C   | 6.5 Domestic Uses of Animals                  | Understand      |
| 33 | A   | 6.5 Domestic Uses of Animals                  | Remember        |
| 34 | D   | 6.6 Commercial Uses and By-Products           | Remember        |
| 35 | A   | 6.6 Commercial Uses and By-Products           | Remember        |
| 36 | D   | 6.6 Commercial Uses and By-Products           | Understand      |

| Q  | Ans      | Curriculum Area                     | Cognitive Level |
|----|----------|-------------------------------------|-----------------|
| 37 | <b>C</b> | 6.6 Commercial Uses and By-Products | Understand      |
| 38 | <b>A</b> | 6.6 Commercial Uses and By-Products | Understand      |
| 39 | <b>C</b> | 6.6 Commercial Uses and By-Products | Apply           |
| 40 | <b>B</b> | 6.6 Commercial Uses and By-Products | Evaluate        |

**Answer Distribution:**

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

**Cognitive Level Distribution:**

Analyse: 3 (7%) | Apply: 4 (10%) | Evaluate: 2 (5%) | Remember: 17 (42%) | Understand: 14 (35%)

**SECTION B: THEORY — MARKING SCHEME****Question 1 [Total: 10 marks]****(a) [4 marks]**

- A - **goat** (1 mark).
- B - **cow / cattle** (1 mark).
- C - **pig** (1 mark).
- D - **fowl (chicken)** (1 mark).
- \*Marking note: Accept 'she-goat', 'nanny goat' for A and 'hen' for D.\*

**(b) [3 marks]**

- A - **ruminant** (1 mark).
- C - **monogastric animal** (1 mark).
- D - **poultry** (accept: poultry, a monogastric bird) (1 mark).
- \*Marking note: 'Monogastric' written for D is accepted, but 'ruminant' is not.\*

**(c) [3 marks]**

- (i) Any ONE of: **B has four legs and D has two**; **B is covered with hair while D has feathers**; **D has a beak, comb and wings**; **B has horns and hooves** (1 mark).
- (ii) Any TWO of: **both are ruminants (chew the cud)**; **both are four-legged mammals covered with hair**; **both have hooves and horns**; **both feed mainly on grass**; **both give birth to live young and suckle them** (1 mark each).
- \*Marking note: In (ii) a difference written in place of a similarity earns no mark.\*

**Question 2 [Total: 12 marks]****(a) [4 marks]**

- P - **fowl** (accept: duck, guinea fowl, turkey) (1 mark).
- Q - **cow** (accept: goat, sheep) (1 mark).
- R - **goat** (accept: sheep, cow) (1 mark).
- S - **cattle** (accept: goats, sheep, pigs, poultry) (1 mark).
- \*Marking note: A wild animal named anywhere in this part earns no mark.\*

**(b) [3 marks]**

- (i) **A useful material obtained from an animal apart from the main product (meat, milk or eggs), such as hide, skin, horn, bone or dung** (1 mark).
- (ii) **R and S** (1 mark) **because the skin is obtained after the animal is slaughtered for meat and the dung is the animal's waste, neither being the main**

product\*\* (1 mark).

- \*Marking note: Only one mark for naming R and S without a reason.\*

**(c)** [3 marks]

- \*\*Heap it and allow it to rot (compost it), turning it from time to time, before spreading\*\* (1 mark).

- Any TWO reasons: \*\*fresh dung scorches the roots of the crop\*\*; \*\*rotting destroys weed seeds and disease germs\*\*; \*\*it removes the smell and the flies\*\*; \*\*the nutrients are then in a form the plant can take up\*\* (1 mark each, maximum 2 marks).

- \*Marking note: 'Dry it in the sun' alone is not accepted as the treatment.\*

**(d)** [2 marks]

- Any TWO, each with a use: \*\*hide or skin\*\* - \*leather for bags, sandals and belts\*; \*\*horns or hooves\*\* - \*combs, buttons, ornaments\*; \*\*bones\*\* - \*bone meal or glue\*; \*\*feathers\*\* - \*stuffing pillows, decoration\* (1 mark each).

- \*Marking note: A by-product named without a use earns no mark.\*

### Question 3 [Total: 10 marks]

**(a)** [4 marks]

- Body covering - \*\*hair / feathers\*\* (1 mark).

- Number of legs - \*\*four / two\*\* (1 mark).

- Type of stomach - \*\*four-chambered (ruminant) stomach / simple stomach with a gizzard\*\* (1 mark).

- Main feed - \*\*grass and crop residues / prepared mash (with grit)\*\* (1 mark).

- \*Marking note: Both halves of a row must be correct before the mark for that row is given.\*

**(b)** [3 marks]

- \*\*The fowl has no teeth and swallows its food whole\*\* (1 mark).

- \*\*The grit in the gizzard grinds the food into small pieces\*\* (1 mark).

- \*\*Cattle chew with their teeth and chew the cud, so they do not need grit\*\* (1 mark).

- \*Marking note: Accept 'the gizzard uses the stones like teeth' for the second mark.\*

**(c)** [3 marks]

- \*\*The pigs\*\* (1 mark).

- \*\*A pig has a simple stomach and cannot digest fibre, so it needs bran and other concentrates, which are costly\*\* (1 mark).

- \*\*Cattle can still use crop residues and browse because the rumen breaks down fibre (or: fowls eat only small quantities each)\*\* (1 mark).

- \*Marking note: A candidate who names the cattle may earn at most 1 mark, for a reasoned point about the scarcity of grass.\*



**Question****(a) [4 marks]**

- (i) Ama - \*\*GH¢1,300\*\* (1 mark); Kojo - \*\*GH¢2,700\*\* (1 mark).
- (ii) Ama - \*\*GH¢800\*\* (1 mark); Kojo - \*\*GH¢700\*\* (1 mark).
- \*Marking note: Follow through: a correct subtraction from a wrong total in (i) earns the mark in (ii).\*

**(b) [3 marks]**

- Any TWO of: \*\*goats are ruminants and feed on grass, browse and crop residues, which cost nothing\*\*; \*\*goats can be grazed or tethered on free pasture\*\*; \*\*broilers are monogastric and need bought, prepared feed daily\*\* (1 mark each).
- \*\*Broilers grow quickly, so Kojo sold several batches of many birds within the year\*\* (1 mark).
- \*Marking note: 'Goats eat less' alone, without the reason, earns no mark.\*

**(c) [3 marks]**

- A clear recommendation (1 mark).
- Any TWO supporting reasons drawn from the table, for example: \*\*the goats gave the higher profit (GH¢800 against GH¢700)\*\*; \*\*the goats needed far less cash for feed (GH¢400 against GH¢2,100)\*\*; \*\*the broilers returned money several times within the year\*\*; \*\*the broilers had larger total sales\*\* (1 mark each).
- \*Marking note: Either enterprise is acceptable; the marks are for the reasons, which must use the figures.\*

**Question 5 [Total: 10 marks]****(a) [4 marks]**

- \*\*Divide into groups and share out the area to be visited\*\* (1 mark).
- \*\*Prepare a record sheet and obtain permission before visiting\*\* (1 mark).
- \*\*Observe and count the animals and ask the keepers about their uses, recording at once\*\* (1 mark).
- \*\*Bring the records together, add up the numbers and present them in a table or chart\*\* (1 mark).
- \*Marking note: Steps given in a workable order; a single step split into two earns one mark only.\*

**(b) [3 marks]**

- Any THREE of: \*\*name of animal\*\*; \*\*group (ruminant, monogastric, poultry)\*\*; \*\*number counted\*\*; \*\*home or farm visited\*\*; \*\*uses of the animal\*\*; \*\*by-products obtained\*\* (1 mark each).
- \*Marking note: 'Information' or 'remarks' is too vague and earns no mark.\*

**(c) [3 marks]**

- Any THREE of: **ask permission and greet the household politely**; **do not enter pens or chase, touch or frighten the animals**; **keep away from dangerous animals such as a bull or a nursing mother**; **wash hands with soap after the visit**; **wear shoes and avoid dung**; **stay in groups and report to the teacher** (1 mark each).

- *Marking note: 'Be careful' or 'behave well' earns no mark without the action.*

**Question 6 [Total: 8 marks]****(a) [3 marks]**

- Any THREE of: **the space and housing available**; **the feed she can get cheaply**; **the money she has**; **the market for the product**; **the time and labour needed**; **the neighbours and the rules of the town** (1 mark each).

- *Marking note: A reason that is not a real consideration, such as 'the colour of the animal', earns no mark.*

**(b) [3 marks]**

- Any THREE of: **it protects the animals from rain, sun, thieves and stray animals**; **it prevents diseases and parasites, so the animals grow well**; **it keeps flies, smells and germs away from the household and the neighbours**; **the droppings can be collected and used as manure**; **the animals do not stray and destroy the neighbours' property** (1 mark each).

- *Marking note: Each reason must be different; 'it keeps them safe' twice earns 1 mark.*

**(c) [2 marks]**

- Any TWO of: **they supply protein food (meat, eggs, milk) and improve nutrition**; **they create employment and income**; **their hides, skins and horns supply raw materials for industry and export**; **their dung provides manure for crops**; **they reduce the money spent on imported meat** (1 mark each).

- *Marking note: A domestic benefit to one family only, such as 'Afia will eat the eggs', does not answer this part.*

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**## CURRICULUM ASSESSMENT BLUEPRINT**

**Strand:** 2: Cycles | **Sub-strand:** 4: Animal Production | **Class:** Basic 7 (JHS 1)

**Content Standard:** **B7/JHS1.2.4.1**

# Climate Change and Green

## ANSWER KEY

### SECTION A: OBJECTIVE TEST

Total Marks: 40

| Q  | Ans | Curriculum Area                                | Cognitive Level |
|----|-----|------------------------------------------------|-----------------|
| 1  | C   | 19.1 Sustainable and Non-Sustainable Energy... | Understand      |
| 2  | B   | 19.1 Sustainable and Non-Sustainable Energy... | Remember        |
| 3  | B   | 19.1 Sustainable and Non-Sustainable Energy... | Remember        |
| 4  | C   | 19.1 Sustainable and Non-Sustainable Energy... | Remember        |
| 5  | D   | 19.1 Sustainable and Non-Sustainable Energy... | Understand      |
| 6  | C   | 19.1 Sustainable and Non-Sustainable Energy... | Understand      |
| 7  | B   | 19.1 Sustainable and Non-Sustainable Energy... | Understand      |
| 8  | A   | 19.1 Sustainable and Non-Sustainable Energy... | Remember        |
| 9  | A   | 19.2 Scientific Ideas Used to Protect the ...  | Remember        |
| 10 | C   | 19.2 Scientific Ideas Used to Protect the ...  | Understand      |
| 11 | D   | 19.2 Scientific Ideas Used to Protect the ...  | Understand      |
| 12 | A   | 19.2 Scientific Ideas Used to Protect the ...  | Evaluate        |
| 13 | D   | 19.2 Scientific Ideas Used to Protect the ...  | Understand      |
| 14 | A   | 19.2 Scientific Ideas Used to Protect the ...  | Understand      |
| 15 | B   | 19.2 Scientific Ideas Used to Protect the ...  | Understand      |
| 16 | A   | 19.2 Scientific Ideas Used to Protect the ...  | Remember        |
| 17 | D   | 19.3 The Greenhouse Effect and Global Warming  | Remember        |
| 18 | D   | 19.3 The Greenhouse Effect and Global Warming  | Remember        |
| 19 | B   | 19.3 The Greenhouse Effect and Global Warming  | Remember        |
| 20 | A   | 19.3 The Greenhouse Effect and Global Warming  | Understand      |
| 21 | A   | 19.3 The Greenhouse Effect and Global Warming  | Understand      |
| 22 | D   | 19.3 The Greenhouse Effect and Global Warming  | Remember        |
| 23 | B   | 19.3 The Greenhouse Effect and Global Warming  | Understand      |
| 24 | D   | 19.3 The Greenhouse Effect and Global Warming  | Analyse         |
| 25 | B   | 19.3 The Greenhouse Effect and Global Warming  | Remember        |
| 26 | B   | 19.4 Minimising Greenhouse Effects and Cop...  | Understand      |
| 27 | C   | 19.4 Minimising Greenhouse Effects and Cop...  | Analyse         |
| 28 | C   | 19.4 Minimising Greenhouse Effects and Cop...  | Analyse         |
| 29 | B   | 19.4 Minimising Greenhouse Effects and Cop...  | Evaluate        |
| 30 | C   | 19.4 Minimising Greenhouse Effects and Cop...  | Evaluate        |
| 31 | C   | 19.4 Minimising Greenhouse Effects and Cop...  | Analyse         |
| 32 | A   | 19.4 Minimising Greenhouse Effects and Cop...  | Evaluate        |
| 33 | D   | 19.4 Minimising Greenhouse Effects and Cop...  | Understand      |
| 34 | B   | 19.5 Local Projects that Sustain Energy, a...  | Understand      |
| 35 | A   | 19.5 Local Projects that Sustain Energy, a...  | Remember        |
| 36 | A   | 19.5 Local Projects that Sustain Energy, a...  | Remember        |

| Q  | Ans | Curriculum Area                               | Cognitive Level |
|----|-----|-----------------------------------------------|-----------------|
| 37 | D   | 19.5 Local Projects that Sustain Energy, a... | Apply           |
| 38 | C   | 19.5 Local Projects that Sustain Energy, a... | Apply           |
| 39 | C   | 19.5 Local Projects that Sustain Energy, a... | Evaluate        |
| 40 | D   | 19.5 Local Projects that Sustain Energy, a... | Understand      |

**Answer Distribution:**

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

**Cognitive Level Distribution:**

Analyse: 4 (10%) | Apply: 2 (5%) | Evaluate: 5 (12%) | Remember: 13 (32%) | Understand: 16 (40%)

## SECTION B: THEORY — MARKING SCHEME

### Question 1 [Total: 12 marks]

**(a)** [4 marks]

- A - **the Sun (solar energy)** (1 mark).
- B - **the wind (moving air)** (1 mark).
- C - **charcoal or firewood (wood fuel)** (1 mark).
- D - **biogas from animal dung or other organic waste** (1 mark).
- **Marking note: 'Fire' is not accepted for C; the fuel must be named.**

**(b)** [4 marks]

- A - **renewable** (1 mark) **because sunlight arrives every day and is not used up** (1 mark).
- C - **not renewable in practice** (1 mark) **because the trees are cut down faster than they can grow again** (1 mark).
- **Marking note: Accept 'renewable only if the trees are replanted' for C, with the reason, for full marks.**

**(c)** [2 marks]

- **D (the biogas plant)** (1 mark).
- **Reason: the cattle dung needed is already available, and the gas gives cooking fuel while the slurry gives manure** (1 mark).
- **Marking note: A letter with no reason earns 1 mark only.**

**(d)** [2 marks]

- Any TWO of: **trees are cut down, destroying the forest and exposing the soil**; **smoke pollutes the air and harms the lungs**; **carbon dioxide released adds to the greenhouse effect** (1 mark each).
- **Marking note: 'It is dirty' is too vague for a mark.**

### Question 2 [Total: 11 marks]

**(a)** [4 marks]

- W - **radiation (heat and light) coming in from the Sun** (1 mark).
- X - **the layer of greenhouse gases in the atmosphere** (1 mark).
- Y - **heat radiated upwards from the warmed Earth's surface** (1 mark).
- Z - **heat trapped and sent back down to the surface** (1 mark).
- **Marking note: 'The sky' is not accepted for X.**

**(b)** [2 marks]

- Any TWO of: **carbon dioxide**; **methane**; **water vapour**; **nitrous oxide**; **chlorofluorocarbons (CFCs)** (1 mark each).
- **Marking note:** Oxygen and nitrogen are not greenhouse gases.\*

**(c)** [2 marks]

- **The gases let the Sun's radiation in but hold back the heat leaving the Earth** (1 mark).
- **As more gas is added, more heat is trapped and the average temperature of the Earth rises** (1 mark).
- **Marking note:** A candidate who writes only 'the Earth becomes hot' earns 1 mark.\*

**(d)** [3 marks]

- Any THREE of: **burning fuels in vehicles, generators and factories**; **burning firewood and charcoal for cooking**; **cutting down and burning forests**; **open burning of rubbish**; **large herds of cattle and flooded rice fields releasing methane** (1 mark each).
- **Marking note:** Two answers that are both 'burning things' count once.\*

**Question 3 [Total: 10 marks]****(a)** [2 marks]

- **A long-term change in the usual pattern of weather (temperature, rainfall, wind) of a place** (1 mark).
- **Measured over many years, not from day to day** (1 mark).
- **Marking note:** A definition of weather alone earns no mark.\*

**(b)** [4 marks]

- **Warming has changed the rainfall pattern, so the rains are late, short or stormy** (1 mark) **and the farmer cannot predict the planting season** (1 mark).
- **Melting ice and the expansion of warm sea water raise the level of the sea** (1 mark) **and stronger waves wash away the shore and flood the village** (1 mark).
- **Marking note:** Credit any sound explanation linked to warming.\*

**(c)** [4 marks]

- Any FOUR of: **plant trees and protect the forest and mangroves**; **harvest water and irrigate in the dry spells**; **plant early-maturing or drought-tolerant varieties and use the weather forecast**; **build sea defence works and avoid building close to the shore**; **use solar lamps and improved stoves**; **stop the open burning of rubbish** (1 mark each).
- **Marking note:** 'Pray for rain' earns no mark.\*

**Question****(a) [4 marks]**

- Any FOUR of: **switch off lights and fans in empty rooms and at closing**; **use LED bulbs in place of filament bulbs**; **use daylight by opening windows and curtains**; **service appliances and keep refrigerator doors shut**; **unplug chargers and idle appliances**; **put up reminder notices** (1 mark each).

- **Marking note: 'Stop using electricity' is not a practical measure.\***

**(b) [3 marks]**

- Any THREE of: **less fuel burnt in thermal stations, so less carbon dioxide and smoke**; **less pressure to build new dams and stations that destroy land and habitats**; **water in the reservoirs is conserved**; **less waste and worn-out equipment to dispose of** (1 mark each).

- **Marking note: 'It saves money' is excluded by the question.\***

**(c) [2 marks]**

- **A clear slogan of not more than eight words calling for energy to be saved** (1 mark).

- **The message explained: everyone must switch off what is not in use, and saving energy protects the environment** (1 mark).

- **Marking note: A slogan longer than eight words loses the first mark.\***

**Question 5 [Total: 10 marks]****(a) [3 marks]**

- **Sunlight passes through the clear cover and is absorbed by the dark inside surface, heating the air** (1 mark).

- **The hot air takes up moisture from the fruit on the trays** (1 mark).

- **The moist air escapes through vents while fresh air is drawn in, and the fruit is kept free of rain, dust and flies** (1 mark).

- **Marking note: A mere statement that 'the Sun dries it' earns 1 mark.\***

**(b) [3 marks]**

- Any THREE materials, each with its correct use, for example: **wood or bamboo - the frame**; **clear polythene or glass - the cover**; **black paint or a black metal sheet - the absorbing surface**; **wire mesh or woven cane - the trays** (1 mark each).

- **Marking note: A material named without its use earns no mark.\***

**(c) [2 marks]**

- **Dry equal masses of the same fruit, one in the dryer and one in the open, at the same time** (1 mark).

- **Weigh both daily, or record the drying time, and compare the loss of water**

and the cleanliness\*\* (1 mark).

- \*Marking note: A test with no comparison earns 1 mark at most.\*

**(d)** [2 marks]

- Any TWO of: \*\*less fruit wasted, so more food and income all year\*\*; \*\*no fuel or electricity burnt, so trees are saved and less carbon dioxide released\*\*; \*\*cleaner, safer dried food\*\*; \*\*new jobs in making dryers and selling dried fruit\*\* (1 mark each).

- \*Marking note: 'It is a good project' earns no mark.\*

## Question 6 [Total: 8 marks]

**(a)** [2 marks]

- \*\*Work that earns a living while protecting or restoring the environment\*\* (1 mark).

- \*\*One sound example, such as raising tree seedlings, installing solar panels or recycling waste\*\* (1 mark).

- \*Marking note: 'Any job done outside' earns no mark.\*

**(b)** [4 marks]

- Any FOUR of: \*\*they absorb carbon dioxide and store carbon\*\*; \*\*they release oxygen\*\*; \*\*their roots hold the soil and reduce erosion and flooding\*\*; \*\*they cool the air and help to bring rain\*\*; \*\*they act as windbreaks and hold back the desert\*\*; \*\*they provide habitats, fruit and timber\*\* (1 mark each).

- \*Marking note: 'Trees are beautiful' earns no mark.\*

**(c)** [2 marks]

- Any TWO of: \*\*plant at the start of the rains and choose suitable species\*\*; \*\*water the seedlings during dry spells\*\*; \*\*fence or guard them against animals and bush fires\*\*; \*\*weed round them and replace those that die\*\*; \*\*make a named person or class responsible for each tree\*\* (1 mark each).

- \*Marking note: 'Plant more trees' does not answer the question.\*

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## ## CURRICULUM ASSESSMENT BLUEPRINT

\*\*Strand:\*\* 5: Humans and the Environment | \*\*Sub-strand:\*\* 4: Climate Change and Green Economy | \*\*Class:\*\* Basic 7 (JHS 1)

\*\*Content Standard:\*\* \*\*B7/JHS1.5.4.1\*\* - Demonstrate understand



# Conversion and Conservation

## ANSWER KEY

### SECTION A: OBJECTIVE TEST

Total Marks: 40

| Q  | Ans | Curriculum Area                             | Cognitive Level |
|----|-----|---------------------------------------------|-----------------|
| 1  | D   | 13.1 The Law of Conservation of Energy      | Remember        |
| 2  | A   | 13.1 The Law of Conservation of Energy      | Remember        |
| 3  | B   | 13.1 The Law of Conservation of Energy      | Remember        |
| 4  | B   | 13.1 The Law of Conservation of Energy      | Understand      |
| 5  | D   | 13.1 The Law of Conservation of Energy      | Apply           |
| 6  | A   | 13.1 The Law of Conservation of Energy      | Understand      |
| 7  | A   | 13.1 The Law of Conservation of Energy      | Analyse         |
| 8  | B   | 13.1 The Law of Conservation of Energy      | Understand      |
| 9  | C   | 13.2 Energy Conversions in Everyday Devices | Understand      |
| 10 | A   | 13.2 Energy Conversions in Everyday Devices | Understand      |
| 11 | A   | 13.2 Energy Conversions in Everyday Devices | Remember        |
| 12 | D   | 13.2 Energy Conversions in Everyday Devices | Understand      |
| 13 | A   | 13.2 Energy Conversions in Everyday Devices | Remember        |
| 14 | B   | 13.2 Energy Conversions in Everyday Devices | Understand      |
| 15 | D   | 13.2 Energy Conversions in Everyday Devices | Understand      |
| 16 | C   | 13.2 Energy Conversions in Everyday Devices | Understand      |
| 17 | A   | 13.2 Energy Conversions in Everyday Devices | Remember        |
| 18 | C   | 13.3 Energy Chains and Simple Calculations  | Understand      |
| 19 | D   | 13.3 Energy Chains and Simple Calculations  | Understand      |
| 20 | C   | 13.3 Energy Chains and Simple Calculations  | Remember        |
| 21 | B   | 13.3 Energy Chains and Simple Calculations  | Apply           |
| 22 | D   | 13.3 Energy Chains and Simple Calculations  | Apply           |
| 23 | D   | 13.3 Energy Chains and Simple Calculations  | Remember        |
| 24 | C   | 13.3 Energy Chains and Simple Calculations  | Analyse         |
| 25 | C   | 13.3 Energy Chains and Simple Calculations  | Understand      |
| 26 | D   | 13.4 Useful Energy and Wasted Energy        | Analyse         |
| 27 | C   | 13.4 Useful Energy and Wasted Energy        | Understand      |
| 28 | B   | 13.4 Useful Energy and Wasted Energy        | Understand      |
| 29 | B   | 13.4 Useful Energy and Wasted Energy        | Understand      |
| 30 | C   | 13.4 Useful Energy and Wasted Energy        | Remember        |
| 31 | C   | 13.4 Useful Energy and Wasted Energy        | Evaluate        |
| 32 | A   | 13.5 Conserving Energy for the Future       | Remember        |
| 33 | B   | 13.5 Conserving Energy for the Future       | Understand      |
| 34 | B   | 13.5 Conserving Energy for the Future       | Understand      |
| 35 | C   | 13.5 Conserving Energy for the Future       | Analyse         |
| 36 | D   | 13.5 Conserving Energy for the Future       | Analyse         |

| Q  | Ans | Curriculum Area                       | Cognitive Level |
|----|-----|---------------------------------------|-----------------|
| 37 | A   | 13.5 Conserving Energy for the Future | Remember        |
| 38 | D   | 13.5 Conserving Energy for the Future | Understand      |
| 39 | B   | 13.5 Conserving Energy for the Future | Understand      |
| 40 | A   | 13.5 Conserving Energy for the Future | Evaluate        |

**Answer Distribution:**

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

**Cognitive Level Distribution:**

Analyse: 5 (12%) | Apply: 3 (7%) | Evaluate: 2 (5%) | Remember: 11 (27%) | Understand: 19 (47%)

## SECTION B: THEORY — MARKING SCHEME

### Question 1 [Total: 12 marks]

**(a)** [4 marks]

- P - **chemical energy** (1 mark).
- Q - **electrical energy** (1 mark).
- R - **light energy** (1 mark).
- S - **heat (thermal) energy** (1 mark).
- **Marking note:** 'Battery energy' or 'current' earns no mark.\*

**(b)** [3 marks]

- **Energy supplied = 60 J** (1 mark).
- **Energy given out = 15 J + 45 J = 60 J** (1 mark).
- **The two totals are equal, so the energy has only been changed in form and none has been destroyed** (1 mark).
- **Marking note:** The sum must be shown; a bare statement of the law earns the third mark only.\*

**(c)** [3 marks]

- (i) **R, the light energy** (1 mark); **because the torch is used in order to give light** (1 mark).
- (ii) **The torch feels warm (the cells or the bulb heat up)** (1 mark).
- **Marking note:** Reason mark only if R is chosen.\*

**(d)** [2 marks]

- **In a closed system no energy enters or leaves without being counted** (1 mark).
- **So all the energy given out can be added up and shown to equal the energy supplied** (1 mark).
- **Marking note:** 'It is covered' or 'it is in a box' alone earns no mark.\*

### Question 2 [Total: 10 marks]

**(a)** [4 marks]

- A - **solar (light) energy to electrical energy** (1 mark).
- B - **electrical energy to heat energy** (1 mark).
- C - **electrical energy to kinetic energy** (1 mark).
- D - **chemical energy to electrical energy to sound energy** (1 mark).
- **Marking note:** Accept 'electrical to sound' for D.\*

**(b)** [3 marks]

- **Useful: kinetic energy of the blades** (1 mark).
- **Wasted: sound energy and heat energy** (1 mark).
- **The wasted energy spreads into the air and surroundings and cannot be used again** (1 mark).
- **Marking note: 'It is destroyed' earns no mark.**

**(c)** [3 marks]

- Any **THREE** of: **sunlight is free and renewable, so no fuel is bought**; **no smoke or carbon dioxide is given off**; **it makes no noise**; **it needs little maintenance** (1 mark each).
- **Marking note: 'It is modern' is not a scientific reason.**

**Question 3 [Total: 10 marks]****(a)** [2 marks]

- **Energy can neither be created nor destroyed** (1 mark).
- **It can only be changed from one form to another (the total remains the same)** (1 mark).
- **Marking note: Both halves are needed for the full 2 marks.**

**(b)** [3 marks]

- **Potential energy of the stored water shown first** (1 mark).
- **Kinetic energy of the falling water and of the turbine shown next** (1 mark).
- **Electrical energy shown last, with the boxes joined by arrows in the correct order** (1 mark).
- **Marking note: A written chain with arrows is accepted in place of boxes.**

**(c)** [3 marks]

- (i) **1,000 J - 850 J** (1 mark) **= 150 J** (1 mark, unit required).
- (ii) **Heat energy (accept sound energy)** (1 mark).
- **Marking note: A correct answer with no unit scores 1 of the 2 marks.**

**(d)** [2 marks]

- **The 150 J is not destroyed; it is changed into heat and sound** (1 mark).
- **Counting every form, the total is still 1,000 J** (1 mark).
- **Marking note: 'The generator is weak' earns no mark.**

**Question 4 [Total: 10 marks]****(a)** [4 marks]

- Microphone - **sound to electrical** (1 mark).
- Phone being charged - **electrical to chemical** (1 mark).
- Candle burning - **chemical to heat and light** (1 mark).

- Green plant - **light (solar) to chemical** (1 mark).
- **Marking note:** A change written the wrong way round earns no mark.\*

**(b)** [3 marks]

- **Plants change light energy into chemical energy stored in food (photosynthesis)** (1 mark).
- **People and animals eat the food and obtain this energy** (1 mark).
- **The energy is released in respiration for growth, movement and warmth (accept: wood and charcoal fuels come from plants)** (1 mark).
- **Marking note:** 'Plants give us oxygen' is true but does not answer the question on energy.\*

**(c)** [3 marks]

- **Materials named: a small motor, dry cells, wires, switch and a paper blade or disc** (1 mark).
- **Method: join them in a complete circuit and close the switch** (1 mark).
- **Observation: the blade spins, showing electrical energy changed into kinetic energy** (1 mark).
- **Marking note:** Any other workable activity, such as a battery-driven toy car, earns full marks.\*

**Question 5 [Total: 10 marks]****(a)** [2 marks]

- **Using energy carefully and avoiding waste** (1 mark).
- **So that the supply lasts longer and is available in the future** (1 mark).
- **Marking note:** Do not confuse this with the LAW of conservation of energy; a statement of the law earns no mark here.\*

**(b)** [4 marks]

- Any FOUR of: **switch off lights, fans and the television in empty rooms**; **use LED lamps in place of filament bulbs**; **dry clothes in the sun**; **iron all the clothes at one time**; **cover cooking pots and open the fridge less often**; **unplug chargers that are not in use** (1 mark each).
- **Marking note:** Two answers that are really the same point count once.\*

**(c)** [4 marks]

- (i) **A smaller electricity bill; appliances and cells last longer** (1 mark).
- (ii) **Less electricity has to be generated, so less fuel is burnt and there are fewer power cuts** (1 mark); **the money saved can be used for other national needs** (1 mark).
- (iii) **Less smoke and carbon dioxide, so cleaner air, less global warming and fewer habitats destroyed** (1 mark).
- **Marking note:** Award the marks under the correct headings.\*

## Question 6

**(a)** [3 marks]

- **Renewable:** replaced naturally and does not run out (1 mark);
- **non-renewable:** exists in a fixed amount and is used up for ever (1 mark).
- (ii) **One correct example of each, for example sunlight and crude oil** (1 mark).
- **Marking note:** Both examples are needed for the single mark in (ii).\*

**(b)** [3 marks]

- **Agree that energy is never destroyed** (1 mark).
- **But it is changed into heat and sound that spread out and can no longer be used** (1 mark).
- **The fuels used to supply useable energy are limited and burning them harms the environment, so saving is still necessary** (1 mark).
- **Marking note:** A reply that only says 'he is wrong' earns no mark.\*

**(c)** [2 marks]

- **Two slogans, each naming a clear energy-saving action or giving a sound reason** (1 mark each).
- **Marking note:** A slogan with no action and no reason, such as 'Save energy', earns no mark.\*

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**## CURRICULUM ASSESSMENT BLUEPRINT**

**Strand:** 4: Forces and Energy | **Sub-strand:** 3: Conversion and Conservation of Energy | **Class:** Basic 7 (JHS 1)

**Content Standard:** **B7/JHS1.4.3.1** - Demonstrate an understanding of the principle of conservation and conversion of energy and their application in real life situations.

> Source: NaCCA Science Common Core Programme Curriculum (B7/J

# Crop Production

## ANSWER KEY

### SECTION A: OBJECTIVE TEST

Total Marks: 40

| Q  | Ans      | Curriculum Area                               | Cognitive Level |
|----|----------|-----------------------------------------------|-----------------|
| 1  | <b>B</b> | 5.1 Plant Nutrients and the Needs of Crops    | Remember        |
| 2  | <b>A</b> | 5.1 Plant Nutrients and the Needs of Crops    | Remember        |
| 3  | <b>D</b> | 5.1 Plant Nutrients and the Needs of Crops    | Understand      |
| 4  | <b>A</b> | 5.1 Plant Nutrients and the Needs of Crops    | Remember        |
| 5  | <b>A</b> | 5.1 Plant Nutrients and the Needs of Crops    | Analyse         |
| 6  | <b>B</b> | 5.1 Plant Nutrients and the Needs of Crops    | Understand      |
| 7  | <b>C</b> | 5.2 Organic Sources of Plant Nutrients        | Remember        |
| 8  | <b>A</b> | 5.2 Organic Sources of Plant Nutrients        | Remember        |
| 9  | <b>D</b> | 5.2 Organic Sources of Plant Nutrients        | Remember        |
| 10 | <b>B</b> | 5.2 Organic Sources of Plant Nutrients        | Remember        |
| 11 | <b>A</b> | 5.2 Organic Sources of Plant Nutrients        | Understand      |
| 12 | <b>C</b> | 5.2 Organic Sources of Plant Nutrients        | Understand      |
| 13 | <b>C</b> | 5.2 Organic Sources of Plant Nutrients        | Analyse         |
| 14 | <b>D</b> | 5.3 Inorganic Sources of Plant Nutrients      | Remember        |
| 15 | <b>B</b> | 5.3 Inorganic Sources of Plant Nutrients      | Remember        |
| 16 | <b>A</b> | 5.3 Inorganic Sources of Plant Nutrients      | Remember        |
| 17 | <b>C</b> | 5.3 Inorganic Sources of Plant Nutrients      | Remember        |
| 18 | <b>D</b> | 5.3 Inorganic Sources of Plant Nutrients      | Understand      |
| 19 | <b>A</b> | 5.3 Inorganic Sources of Plant Nutrients      | Understand      |
| 20 | <b>B</b> | 5.3 Inorganic Sources of Plant Nutrients      | Analyse         |
| 21 | <b>A</b> | 5.4 Comparing Organic and Inorganic Sources   | Understand      |
| 22 | <b>C</b> | 5.4 Comparing Organic and Inorganic Sources   | Understand      |
| 23 | <b>D</b> | 5.4 Comparing Organic and Inorganic Sources   | Apply           |
| 24 | <b>D</b> | 5.4 Comparing Organic and Inorganic Sources   | Understand      |
| 25 | <b>C</b> | 5.4 Comparing Organic and Inorganic Sources   | Evaluate        |
| 26 | <b>B</b> | 5.4 Comparing Organic and Inorganic Sources   | Analyse         |
| 27 | <b>D</b> | 5.5 Physical Characteristics of the Nutrie... | Remember        |
| 28 | <b>C</b> | 5.5 Physical Characteristics of the Nutrie... | Remember        |
| 29 | <b>B</b> | 5.5 Physical Characteristics of the Nutrie... | Remember        |
| 30 | <b>D</b> | 5.5 Physical Characteristics of the Nutrie... | Understand      |
| 31 | <b>C</b> | 5.5 Physical Characteristics of the Nutrie... | Understand      |
| 32 | <b>C</b> | 5.5 Physical Characteristics of the Nutrie... | Analyse         |
| 33 | <b>D</b> | 5.5 Physical Characteristics of the Nutrie... | Analyse         |
| 34 | <b>A</b> | 5.6 Applying Plant Nutrients in the Field ... | Remember        |
| 35 | <b>B</b> | 5.6 Applying Plant Nutrients in the Field ... | Remember        |
| 36 | <b>B</b> | 5.6 Applying Plant Nutrients in the Field ... | Understand      |

| Q  | Ans      | Curriculum Area                               | Cognitive Level |
|----|----------|-----------------------------------------------|-----------------|
| 37 | <b>A</b> | 5.6 Applying Plant Nutrients in the Field ... | Remember        |
| 38 | <b>B</b> | 5.6 Applying Plant Nutrients in the Field ... | Understand      |
| 39 | <b>C</b> | 5.6 Applying Plant Nutrients in the Field ... | Apply           |
| 40 | <b>D</b> | 5.6 Applying Plant Nutrients in the Field ... | Evaluate        |

**Answer Distribution:**

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

**Cognitive Level Distribution:**

Analyse: 6 (15%) | Apply: 2 (5%) | Evaluate: 2 (5%) | Remember: 17 (42%) | Understand: 13 (32%)



## SECTION B: THEORY — MARKING SCHEME

### Question 1 [Total: 10 marks]

**(a)** [4 marks]

- W - **organic** (1 mark).
- X - **inorganic** (1 mark).
- Y - **organic** (1 mark).
- Z - **inorganic** (1 mark).
- \*Marking note: 'Natural' and 'artificial' are accepted for organic and inorganic respectively.\*

**(b)** [4 marks]

- W - any TWO of: **dark brown or black**; **crumbly or loose**; **bulky and heavy**; **moist**; **earthy-smelling** (1 mark each).
- X - any TWO of: **small hard granules (or crystals)**; **dry**; **light in colour (white, grey or pinkish)**; **sold in a sealed sack**; **concentrated, so little is needed** (1 mark each).
- \*Marking note: A characteristic that is not physical, such as 'it is cheap', earns no mark.\*

**(c)** [2 marks]

- **W (the compost)** (1 mark); accept Y (poultry manure) with the same reason.
- **Because organic manures are bulky and contain only a small proportion of nutrients, while inorganic fertilizers are concentrated** (1 mark).
- \*Marking note: The reason must mention the low concentration or the bulkiness of the organic material.\*

### Question 2 [Total: 12 marks]

**(a)** [4 marks]

- Any TWO organic: **compost**; **farmyard (cattle, sheep or goat) manure**; **poultry droppings**; **green manure**; **household or market waste** (1 mark each).
- Any TWO inorganic: **NPK**; **sulphate of ammonia**; **urea**; **superphosphate**; **muriate of potash**; **liquid (foliar) fertilizer** (1 mark each).
- \*Marking note: 'Fertilizer' written alone earns no mark; a named type is required.\*

**(b)** [4 marks]

- Source - **\*\*remains of plants and animals / manufactured from minerals in a factory\*\*** (1 mark).
- Quantity - **\*\*bulky, large quantities needed / concentrated, small quantities needed\*\*** (1 mark).
- Speed of action - **\*\*slow release over a long time / quick action\*\*** (1 mark).
- Effect on the soil - **\*\*adds humus and improves soil structure / adds no humus and may spoil the soil if used alone\*\*** (1 mark).
- **\*Marking note: Both halves of a row must be filled in correctly before the mark for that row is given.\***

**(c)** [2 marks]

- **\*\*No, it would not feed the plot as well\*\*** (1 mark).
- **\*\*The manure is bulky and contains a much smaller proportion of nutrients, so a far larger quantity is needed\*\*** (1 mark).
- **\*Marking note: No mark for the reason if the candidate answers 'yes'.\***

**(d)** [2 marks]

- Organic - any ONE of: **\*\*cheap or free\*\***; **\*\*made from waste on the compound\*\***; **\*\*adds humus and improves the soil\*\***; **\*\*releases nutrients slowly\*\*** (1 mark).
- Inorganic - any ONE of: **\*\*concentrated and easy to carry\*\***; **\*\*acts quickly\*\***; **\*\*supplies known amounts of named nutrients\*\*** (1 mark).
- **\*Marking note: An advantage written twice in different words earns 1 mark only.\***

**Question 3 [Total: 10 marks]****(a)** [3 marks]

- P - **\*\*broadcasting\*\*** (1 mark).
- Q - **\*\*ring application\*\*** (accept: circle or spot application) (1 mark).
- R - **\*\*applying liquid fertilizer by watering at the base of the plant\*\*** (accept: liquid or soil drench application) (1 mark).
- **\*Marking note: 'Foliar spraying' is not accepted for R; the liquid is poured at the base, not sprayed on the leaves.\***

**(b)** [2 marks]

- **\*\*P (broadcasting)\*\*** (1 mark).
- **\*\*The plants are many and close together, so the whole bed must be covered evenly; ringing each plant would be slow and would disturb them\*\*** (1 mark).
- **\*Marking note: No mark for the reason if Q is chosen.\***

**(c) [2 marks]**

- **Fertilizer against the stem would scorch (burn) the stem and roots** (1 mark).
- **At a short distance it lies over the feeding roots, which can still take up the nutrients** (1 mark).
- **Marking note: 'It will spoil the plant' alone earns 1 mark.**

**(d) [3 marks]**

- Any THREE of: **use only the measured quantity**; **apply when the soil is moist, not in a dry, hot afternoon**; **do not broadcast on a windy day**; **keep the fertilizer off the leaves and stems**; **use gloves or a scoop and wash the hands afterwards**; **keep fertilizer away from food and drinking water** (1 mark each).
- **Marking note: 'Be careful' earns no mark.**

**Question 4 [Total: 10 marks]****(a) [4 marks]**

- **Clear a shaded space or dig a shallow pit** (1 mark).
- **Heap the materials in layers, mixing dry and fresh material, with a little soil between the layers** (1 mark).
- **Water the heap to keep it damp and cover it** (1 mark).
- **Turn the heap regularly (every one to two weeks) until it is well rotted** (1 mark).
- **Marking note: The steps must be in a workable order; a bare list of materials earns no mark.**

**(b) [3 marks]**

- Any THREE of: **dark brown or black in colour**; **crumbly, with the original materials no longer recognisable**; **earthy smell, not a foul smell**; **the heap has cooled down and shrunk in size** (1 mark each).
- **Marking note: 'It looks like soil' is accepted once, for the colour or the crumbly texture, but not for both.**

**(c) [3 marks]**

- Any THREE of: **it costs nothing and uses waste from the compound**; **it adds humus and improves the soil**; **it releases nutrients slowly and is not easily washed away**; **there is no risk of scorching the crop**; **it keeps the compound clean** (1 mark each).
- **Marking note: Candidates who argue that NPK should also be used in small amounts should not be penalised, provided three reasons for compost are given.**

**Question 5****(a)** [3 marks]

- (i) **Bed 3** (1 mark).
- (ii) **7 kg** (1 mark).
- (iii) **Bed 1** (1 mark).
- *Marking note: Accept 7 without the unit in (ii).*

**(b)** [3 marks]

- **Too much fertilizer was applied** (1 mark).
- **The excess made the soil solution too strong and scorched (burnt) the roots and leaves** (1 mark).
- **The damaged plants grew poorly, so the yield fell** (1 mark).
- *Marking note: Accept 'the plants were burnt by the fertilizer' for 2 marks if the idea of excess is also stated.*

**(c)** [2 marks]

- Any TWO of: **the same size and type of bed and soil**; **the same crop**; **the same number of plants**; **the same watering and weeding**; **the same length of time** (1 mark each).
- *Marking note: The treatment itself is the variable being tested and must not be given as an answer.*

**(d)** [2 marks]

- **Use poultry manure worked into the soil together with the correct measured amount of NPK** (accept: use the measured amount of NPK only) (1 mark).
- **Reason drawn from the table: Bed 3 (correct amount) gave the best yield and Bed 4 shows that extra fertilizer lowers the yield; the manure also improves the soil** (1 mark).
- *Marking note: The reason must refer to the results in the table.*

**Question 6 [Total: 8 marks]****(a)** [3 marks]

- Any THREE, each with its reason: **store in a dry place** - *it absorbs moisture and cakes*; **keep the bag sealed and off the floor** - *to keep out moisture*; **store away from food, water and children** - *the chemicals are poisonous*; **keep it out of the sun and away from fire** - *heat spoils it and some fertilizers are a fire risk* (1 mark each).
- *Marking note: A rule given without its reason earns no mark.*

**(b)** [3 marks]

- **In dry soil the fertilizer cannot dissolve, so the plant cannot take it up** (1 mark).
- **The plants may be scorched, and some of the nutrient is lost in the heat** (1

mark).

- **Apply when the soil is moist - after a light rain or watering, in the cool of the morning or evening** (1 mark).

- **Marking note:** The better time must be stated for the third mark.\*

**(c)** [2 marks]

- Any TWO of: **read the label and dilute it with the stated volume of water**; **apply in the cool of the morning or evening**; **spread it evenly and avoid spraying into the wind**; **wash the sprayer and his hands after use**; **do not apply just before harvesting** (1 mark each).

- **Marking note:** 'Pour it on the plants' earns no mark; the idea of diluting or of measuring must appear in at least one answer.\*

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## ## CURRICULUM ASSESSMENT BLUEPRINT

**Strand:** 2: Cycles | **Sub-strand:** 3: Crop Production | **Class:** Basic 7 (JHS 1)

**Content Standard:** **B7/JHS1.2.3.1** - Demonstrat

# Ecosystem

## ANSWER KEY

### SECTION A: OBJECTIVE TEST

Total Marks: 40

| Q  | Ans | Curriculum Area                               | Cognitive Level |
|----|-----|-----------------------------------------------|-----------------|
| 1  | C   | 9.1 The Concept of an Ecosystem               | Remember        |
| 2  | C   | 9.1 The Concept of an Ecosystem               | Understand      |
| 3  | B   | 9.1 The Concept of an Ecosystem               | Remember        |
| 4  | D   | 9.1 The Concept of an Ecosystem               | Remember        |
| 5  | A   | 9.1 The Concept of an Ecosystem               | Remember        |
| 6  | C   | 9.1 The Concept of an Ecosystem               | Analyse         |
| 7  | A   | 9.1 The Concept of an Ecosystem               | Understand      |
| 8  | B   | 9.2 Terrestrial, Aquatic and Arboreal Ecos... | Remember        |
| 9  | D   | 9.2 Terrestrial, Aquatic and Arboreal Ecos... | Remember        |
| 10 | B   | 9.2 Terrestrial, Aquatic and Arboreal Ecos... | Remember        |
| 11 | B   | 9.2 Terrestrial, Aquatic and Arboreal Ecos... | Remember        |
| 12 | A   | 9.2 Terrestrial, Aquatic and Arboreal Ecos... | Understand      |
| 13 | B   | 9.2 Terrestrial, Aquatic and Arboreal Ecos... | Understand      |
| 14 | D   | 9.2 Terrestrial, Aquatic and Arboreal Ecos... | Understand      |
| 15 | D   | 9.3 Biotic Components of an Ecosystem         | Remember        |
| 16 | B   | 9.3 Biotic Components of an Ecosystem         | Remember        |
| 17 | D   | 9.3 Biotic Components of an Ecosystem         | Understand      |
| 18 | C   | 9.3 Biotic Components of an Ecosystem         | Understand      |
| 19 | A   | 9.3 Biotic Components of an Ecosystem         | Understand      |
| 20 | D   | 9.3 Biotic Components of an Ecosystem         | Analyse         |
| 21 | C   | 9.4 Abiotic Components of an Ecosystem        | Remember        |
| 22 | C   | 9.4 Abiotic Components of an Ecosystem        | Remember        |
| 23 | B   | 9.4 Abiotic Components of an Ecosystem        | Understand      |
| 24 | D   | 9.4 Abiotic Components of an Ecosystem        | Understand      |
| 25 | D   | 9.4 Abiotic Components of an Ecosystem        | Understand      |
| 26 | A   | 9.4 Abiotic Components of an Ecosystem        | Analyse         |
| 27 | C   | 9.5 Organisms of Different Ecosystems         | Understand      |
| 28 | B   | 9.5 Organisms of Different Ecosystems         | Understand      |
| 29 | C   | 9.5 Organisms of Different Ecosystems         | Understand      |
| 30 | D   | 9.5 Organisms of Different Ecosystems         | Analyse         |
| 31 | A   | 9.5 Organisms of Different Ecosystems         | Remember        |
| 32 | A   | 9.5 Organisms of Different Ecosystems         | Analyse         |
| 33 | B   | 9.5 Organisms of Different Ecosystems         | Analyse         |
| 34 | A   | 9.6 Interactions and Interdependence in an... | Remember        |
| 35 | A   | 9.6 Interactions and Interdependence in an... | Understand      |
| 36 | D   | 9.6 Interactions and Interdependence in an... | Understand      |
| 37 | C   | 9.6 Interactions and Interdependence in an... | Remember        |

| Q  | Ans      | Curriculum Area                               | Cognitive Level |
|----|----------|-----------------------------------------------|-----------------|
| 38 | <b>A</b> | 9.6 Interactions and Interdependence in an... | Understand      |
| 39 | <b>C</b> | 9.6 Interactions and Interdependence in an... | Analyse         |
| 40 | <b>B</b> | 9.6 Interactions and Interdependence in an... | Evaluate        |

**Answer Distribution:**

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

**Cognitive Level Distribution:**

Analyse: 7 (17%) | Evaluate: 1 (2%) | Remember: 15 (37%) | Understand: 17 (42%)

## SECTION B: THEORY — MARKING SCHEME

### Question 1 [Total: 10 marks]

**(a)** [5 marks]

- A - **the Sun (sunlight)** (1 mark).
- B - **a water plant** (accept: reed, water weed) (1 mark).
- C - **a fish** (1 mark).
- D - **the mud and stones at the bottom of the pond** (accept: soil, the pond bed) (1 mark).
- E - **a bird** (accept: a wading bird such as a heron or an egret) (1 mark).
- **Marking note:** A general answer such as 'plant' for B and 'animal' for C or E is accepted at this level.\*

**(b)** [3 marks]

- Any TWO of **B**, **C** and **E** as biotic (1 mark each).
- Either **A** or **D** as abiotic (1 mark).
- **Marking note:** Letters alone are enough; names need not be repeated.\*

**(c)** [2 marks]

- **Water plant to fish to bird** (1 mark).
- **Arrows drawn in the correct direction, from the eaten to the eater** (1 mark).
- **Marking note:** A chain written the wrong way round earns the first mark only if the three organisms are correct.\*

### Question 2 [Total: 12 marks]

**(a)** [3 marks]

- P - **terrestrial** (accept: land ecosystem) (1 mark).
- Q - **aquatic** (accept: water ecosystem) (1 mark).
- R - **arboreal** (accept: tree ecosystem) (1 mark).
- **Marking note:** 'Forest' written for R earns no mark; the category wanted is arboreal.\*

**(b)** [3 marks]

- Terrestrial - any ONE named Ghanaian land area, for example **the northern savanna grassland** (1 mark).
- Aquatic - any ONE named Ghanaian water body, for example **the Volta Lake** (1 mark).
- Arboreal - **the life on the branches of a named tall tree, such as a silk-cotton tree** (1 mark).
- **Marking note:** An example placed under the wrong category earns no mark.\*



**(c) [4 marks]**

- Q - a named water organism, for example **tilapia** (1 mark), with **gills** for breathing in water, fins for swimming or a streamlined body (1 mark).
- R - a named tree-living organism, for example **a monkey, a squirrel or a bird** (1 mark), with **limbs or a tail** for gripping branches, claws for climbing or wings for flying (1 mark).
- **Marking note:** The feature must match the organism named; a feature with no organism earns no mark.\*

**(d) [2 marks]**

- **The fish breathes through gills, which work only in water** (1 mark).
- **On land the gills dry up, so the fish cannot take in oxygen and it suffocates** (1 mark).
- **Marking note:** 'There is no water' alone earns 1 mark; the gills must be mentioned for the second.\*

**Question 3 [Total: 10 marks]****(a) [4 marks]**

- (i) **A living part of the ecosystem** (1 mark) **plus a correct example** (1 mark).
- (ii) **A non-living part of the ecosystem** (1 mark) **plus a correct example** (1 mark).
- **Marking note:** An example that does not match the definition earns no mark.\*

**(b) [3 marks]**

- Grass - **producer, makes its own food using sunlight** (1 mark).
- Grasshopper - **consumer, feeds on plants** (1 mark).
- Fungus - **decomposer, breaks down dead material and returns nutrients to the soil** (1 mark).
- **Marking note:** The name of the role alone is enough for the mark.\*

**(c) [3 marks]**

- **The grass makes its own food from sunlight, so food is produced inside the ecosystem** (1 mark).
- **The grasshopper and other consumers feed on the grass** (1 mark).
- **The fungus breaks down the dead material and returns the nutrients to the soil for the grass to use again** (1 mark).
- **Marking note:** The return of nutrients is essential for the third mark.\*

**Question 4****(a)** [3 marks]

- \*\*The waste killed the water plants\*\* (1 mark).
- \*\*The fish lost their food (and the polluted water also harmed them), so their numbers fell\*\* (1 mark).
- \*\*The birds, which feed on the fish, had no food and left\*\* (1 mark).
- \*Marking note: The chain must run plants to fish to birds; a single sentence naming pollution earns 1 mark only.\*

**(b)** [2 marks]

- \*\*The components are linked and depend on one another (interdependence)\*\* (1 mark).
- \*\*A change in one part affects the other parts, including people\*\* (1 mark).
- \*Marking note: 'Pollution is bad' earns no mark.\*

**(c)** [3 marks]

- Any THREE of: \*\*stop the untreated waste or require the factory to treat it\*\*;
- \*\*enforce by-laws and punish offenders\*\*;
- \*\*stop the dumping of refuse at the lagoon\*\*;
- \*\*replant the plants around the lagoon\*\*;
- \*\*close the lagoon to fishing for a season\*\*;
- \*\*educate the community\*\* (1 mark each).
- \*Marking note: 'Pray for the lagoon' and other non-scientific measures earn no mark.\*

**(d)** [2 marks]

- Any ONE of: \*\*competition\*\*;
- \*\*pollination\*\*;
- \*\*shelter or nesting\*\*;
- \*\*seed dispersal\*\* (1 mark).
- \*\*A correct example of the interaction named\*\* (1 mark).
- \*Marking note: Predation is a feeding relationship and earns no mark here.\*

**Question 5 [Total: 10 marks]****(a)** [4 marks]

- Any FOUR, each with a correct reason: \*\*water - the medium in which the organisms live\*\*;
- \*\*sand or gravel - holds the roots of the plants\*\*;
- \*\*water plants - make food and give out oxygen\*\*;
- \*\*small fish or snails - the consumers\*\*;
- \*\*light - needed by the plants to make food\*\*;
- \*\*stones - shelter for the animals\*\* (1 mark each).
- \*Marking note: An item named with no reason earns no mark.\*

**(b)** [2 marks]

- \*\*Biotic items correctly grouped\*\* (1 mark).
- \*\*Abiotic items correctly grouped\*\* (1 mark).
- \*Marking note: One item in the wrong group loses the mark for that group only.\*

**(c)** [2 marks]

- Any TWO of: **remove dead material and uneaten food**; **top up or partly change the water**; **feed the fish lightly**; **keep the tank in the light but out of the hot midday Sun**; **check the plants** (1 mark each).
- **Marking note: 'Clean it' alone is too vague.**

**(d)** [2 marks]

- A clear position, for example **I agree, the tank is a model rather than a true ecosystem** (1 mark).
- **Reason: a true ecosystem is self-sustaining, while the tank is too small and needs food to be added and waste removed** (1 mark).
- **Marking note: Either position may be taken; the mark is for the reason, which must mention self-sustaining or outside help.**

**Question 6 [Total: 8 marks]****(a)** [3 marks]

- Any THREE of: **animals lose shelter and food and move away or die**; **the soil is eroded and loses fertility**; **the area becomes hotter and drier**; **streams dry up**; **shade-loving plants die** (1 mark each).
- **Marking note: 'The forest is finished' is a restatement and earns no mark.**

**(b)** [3 marks]

- Any THREE of: **timber, firewood, fruits, bush meat or herbs**; **clean air and oxygen**; **protection of the soil from erosion**; **water and rainfall**; **shade and a cooler climate**; **a place for study and tourism** (1 mark each).
- **Marking note: Two answers about wood and firewood count as one.**

**(c)** [2 marks]

- **A workable activity, for example raising and planting seedlings of local trees at the start of the rains** (1 mark).
- **A way of judging success, for example counting the surviving seedlings or noting the return of grasses, insects and birds** (1 mark).
- **Marking note: An activity with no way of judging success earns 1 mark only.**

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**## CURRICULUM ASSESSMENT BLUEPRINT****Strand:** 3: Systems | **Sub-strand:** 3: Ecosystem | **Class:** Basic 7 (JHS 1)**Content Standard:** **B7/JHS1.3.3.1** - Recognise the components of and interdependences in an ecosystem, and appreciate their inte

# Electricity and Electr

## ANSWER KEY

### SECTION A: OBJECTIVE TEST

Total Marks: 40

| Q  | Ans | Curriculum Area                               | Cognitive Level |
|----|-----|-----------------------------------------------|-----------------|
| 1  | D   | 12.1 Generation of Thermal and Nuclear Ele... | Remember        |
| 2  | C   | 12.1 Generation of Thermal and Nuclear Ele... | Understand      |
| 3  | C   | 12.1 Generation of Thermal and Nuclear Ele... | Remember        |
| 4  | D   | 12.1 Generation of Thermal and Nuclear Ele... | Remember        |
| 5  | C   | 12.1 Generation of Thermal and Nuclear Ele... | Understand      |
| 6  | A   | 12.1 Generation of Thermal and Nuclear Ele... | Remember        |
| 7  | B   | 12.1 Generation of Thermal and Nuclear Ele... | Understand      |
| 8  | B   | 12.1 Generation of Thermal and Nuclear Ele... | Analyse         |
| 9  | C   | 12.2 Effects of Electricity Generation on ... | Remember        |
| 10 | A   | 12.2 Effects of Electricity Generation on ... | Understand      |
| 11 | B   | 12.2 Effects of Electricity Generation on ... | Understand      |
| 12 | A   | 12.2 Effects of Electricity Generation on ... | Remember        |
| 13 | B   | 12.2 Effects of Electricity Generation on ... | Understand      |
| 14 | D   | 12.2 Effects of Electricity Generation on ... | Evaluate        |
| 15 | B   | 12.2 Effects of Electricity Generation on ... | Understand      |
| 16 | B   | 12.3 Basic Electronic Components and Their... | Remember        |
| 17 | A   | 12.3 Basic Electronic Components and Their... | Remember        |
| 18 | A   | 12.3 Basic Electronic Components and Their... | Understand      |
| 19 | D   | 12.3 Basic Electronic Components and Their... | Remember        |
| 20 | C   | 12.3 Basic Electronic Components and Their... | Remember        |
| 21 | A   | 12.3 Basic Electronic Components and Their... | Remember        |
| 22 | C   | 12.3 Basic Electronic Components and Their... | Remember        |
| 23 | D   | 12.3 Basic Electronic Components and Their... | Understand      |
| 24 | A   | 12.3 Basic Electronic Components and Their... | Apply           |
| 25 | C   | 12.4 The P-N Junction Diode                   | Remember        |
| 26 | B   | 12.4 The P-N Junction Diode                   | Understand      |
| 27 | D   | 12.4 The P-N Junction Diode                   | Remember        |
| 28 | C   | 12.4 The P-N Junction Diode                   | Analyse         |
| 29 | B   | 12.4 The P-N Junction Diode                   | Remember        |
| 30 | C   | 12.5 Assembling a Simple Electronic Circuit   | Understand      |
| 31 | B   | 12.5 Assembling a Simple Electronic Circuit   | Understand      |
| 32 | A   | 12.5 Assembling a Simple Electronic Circuit   | Remember        |
| 33 | A   | 12.5 Assembling a Simple Electronic Circuit   | Understand      |
| 34 | D   | 12.5 Assembling a Simple Electronic Circuit   | Apply           |
| 35 | D   | 12.5 Assembling a Simple Electronic Circuit   | Understand      |
| 36 | C   | 12.6 How the Components Affect One Another    | Analyse         |
| 37 | D   | 12.6 How the Components Affect One Another    | Analyse         |

| Q  | Ans      | Curriculum Area                            | Cognitive Level |
|----|----------|--------------------------------------------|-----------------|
| 38 | <b>D</b> | 12.6 How the Components Affect One Another | Analyse         |
| 39 | <b>A</b> | 12.6 How the Components Affect One Another | Understand      |
| 40 | <b>B</b> | 12.6 How the Components Affect One Another | Understand      |

**Answer Distribution:**

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

**Cognitive Level Distribution:**

Analyse: 5 (12%) | Apply: 2 (5%) | Evaluate: 1 (2%) | Remember: 16 (40%) | Understand: 16 (40%)

**SECTION B: THEORY — MARKING SCHEME****Question 1 [Total: 12 marks]****(a)** [4 marks]

- W - **battery** (two dry cells in series) (1 mark).
- X - **switch** (1 mark).
- Y - **resistor** (1 mark).
- Z - **LED (light-emitting diode)** (1 mark).
- \*Marking note: 'Bulb' or 'lamp' written for Z earns no mark.\*

**(b)** [4 marks]

- W - **supplies the electrical energy (drives the current)** (1 mark).
- X - **completes or breaks the circuit, switching the current on and off** (1 mark).
- Y - **limits the current and so protects the LED** (1 mark).
- Z - **gives out light when current passes through it** (1 mark).
- \*Marking note: 'It carries current' is true of every part and earns no mark.\*

**(c)** [2 marks]

- **No, the LED is not lit** (1 mark).
- **The switch is open, so the circuit is incomplete and no current flows** (1 mark).
- \*Marking note: No mark for the reason if the candidate says the LED is lit.\*

**(d)** [2 marks]

- **Each dry cell supplies 1.5 V** (1 mark).
- **Two cells joined in series add up:  $1.5 + 1.5 = 3 \text{ V}$**  (1 mark).
- \*Marking note: The word 'series' or a correct description of it is needed for the second mark.\*

**Question 2 [Total: 10 marks]****(a)** [4 marks]

- A - **furnace or boiler** (1 mark).
- B - **turbine** (1 mark).
- C - **generator** (1 mark).
- D - **transmission pylon (power lines)** (1 mark).
- \*Marking note: 'Chimney' written for A earns no mark; the chimney only carries away the waste gases.\*

**(b)** [3 marks]

- **A: chemical energy of the fuel to heat energy** (1 mark).
- **B: heat energy of the steam to kinetic energy of the turbine** (1 mark).
- **C: kinetic energy to electrical energy** (1 mark).
- **Marking note: The three changes must be in the right order.**

**(c)** [3 marks]

- Any THREE of: **the heat comes from splitting uranium nuclei in a reactor, not from burning fuel; no smoke or carbon dioxide is given off; a very small mass of fuel gives a great deal of energy; it produces radioactive waste; it is far more costly to build** (1 mark each).
- **Marking note: A point that is true of BOTH stations, such as 'it uses a turbine', earns no mark.**

**Question 3 [Total: 10 marks]****(a)** [4 marks]

- Any FOUR of: **carbon dioxide adds to global warming; smoke and soot cause breathing diseases; sulphur dioxide causes acid rain; hot water kills fish in the river; oil spills pollute land and water; noise pollution; a non-renewable fuel is used up** (1 mark each).
- **Marking note: Two answers that both say 'air pollution' count as one.**

**(b)** [3 marks]

- Any THREE of: **fit filters or scrubbers to the chimneys; use cleaner fuel such as natural gas; use more solar, wind and hydro power; cool the water before returning it; site stations away from homes and plant trees; save electricity so that less is generated** (1 mark each).
- **Marking note: 'Stop making electricity' is not a practical measure and earns no mark.**

**(c)** [3 marks]

- **Agree that it gives off no smoke and no carbon dioxide** (1 mark).
- **But it produces radioactive waste that stays dangerous for a very long time** (1 mark).
- **And an accident or leak would release harmful radiation (accept: warm water still harms aquatic life)** (1 mark).
- **Marking note: A reply that merely says 'he is wrong' earns no mark.**

**Question 4 [Total: 10 marks]**

**(a)** [4 marks]

- **\*\*Correct symbol for the two cells in series\*\*** (1 mark).
- **\*\*Correct symbols for the switch and the resistor\*\*** (1 mark).
- **\*\*Correct symbol for the LED (triangle and bar)\*\*** (1 mark).
- **\*\*A complete single loop with all four components in series and labelled\*\*** (1 mark).
- **\*Marking note: A circuit with a gap anywhere, other than at the switch, loses the completeness mark.\***

**(b)** [3 marks]

- **\*\*The longer leg (positive leg or anode) is joined towards the positive terminal of the battery\*\*** (1 mark).
- **\*\*Connected this way the LED is forward biased and lights\*\*** (1 mark).
- **\*\*Turned round it is reverse biased, no current flows and it does not light\*\*** (1 mark).
- **\*Marking note: 'It will burn out' is wrong for a reversed LED at this voltage and earns no mark.\***

**(c)** [3 marks]

- (i) **\*\*The circuit is complete, current flows and the LED lights\*\*** (1 mark).
- (ii) **\*\*The path is broken, the current stops and the LED goes out\*\*** (1 mark).
- **\*\*The resistor limits the current and protects the LED\*\*** (1 mark).
- **\*Marking note: 'The electricity rests' earns no mark.\***

**Question 5 [Total: 10 marks]****(a)** [4 marks]

- Resistor - **\*\*limits or controls the current\*\*** (1 mark).
- Capacitor - **\*\*stores charge and releases it later\*\*** (1 mark).
- Diode - **\*\*allows current in one direction only\*\*** (1 mark).
- Inductor - **\*\*opposes a change in the current (stores energy in a magnetic field)\*\*** (1 mark).
- **\*Marking note: A function written against the wrong component earns no mark.\***

**(b)** [3 marks]

- (i) **\*\*A component made by joining a P region and an N region of semiconductor, which passes current one way only\*\*** (1 mark).
- (ii) **\*\*The P region and the N region\*\*** (1 mark); **\*\*the P region is joined to the positive terminal\*\*** (1 mark).
- **\*Marking note: Accept 'positive region' for P and 'negative region' for N.\***



**(c) [3 marks]**

- Any THREE of: **unplug the gadget and remove the cells first**; **never open a gadget connected to the mains; leave mains parts to the teacher**; **work under supervision with the proper tools**; **keep the parts safely and sketch the arrangement before removing anything**; **handle sharp edges with care and wash hands afterwards** (1 mark each).

- **Marking note:** 'Be careful' alone is too vague for a mark.\*

**Question 6 [Total: 8 marks]****(a) [3 marks]**

- **The LED becomes dimmer** (1 mark).

- **The total resistance of the circuit increases** (1 mark).

- **So the current falls, and the brightness depends on the current** (1 mark).

- **Marking note:** No reason marks if the candidate says the LED becomes brighter.\*

**(b) [3 marks]**

- **The LED did not light** (1 mark).

- **The diode was reverse biased and a diode conducts in one direction only** (1 mark).

- **So no current flowed round the circuit** (1 mark).

- **Marking note:** 'The diode was spoilt' earns no mark.\*

**(c) [2 marks]**

- **The components depend on one another; a change in one affects the working of the others** (1 mark).

- **Example: the resistor fixes the current, and the current decides the brightness of the LED (or protects it from damage)** (1 mark).

- **Marking note:** A definition with no example earns 1 mark only.\*

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**## CURRICULUM ASSESSMENT BLUEPRINT**

**Strand:** 4: Forces and Energy | **Sub-strand:** 2: Electricity and Electronics |  
**Class:** Basic 7 (JHS 1)

**Content Standard:** **B7/JHS1.4.2.1** - Demonstrate understanding of forms of electricity, its generation and effects on the environment

# Energy

## ANSWER KEY

### SECTION A: OBJECTIVE TEST

Total Marks: 40

| Q  | Ans | Curriculum Area                               | Cognitive Level |
|----|-----|-----------------------------------------------|-----------------|
| 1  | D   | 11.1 Forms of Energy                          | Remember        |
| 2  | B   | 11.1 Forms of Energy                          | Remember        |
| 3  | B   | 11.1 Forms of Energy                          | Remember        |
| 4  | D   | 11.1 Forms of Energy                          | Remember        |
| 5  | D   | 11.1 Forms of Energy                          | Remember        |
| 6  | A   | 11.1 Forms of Energy                          | Remember        |
| 7  | B   | 11.1 Forms of Energy                          | Remember        |
| 8  | D   | 11.1 Forms of Energy                          | Understand      |
| 9  | D   | 11.2 Potential, Kinetic and Mechanical Energy | Remember        |
| 10 | A   | 11.2 Potential, Kinetic and Mechanical Energy | Understand      |
| 11 | C   | 11.2 Potential, Kinetic and Mechanical Energy | Understand      |
| 12 | C   | 11.2 Potential, Kinetic and Mechanical Energy | Understand      |
| 13 | A   | 11.2 Potential, Kinetic and Mechanical Energy | Understand      |
| 14 | C   | 11.2 Potential, Kinetic and Mechanical Energy | Understand      |
| 15 | A   | 11.2 Potential, Kinetic and Mechanical Energy | Understand      |
| 16 | C   | 11.3 Daily Applications of Energy             | Understand      |
| 17 | A   | 11.3 Daily Applications of Energy             | Understand      |
| 18 | A   | 11.3 Daily Applications of Energy             | Understand      |
| 19 | C   | 11.3 Daily Applications of Energy             | Understand      |
| 20 | A   | 11.3 Daily Applications of Energy             | Apply           |
| 21 | B   | 11.3 Daily Applications of Energy             | Analyse         |
| 22 | B   | 11.3 Daily Applications of Energy             | Understand      |
| 23 | A   | 11.4 Calculating Potential and Kinetic Energy | Remember        |
| 24 | D   | 11.4 Calculating Potential and Kinetic Energy | Remember        |
| 25 | C   | 11.4 Calculating Potential and Kinetic Energy | Apply           |
| 26 | B   | 11.4 Calculating Potential and Kinetic Energy | Apply           |
| 27 | D   | 11.4 Calculating Potential and Kinetic Energy | Analyse         |
| 28 | D   | 11.4 Calculating Potential and Kinetic Energy | Analyse         |
| 29 | B   | 11.5 Transfer of Heat                         | Remember        |
| 30 | B   | 11.5 Transfer of Heat                         | Remember        |
| 31 | C   | 11.5 Transfer of Heat                         | Remember        |
| 32 | B   | 11.5 Transfer of Heat                         | Understand      |
| 33 | C   | 11.5 Transfer of Heat                         | Understand      |
| 34 | C   | 11.5 Transfer of Heat                         | Analyse         |
| 35 | B   | 11.6 Characteristics of Light                 | Understand      |
| 36 | D   | 11.6 Characteristics of Light                 | Apply           |
| 37 | D   | 11.6 Characteristics of Light                 | Remember        |

| Q  | Ans      | Curriculum Area               | Cognitive Level |
|----|----------|-------------------------------|-----------------|
| 38 | <b>A</b> | 11.6 Characteristics of Light | Understand      |
| 39 | <b>A</b> | 11.6 Characteristics of Light | Remember        |
| 40 | <b>C</b> | 11.6 Characteristics of Light | Analyse         |

**Answer Distribution:**

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

**Cognitive Level Distribution:**

Analyse: 5 (12%) | Apply: 4 (10%) | Remember: 15 (37%) | Understand: 16 (40%)

# Energy

## SECTION B: THEORY — MARKING SCHEME

### Question 1 [Total: 12 marks]

(a) [2 marks]

- A - **potential energy** (1 mark).
- C - **kinetic energy** (1 mark).
- \*Marking note: 'Gravitational potential energy' is accepted for A.\*

(b) [3 marks]

- **The potential energy decreases as the height falls** (1 mark).
- **The kinetic energy increases as the speed rises (at B the ball has both)** (1 mark).
- **The potential energy lost is changed into kinetic energy, so the total mechanical energy stays the same** (1 mark).
- \*Marking note: 'The ball moves faster' alone earns 1 mark only.\*

(c) [4 marks]

- (i)  **$PE = mgh = 0.5 \times 10 \times 3.2$**  (1 mark)  **$= 16 \text{ J}$**  (1 mark).
- (ii)  **$KE = 16 \text{ J}$ , so  $\frac{1}{2} \times 0.5 \times v^2 = 16$ ,  $v^2 = 64$**  (1 mark)  **$v = 8 \text{ m/s}$**  (1 mark).
- \*Marking note: Award the method marks for a correct formula and correct substitution even if the arithmetic is wrong.\*

(d) [3 marks]

- **There is friction between the ball and the slope, and air resistance** (1 mark).
- **Some of the energy is used to do work against friction, so less becomes kinetic energy** (1 mark).
- **The missing energy becomes heat (and a little sound); it is not destroyed** (1 mark).
- \*Marking note: 'The energy is lost' without naming heat or sound earns no mark for the third point.\*

### Question 2 [Total: 10 marks]

(a) [2 marks]

- (i) **The ability (capacity) to do work** (1 mark).
- (ii) **The joule (J)** (1 mark).
- \*Marking note: 'Power' or 'watt' earns no mark in (ii).\*

**(b)** [4 marks]

- Any FOUR forms, each with a correct source, for example **heat - a cooking fire**; **light - a lamp**; **sound - a drum**; **solar - the Sun**; **electrical - a dry cell**; **chemical - food or charcoal**; **nuclear - a nuclear reactor** (1 mark each).
- *Marking note: A form named without a source earns no mark.*

**(c)** [4 marks]

- Electric bulb - **electrical to light (and heat)** (1 mark).
- Dry cell - **chemical to electrical** (1 mark).
- Electric fan - **electrical to kinetic (mechanical)** (1 mark).
- Solar panel - **solar to electrical** (1 mark).
- *Marking note: The change must be written as one form to another; a single word earns no mark.*

**Question 3 [Total: 10 marks]****(a)** [2 marks]

- (i) **The sum of the potential energy and the kinetic energy of a body** (1 mark).
- (ii) **PE = mgh**, with m = mass, g = acceleration due to gravity and h = height (1 mark).
- *Marking note: The expression must be accompanied by the meanings of the letters.*

**(b)** [4 marks]

- (i) **KE =  $\frac{1}{2} \times 2\,000 \times 10^2$**  (1 mark) **= 100 000 J** (1 mark).
- (ii) **KE =  $\frac{1}{2} \times 2\,000 \times 20^2$**  (1 mark) **= 400 000 J** (1 mark).
- *Marking note: Accept 100 kJ and 400 kJ. A missing unit loses the final mark of the part only.*

**(c)** [2 marks]

- **Doubling the speed makes the kinetic energy four times as large, since KE depends on the square of the speed** (1 mark).
- **The lorry then needs a much longer distance to stop and does far more damage in a crash** (1 mark).
- *Marking note: 'Speeding is bad' without reference to the energy earns no mark.*

**(d)** [2 marks]

- **The mass of the body** (1 mark).
- **The height above the ground** (accept: the acceleration due to gravity) (1 mark).
- *Marking note: 'Speed' is a factor for kinetic energy and earns no mark here.*

**Question 4****(a)** [3 marks]

- **Pin P** (1 mark).
- **Metal is a good conductor, so heat travels quickly along the rod to the far end** (1 mark).
- **The wax at that end melts first and releases the pin (the other two rods are poor conductors)** (1 mark).
- **Marking note: No reason marks if the wrong pin is named.**

**(b)** [2 marks]

- (i) **Conduction** (1 mark).
- (ii) **Convection** (1 mark).
- **Marking note: Radiation written for (i) earns no mark.**

**(c)** [2 marks]

- Any TWO of: **the same length and thickness of rod**; **the same depth in the same hot water (same temperature)**; **the same amount of wax**; **pins of the same size, started at the same time** (1 mark each).
- **Marking note: 'The material' must NOT be given; that is what is being changed.**

**(d)** [3 marks]

- **Radiation** (1 mark).
- **It needs no material medium; it can travel through empty space (a vacuum)** (1 mark).
- **Example: heat from the Sun reaching the Earth, or warmth felt in front of a coal pot** (1 mark).
- **Marking note: 'It does not need water' is not enough for the second mark.**

**Question 5 [Total: 10 marks]****(a)** [4 marks]

- **Stand the three cards one behind the other and line up the holes, using the thread** (1 mark).
- **Place the lighted candle in line at one end and look through the nearest hole** (1 mark).
- **Observation: the flame is seen when the holes are in a straight line** (1 mark).
- **Observation: the flame cannot be seen when one card is moved out of line** (1 mark).
- **Marking note: Both observations are needed; the second is the one that proves the point.**

**(b)** [4 marks]

- **Reflection: light bounces back when it falls on a surface** (1 mark), **example: an image in a mirror or in still water** (1 mark).
- **Refraction: light bends as it passes from one transparent medium into another** (1 mark), **example: a pencil looking bent in a glass of water** (1 mark).
- **Marking note: An example that belongs to the other effect earns no mark.**

**(c)** [2 marks]

- (i) **Dispersion** (1 mark).
- (ii) **A glass prism** (1 mark).
- **Marking note: 'A mirror' earns no mark in (ii).**

**Question 6 [Total: 8 marks]****(a)** [3 marks]

- **Solar energy to electrical energy** (1 mark).
- Any TWO of: **Ghana has strong sunshine for most of the year**; **sunlight is free, so running costs are low**; **the panels work when the mains supply fails**; **they are clean, quiet and give off no smoke** (1 mark each).
- **Marking note: 'It is modern' earns no mark.**

**(b)** [3 marks]

- Any THREE of: **switch off lights and fans in empty rooms**; **use energy-saving (LED) bulbs**; **use daylight instead of lights**; **service the appliances**; **appoint class energy monitors**; **cook with a lid on the pot on an efficient stove** (1 mark each).
- **Marking note: Two answers about switching things off count as one.**

**(c)** [2 marks]

- **The water at the bottom is heated, expands, becomes less dense and rises, while cooler water sinks to take its place** (1 mark).
- **These convection currents carry the heat through the whole pot of water** (1 mark).
- **Marking note: The word convection alone, with no mention of the rising and sinking water, earns 1 mark.**

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**## CURRICULUM ASSESSMENT BLUEPRINT**

**Strand:** 4: Forces and Energy | **Sub-strand:** 1: Energy | **Class:** Basic 7 (JHS 1)

**\*\*Content Standard:\*\*** **\*\*B7/JHS1.4.1.1\*\*** - Demonstrate an understanding of forms of energy and their daily applications. **\*\*B7/JHS1.4.1.2\***





# Farming Systems

## ANSWER KEY

### SECTION A: OBJECTIVE TEST

Total Marks: 40

| Q  | Ans | Curriculum Area                               | Cognitive Level |
|----|-----|-----------------------------------------------|-----------------|
| 1  | A   | 10.1 Meaning and Types of Farming Systems     | Remember        |
| 2  | A   | 10.1 Meaning and Types of Farming Systems     | Understand      |
| 3  | D   | 10.1 Meaning and Types of Farming Systems     | Understand      |
| 4  | B   | 10.1 Meaning and Types of Farming Systems     | Remember        |
| 5  | A   | 10.1 Meaning and Types of Farming Systems     | Remember        |
| 6  | D   | 10.1 Meaning and Types of Farming Systems     | Remember        |
| 7  | D   | 10.2 Land Rotation                            | Remember        |
| 8  | B   | 10.2 Land Rotation                            | Remember        |
| 9  | B   | 10.2 Land Rotation                            | Remember        |
| 10 | D   | 10.2 Land Rotation                            | Understand      |
| 11 | D   | 10.2 Land Rotation                            | Understand      |
| 12 | B   | 10.2 Land Rotation                            | Understand      |
| 13 | C   | 10.2 Land Rotation                            | Analyse         |
| 14 | A   | 10.3 Crop Rotation                            | Remember        |
| 15 | B   | 10.3 Crop Rotation                            | Understand      |
| 16 | B   | 10.3 Crop Rotation                            | Remember        |
| 17 | A   | 10.3 Crop Rotation                            | Understand      |
| 18 | C   | 10.3 Crop Rotation                            | Understand      |
| 19 | B   | 10.3 Crop Rotation                            | Analyse         |
| 20 | A   | 10.3 Crop Rotation                            | Apply           |
| 21 | A   | 10.4 Mixed Cropping                           | Apply           |
| 22 | C   | 10.4 Mixed Cropping                           | Understand      |
| 23 | A   | 10.4 Mixed Cropping                           | Understand      |
| 24 | C   | 10.4 Mixed Cropping                           | Understand      |
| 25 | D   | 10.4 Mixed Cropping                           | Analyse         |
| 26 | A   | 10.4 Mixed Cropping                           | Analyse         |
| 27 | C   | 10.5 Mixed Farming                            | Apply           |
| 28 | B   | 10.5 Mixed Farming                            | Understand      |
| 29 | D   | 10.5 Mixed Farming                            | Understand      |
| 30 | B   | 10.5 Mixed Farming                            | Understand      |
| 31 | B   | 10.5 Mixed Farming                            | Understand      |
| 32 | C   | 10.5 Mixed Farming                            | Analyse         |
| 33 | C   | 10.6 Organic Farming and the Choice of a S... | Remember        |
| 34 | D   | 10.6 Organic Farming and the Choice of a S... | Remember        |
| 35 | C   | 10.6 Organic Farming and the Choice of a S... | Understand      |
| 36 | D   | 10.6 Organic Farming and the Choice of a S... | Understand      |

| Q  | Ans      | Curriculum Area                               | Cognitive Level |
|----|----------|-----------------------------------------------|-----------------|
| 37 | <b>A</b> | 10.6 Organic Farming and the Choice of a S... | Understand      |
| 38 | <b>C</b> | 10.6 Organic Farming and the Choice of a S... | Analyse         |
| 39 | <b>C</b> | 10.6 Organic Farming and the Choice of a S... | Evaluate        |
| 40 | <b>D</b> | 10.6 Organic Farming and the Choice of a S... | Evaluate        |

**Answer Distribution:**

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

**Cognitive Level Distribution:**

Analyse: 6 (15%) | Apply: 3 (7%) | Evaluate: 2 (5%) | Remember: 11 (27%) | Understand: 18 (45%)

## SECTION B: THEORY — MARKING SCHEME

### Question 1 [Total: 10 marks]

**(a)** [3 marks]

- A - **\*\*mixed cropping\*\*** (1 mark).
- B - **\*\*mixed farming\*\*** (1 mark).
- C - **\*\*land rotation\*\*** (accept: shifting cultivation) (1 mark).
- **\*Marking note: Mixed farming written for A, or mixed cropping for B, earns no mark; the two must be told apart.\***

**(b)** [4 marks]

- Any TWO advantages: **\*\*income from two sources\*\***; **\*\*manure for the crops\*\***; **\*\*crop residues as feed\*\***; **\*\*land and labour used all year\*\*** (1 mark each).
- Any TWO difficulties: **\*\*more capital needed\*\***; **\*\*more labour and skill needed\*\***; **\*\*animals may stray into the crops\*\***; **\*\*animal diseases must be controlled\*\*** (1 mark each).
- **\*Marking note: A general answer such as 'it is good' earns no mark.\***

**(c)** [3 marks]

- **\*\*The plots have lost their fertility after cropping\*\*** (1 mark).
- **\*\*The bush that grows and decays on them returns nutrients and restores the fertility\*\*** (1 mark).
- **\*\*Reason: population growth has reduced the land available, so fallow periods are too short\*\*** (accept: land is now scarce or expensive) (1 mark).
- **\*Marking note: 'He is lazy' earns no mark.\***

### Question 2 [Total: 12 marks]

**(a)** [2 marks]

- (i) **\*\*Crop rotation\*\*** (1 mark).
- (ii) **\*\*Different crops follow one another in a planned order on the same land, and the plan is repeated\*\*** (1 mark).
- **\*Marking note: 'Land rotation' earns no mark; the land does not change.\***

**(b)** [3 marks]

- **\*\*Groundnut is a legume\*\*** (1 mark).
- **\*\*Bacteria in the nodules on its roots fix nitrogen from the air\*\*** (1 mark).
- **\*\*The soil is therefore enriched for the crop that follows (and the farmer saves on fertilizer)\*\*** (1 mark).
- **\*Marking note: 'It is a good food' alone earns no mark in this part.\***

**(c) [4 marks]**

- Any FOUR of: **soil fertility is maintained**; **pests and diseases are controlled**; **weeds are kept down**; **the soil structure is improved by different rooting depths**; **the resting year lets the land recover**; **a variety of food and income** (1 mark each).

- **Marking note:** Two answers that both say 'the soil becomes rich' count as one.\*

**(d) [3 marks]**

- **A plan naming all four in a sensible order over four years** (2 marks; 1 mark if one item is missing or out of place).

- **A sound reason, for example the legume follows the heavy feeder to restore nitrogen, or the resting year comes last before the plan is repeated** (1 mark).

- **Marking note:** Any order is accepted provided the reason given supports it.\*

**Question 3 [Total: 10 marks]****(a) [3 marks]**

- Opoku - **land rotation** (accept: shifting cultivation) (1 mark).

- Mariama - **crop rotation** (1 mark).

- Yaw - **mixed cropping** (1 mark).

- **Marking note:** The names must match the descriptions exactly.\*

**(b) [4 marks]**

- Opoku - **the piece of land changes (the crop may stay the same)** (1 mark).

- Mariama - **the crop changes, the land stays the same** (1 mark).

- Opoku - **a large area of land is needed** (1 mark).

- Mariama - **only a small area is needed** (1 mark).

- **Marking note:** A row left blank earns no mark; both columns must be completed.\*

**(c) [3 marks]**

- Any TWO of: **several foods from one small plot**; **insurance if one crop fails**; **the soil is covered, so less erosion and fewer weeds**; **the legume adds nitrogen** (1 mark each).

- Any ONE of: **the crops compete for light, water and nutrients**; **machines cannot easily be used**; **the crops ripen at different times, so more labour is needed** (1 mark).

- **Marking note:** A difficulty written as an advantage earns no mark.\*

**Question 4 [Total: 10 marks]**

**(a)** [2 marks]

- **Mixed farming** (1 mark).
- **Reason: she keeps animals and grows crops on the same holding at the same time** (1 mark).
- **Marking note: 'Mixed cropping' earns no mark.**

**(b)** [4 marks]

- Fowls to crops - any TWO of: **droppings used as manure**; **they eat insect pests**; **they scratch out weed seeds** (1 mark each).
- Crops to fowls - any TWO of: **maize grain and bran as feed**; **green leaves and crop residues as feed**; **stalks used as litter or for the house** (1 mark each).
- **Marking note: The same point written twice in different words counts once.**

**(c)** [2 marks]

- Any TWO of: **compost or allow the droppings to rot before use**; **dig them into the soil and keep them off the leaves**; **do not apply shortly before harvesting leafy vegetables**; **wash hands and wash the vegetables well** (1 mark each).
- **Marking note: 'Use gloves' is accepted as one answer.**

**(d)** [2 marks]

- Any TWO of: **stop using artificial fertilizer**; **stop using chemical pesticides or weedicides and use natural methods**; **check that the poultry feed and any water running onto the plot are free of chemicals**; **keep records and obtain proper certification** (1 mark each).
- **Marking note: 'Grow more vegetables' earns no mark.**

**Question 5 [Total: 10 marks]****(a)** [2 marks]

- **Farming without artificial fertilizers and chemical pesticides or weedicides** (1 mark).
- **Using compost, manure and natural methods of pest and weed control instead** (1 mark).
- **Marking note: 'Farming without chemicals' alone earns 1 mark.**

**(b)** [4 marks]

- Any FOUR of: **apply compost or manure**; **practise crop rotation with legumes**; **mulch the beds**; **use neem extract, ash or other natural pest control**; **hand weed and hand pick pests**; **encourage natural enemies**; **use resistant varieties** (1 mark each).
- **Marking note: 'Do not use chemicals' is a definition, not a practice, and earns no mark here.**

**(c) [4 marks]**

- Any TWO merits: **food free from chemical residues**; **soil life and structure protected**; **no pollution of water bodies**; **less money spent on chemicals**; **higher price for the produce** (1 mark each).
- Any TWO demerits: **lower yields**; **much more labour**; **pests and diseases harder to control quickly**; **certification costs money and takes time** (1 mark each).
- **Marking note: A merit written in the demerit column earns no mark.**

**Question 6 [Total: 8 marks]****(a) [4 marks]**

- **A named system, for example mixed cropping (accept mixed farming or crop rotation, if well defended)** (1 mark).
- Any THREE reasons that fit the farmer's situation: **the land is small, so no land can be left fallow for long**; **several foods from one plot and food spread through the year**; **insurance if one crop fails**; **a legume keeps the soil fertile without costly fertilizer**; **the work can be done by hand** (1 mark each).
- **Marking note: The reasons must refer to this farmer's land, money or family; general praise of the system earns no mark.**

**(b) [2 marks]**

- **Land rotation needs a large area, with several plots lying fallow at a time** (1 mark).
- **On two acres the fallow would be too short to restore fertility, and the resting land would feed nobody** (1 mark).
- **Marking note: 'It is an old system' earns no mark.**

**(c) [2 marks]**

- Any TWO of: **a demonstration plot in the school garden shown to the farmers**; **an open day, durbar or exhibition with posters**; **simple leaflets or posters in the local language**; **a talk on local radio** (1 mark each).
- **Marking note: 'Tell them' is too vague for a mark.**

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**## CURRICULUM ASSESSMENT BLUEPRINT**

**Strand:** 3: Systems | **Sub-strand:** 4: Farming Systems | **Class:** Basic 7 (JHS 1)

**Content Standard:** **B7/JHS1.3.4.1** - Demonstrate an understanding of the differences among the various farming systems: Land Rotation, Crop Rotation, Mixed Cropping, Mixed Farmi

# Force and Motion

## ANSWER KEY

### SECTION A: OBJECTIVE TEST

Total Marks: 40

| Q  | Ans | Curriculum Area                               | Cognitive Level |
|----|-----|-----------------------------------------------|-----------------|
| 1  | C   | 14.1 Forces, Balanced and Unbalanced Force... | Remember        |
| 2  | A   | 14.1 Forces, Balanced and Unbalanced Force... | Remember        |
| 3  | D   | 14.1 Forces, Balanced and Unbalanced Force... | Understand      |
| 4  | D   | 14.1 Forces, Balanced and Unbalanced Force... | Understand      |
| 5  | A   | 14.1 Forces, Balanced and Unbalanced Force... | Understand      |
| 6  | B   | 14.1 Forces, Balanced and Unbalanced Force... | Apply           |
| 7  | C   | 14.1 Forces, Balanced and Unbalanced Force... | Remember        |
| 8  | D   | 14.1 Forces, Balanced and Unbalanced Force... | Analyse         |
| 9  | A   | 14.2 Newton's First Law of Motion             | Remember        |
| 10 | B   | 14.2 Newton's First Law of Motion             | Remember        |
| 11 | D   | 14.2 Newton's First Law of Motion             | Understand      |
| 12 | B   | 14.2 Newton's First Law of Motion             | Understand      |
| 13 | C   | 14.2 Newton's First Law of Motion             | Understand      |
| 14 | C   | 14.2 Newton's First Law of Motion             | Analyse         |
| 15 | B   | 14.2 Newton's First Law of Motion             | Remember        |
| 16 | A   | 14.3 Applications of Newton's First Law in... | Understand      |
| 17 | A   | 14.3 Applications of Newton's First Law in... | Understand      |
| 18 | D   | 14.3 Applications of Newton's First Law in... | Analyse         |
| 19 | B   | 14.3 Applications of Newton's First Law in... | Analyse         |
| 20 | A   | 14.3 Applications of Newton's First Law in... | Understand      |
| 21 | D   | 14.3 Applications of Newton's First Law in... | Analyse         |
| 22 | B   | 14.4 Magnets: Their Behaviour and Uses        | Remember        |
| 23 | D   | 14.4 Magnets: Their Behaviour and Uses        | Remember        |
| 24 | C   | 14.4 Magnets: Their Behaviour and Uses        | Remember        |
| 25 | C   | 14.4 Magnets: Their Behaviour and Uses        | Remember        |
| 26 | A   | 14.4 Magnets: Their Behaviour and Uses        | Understand      |
| 27 | B   | 14.4 Magnets: Their Behaviour and Uses        | Analyse         |
| 28 | C   | 14.4 Magnets: Their Behaviour and Uses        | Understand      |
| 29 | A   | 14.4 Magnets: Their Behaviour and Uses        | Understand      |
| 30 | B   | 14.5 Simple Machines and the Classes of Le... | Remember        |
| 31 | D   | 14.5 Simple Machines and the Classes of Le... | Understand      |
| 32 | C   | 14.5 Simple Machines and the Classes of Le... | Remember        |
| 33 | D   | 14.5 Simple Machines and the Classes of Le... | Understand      |
| 34 | A   | 14.5 Simple Machines and the Classes of Le... | Remember        |
| 35 | C   | 14.5 Simple Machines and the Classes of Le... | Apply           |
| 36 | A   | 14.6 Work Input, Work Output and Efficiency   | Remember        |

| Q  | Ans | Curriculum Area                             | Cognitive Level |
|----|-----|---------------------------------------------|-----------------|
| 37 | D   | 14.6 Work Input, Work Output and Efficiency | Understand      |
| 38 | B   | 14.6 Work Input, Work Output and Efficiency | Apply           |
| 39 | B   | 14.6 Work Input, Work Output and Efficiency | Understand      |
| 40 | C   | 14.6 Work Input, Work Output and Efficiency | Understand      |

**Answer Distribution:**

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

**Cognitive Level Distribution:**

Analyse: 6 (15%) | Apply: 3 (7%) | Remember: 14 (35%) | Understand: 17 (42%)



**SECTION B: THEORY — MARKING SCHEME****Question 1 [Total: 10 marks]****(a)** [2 marks]

- (i) **A push or a pull acting on a body** (1 mark).
- (ii) **The newton (N)** (1 mark).
- **Marking note: 'Energy used' earns no mark for (i).**

**(b)** [3 marks]

- (i) **500 N - 420 N** (1 mark) **= 80 N** (1 mark, unit required).
- (ii) **Towards the Blue House team (the direction of the larger force)** (1 mark).
- **Marking note: An answer of 920 N shows that the directions have been ignored; no marks.**

**(c)** [3 marks]

- **Balanced: equal forces in opposite directions, resultant zero** (1 mark).
- **Unbalanced: the forces do not cancel; a resultant force remains** (1 mark).
- **A body at rest stays at rest under balanced forces but starts to move under unbalanced forces** (1 mark).
- **Marking note: Credit a clear labelled sketch that makes the same points.**

**(d)** [2 marks]

- **The tendency of a body to resist a change in its state of rest or of motion** (1 mark).
- **It is measured by the mass of the body** (1 mark).
- **Marking note: 'Weight' is not accepted for the second mark.**

**Question 2 [Total: 8 marks]****(a)** [2 marks]

- **A body continues in its state of rest or of uniform motion in a straight line** (1 mark).
- **unless an unbalanced (external) force acts on it** (1 mark).
- **Marking note: Only one mark if the state of uniform motion is left out.**

**(b)** [3 marks]

- (i) **The ball was at rest and the push was an unbalanced force that set it moving** (1 mark).
- (ii) **The moving ball would have continued in a straight line** (1 mark); **the block (and friction) exerted an unbalanced force that stopped it** (1 mark).

- \*Marking note: 'It became tired' earns no mark.\*

**(c)** [3 marks]

- \*\*The statement is not correct\*\* (1 mark).
- \*\*A body keeps moving at constant velocity when no unbalanced force acts on it\*\* (1 mark).
- \*\*A force is needed on Earth only to overcome friction and air resistance\*\* (1 mark).
- \*Marking note: No explanation marks if the candidate agrees with the statement.\*

**Question 3 [Total: 10 marks]**

**(a)** [3 marks]

- \*\*The passengers were moving forward with the vehicle\*\* (1 mark).
- \*\*The braking force stopped the vehicle\*\* (1 mark).
- \*\*The passengers' bodies continued moving forward because of inertia\*\* (1 mark).
- \*Marking note: The word inertia, or a correct description of it, is needed for the third mark.\*

**(b)** [2 marks]

- \*\*It provides the backward force that stops the passenger with the vehicle\*\* (1 mark).
- \*\*So the passenger is held in the seat and is not thrown forward and injured\*\* (1 mark).
- \*Marking note: 'It makes the person comfortable' earns no mark.\*

**(c)** [3 marks]

- Any THREE of: \*\*passengers thrown backwards when a vehicle starts off\*\*;  
\*\*tightening a hoe or cutlass blade by knocking the handle\*\*;  
\*\*beating dust out of a mat\*\*;  
\*\*pulling a cloth from under a glass\*\*;  
\*\*shaking water off an umbrella\*\* (1 mark each).
- \*Marking note: Three versions of the same braking example count once.\*

**(d)** [2 marks]

- Any TWO of: \*\*it explains the need for seat belts and other safety measures, so preventing injury\*\*;  
\*\*it guides safe driving, such as keeping a safe distance\*\*;  
\*\*it is used in designing vehicles, machines and tools\*\* (1 mark each).
- \*Marking note: 'It is in the syllabus' earns no mark.\*

**Question****(a)** [4 marks]

- (i) **P is a north pole** (1 mark); **Q is a north pole** (1 mark); **because the magnets repelled each other and like poles repel (accept: the other ends are marked S)** (1 mark).
- (ii) **North pole** (1 mark).
- **Marking note: Accept 'north-seeking pole'.**

**(b)** [3 marks]

- **Like poles repel each other** (1 mark).
- **Unlike poles attract each other (a magnet attracts iron)** (1 mark).
- **A freely suspended magnet comes to rest in the north-south direction** (1 mark).
- **Marking note: 'It is strong' is not a characteristic.**

**(c)** [2 marks]

- **Bring each bar near a known suspended magnet (or hang each bar freely)** (1 mark).
- **The bar that repels is the magnet, since repulsion is the sure test (accept: the bar that comes to rest north-south is the magnet)** (1 mark).
- **Marking note: An answer based on attraction alone earns no second mark.**

**(d)** [3 marks]

- Any THREE of: **in a compass for finding direction**; **in loudspeakers, earphones and phones**; **in electric motors and generators**; **in fridge and cupboard door catches**; **in cranes for lifting scrap iron**; **for separating iron from other waste** (1 mark each).
- **Marking note: Two answers naming devices that work in the same way, such as a radio and an earphone, count once.**

**Question 5 [Total: 12 marks]****(a)** [3 marks]

- (i) **A device that makes work easier to do** (1 mark).
- (ii) Any TWO of: **inclined plane, pulley, wheel and axle, screw, wedge** (1 mark each).
- **Marking note: A lever named in (ii) earns no mark, as it is excluded by the question.**

**(b)** [3 marks]

- A - **first-class lever** (1 mark).
- B - **second-class lever** (1 mark).
- C - **third-class lever** (1 mark).

- \*Marking note: Accept 'first order', 'second order' and 'third order'.\*

**(c)** [3 marks]

- \*\*Fulcrum: at the wheel (the axle at the front)\*\* (1 mark).
- \*\*Load: in the tray, between the wheel and the handles\*\* (1 mark).
- \*\*Effort: at the handles, where the hands lift\*\* (1 mark).
- \*Marking note: Marks are for the positions, not merely for the words.\*

**(d)** [3 marks]

- \*\*The load moves a greater distance (and faster) than the effort\*\* (1 mark).
- \*\*The tool gives a delicate, well-controlled grip or movement\*\* (1 mark).
- \*\*One other example: a broom, a fishing rod, a spade or the human forearm\*\* (1 mark).
- \*Marking note: A wheelbarrow or a see-saw named as the example earns no mark.\*

### Question 6 [Total: 8 marks]

**(a)** [2 marks]

- \*\*Work input: the work done by the effort on the machine (the energy supplied to it)\*\* (1 mark).
- \*\*Work output: the useful work done by the machine on the load\*\* (1 mark).
- \*Marking note: The word 'useful' is expected in the second answer.\*

**(b)** [3 marks]

- \*\*Efficiency = (work output / work input) x 100%\*\* (1 mark).
- \*\*= (1,800 / 2,400) x 100%\*\* (1 mark).
- \*\*= 75%\*\* (1 mark, percentage sign required).
- \*Marking note: A correct answer with no working earns 2 of the 3 marks.\*

**(c)** [3 marks]

- (i) \*\*600 J was used in overcoming friction and in lifting the moving parts of the machine\*\* (1 mark); \*\*it appears as heat (and sound), so it is wasted but not destroyed\*\* (1 mark).
- (ii) \*\*Oil the moving parts to reduce friction (accept: use lighter or smoother parts)\*\* (1 mark).
- \*Marking note: 'The machine is old' earns no mark.\*

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### ## CURRICULUM ASSESSMENT BLUEPRINT

\*\*Strand:\*\* 4: Forces and Energy | \*\*Sub-strand:\*\* 4: Force and Motion |  
 \*\*Class:\*\* Basic 7 (JHS 1)

**Content Standard:** **B7/JHS1.4.4.1** - Examine the concept of motion, Newton's first law of motion, magnetic force

# Human Body Digest

## ANSWER KEY

### SECTION A: OBJECTIVE TEST

Total Marks: 40

| Q  | Ans | Curriculum Area                               | Cognitive Level |
|----|-----|-----------------------------------------------|-----------------|
| 1  | C   | 7.1 The Concept of Food and the Need to Eat   | Remember        |
| 2  | B   | 7.1 The Concept of Food and the Need to Eat   | Understand      |
| 3  | C   | 7.1 The Concept of Food and the Need to Eat   | Remember        |
| 4  | B   | 7.1 The Concept of Food and the Need to Eat   | Understand      |
| 5  | D   | 7.1 The Concept of Food and the Need to Eat   | Understand      |
| 6  | A   | 7.1 The Concept of Food and the Need to Eat   | Understand      |
| 7  | C   | 7.1 The Concept of Food and the Need to Eat   | Analyse         |
| 8  | A   | 7.2 The Parts of the Alimentary Canal         | Remember        |
| 9  | A   | 7.2 The Parts of the Alimentary Canal         | Remember        |
| 10 | D   | 7.2 The Parts of the Alimentary Canal         | Remember        |
| 11 | C   | 7.2 The Parts of the Alimentary Canal         | Remember        |
| 12 | B   | 7.2 The Parts of the Alimentary Canal         | Remember        |
| 13 | D   | 7.2 The Parts of the Alimentary Canal         | Understand      |
| 14 | B   | 7.2 The Parts of the Alimentary Canal         | Remember        |
| 15 | D   | 7.2 The Parts of the Alimentary Canal         | Understand      |
| 16 | A   | 7.3 What Happens to Food at Each Stage of ... | Understand      |
| 17 | D   | 7.3 What Happens to Food at Each Stage of ... | Remember        |
| 18 | C   | 7.3 What Happens to Food at Each Stage of ... | Remember        |
| 19 | A   | 7.3 What Happens to Food at Each Stage of ... | Understand      |
| 20 | A   | 7.3 What Happens to Food at Each Stage of ... | Analyse         |
| 21 | D   | 7.3 What Happens to Food at Each Stage of ... | Remember        |
| 22 | C   | 7.3 What Happens to Food at Each Stage of ... | Understand      |
| 23 | C   | 7.3 What Happens to Food at Each Stage of ... | Understand      |
| 24 | B   | 7.3 What Happens to Food at Each Stage of ... | Analyse         |
| 25 | A   | 7.4 End Products of Digestion                 | Remember        |
| 26 | B   | 7.4 End Products of Digestion                 | Remember        |
| 27 | C   | 7.4 End Products of Digestion                 | Remember        |
| 28 | C   | 7.4 End Products of Digestion                 | Apply           |
| 29 | A   | 7.4 End Products of Digestion                 | Apply           |
| 30 | B   | 7.4 End Products of Digestion                 | Understand      |
| 31 | B   | 7.4 End Products of Digestion                 | Understand      |
| 32 | D   | 7.5 Absorption and the Use of Digested Food   | Remember        |
| 33 | B   | 7.5 Absorption and the Use of Digested Food   | Understand      |
| 34 | B   | 7.5 Absorption and the Use of Digested Food   | Remember        |
| 35 | D   | 7.5 Absorption and the Use of Digested Food   | Understand      |
| 36 | C   | 7.5 Absorption and the Use of Digested Food   | Analyse         |
| 37 | A   | 7.6 Food Tests                                | Remember        |

| Q  | Ans      | Curriculum Area | Cognitive Level |
|----|----------|-----------------|-----------------|
| 38 | <b>D</b> | 7.6 Food Tests  | Remember        |
| 39 | <b>A</b> | 7.6 Food Tests  | Remember        |
| 40 | <b>D</b> | 7.6 Food Tests  | Apply           |

**Answer Distribution:**

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

**Cognitive Level Distribution:**

Analyse: 4 (10%) | Apply: 3 (7%) | Remember: 19 (47%) | Understand: 14 (35%)

## SECTION B: THEORY — MARKING SCHEME

### Question 1 [Total: 10 marks]

**(a)** [6 marks]

- A - **mouth** (1 mark).
  - B - **oesophagus** (accept: gullet) (1 mark).
  - C - **stomach** (1 mark).
  - D - **small intestine** (1 mark).
  - E - **large intestine** (accept: colon) (1 mark).
  - F - **anus** (1 mark).
- \*Marking note: 'Windpipe' or 'trachea' written for B earns no mark; that tube carries air, not food.\*

**(b)** [2 marks]

- D - **completes digestion and absorbs the digested food into the blood** (1 mark).
  - E - **absorbs water from the undigested material (and stores faeces)** (1 mark).
- \*Marking note: 'It digests food' alone is too vague for E.\*

**(c)** [2 marks]

- **A, B, C, D, E, F** (2 marks; 1 mark if one letter is out of place or missing).
- \*Marking note: No mark if the list begins anywhere other than A.\*

### Question 2 [Total: 12 marks]

**(a)** [4 marks]

- (i) **A substance taken into the body** (1 mark) **which provides energy, builds and repairs the body or protects it from disease** (1 mark).
  - (ii) Any FOUR of: **carbohydrates**; **proteins**; **fats and oils**; **vitamins**; **mineral salts**; **water** (half a mark each, 2 marks for four correct).
- \*Marking note: 'Something we eat' alone earns 1 mark in (i).\*

**(b)** [4 marks]

- Any FOUR of: **to supply energy**; **for growth**; **to repair worn-out or damaged parts**; **to protect the body from disease (keep healthy)**; **to keep the body warm**; **to supply water and roughage for the proper working of the body** (1 mark each).
- \*Marking note: 'To become fat' earns no mark.\*



**(c) [4 marks]**

- Any THREE differences: **thin, with ribs showing / normal weight**; **dull hair, dry skin, sunken eyes / bright eyes and healthy skin**; **weak and tired / active and playful**; **often ill and heals slowly / resists disease** (1 mark each).
- **The child should be given regular balanced meals and taken to a health centre** (1 mark).
- **Marking note: Marks are given for the comparison; 'he is sick' without a visible feature earns no mark.**

**Question 3 [Total: 10 marks]****(a) [4 marks]**

- Mouth - **chewing by the teeth, moistening with saliva and the start of starch digestion** (1 mark).
- Stomach - **churning with acid and enzymes; protein digestion begins** (1 mark).
- Small intestine - **digestion is completed and the digested food is absorbed into the blood** (1 mark).
- Large intestine - **water is absorbed and the waste is stored as faeces** (1 mark).
- **Marking note: A row that merely says 'food passes through' earns no mark.**

**(b) [3 marks]**

- **Physical digestion: food is broken into smaller pieces without a change in the substance** (1 mark).
- **Chemical digestion: the food substances are changed into simpler ones by enzymes** (1 mark).
- **One correct example of each from the breakfast** (1 mark).
- **Marking note: Both examples are needed for the third mark.**

**(c) [3 marks]**

- **She cannot chew, so large pieces of food are swallowed** (1 mark).
- **Large pieces give the enzymes a small surface to act on, so digestion is slow and incomplete** (1 mark).
- **Advice: eat soft, mashed or well cooked food, eat slowly, or see a dentist** (1 mark).
- **Marking note: The idea of surface area or of slower enzyme action is needed for the second mark.**

**Question 4 [Total: 10 marks]**

**(a)** [4 marks]

- A continuous tube from mouth to anus, drawn large and in pencil (1 mark).
- Any THREE further parts correctly placed and clearly named, with label lines touching the parts (1 mark each).
- \*Marking note: Marks are for correct position and labelling, not for beauty; a label line that does not touch its part earns no mark.\*

**(b)** [3 marks]

- \*\*The statement is not correct\*\* (1 mark).
- \*\*Food is pushed along by waves of muscular squeezing in the wall of the canal\*\* (1 mark).
- \*\*This works in any position, so a person lying down can still swallow (and food travels upwards in part of the large intestine)\*\* (1 mark).
- \*Marking note: No mark for the explanation if the candidate agrees with the pupil.\*

**(c)** [3 marks]

- \*\*It is made up of several different organs joined together\*\* (1 mark).
- \*\*Each organ does a special piece of work\*\* (1 mark).
- \*\*The organs work together in order for one purpose, the digestion and absorption of food\*\* (1 mark).
- \*Marking note: Accept 'other organs such as the liver help it' for the third mark.\*

**Question 5 [Total: 10 marks]****(a)** [3 marks]

- (i) \*\*starch to glucose (simple sugar)\*\* (1 mark).
- (ii) \*\*protein to amino acids\*\* (1 mark).
- (iii) \*\*oils and fats to fatty acids and glycerol\*\* (1 mark).
- \*Marking note: The arrow or the word 'gives' must be shown; a bare list of products earns 1 mark in total.\*

**(b)** [3 marks]

- (i) \*\*The passing of digested food through the wall of the alimentary canal into the blood\*\* (1 mark).
- (ii) \*\*The small intestine\*\* (1 mark).
- (iii) \*\*By the blood, through the blood vessels\*\* (1 mark).
- \*Marking note: 'Digestion' written for (i) earns no mark.\*

**(c)** [2 marks]

- \*\*They increase the surface area of the inner wall\*\* (1 mark).
- \*\*So more digested food is absorbed, and absorbed more quickly\*\* (1 mark).
- \*Marking note: 'They catch the food' earns no mark.\*

**(d)** [2 marks]

- (i) **To release energy in the cells during respiration** (1 mark).
- (ii) **To build new tissue for growth and for the repair of worn-out parts** (1 mark).
- **Marking note: 'For strength' is accepted for (i) only if energy is mentioned.**

**Question 6 [Total: 8 marks]****(a)** [4 marks]

- P - **starch** (1 mark).
- Q - **glucose (a simple or reducing sugar)** (1 mark).
- R - **protein** (1 mark).
- S - **oil or fat** (1 mark).
- **Marking note: 'Carbohydrate' is accepted for P and Q only if the candidate also writes starch or sugar.**

**(b)** [2 marks]

- P - any ONE of: **boiled yam**; **cassava**; **plantain**; **cocoyam**; **rice**; **kenkey or banku**; **bread** (1 mark).
- R - any ONE of: **egg white**; **meat**; **fish**; **beans**; **groundnut**; **milk** (1 mark).
- **Marking note: A food named for the wrong test earns no mark.**

**(c)** [2 marks]

- **The enzyme in saliva changed some of the starch into sugar** (1 mark).
- **Some starch was left, so the iodine test was still positive but weaker** (1 mark).
- **Marking note: 'The bread was wet' earns no mark; the action of saliva must be given.**

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**## CURRICULUM ASSESSMENT BLUEPRINT**

**Strand:** 3: Systems | **Sub-strand:** 1: The Human Body System | **Class:** Basic 7 (JHS 1)

**Content Standard:** **B7/JHS1.3.1.1** - Demonstrate understanding of the concept of food, the process of digestion and its importance in humans.

> Source: NaCCA Science Common Core Programme Curriculum (B7/

# Human Health

## ANSWER KEY

### SECTION A: OBJECTIVE TEST

Total Marks: 40

| Q  | Ans | Curriculum Area                               | Cognitive Level |
|----|-----|-----------------------------------------------|-----------------|
| 1  | A   | 17.1 Food Nutrients and Their Uses in the ... | Remember        |
| 2  | B   | 17.1 Food Nutrients and Their Uses in the ... | Understand      |
| 3  | D   | 17.1 Food Nutrients and Their Uses in the ... | Remember        |
| 4  | A   | 17.1 Food Nutrients and Their Uses in the ... | Understand      |
| 5  | C   | 17.1 Food Nutrients and Their Uses in the ... | Understand      |
| 6  | B   | 17.1 Food Nutrients and Their Uses in the ... | Remember        |
| 7  | C   | 17.1 Food Nutrients and Their Uses in the ... | Remember        |
| 8  | A   | 17.1 Food Nutrients and Their Uses in the ... | Remember        |
| 9  | A   | 17.2 Deficiency Diseases and the Nutrients... | Remember        |
| 10 | B   | 17.2 Deficiency Diseases and the Nutrients... | Remember        |
| 11 | D   | 17.2 Deficiency Diseases and the Nutrients... | Remember        |
| 12 | B   | 17.2 Deficiency Diseases and the Nutrients... | Remember        |
| 13 | D   | 17.2 Deficiency Diseases and the Nutrients... | Remember        |
| 14 | C   | 17.2 Deficiency Diseases and the Nutrients... | Remember        |
| 15 | A   | 17.2 Deficiency Diseases and the Nutrients... | Remember        |
| 16 | D   | 17.2 Deficiency Diseases and the Nutrients... | Analyse         |
| 17 | C   | 17.3 Symptoms, Effects and Prevention of D... | Remember        |
| 18 | B   | 17.3 Symptoms, Effects and Prevention of D... | Remember        |
| 19 | A   | 17.3 Symptoms, Effects and Prevention of D... | Understand      |
| 20 | C   | 17.3 Symptoms, Effects and Prevention of D... | Apply           |
| 21 | D   | 17.3 Symptoms, Effects and Prevention of D... | Apply           |
| 22 | B   | 17.3 Symptoms, Effects and Prevention of D... | Apply           |
| 23 | A   | 17.3 Symptoms, Effects and Prevention of D... | Evaluate        |
| 24 | D   | 17.4 The Nature and Transmission of Viral ... | Remember        |
| 25 | A   | 17.4 The Nature and Transmission of Viral ... | Remember        |
| 26 | B   | 17.4 The Nature and Transmission of Viral ... | Understand      |
| 27 | C   | 17.4 The Nature and Transmission of Viral ... | Analyse         |
| 28 | B   | 17.4 The Nature and Transmission of Viral ... | Understand      |
| 29 | A   | 17.4 The Nature and Transmission of Viral ... | Understand      |
| 30 | C   | 17.4 The Nature and Transmission of Viral ... | Remember        |
| 31 | D   | 17.4 The Nature and Transmission of Viral ... | Analyse         |
| 32 | C   | 17.5 Symptoms, Effects and Prevention of C... | Remember        |
| 33 | B   | 17.5 Symptoms, Effects and Prevention of C... | Apply           |
| 34 | A   | 17.5 Symptoms, Effects and Prevention of C... | Understand      |
| 35 | B   | 17.5 Symptoms, Effects and Prevention of C... | Understand      |
| 36 | D   | 17.5 Symptoms, Effects and Prevention of C... | Analyse         |
| 37 | C   | 17.6 Roles of the Individual, the Communit... | Evaluate        |

| Q  | Ans | Curriculum Area                               | Cognitive Level |
|----|-----|-----------------------------------------------|-----------------|
| 38 | D   | 17.6 Roles of the Individual, the Communit... | Apply           |
| 39 | C   | 17.6 Roles of the Individual, the Communit... | Remember        |
| 40 | D   | 17.6 Roles of the Individual, the Communit... | Create          |

**Answer Distribution:**

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

**Cognitive Level Distribution:**

Analyse: 4 (10%) | Apply: 5 (12%) | Create: 1 (2%) | Evaluate: 2 (5%) | Remember: 19 (47%) | Understand: 9

## SECTION B: THEORY — MARKING SCHEME

### Question 1 [Total: 12 marks]

**(a)** [3 marks]

- A - **protein** (1 mark).
- B - **vitamin C (ascorbic acid)** (1 mark).
- C - **vitamin A (accept: fats and oils)** (1 mark).
  - \*Marking note: 'Vitamins' alone for B or C earns no mark; the particular vitamin is required.\*

**(b)** [3 marks]

- **Protein - growth and repair of worn-out tissues** (1 mark).
- **Vitamin C - keeps the gums, skin and blood vessels healthy; helps wounds to heal** (1 mark).
- **Vitamin A - needed for vision, especially in dim light, and for healthy skin** (1 mark).
- \*Marking note: 'It makes the body strong' is too vague for a mark.\*

**(c)** [3 marks]

- A - **kwashiorkor** (1 mark).
- B - **scurvy** (1 mark).
- C - **night blindness** (1 mark).
- \*Marking note: Accept 'protein-energy malnutrition' for A.\*

**(d)** [3 marks]

- **Carbohydrates - one local example such as yam or kenkey** (1 mark).
- **Fats and oils - one local example such as palm oil or groundnut** (1 mark).
- **Vitamins and mineral salts - one local example such as kontomire or oranges (accept water and roughage)** (1 mark).
- \*Marking note: A class named without an example earns no mark.\*

### Question 2 [Total: 10 marks]

**(a)** [3 marks]

- First child - **kwashiorkor** (1 mark).
- Second child - **rickets** (1 mark).
- Third child - **anaemia** (1 mark).
- \*Marking note: Mark each diagnosis separately.\*

**(b)** [3 marks]

- First child - **protein** (1 mark).
- Second child - **vitamin D (and calcium)** (1 mark).
- Third child - **iron** (1 mark).
- \*Marking note: The nutrient must match the disease named in (a).\*

**(c)** [4 marks]

- **Two protein-rich foods, such as fish, beans, eggs, groundnuts or milk** (1 mark each).
- **Two further effects, such as stunted growth, weakness, low resistance to infection, poor mental development or even death** (1 mark each).
- \*Marking note: A food that is chiefly a carbohydrate, such as kenkey, earns no mark.\*

**Question 3 [Total: 8 marks]****(a)** [4 marks]

- Scurvy - **symptom: bleeding, swollen gums (accept slow healing of wounds)** (1 mark); **prevention: eat fresh fruits and vegetables rich in vitamin C** (1 mark).
- Goitre - **symptom: swelling in front of the neck** (1 mark); **prevention: use iodated salt; eat sea fish** (1 mark).
- \*Marking note: A cause written in the symptom column earns no mark.\*

**(b)** [2 marks]

- **Sunlight on the skin helps the body to make vitamin D** (1 mark).
- **Vitamin D helps the body to absorb calcium, which hardens the bones** (1 mark).
- \*Marking note: 'The sun is good for the body' earns no mark.\*

**(c)** [2 marks]

- Any TWO of: **the foods grown are mostly starchy staples, so meals are not balanced**; **the protein foods and vegetables are sold instead of eaten**; **ignorance of what each food supplies**; **poor cooking habits destroy the vitamins**; **customs that deny children certain foods** (1 mark each).
- \*Marking note: 'They are poor' alone earns no mark without an explanation.\*

**Question 4 [Total: 10 marks]****(a)** [3 marks]

- (i) **A very small agent that can multiply only inside the living cells of a host** (1 mark).
- (ii) Any TWO of: **they spread easily, often before symptoms appear**;

**\*\*antibiotics do not cure them\*\***; **\*\*the viruses change into new forms\*\***; **\*\*there is often no specific medicine\*\*** (1 mark each).

- **\*Marking note: 'It is a germ' alone is too vague for (i).\***

**(b) [4 marks]**

- (i) **\*\*In droplets from coughing, sneezing or talking, breathed in or picked up from surfaces on the hands\*\*** (1 mark).

- (ii) **\*\*Infected people travel to markets, funerals and other gatherings in nearby communities\*\*** (1 mark).

- (iii) **\*\*Travellers carry it by air, road or sea, often before showing symptoms\*\*** (1 mark).

- **\*\*One correct symptom named, such as fever, dry cough or loss of taste and smell\*\*** (1 mark).

- **\*Marking note: 'By air' alone for (iii) is not enough; the travel of infected people must be clear.\***

**(c) [3 marks]**

- (i) **\*\*It spread to very many countries and continents at the same time\*\*** (1 mark); **\*\*and affected a very large number of people, beyond local control\*\*** (1 mark).

- (ii) **\*\*Ebola spreads by direct contact with body fluids, not chiefly by droplets in the air\*\*** (1 mark).

- **\*Marking note: 'Many people died' alone does not define a pandemic.\***

**Question 5 [Total: 12 marks]**

**(a) [4 marks]**

- A - **\*\*washing the hands with soap under running water\*\*** (1 mark).

- B - **\*\*wearing a nose mask\*\*** (1 mark).

- C - **\*\*keeping a social (physical) distance\*\*** (1 mark).

- D - **\*\*vaccination or immunisation\*\*** (1 mark).

- **\*Marking note: 'Injection' is accepted for D only if vaccination is clearly meant.\***

**(b) [4 marks]**

- A - **\*\*removes and destroys virus particles on the hands\*\*** (1 mark).

- B - **\*\*traps the droplets breathed out and keeps out those from others\*\*** (1 mark).

- C - **\*\*droplets fall to the ground before reaching another person\*\*** (1 mark).

- D - **\*\*the body makes antibodies in advance, so the person is protected\*\*** (1 mark).

- **\*Marking note: A repeat of the name of the measure, with no explanation, earns no mark.\***



**(c)** [2 marks]

- Any TWO of: **cover the nose and mouth when coughing or sneezing**; **avoid crowded places**; **use hand sanitiser**; **isolate or quarantine infected persons**; **avoid handshaking and touching the face**; **keep rooms well ventilated** (1 mark each).
- *Marking note: A measure already shown in Fig. 2 earns no mark.*

**(d)** [2 marks]

- Any TWO of: **schools close and learning is lost**; **businesses close, people lose their incomes and the economy suffers**; **hospitals are overwhelmed so other patients suffer**; **travel, trade and social gatherings are restricted** (1 mark each).
- *Marking note: Two answers that both say 'people suffer' count once.*

**Question 6 [Total: 8 marks]****(a)** [3 marks]

- (i) **Observe the safety measures and report symptoms early** (1 mark).
- (ii) **Educate and support one another, avoid stigma and help the health workers** (1 mark).
- (iii) **Provide testing, treatment, vaccines and accurate information, and enforce regulations** (1 mark).
- *Marking note: A role placed at the wrong level earns no mark.*

**(b)** [2 marks]

- **People hide their illness instead of reporting it, so cases are found late and spread unchecked** (1 mark).
- **The family is left without food and support and must move about to find help, spreading the disease** (1 mark).
- *Marking note: 'It is unkind' alone earns no mark; the effect on the spread of disease is required.*

**(c)** [3 marks]

- Any THREE of: **a clear heading naming the disease**; **the main symptoms**; **how the disease is spread**; **the preventive measures, with simple pictures**; **what to do and where to report, with a contact number**; **large, simple lettering in a language the people read** (1 mark each).
- *Marking note: 'Make it beautiful' earns no mark.*

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**## CURRICULUM ASSESSMENT BLUEPRINT****Strand:** 5: Humans and the Environment | **Sub-strand:** 2: Human Health |

**\*\*Class:\*\*** Basic 7 (JHS 1)

**\*\*Content Standard:\*\*** **\*\*B7/JHS1.5.2.1\*\*** - Demonstrate knowledge of common deficiency diseases of humans,

|

# Life Cycle Housefly

## ANSWER KEY

### SECTION A: OBJECTIVE TEST

Total Marks: 40

| Q  | Ans | Curriculum Area                                        | Cognitive Level |
|----|-----|--------------------------------------------------------|-----------------|
| 1  | A   | 4.1 Life Cycles and Metamorphosis                      | Remember        |
| 2  | C   | 4.1 Life Cycles and Metamorphosis                      | Remember        |
| 3  | D   | 4.1 Life Cycles and Metamorphosis                      | Understand      |
| 4  | C   | 4.1 Life Cycles and Metamorphosis                      | Understand      |
| 5  | A   | 4.1 Life Cycles and Metamorphosis                      | Analyse         |
| 6  | C   | 4.2 The Stages of the Life Cycle of the Housefly       | Remember        |
| 7  | B   | 4.2 The Stages of the Life Cycle of the Housefly       | Remember        |
| 8  | D   | 4.2 The Stages of the Life Cycle of the Housefly       | Remember        |
| 9  | B   | 4.2 The Stages of the Life Cycle of the Housefly       | Remember        |
| 10 | B   | 4.2 The Stages of the Life Cycle of the Housefly       | Remember        |
| 11 | C   | 4.2 The Stages of the Life Cycle of the Housefly       | Understand      |
| 12 | B   | 4.2 The Stages of the Life Cycle of the Housefly       | Remember        |
| 13 | D   | 4.2 The Stages of the Life Cycle of the Housefly       | Remember        |
| 14 | C   | 4.2 The Stages of the Life Cycle of the Housefly       | Analyse         |
| 15 | D   | 4.3 Features of Each Stage of the Housefly             | Remember        |
| 16 | D   | 4.3 Features of Each Stage of the Housefly             | Understand      |
| 17 | B   | 4.3 Features of Each Stage of the Housefly             | Understand      |
| 18 | A   | 4.3 Features of Each Stage of the Housefly             | Understand      |
| 19 | D   | 4.3 Features of Each Stage of the Housefly             | Analyse         |
| 20 | A   | 4.3 Features of Each Stage of the Housefly             | Analyse         |
| 21 | A   | 4.4 Feeding Habits and Breeding Places of the Housefly | Remember        |
| 22 | C   | 4.4 Feeding Habits and Breeding Places of the Housefly | Understand      |
| 23 | B   | 4.4 Feeding Habits and Breeding Places of the Housefly | Remember        |
| 24 | D   | 4.4 Feeding Habits and Breeding Places of the Housefly | Understand      |
| 25 | C   | 4.4 Feeding Habits and Breeding Places of the Housefly | Remember        |
| 26 | D   | 4.4 Feeding Habits and Breeding Places of the Housefly | Understand      |
| 27 | B   | 4.4 Feeding Habits and Breeding Places of the Housefly | Analyse         |
| 28 | B   | 4.5 The Housefly as a Menace to Humans                 | Remember        |
| 29 | A   | 4.5 The Housefly as a Menace to Humans                 | Remember        |
| 30 | B   | 4.5 The Housefly as a Menace to Humans                 | Remember        |
| 31 | C   | 4.5 The Housefly as a Menace to Humans                 | Understand      |
| 32 | D   | 4.5 The Housefly as a Menace to Humans                 | Analyse         |
| 33 | D   | 4.5 The Housefly as a Menace to Humans                 | Evaluate        |
| 34 | A   | 4.5 The Housefly as a Menace to Humans                 | Understand      |
| 35 | C   | 4.6 Reducing the Housefly Menace in the Community      | Understand      |
| 36 | A   | 4.6 Reducing the Housefly Menace in the Community      | Understand      |

| Q  | Ans | Curriculum Area                               | Cognitive Level |
|----|-----|-----------------------------------------------|-----------------|
| 37 | A   | 4.6 Reducing the Housefly Menace in the Co... | Understand      |
| 38 | C   | 4.6 Reducing the Housefly Menace in the Co... | Analyse         |
| 39 | B   | 4.6 Reducing the Housefly Menace in the Co... | Evaluate        |
| 40 | A   | 4.6 Reducing the Housefly Menace in the Co... | Evaluate        |

**Answer Distribution:**

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

**Cognitive Level Distribution:**

Analyse: 7 (17%) | Evaluate: 3 (7%) | Remember: 16 (40%) | Understand: 14 (35%)

**SECTION B: THEORY — MARKING SCHEME****Question 1 [Total: 10 marks]****(a)** [4 marks]

- A - **larva** (accept: maggot) (1 mark).
- B - **adult** (adult housefly) (1 mark).
- C - **egg** (eggs) (1 mark).
- D - **pupa** (1 mark).
- \*Marking note: 'Caterpillar' or 'nymph' written for A earns no mark.\*

**(b)** [2 marks]

- **C, A, D, B** (2 marks; 1 mark if only one letter is out of place).
- \*Marking note: Accept the same order written in a circle, for example A, D, B, C, provided the sequence is unbroken.\*

**(c)** [2 marks]

- A - **the larva feeds on the rotting matter and grows quickly** (1 mark).
- D - **the insect rests inside a hard case, does not feed, and changes into the adult** (1 mark).
- \*Marking note: 'It sleeps' alone is not accepted for D; the change into the adult must be shown.\*

**(d)** [2 marks]

- **The adult lays eggs, which start the same stages again** (1 mark).
- **The same stages follow one another in the same order over and over again** (1 mark).
- \*Marking note: A candidate who writes only 'because it is round' earns no mark.\*

**Question 2 [Total: 12 marks]****(a)** [3 marks]

- J - **food left uncovered (open) on a table, with flies on it** (1 mark).
- K - **a heap of refuse (rubbish) with flies around it** (1 mark).
- L - **food covered with a net cover, with no flies on it** (1 mark).
- \*Marking note: The idea of 'uncovered' at J and 'covered' at L must be clear.\*

**(b)** [3 marks]

- **K provides food and a breeding place for the flies** (1 mark).
- **J is open and easy to reach, and the smell attracts the flies** (1 mark).
- **The net at L keeps the flies off the food although they can still smell it** (1 mark).

mark).

- \*Marking note: Accept 'the flies cannot pass through the holes of the net' for the third mark.\*

**(c)** [3 marks]

- Any THREE of: \*\*they transfer disease germs to the food and cause diseases such as dysentery, diarrhoea, cholera or typhoid fever\*\*; \*\*they contaminate the food and cause food poisoning\*\*; \*\*they are a nuisance in the home and environment\*\*; \*\*they spoil the food so that it must be thrown away\*\* (1 mark each).

- \*Marking note: Malaria earns no mark; it is spread by the mosquito.\*

**(d)** [3 marks]

- Any THREE of: \*\*keep all food covered until it is eaten\*\*; \*\*clear the refuse heap and use a bin with a tight lid\*\*; \*\*dispose of refuse at the proper dumping site regularly\*\*; \*\*sweep the compound and keep gutters clean and dry\*\*; \*\*clear away food remains and wash plates at once\*\* (1 mark each).

- \*Marking note: 'Kill all the flies' is not accepted as a lasting change.\*

### Question 3 [Total: 10 marks]

**(a)** [4 marks]

- Egg - \*\*in rotting rubbish, manure, faeces or decaying food\*\* (1 mark).  
- Larva - \*\*in the same rotting matter, where it feeds\*\* (1 mark).  
- Pupa - \*\*in the drier part of the rubbish or in the soil beneath it\*\* (1 mark).  
- Adult - \*\*flying about the compound, on refuse, faeces and uncovered food\*\* (1 mark).

- \*Marking note: 'In dirty places' may be credited once only.\*

**(b)** [3 marks]

- \*\*The warm, damp conditions suit the eggs and larvae\*\* (1 mark).  
- \*\*Rotting rubbish and waste, which are food and breeding places, are plentiful\*\* (1 mark).  
- \*\*The life cycle is completed in only one to three weeks and many eggs are laid each time\*\* (1 mark).

- \*Marking note: Accept 'the flies breed faster because it is warm and wet' for 1 mark only.\*

**(c)** [3 marks]

- Any THREE of: \*\*the eggs and larvae cannot move away, so they are easy to destroy\*\*; \*\*killing adults does not stop the ones already breeding in the refuse from hatching\*\*; \*\*removing the breeding places prevents new flies altogether, so the control lasts\*\*; \*\*adults can fly in from other compounds\*\*; \*\*cleaning needs no chemicals, so it is cheap and safe\*\* (1 mark each).

- \*Marking note: A candidate who merely repeats 'it is better' without a reason

earns no mark.\*

#### Question 4 [Total: 10 marks]

(a) [3 marks]

- \*\*It settles on the food and pours out saliva onto it\*\* (1 mark).
- \*\*The saliva dissolves or softens the food\*\* (1 mark).
- \*\*The fly sucks up the liquid food through its sponge-like mouth parts\*\* (1 mark).
- \*Marking note: 'It bites the food' earns no mark; the housefly has no biting mouth parts.\*

(b) [3 marks]

- \*\*Germs from the gutter stick to the hairy legs, body and mouth parts of the fly\*\* (1 mark).
- \*\*The fly lands on the kenkey and leaves the germs on it (it may also vomit or pass waste on the food)\*\* (1 mark).
- \*\*The customer eats the contaminated food and takes in the germs\*\* (1 mark).
- \*Marking note: The three steps must be in a sensible order for full marks.\*

(c) [2 marks]

- Any TWO of: \*\*dysentery\*\*; \*\*diarrhoea\*\*; \*\*cholera\*\*; \*\*typhoid fever\*\*; \*\*food poisoning\*\* (1 mark each).
- \*Marking note: Malaria earns no mark.\*

(d) [2 marks]

- Any TWO of: \*\*insecticide on or near food can poison the customers\*\*; \*\*it kills only the flies present at the time\*\*; \*\*the gutter keeps breeding new flies, so they return at once\*\* (1 mark each, maximum 1 mark if no alternative is given).
- \*\*She should clear the gutter and keep the food in a covered glass case or under a net cover\*\* (1 mark).
- \*Marking note: Full marks need both a reason and the better practice.\*

#### Question 5 [Total: 10 marks]

(a) [3 marks]

- Any THREE of: \*\*a clean-up of refuse heaps and choked gutters\*\*; \*\*refuse containers with tight lids, emptied regularly\*\*; \*\*keeping the school toilets clean and covered\*\*; \*\*asking vendors to keep food covered\*\*; \*\*keeping the compound swept and dry\*\* (1 mark each).
- \*Marking note: A bare list is enough; a short description is not required for the mark but should be encouraged.\*

**(b)** [3 marks]

- (i) Any TWO materials, for example **a plastic bottle** and **bait** such as rotting fruit, fish or sugar water (1 mark for both).
- (ii) **The smell of the bait attracts the flies into the bottle through a narrow opening** (1 mark); **the flies cannot find their way out and are trapped** (1 mark).
- **Marking note:** Accept any workable trap, for example a bowl of soapy water with bait covered by a perforated lid.\*

**(c)** [2 marks]

- **Count the flies (in a trap or at a fixed place) before the intervention and again some weeks after it** (1 mark).
- **Compare the two counts; fewer flies afterwards shows that the intervention worked** (accept: check whether fewer pupils report sick with diarrhoea) (1 mark).
- **Marking note:** A candidate who writes only 'look at the place' earns no mark; some form of comparison is needed.\*

**(d)** [2 marks]

- Any TWO short, correct messages about covering food, clearing refuse, keeping gutters clean, using and covering latrines or washing hands (1 mark each).
- **Marking note:** The message must be about reducing flies or protecting food; a general greeting earns no mark.\*

**Question 6 [Total: 8 marks]****(a)** [2 marks]

- (i) **Home W** (1 mark).
- (ii) **39** (1 mark).
- **Marking note:** Accept 39 flies written with or without the word 'flies'.\*

**(b)** [3 marks]

- **Homes with open refuse (W and Y) caught many more flies than homes with covered refuse (X and Z)** (1 mark).
- **Open refuse gives the flies food and a breeding place** (1 mark).
- **Emptying often also matters: X, emptied daily, caught fewer flies than Z, emptied weekly, because refuse left standing allows the eggs to hatch** (1 mark).
- **Marking note:** Candidates must quote at least one pair of figures from the table for the first mark.\*

**(c)** [3 marks]

- (i) **Keep refuse in a covered bin and empty it daily** (1 mark).
- (ii) Any TWO of: **only four homes were tested**; **the count was made on one day only**; **other differences between the homes, such as cleanliness or a nearby dump, were not controlled**; **the traps may not catch the same share of**



the flies in every home\*\* (1 mark each).

- \*Marking note: 'They did not count well' earns no mark without a reason.\*

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## ## CURRICULUM ASSESSMENT BLUEPRINT

**\*\*Strand:\*\*** 2: Cycles | **\*\*Sub-strand:\*\*** 2: Life Cycle of Organisms | **\*\*Class:\*\***  
Basic 7 (JHS 1)

**\*\*Content Standard:\*\*** **\*\*B7/JHS1.2.2.1\***

# Living Cells

## ANSWER KEY

### SECTION A: OBJECTIVE TEST

Total Marks: 40

| Q  | Ans | Curriculum Area                               | Cognitive Level |
|----|-----|-----------------------------------------------|-----------------|
| 1  | B   | 2.1 Cells as the Units of Living Things       | Remember        |
| 2  | A   | 2.1 Cells as the Units of Living Things       | Remember        |
| 3  | D   | 2.1 Cells as the Units of Living Things       | Apply           |
| 4  | B   | 2.1 Cells as the Units of Living Things       | Understand      |
| 5  | B   | 2.1 Cells as the Units of Living Things       | Remember        |
| 6  | D   | 2.1 Cells as the Units of Living Things       | Remember        |
| 7  | A   | 2.2 The Structure of the Animal Cell          | Remember        |
| 8  | C   | 2.2 The Structure of the Animal Cell          | Remember        |
| 9  | D   | 2.2 The Structure of the Animal Cell          | Remember        |
| 10 | C   | 2.2 The Structure of the Animal Cell          | Remember        |
| 11 | B   | 2.2 The Structure of the Animal Cell          | Understand      |
| 12 | C   | 2.2 The Structure of the Animal Cell          | Remember        |
| 13 | A   | 2.2 The Structure of the Animal Cell          | Analyse         |
| 14 | C   | 2.2 The Structure of the Animal Cell          | Apply           |
| 15 | A   | 2.3 Functions of the Organelles of the Ani... | Remember        |
| 16 | C   | 2.3 Functions of the Organelles of the Ani... | Understand      |
| 17 | A   | 2.3 Functions of the Organelles of the Ani... | Remember        |
| 18 | B   | 2.3 Functions of the Organelles of the Ani... | Understand      |
| 19 | D   | 2.3 Functions of the Organelles of the Ani... | Remember        |
| 20 | C   | 2.3 Functions of the Organelles of the Ani... | Analyse         |
| 21 | C   | 2.3 Functions of the Organelles of the Ani... | Analyse         |
| 22 | D   | 2.4 The Structure of the Plant Cell           | Remember        |
| 23 | A   | 2.4 The Structure of the Plant Cell           | Remember        |
| 24 | D   | 2.4 The Structure of the Plant Cell           | Remember        |
| 25 | D   | 2.4 The Structure of the Plant Cell           | Remember        |
| 26 | C   | 2.4 The Structure of the Plant Cell           | Remember        |
| 27 | D   | 2.4 The Structure of the Plant Cell           | Understand      |
| 28 | B   | 2.4 The Structure of the Plant Cell           | Analyse         |
| 29 | C   | 2.5 Functions of the Organelles of the Pla... | Understand      |
| 30 | A   | 2.5 Functions of the Organelles of the Pla... | Remember        |
| 31 | B   | 2.5 Functions of the Organelles of the Pla... | Understand      |
| 32 | A   | 2.5 Functions of the Organelles of the Pla... | Analyse         |
| 33 | A   | 2.5 Functions of the Organelles of the Pla... | Understand      |
| 34 | B   | 2.5 Functions of the Organelles of the Pla... | Apply           |
| 35 | D   | 2.6 Comparing Plant and Animal Cells          | Remember        |
| 36 | B   | 2.6 Comparing Plant and Animal Cells          | Understand      |
| 37 | B   | 2.6 Comparing Plant and Animal Cells          | Apply           |

| Q  | Ans      | Curriculum Area                      | Cognitive Level |
|----|----------|--------------------------------------|-----------------|
| 38 | <b>A</b> | 2.6 Comparing Plant and Animal Cells | Analyse         |
| 39 | <b>D</b> | 2.6 Comparing Plant and Animal Cells | Create          |
| 40 | <b>C</b> | 2.6 Comparing Plant and Animal Cells | Evaluate        |

**Answer Distribution:**

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

**Cognitive Level Distribution:**

Analyse: 6 (15%) | Apply: 4 (10%) | Create: 1 (2%) | Evaluate: 1 (2%) | Remember: 19 (47%) | Understand: 9

**SECTION B: THEORY — MARKING SCHEME****Question 1 [Total: 10 marks]****(a) [5 marks]**

- A - **cell membrane** (1 mark).
- B - **cytoplasm** (1 mark).
- C - **nucleus** (1 mark).
- D - **mitochondrion** (1 mark).
- E - **vacuole** (1 mark).
- **Marking note:** 'Cell wall' written for A earns no mark; an animal cell has no cell wall.\*

**(b) [3 marks]**

- A - **controls what enters and leaves the cell** (accept: holds the cell together) (1 mark).
- C - **controls the activities of the cell** (1 mark).
- D - **releases energy from food (respiration)** (1 mark).
- **Marking note:** 'It protects the cell' alone is not accepted for C.\*

**(c) [2 marks]**

- Any TWO of: **there is no cell wall and the shape is irregular**; **there are no chloroplasts**; **there is no large central vacuole**; **the nucleus is not pushed to one side** (1 mark each).
- **Marking note:** The reason must be visible in the figure.\*

**Question 2 [Total: 12 marks]****(a) [4 marks]**

- (i) **The smallest unit of a living organism** (1 mark) **that can carry out the activities of life / from which all living things are built** (1 mark).
- (ii) Any TWO of: **both plants and animals are made of cells**; **the cell is the basic unit of all living things**; **cells are too small to be seen without a microscope** (1 mark each).
- **Marking note:** 'A small thing in the body' earns no mark in (i).\*

**(b) [4 marks]**

- Cytoplasm - **where the organelles lie and most reactions of the cell take place** (1 mark).
- Nucleus - **controls the activities of the cell** (1 mark).
- Mitochondrion - **releases energy from food** (1 mark).
- Ribosome - **makes proteins** (1 mark).
- **Marking note:** The table must be copied; one correct function each is

enough.\*

**(c) [4 marks]**

- **A rounded cell with an irregular outline and no cell wall** (1 mark).
- **Four correct parts shown and named** - cell membrane, cytoplasm, nucleus, mitochondrion (accept vacuole or ribosome) (2 marks; 1 mark if only two or three are correct).
- **Each label written outside the drawing with a straight line touching the part named** (1 mark).
- **Marking note:** A drawing that includes a cell wall or chloroplasts loses the first mark.\*

**Question 3 [Total: 10 marks]**

**(a) [5 marks]**

- P - **cell wall** (1 mark).
- Q - **cell membrane** (1 mark).
- R - **nucleus** (1 mark).
- S - **chloroplast** (1 mark).
- T - **vacuole (large central vacuole)** (1 mark).
- **Marking note:** P and Q must not be interchanged; the cell wall is the outer line.\*

**(b) [3 marks]**

- P - **gives the cell a fixed shape and gives support and protection** (1 mark).
- S - **makes food for the plant by photosynthesis / holds the chlorophyll** (1 mark).
- T - **stores cell sap and keeps the cell firm** (1 mark).
- **Marking note:** 'It protects the cell' alone is not enough for P; the idea of shape or support is needed.\*

**(c) [2 marks]**

- (i) **The root is in the soil where there is no light, so no food can be made there** (1 mark).
- (ii) **Food made in the leaves is carried down to the root through the stem** (1 mark).
- **Marking note:** 'Because it is not green' is accepted for (i) only if the absence of light is also stated.\*

**Question 4 [Total: 10 marks]**

**(a)** [4 marks]

- Cell wall - **present / absent** (1 mark).
- Chloroplast - **present / absent** (1 mark).
- Vacuole - **present in both, large and central in the plant cell but small in the animal cell** (1 mark).
- Nucleus - **present / present** (1 mark).
- **Marking note:** For the vacuole row, both 'present' entries and the difference are needed for the mark.\*

**(b)** [3 marks]

- **The plant cell has a firm cell wall made of cellulose** (1 mark).
- **The wall is rigid and holds the cell in a fixed shape** (1 mark).
- **The animal cell has only a thin flexible membrane, so its shape is irregular** (1 mark).
- **Marking note:** A candidate who writes only 'the plant cell has a cell wall' earns 1 mark.\*

**(c)** [3 marks]

- **The pupil is not correct; a nucleus and cytoplasm occur in both kinds of cell** (1 mark).
- Any TWO of: **a cell wall**; **chloroplasts**; **a large central vacuole**; **a regular box-like shape** (1 mark each).
- **Marking note:** No mark for the features if the candidate says the pupil is correct.\*

**Question 5 [Total: 10 marks]****(a)** [4 marks]

- Cell membrane - **rubber bag; thin and flexible and holds everything in** (1 mark).
- Cytoplasm - **starch paste; jelly-like and fills the cell** (1 mark).
- Nucleus - **one stone or large seed; rounded, firm and single** (1 mark).
- Mitochondrion - **maize grains; small, oval and several** (1 mark).
- **Marking note:** A material named without a reason earns no mark. Accept any sensible alternative use of the materials listed.\*

**(b)** [4 marks]

- (i) **The wooden box** (1 mark); **it is firm, straight-sided and keeps a fixed box-like shape** (1 mark).
- (ii) Any TWO of: **chloroplasts (green leaf pieces)**; **a large central vacuole**; **the nucleus pushed to one side** (1 mark each).
- **Marking note:** Naming the same part twice in (ii) scores once.\*

**(c) [2 marks]**

- Any TWO of: **\*\*the model is far larger than the real cell and gives a wrong idea of size\*\***; **\*\*the materials are not alive and carry out none of the activities of a cell\*\***; **\*\*some organelles are too small or too many to be shown\*\***; **\*\*the model is fixed while the contents of a living cell move\*\*** (1 mark each).
- **\*Marking note: 'It is not real' alone earns no mark; the limitation must be explained.\***

**Question 6 [Total: 8 marks]****(a) [4 marks]**

- (i) **\*\*The chloroplasts could make no food in the dark\*\*** (1 mark); **\*\*the chlorophyll broke down and was not replaced, so the green colour faded\*\*** (1 mark).
- (ii) **\*\*The vacuoles lost water (cell sap)\*\*** (1 mark); **\*\*so the cells were no longer firm against their cell walls and the plant drooped\*\*** (1 mark).
- **\*Marking note: An answer in terms of the whole plant only, with no mention of cells, earns at most 1 mark in each part.\***

**(b) [2 marks]**

- **\*\*Water the plant (and return it to the light)\*\*** (1 mark).
- **\*\*The vacuoles fill with water again and the cells become firm\*\*** (1 mark).
- **\*Marking note: The second mark needs the vacuole or the cell, not just 'the plant drinks water'.\***

**(c) [2 marks]**

- **\*\*The animal cell has no cell wall to resist the water entering\*\*** (1 mark).
- **\*\*So it swells and may burst instead of becoming firm, while the plant cell is held by its wall\*\*** (1 mark).
- **\*Marking note: Accept 'the membrane is thin and flexible' for the first mark.\***

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**## CURRICULUM ASSESSMENT BLUEPRINT**

**\*\*Strand:\*\* 1: Diversity of Matter | **\*\*Sub-strand:\*\* 2: Living Cells | **\*\*Class:\*\* Basic 7 (JHS 1)******

**\*\*Content Standard:\*\* **\*\*B7/JHS1.1.2.1\*\*** - Demonstrate understanding of the structure of organisms and functions of cells in living systems.**

**> Source: NaCCA Science Common Core Programme C**

# Materials

## ANSWER KEY

### SECTION A: OBJECTIVE TEST

Total Marks: 40

| Q  | Ans | Curriculum Area                               | Cognitive Level |
|----|-----|-----------------------------------------------|-----------------|
| 1  | C   | 1.1 Materials and Their Properties            | Remember        |
| 2  | D   | 1.1 Materials and Their Properties            | Remember        |
| 3  | B   | 1.1 Materials and Their Properties            | Apply           |
| 4  | B   | 1.1 Materials and Their Properties            | Understand      |
| 5  | A   | 1.1 Materials and Their Properties            | Understand      |
| 6  | C   | 1.1 Materials and Their Properties            | Remember        |
| 7  | B   | 1.1 Materials and Their Properties            | Understand      |
| 8  | A   | 1.2 Solids, Liquids and Gases Compared        | Remember        |
| 9  | D   | 1.2 Solids, Liquids and Gases Compared        | Understand      |
| 10 | B   | 1.2 Solids, Liquids and Gases Compared        | Analyse         |
| 11 | D   | 1.2 Solids, Liquids and Gases Compared        | Remember        |
| 12 | A   | 1.2 Solids, Liquids and Gases Compared        | Understand      |
| 13 | B   | 1.2 Solids, Liquids and Gases Compared        | Analyse         |
| 14 | B   | 1.2 Solids, Liquids and Gases Compared        | Analyse         |
| 15 | C   | 1.3 The Importance of Liquids to Life         | Remember        |
| 16 | C   | 1.3 The Importance of Liquids to Life         | Understand      |
| 17 | A   | 1.3 The Importance of Liquids to Life         | Understand      |
| 18 | A   | 1.3 The Importance of Liquids to Life         | Apply           |
| 19 | C   | 1.3 The Importance of Liquids to Life         | Evaluate        |
| 20 | B   | 1.3 The Importance of Liquids to Life         | Understand      |
| 21 | A   | 1.3 The Importance of Liquids to Life         | Evaluate        |
| 22 | D   | 1.4 The Importance of Solids to Life          | Remember        |
| 23 | D   | 1.4 The Importance of Solids to Life          | Remember        |
| 24 | A   | 1.4 The Importance of Solids to Life          | Understand      |
| 25 | C   | 1.4 The Importance of Solids to Life          | Remember        |
| 26 | C   | 1.4 The Importance of Solids to Life          | Understand      |
| 27 | B   | 1.4 The Importance of Solids to Life          | Evaluate        |
| 28 | A   | 1.5 The First Twenty Elements and Their Sy... | Remember        |
| 29 | D   | 1.5 The First Twenty Elements and Their Sy... | Remember        |
| 30 | A   | 1.5 The First Twenty Elements and Their Sy... | Remember        |
| 31 | D   | 1.5 The First Twenty Elements and Their Sy... | Remember        |
| 32 | C   | 1.5 The First Twenty Elements and Their Sy... | Remember        |
| 33 | A   | 1.5 The First Twenty Elements and Their Sy... | Understand      |
| 34 | C   | 1.5 The First Twenty Elements and Their Sy... | Apply           |
| 35 | D   | 1.6 Metals, Non-Metals and Noble Gases in ... | Understand      |
| 36 | B   | 1.6 Metals, Non-Metals and Noble Gases in ... | Remember        |
| 37 | D   | 1.6 Metals, Non-Metals and Noble Gases in ... | Understand      |



| Q  | Ans      | Curriculum Area                               | Cognitive Level |
|----|----------|-----------------------------------------------|-----------------|
| 38 | <b>B</b> | 1.6 Metals, Non-Metals and Noble Gases in ... | Understand      |
| 39 | <b>C</b> | 1.6 Metals, Non-Metals and Noble Gases in ... | Analyse         |
| 40 | <b>D</b> | 1.6 Metals, Non-Metals and Noble Gases in ... | Analyse         |

**Answer Distribution:**

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

**Cognitive Level Distribution:**

Analyse: 5 (12%) | Apply: 3 (7%) | Evaluate: 3 (7%) | Remember: 15 (37%) | Understand: 14 (35%)

**SECTION B: THEORY — MARKING SCHEME****Question 1 [Total: 10 marks]****(a)** [3 marks]

- P - **solid** (1 mark).
- Q - **liquid** (1 mark).
- R - **gas** (1 mark).
- **Marking note:** One mark for each correct state; no half marks.\*

**(b)** [3 marks]

- P - **the particles are closely packed in a regular pattern and the material keeps its own shape** (1 mark).
- Q - **the particles are close together but irregular and there is a flat surface; the material takes the shape of the bottom of the jar** (1 mark).
- R - **the particles are far apart, moving, and spread through the whole jar** (1 mark).
- **Marking note:** The reason must come from the figure; 'because it is a solid' earns no mark.\*

**(c)** [2 marks]

- (i) **It escapes and spreads out into the room** (1 mark).
- (ii) **The particles of a gas move freely in all directions, so a gas spreads to fill the space available** (1 mark).
- **Marking note:** 'It finishes' alone earns no mark in (i).\*

**(d)** [2 marks]

- P - any solid, e.g. **salt, sugar, charcoal, stone, cooking pot** (1 mark).
- Q - any liquid, e.g. **water, palm oil, kerosene, milk** (1 mark).
- **Marking note:** The example must be found in a home; 'iron ore' is not accepted.\*

**Question 2 [Total: 12 marks]****(a)** [6 marks]

- Solids - **chalk, river sand** (1 mark each).
- Liquids - **borehole water, groundnut oil** (1 mark each).
- Gases - **cooking gas, air in the balloon** (1 mark each).
- **Marking note:** A material entered under the wrong heading earns no mark; the table must be copied.\*

**(b) [3 marks]**

- (i) **Fixed shape and fixed volume** (1 mark).
- (ii) **No fixed shape, but fixed volume** (1 mark).
- (iii) **Neither fixed shape nor fixed volume** (1 mark).
  - *Marking note: Both halves of the answer are needed for the mark in each case.*

**(c) [3 marks]**

- **Push an inverted empty glass (or blow up a balloon) into a bucket of water** (1 mark).
- **Observation: the water does not fill the glass and the inside remains dry (the balloon swells)** (1 mark).
- **Conclusion: the air (gas) in the glass occupies the space** (1 mark).
  - *Marking note: An activity with no stated observation earns 1 mark only.*

**Question 3 [Total: 10 marks]****(a) [4 marks]**

- Any FOUR of: **drinking**; **cooking**; **bathing or washing the body**; **washing clothes and utensils**; **cleaning the house and compound**; **watering plants**; **giving to livestock** (1 mark each).
  - *Marking note: Two answers that mean the same thing, such as 'washing' and 'washing clothes', score once.*

**(b) [2 marks]**

- Any TWO of: **helps in digestion**; **carries digested food and oxygen round the body in the blood**; **removes waste as urine and sweat**; **controls body temperature** (1 mark each).
  - *Marking note: Uses outside the body earn no mark here.*

**(c) [2 marks]**

- Any TWO of: **water is scarce, especially in the dry season**; **it costs time, labour and money to fetch**; **it cannot be replaced at once when wasted**; **uncovered or wasted water breeds mosquitoes and causes disease**; **all life in the village depends on it** (1 mark each).
  - *Marking note: Two distinct reasons are required.*

**(d) [2 marks]**

- Liquid: **palm oil, groundnut oil, kerosene, petrol or diesel** (1 mark).
- Use: **cooking** / **lighting a lamp or stove** / **fuel for vehicles and machines** (1 mark).
  - *Marking note: The use must match the liquid named.*

## Question 4

**(a)** [4 marks]

- Wood - **furniture, doors or roofing timber** (1 mark).
- Cement blocks - **building walls or foundations** (1 mark).
- Iron - **cutlasses, hoes, nails or burglar bars** (1 mark).
- Glass - **windows, louvre blades, bottles or mirrors** (1 mark).
- **Marking note:** The table must be copied; one correct use per material is enough.\*

**(b)** [4 marks]

- (i) **Iron is hard and strong** (1 mark); **so it takes and keeps a sharp edge without breaking or bending** (1 mark).
- (ii) **Wood is light but strong** (1 mark); **and it is easily cut, shaped and smoothed with simple tools** (1 mark).
- **Marking note:** A property without the reason earns 1 mark only.\*

**(c)** [2 marks]

- Any TWO of: **measure carefully and avoid wastage**; **keep and use off-cuts**; **repair and re-use old articles**; **send scrap iron and glass for recycling**; **plant trees to replace the timber used** (1 mark each).
- **Marking note:** A measure that does not preserve material, such as 'work faster', earns no mark.\*

**Question 5 [Total: 10 marks]****(a)** [4 marks]

- W - **magnesium** (1 mark).
- X - **sulphur** (1 mark).
- Y - **argon** (1 mark).
- Z - **potassium** (1 mark).
- **Marking note:** Accept the correct chemical symbol written in place of the name.\*

**(b)** [2 marks]

- (i) **Y** (1 mark).
- (ii) **X** (1 mark).
- **Marking note:** The letter alone is enough; no mark for naming a different element.\*

**(c)** [2 marks]

- (i) **Beryllium (Be)** or **calcium (Ca)** (1 mark).
- (ii) **They have similar chemical properties** (accept: the same number of outermost electrons) (1 mark).
- **Marking note:** Only one element need be named in (i).\*

**(d)** [2 marks]

- Rule: **in order of increasing atomic number** (1 mark).
- Evidence: **the numbers in the boxes increase by one from each element to the next across a period and on to the next period** (1 mark).
- **Marking note:** 'In order of their mass' earns no mark.\*

**Question 6 [Total: 8 marks]****(a)** [4 marks]

- Carbon - **C** (1 mark).
- Sodium - **Na** (1 mark).
- Aluminium - **Al** (1 mark).
- Helium - **He** (1 mark).
- **Marking note:** A symbol written wholly in capitals, such as NA or AL, earns no mark.\*

**(b)** [2 marks]

- **Place the sample in a simple circuit with a cell and a bulb** (accept: hammer it gently, or rub it) (1 mark).
- **If the bulb lights, the sample conducts electricity and is a metal** (accept: it flattens into a sheet instead of breaking; it is shiny) (1 mark).
- **Marking note:** A test with no stated result earns 1 mark only.\*

**(c)** [2 marks]

- Any TWO differences, each stated for both sides: **metals conduct heat and electricity, non-metals generally do not**; **metals are shiny, non-metals are dull**; **metals can be beaten into sheets and drawn into wires, non-metals are brittle**; **most metals are solids with high melting points, non-metals may be solids, liquids or gases** (1 mark each).
- **Marking note:** A statement about metals only, with nothing said about non-metals, earns no mark.\*

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**## CURRICULUM ASSESSMENT BLUEPRINT**

**Strand:** 1: Diversity of Matter | **Sub-strand:** 1: Materials | **Class:** Basic 7 (JHS 1)

\*\*

# Science and Industry

## ANSWER KEY

### SECTION A: OBJECTIVE TEST

Total Marks: 40

| Q  | Ans | Curriculum Area                               | Cognitive Level |
|----|-----|-----------------------------------------------|-----------------|
| 1  | D   | 18.1 Careers in Science and What Each Work... | Remember        |
| 2  | C   | 18.1 Careers in Science and What Each Work... | Remember        |
| 3  | D   | 18.1 Careers in Science and What Each Work... | Remember        |
| 4  | B   | 18.1 Careers in Science and What Each Work... | Remember        |
| 5  | A   | 18.1 Careers in Science and What Each Work... | Remember        |
| 6  | B   | 18.1 Careers in Science and What Each Work... | Remember        |
| 7  | B   | 18.1 Careers in Science and What Each Work... | Remember        |
| 8  | D   | 18.1 Careers in Science and What Each Work... | Understand      |
| 9  | D   | 18.1 Careers in Science and What Each Work... | Remember        |
| 10 | C   | 18.2 Ghanaian Scientists and Science Educa... | Remember        |
| 11 | A   | 18.2 Ghanaian Scientists and Science Educa... | Remember        |
| 12 | D   | 18.2 Ghanaian Scientists and Science Educa... | Remember        |
| 13 | A   | 18.2 Ghanaian Scientists and Science Educa... | Remember        |
| 14 | C   | 18.2 Ghanaian Scientists and Science Educa... | Understand      |
| 15 | B   | 18.2 Ghanaian Scientists and Science Educa... | Evaluate        |
| 16 | C   | 18.2 Ghanaian Scientists and Science Educa... | Understand      |
| 17 | D   | 18.2 Ghanaian Scientists and Science Educa... | Remember        |
| 18 | C   | 18.3 Internationally Recognised Scientists... | Remember        |
| 19 | A   | 18.3 Internationally Recognised Scientists... | Understand      |
| 20 | B   | 18.3 Internationally Recognised Scientists... | Remember        |
| 21 | B   | 18.3 Internationally Recognised Scientists... | Remember        |
| 22 | C   | 18.3 Internationally Recognised Scientists... | Remember        |
| 23 | B   | 18.3 Internationally Recognised Scientists... | Understand      |
| 24 | D   | 18.3 Internationally Recognised Scientists... | Evaluate        |
| 25 | A   | 18.3 Internationally Recognised Scientists... | Apply           |
| 26 | B   | 18.4 The Impact of Science, Technology and... | Remember        |
| 27 | C   | 18.4 The Impact of Science, Technology and... | Understand      |
| 28 | D   | 18.4 The Impact of Science, Technology and... | Understand      |
| 29 | C   | 18.4 The Impact of Science, Technology and... | Analyse         |
| 30 | A   | 18.4 The Impact of Science, Technology and... | Understand      |
| 31 | C   | 18.4 The Impact of Science, Technology and... | Understand      |
| 32 | B   | 18.4 The Impact of Science, Technology and... | Analyse         |
| 33 | D   | 18.4 The Impact of Science, Technology and... | Remember        |
| 34 | C   | 18.5 Science and Technology Careers Ghana ... | Evaluate        |
| 35 | A   | 18.5 Science and Technology Careers Ghana ... | Understand      |
| 36 | B   | 18.5 Science and Technology Careers Ghana ... | Evaluate        |
| 37 | A   | 18.5 Science and Technology Careers Ghana ... | Apply           |

| Q  | Ans      | Curriculum Area                               | Cognitive Level |
|----|----------|-----------------------------------------------|-----------------|
| 38 | <b>A</b> | 18.5 Science and Technology Careers Ghana ... | Apply           |
| 39 | <b>D</b> | 18.5 Science and Technology Careers Ghana ... | Apply           |
| 40 | <b>A</b> | 18.5 Science and Technology Careers Ghana ... | Understand      |

**Answer Distribution:**

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

**Cognitive Level Distribution:**

Analyse: 2 (5%) | Apply: 4 (10%) | Evaluate: 4 (10%) | Remember: 19 (47%) | Understand: 11 (27%)

**SECTION B: THEORY — MARKING SCHEME****Question 1 [Total: 12 marks]****(a)** [4 marks]

- A - **\*\*laboratory scientist or technician (accept biologist, chemist)\*\*** (1 mark).
- B - **\*\*doctor or nurse (accept any named health worker)\*\*** (1 mark).
- C - **\*\*agricultural scientist (accept agronomist, soil scientist)\*\*** (1 mark).
- D - **\*\*civil engineer (accept architect, surveyor)\*\*** (1 mark).
- **\*Marking note: 'He works in a laboratory' is not the name of a career.\***

**(b)** [4 marks]

- A - **\*\*tests samples so that disease is detected and food and water are safe\*\*** (1 mark).
- B - **\*\*treats the sick and immunises people, saving lives\*\*** (1 mark).
- C - **\*\*improves crops and methods, so more food is produced\*\*** (1 mark).
- D - **\*\*designs safe roads, bridges, dams and buildings\*\*** (1 mark).
- **\*Marking note: 'He earns money' is not a benefit to the people.\***

**(c)** [2 marks]

- **\*\*Integrated science and mathematics (accept English for one)\*\*** (1 mark).
- **\*\*Reason: science supplies the knowledge and mathematics the measurement and calculation that all these careers require\*\*** (1 mark).
- **\*Marking note: Subjects named with no reason earn 1 mark only.\***

**(d)** [2 marks]

- **\*\*A scientist carries out research and makes discoveries in a field of science\*\*** (1 mark).
- **\*\*A science educator studies and improves the teaching and learning of science (trains teachers, writes books, develops the curriculum)\*\*** (1 mark).
- **\*Marking note: 'One is cleverer than the other' earns no mark.\***

**Question 2 [Total: 10 marks]****(a)** [3 marks]

- (i) **\*\*Alexander Fleming\*\*** (1 mark).
- (ii) **\*\*Penicillin\*\*** (1 mark).
- (iii) **\*\*A substance from the mould had killed or stopped the growth of the bacteria\*\*** (1 mark).
- **\*Marking note: 'Antibiotic' alone is accepted for (ii) only if penicillin is named.\***



**(b)** [3 marks]

- Q - **the substance is tested and studied in the laboratory and the results recorded** (1 mark).
- R - **the medicine is manufactured in a factory and packed** (1 mark).
- S - **the patient is treated and recovers** (1 mark).
- **Marking note: Credit any clear wording of each stage.**

**(c)** [2 marks]

- **Science makes the discovery and tests it** (1 mark).
- **Industry manufactures it in large quantities so that it reaches the people** (1 mark).
- **Marking note: A general statement with no reference to the stages earns 1 mark at most.**

**(d)** [2 marks]

- Any TWO of: **keen observation**; **curiosity and an open mind**; **patience and careful follow-up work**; **careful recording and reporting of findings** (1 mark each).
- **Marking note: 'He was clever' is too vague for a mark.**

**Question 3 [Total: 10 marks]****(a)** [4 marks]

- Allotey - **mathematics and physics** (1 mark).
- Oduro - **food science (food and crop processing)** (1 mark).
- Addy - **biochemistry and the biological sciences** (1 mark).
- Anamuah-Mensah - **science education** (1 mark).
- **Marking note: A field written against the wrong person earns no mark.**

**(b)** [3 marks]

- **Prof. Theophilus Ossei-Anto** (1 mark); **Prof. Christian Anthony-Krueger** (1 mark).
- **They improve the teaching and learning of science, so that more pupils understand it and take up science careers** (1 mark).
- **Marking note: Any other correctly named Ghanaian science educator is accepted.**

**(c)** [3 marks]

- Any THREE of: **it shows that Ghanaians can succeed in science**; **it gives models the pupils can relate to**; **it shows science applied to Ghana's own problems**; **it builds national pride and encourages pupils to serve their country** (1 mark each).
- **Marking note: Two answers that both say 'to encourage us' count once.**

**Question 4****(a)** [3 marks]

- (i) **One sound home example, such as a refrigerator or a mobile phone** (1 mark).
- (ii) **One sound school example, such as computers or laboratory apparatus** (1 mark).
- (iii) **One sound community example, such as treated water, electricity or mobile money** (1 mark).
- **Marking note: The same example may not be used twice.**

**(b)** [3 marks]

- Any THREE of: **people communicate quickly over long distances**; **farmers and traders check market prices before travelling**; **money is sent and received by mobile money**; **emergencies can be reported at once**; **information for learning becomes available** (1 mark each).
- **Marking note: 'It is nice to have' earns no mark.**

**(c)** [2 marks]

- Any TWO of: **air and water pollution from factories and vehicles**; **plastic and electronic waste**; **loss of jobs to machines**; **too much screen time harming study and health**; **destruction of land and rivers by mining machinery** (1 mark each).
- **Marking note: A harmful effect must be stated, not merely a device named.**

**(d)** [2 marks]

- **Putting a new or improved idea into practical use to solve a problem** (1 mark).
- **One sound Ghanaian example, such as mobile money, delivery drones or solar lamps** (1 mark).
- **Marking note: A copy of an existing article is not an innovation.**

**Question 5 [Total: 10 marks]****(a)** [4 marks]

- **First career named** (1 mark) **with a sound reason** (1 mark).
- **Second career named** (1 mark) **with a sound reason** (1 mark).
- **Marking note: A career named with no reason earns 1 mark only.**

**(b)** [3 marks]

- Any THREE of: **study science and mathematics seriously and ask questions**; **take part in practical work and the science club**; **read about science and about scientists**; **keep a portfolio of findings**; **visit places of work and talk to scientists** (1 mark each).
- **Marking note: 'Pray to become a scientist' earns no mark.**

**(c) [3 marks]**

- **The belief is wrong; science needs observation and hard work, which do not depend on sex** (1 mark).
- **Evidence: named Ghanaian women scientists, such as Prof. Ibok Oduro or Prof. Ewurama Addy (accept women doctors, engineers and so on)** (1 mark).
- **Excluding girls wastes half the nation's talent, so girls should be encouraged equally** (1 mark).
- **Marking note: A reply that agrees with the belief earns no mark.**

**Question 6 [Total: 8 marks]****(a) [2 marks]**

- **An organised collection of one's own work and evidence on a topic** (1 mark).
- **One reason: it keeps the evidence in one place for presentation and shows how the work has developed** (1 mark).
- **Marking note: 'A bag for books' earns no mark.**

**(b) [4 marks]**

- Any FOUR of: **notes from books and the internet**; **photographs or drawings of technology in use**; **records of interviews with people at work**; **newspaper cuttings and leaflets**; **tables or charts of data the class collected**; **a written report with conclusions** (1 mark each).
- **Marking note: 'Everything about science' earns no mark.**

**(c) [2 marks]**

- Any TWO of: **speak clearly and in an orderly sequence, keeping to time**; **show charts, pictures and specimens as evidence**; **share the work among the members**; **practise beforehand and be ready for questions** (1 mark each).
- **Marking note: 'Wear fine clothes' earns no mark.**

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**## CURRICULUM ASSESSMENT BLUEPRINT**

**Strand:** 5: Humans and the Environment | **Sub-strand:** 3: Science and Industry | **Class:** Basic 7 (JHS 1)

**Content Standard:** **B7/JHS1.5.3.1** - Realise how careers in science can improve human life, and research about Ghanaian and internationally recognised

# Solar System

## ANSWER KEY

### SECTION A: OBJECTIVE TEST

Total Marks: 40

| Q  | Ans | Curriculum Area                            | Cognitive Level |
|----|-----|--------------------------------------------|-----------------|
| 1  | D   | 8.1 The Solar System and the Sun           | Remember        |
| 2  | A   | 8.1 The Solar System and the Sun           | Remember        |
| 3  | B   | 8.1 The Solar System and the Sun           | Remember        |
| 4  | D   | 8.1 The Solar System and the Sun           | Understand      |
| 5  | D   | 8.1 The Solar System and the Sun           | Understand      |
| 6  | C   | 8.1 The Solar System and the Sun           | Understand      |
| 7  | B   | 8.1 The Solar System and the Sun           | Analyse         |
| 8  | D   | 8.2 The Inner Planets and Their Properties | Remember        |
| 9  | B   | 8.2 The Inner Planets and Their Properties | Remember        |
| 10 | B   | 8.2 The Inner Planets and Their Properties | Remember        |
| 11 | C   | 8.2 The Inner Planets and Their Properties | Understand      |
| 12 | D   | 8.2 The Inner Planets and Their Properties | Remember        |
| 13 | D   | 8.2 The Inner Planets and Their Properties | Remember        |
| 14 | A   | 8.2 The Inner Planets and Their Properties | Remember        |
| 15 | C   | 8.2 The Inner Planets and Their Properties | Understand      |
| 16 | C   | 8.2 The Inner Planets and Their Properties | Understand      |
| 17 | D   | 8.3 The Galaxy and the Milky Way           | Remember        |
| 18 | A   | 8.3 The Galaxy and the Milky Way           | Remember        |
| 19 | A   | 8.3 The Galaxy and the Milky Way           | Understand      |
| 20 | C   | 8.3 The Galaxy and the Milky Way           | Understand      |
| 21 | B   | 8.3 The Galaxy and the Milky Way           | Analyse         |
| 22 | C   | 8.3 The Galaxy and the Milky Way           | Analyse         |
| 23 | A   | 8.4 The Paths and Movements of the Planets | Remember        |
| 24 | B   | 8.4 The Paths and Movements of the Planets | Remember        |
| 25 | A   | 8.4 The Paths and Movements of the Planets | Remember        |
| 26 | D   | 8.4 The Paths and Movements of the Planets | Remember        |
| 27 | C   | 8.4 The Paths and Movements of the Planets | Understand      |
| 28 | A   | 8.4 The Paths and Movements of the Planets | Analyse         |
| 29 | B   | 8.4 The Paths and Movements of the Planets | Remember        |
| 30 | A   | 8.5 The Planets Mercury and Venus          | Understand      |
| 31 | D   | 8.5 The Planets Mercury and Venus          | Analyse         |
| 32 | B   | 8.5 The Planets Mercury and Venus          | Apply           |
| 33 | B   | 8.5 The Planets Mercury and Venus          | Understand      |
| 34 | C   | 8.5 The Planets Mercury and Venus          | Understand      |
| 35 | A   | 8.5 The Planets Mercury and Venus          | Understand      |
| 36 | D   | 8.5 The Planets Mercury and Venus          | Analyse         |
| 37 | A   | 8.6 Making a Model of the Solar System     | Understand      |

| Q  | Ans      | Curriculum Area                        | Cognitive Level |
|----|----------|----------------------------------------|-----------------|
| 38 | <b>C</b> | 8.6 Making a Model of the Solar System | Apply           |
| 39 | <b>C</b> | 8.6 Making a Model of the Solar System | Evaluate        |
| 40 | <b>B</b> | 8.6 Making a Model of the Solar System | Apply           |

**Answer Distribution:**

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

**Cognitive Level Distribution:**

Analyse: 6 (15%) | Apply: 3 (7%) | Evaluate: 1 (2%) | Remember: 16 (40%) | Understand: 14 (35%)

**SECTION B: THEORY — MARKING SCHEME****Question 1 [Total: 10 marks]****(a)** [4 marks]

- W - **Mercury** (1 mark).
- X - **Venus** (1 mark).
- Y - **Earth** (1 mark).
- Z - **Mars** (1 mark).
- **Marking note:** The four names must be in the correct order from the Sun; a correct name in the wrong position earns no mark.\*

**(b)** [2 marks]

- **The inner planets** (accept: the rocky or terrestrial planets) (1 mark).
- Any ONE of: **they have solid rocky surfaces**; **they are small**; **they are close to the Sun**; **they have few or no moons** (1 mark).
- **Marking note:** 'They are hot' alone is too vague.\*

**(c)** [2 marks]

- **Y, the Earth** (1 mark).
- Any ONE of: **it is neither too hot nor too cold**; **it has liquid water**; **it has air (oxygen) to breathe** (1 mark).
- **Marking note:** 'Because people live there' is a restatement and earns no mark for the reason.\*

**(d)** [2 marks]

- **Z is farther from the Sun, so its path (orbit) is much longer** (1 mark).
- **It also travels more slowly than W, so it takes much more time to complete one round** (1 mark).
- **Marking note:** The idea of a longer path is essential for the first mark.\*

**Question 2 [Total: 10 marks]****(a)** [3 marks]

- (i) **A very large group (collection) of stars, dust and gas** (1 mark) **held together by gravity** (1 mark).
- (ii) **The Milky Way** (1 mark).
- **Marking note:** 'Many stars' alone earns 1 mark in (i).\*

**(b)** [3 marks]

- **The light of the Milky Way is faint** (1 mark).
- **The bright lights of the town brighten the sky and hide it, while the village is dark** (1 mark).

- **Suggestion: travel to a dark place outside the town, or switch off the compound lights and observe on a moonless night** (1 mark).

- **Marking note: 'Too many buildings in Accra' earns no mark.**

**(c)** [2 marks]

- **Planet, solar system, galaxy** (1 mark).

- **Reason: the planet is one member of the solar system, and the solar system is one small part of the galaxy** (1 mark).

- **Marking note: No mark for the reason if the order is wrong.**

**(d)** [2 marks]

- Any TWO of: **a star gives out its own light and heat, a planet reflects light; a star is far larger and hotter; stars twinkle, planets shine steadily; a planet moves round a star** (1 mark each).

- **Marking note: 'Stars are far away and planets are near' is too weak on its own.**

### Question 3 [Total: 10 marks]

**(a)** [4 marks]

- P - **the axis (of the planet)** (1 mark).

- Q - **rotation** (accept: the planet turning or spinning on its own axis) (1 mark).

- R - **revolution** (accept: the planet moving round the Sun) (1 mark).

- S - **the orbit** (accept: the elliptical path of the planet) (1 mark).

- **Marking note: Rotation written for R, or revolution for Q, earns no mark.**

**(b)** [2 marks]

- Q - **day and night** (1 mark).

- R - **the year** (accept: the seasons) (1 mark).

- **Marking note: Day and night written for R earns no mark.**

**(c)** [2 marks]

- **Elliptical (an ellipse)** (1 mark).

- **The distance between the planet and the Sun changes as the planet moves; it is nearer at some times and farther at others** (1 mark).

- **Marking note: 'The planet is always the same distance from the Sun' is wrong and earns no mark.**

**(d)** [2 marks]

- **The gravitational pull of the Sun holds the planet and stops it flying away** (1 mark).

- **The planet is also moving forward fast along its path, so it keeps going round instead of falling in** (1 mark).

- \*Marking note: Gravity alone earns 1 mark; the forward motion is needed for the second.\*

#### Question 4 [Total: 12 marks]

**(a)** [4 marks]

- Mercury - \*\*first (nearest) planet from the Sun\*\* (1 mark).
- Venus - \*\*second planet from the Sun\*\* (1 mark).
- Mercury - any ONE of: \*\*the smallest inner planet\*\*; \*\*has almost no atmosphere\*\*; \*\*very hot by day and very cold by night\*\*; \*\*has the shortest year\*\* (1 mark).
- Venus - any ONE of: \*\*the hottest planet\*\*; \*\*has a very thick atmosphere\*\*; \*\*nearly the same size as the Earth\*\*; \*\*seen as the morning or evening star\*\* (1 mark).
- \*Marking note: A property given for the wrong planet earns no mark.\*

**(b)** [3 marks]

- \*\*Venus has a very thick atmosphere\*\* (1 mark).
- \*\*The atmosphere traps the heat so that it cannot escape, and the heat builds up\*\* (1 mark).
- \*\*Mercury has almost no atmosphere, so its heat escapes quickly into space\*\* (1 mark).
- \*Marking note: The trapping of heat must be stated; 'Venus is bigger' earns no mark.\*

**(c)** [3 marks]

- (i)  $365 / 88$  (1 mark)  $= 4.1$ , which is about 4 times (1 mark).
- (ii) \*\*A year on Mercury is much shorter than an Earth year (about a quarter as long)\*\* (1 mark).
- \*Marking note: Award the second mark for 4 even if the division is not shown, but the working mark is then lost.\*

**(d)** [2 marks]

- \*\*It is seen shining brightly just before sunrise or just after sunset\*\* (1 mark).
- \*\*It is not a star but a planet, which shines by reflecting the light of the Sun\*\* (1 mark).
- \*Marking note: The second mark requires the idea of reflected light or of Venus being a planet.\*

#### Question 5 [Total: 10 marks]

**(a)** [3 marks]

- Any THREE materials, each with a correct use, for example: \*\*a large ball of clay for the Sun\*\*; \*\*smaller balls, beads or seeds for the planets\*\*; \*\*loops of



wire or lines on cardboard for the paths\*\*; \*\*thread or sticks to hold the planets\*\* (1 mark each).

- \*Marking note: A material named with no use, or a use with no material, earns no mark.\*

**(b)** [4 marks]

- Any FOUR steps in a workable order: \*\*fix the largest ball at the centre as the Sun\*\*; \*\*make four planets of different sizes\*\*; \*\*make four oval paths of increasing size round the Sun\*\*; \*\*place each planet on its own path in the correct order\*\*; \*\*label the bodies and state that the model is not to scale\*\* (1 mark each).

- \*Marking note: Steps given out of a workable order lose 1 mark overall.\*

**(c)** [3 marks]

- Any TWO of: \*\*the order of the planets from the Sun\*\*; \*\*that each planet has its own path\*\*; \*\*the oval shape of the paths\*\*; \*\*that the Sun is at the centre\*\* (1 mark each).

- Any ONE of: \*\*the true sizes of the bodies\*\*; \*\*the true distances between them\*\*; \*\*the real speeds of the planets\*\*; \*\*that the paths are not solid rings\*\* (1 mark).

- \*Marking note: 'It is not real' is too vague for the third mark.\*

## Question 6 [Total: 8 marks]

**(a)** [3 marks]

- \*\*The pupil is not correct\*\* (1 mark).

- \*\*The Earth rotates on its own axis once in about 24 hours\*\* (1 mark).

- \*\*The half facing the Sun has day and the half turned away has night, so the Sun only appears to move\*\* (1 mark).

- \*Marking note: A candidate who agrees with the pupil can earn no mark in this part.\*

**(b)** [3 marks]

- Any THREE of: \*\*it supplies light for seeing and for photosynthesis\*\*; \*\*it supplies heat and keeps the Earth warm\*\*; \*\*it drives the water cycle and the winds\*\*; \*\*its gravity holds the planets in their paths\*\*; \*\*it supplies energy for solar panels\*\* (1 mark each).

- \*Marking note: 'It makes the day bright' and 'it gives light' count as one answer only.\*

**(c)** [2 marks]

- \*\*Shine a torch (the Sun) on a globe or ball marked to show Ghana, in a darkened room\*\* (1 mark).

- \*\*Keep the torch still and turn the globe slowly; the lit half has day, the dark

half night, and the mark passes from one to the other\*\* (1 mark).

- \*Marking note: No mark for an activity in which the torch is moved round the globe.\*

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## ## CURRICULUM ASSESSMENT BLUEPRINT

\*\*Strand:\*\* 3: Systems | \*\*Sub-strand:\*\* 2: The Solar System | \*\*Class:\*\* Basic 7 (JHS 1)

\*\*Content Standard:\*\* \*\*B7/JHS1.3.2.1\*\* - Demonstrate knowledge of the inner planets of the solar system and understand their mov

# Understanding the Environment

## ANSWER KEY

### SECTION A: OBJECTIVE TEST

Total Marks: 40

| Q  | Ans | Curriculum Area                               | Cognitive Level |
|----|-----|-----------------------------------------------|-----------------|
| 1  | C   | 20.1 Land Forms in Ghana and the Living Th... | Remember        |
| 2  | D   | 20.1 Land Forms in Ghana and the Living Th... | Remember        |
| 3  | A   | 20.1 Land Forms in Ghana and the Living Th... | Remember        |
| 4  | D   | 20.1 Land Forms in Ghana and the Living Th... | Remember        |
| 5  | A   | 20.1 Land Forms in Ghana and the Living Th... | Remember        |
| 6  | B   | 20.1 Land Forms in Ghana and the Living Th... | Understand      |
| 7  | B   | 20.1 Land Forms in Ghana and the Living Th... | Understand      |
| 8  | D   | 20.1 Land Forms in Ghana and the Living Th... | Understand      |
| 9  | C   | 20.2 Characteristics that Enable Animals t... | Remember        |
| 10 | A   | 20.2 Characteristics that Enable Animals t... | Understand      |
| 11 | B   | 20.2 Characteristics that Enable Animals t... | Understand      |
| 12 | C   | 20.2 Characteristics that Enable Animals t... | Understand      |
| 13 | D   | 20.2 Characteristics that Enable Animals t... | Understand      |
| 14 | A   | 20.2 Characteristics that Enable Animals t... | Understand      |
| 15 | C   | 20.2 Characteristics that Enable Animals t... | Analyse         |
| 16 | D   | 20.2 Characteristics that Enable Animals t... | Analyse         |
| 17 | D   | 20.3 Characteristics that Enable Plants to... | Understand      |
| 18 | B   | 20.3 Characteristics that Enable Plants to... | Understand      |
| 19 | B   | 20.3 Characteristics that Enable Plants to... | Remember        |
| 20 | D   | 20.3 Characteristics that Enable Plants to... | Understand      |
| 21 | B   | 20.3 Characteristics that Enable Plants to... | Understand      |
| 22 | C   | 20.3 Characteristics that Enable Plants to... | Analyse         |
| 23 | B   | 20.3 Characteristics that Enable Plants to... | Analyse         |
| 24 | A   | 20.3 Characteristics that Enable Plants to... | Apply           |
| 25 | D   | 20.4 Associations among Plants and Animals    | Remember        |
| 26 | C   | 20.4 Associations among Plants and Animals    | Understand      |
| 27 | D   | 20.4 Associations among Plants and Animals    | Remember        |
| 28 | B   | 20.4 Associations among Plants and Animals    | Apply           |
| 29 | D   | 20.4 Associations among Plants and Animals    | Apply           |
| 30 | B   | 20.4 Associations among Plants and Animals    | Apply           |
| 31 | C   | 20.4 Associations among Plants and Animals    | Apply           |
| 32 | C   | 20.4 Associations among Plants and Animals    | Analyse         |
| 33 | A   | 20.4 Associations among Plants and Animals    | Remember        |
| 34 | A   | 20.5 Mechanisms for Survival, Field Study ... | Remember        |
| 35 | B   | 20.5 Mechanisms for Survival, Field Study ... | Understand      |
| 36 | A   | 20.5 Mechanisms for Survival, Field Study ... | Remember        |

| Q  | Ans | Curriculum Area                               | Cognitive Level |
|----|-----|-----------------------------------------------|-----------------|
| 37 | A   | 20.5 Mechanisms for Survival, Field Study ... | Remember        |
| 38 | C   | 20.5 Mechanisms for Survival, Field Study ... | Apply           |
| 39 | A   | 20.5 Mechanisms for Survival, Field Study ... | Evaluate        |
| 40 | C   | 20.5 Mechanisms for Survival, Field Study ... | Analyse         |

**Answer Distribution:**

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

**Cognitive Level Distribution:**

Analyse: 6 (15%) | Apply: 6 (15%) | Evaluate: 1 (2%) | Remember: 13 (32%) | Understand: 14 (35%)

## SECTION B: THEORY — MARKING SCHEME

### Question 1 [Total: 12 marks]

**(a)** [4 marks]

- A - **mountain** (accept mountain peak) (1 mark).
- B - **valley** (1 mark).
- C - **plateau** (accept table land) (1 mark).
- D - **plain** (1 mark).
- \*Marking note: 'Hill' is accepted for A only if the candidate describes it as high land rising to a peak.\*

**(b)** [2 marks]

- C - **the Kwahu Plateau** (accept any correctly named Ghanaian plateau) (1 mark).
- D - **the Accra Plains or the Afram Plains** (accept any correctly named Ghanaian plain) (1 mark).
- \*Marking note: A named place that is not of the land form asked for earns no mark.\*

**(c)** [4 marks]

- Any TWO plants of the plains: **grasses** (elephant grass, spear grass); **baobab**; **shea tree**; **dawadawa**; **acacia** (1 mark each).
- Any TWO animals of the plains: **cattle, sheep or goats**; **antelope**; **grasscutter**; **guinea fowl**; **grasshopper**; **egret** (1 mark each).
- \*Marking note: A forest tree such as the odum is not accepted as a plant of the open plain.\*

**(d)** [2 marks]

- Any TWO of: **the soil on the slopes is thin, stony and poor, so few plants can grow**; **it is colder and windier high up**; **water drains away quickly from the steep slope**; **with few plants there is little food and shelter for animals**; **the steep rocky ground is difficult to move about on** (1 mark each).
- \*Marking note: 'Because it is a mountain' earns no mark.\*

### Question 2 [Total: 10 marks]

**(a)** [2 marks]

- **A feature of the body, or a habit, that helps a living thing to survive where it lives** (1 mark).
- **One correct example** (1 mark).
- \*Marking note: An example alone, without the meaning, earns 1 mark.\*

**(b) [6 marks]**

- Goat - **strong legs with hard split hooves** (1 mark) - **grip the rock, so it does not slip on the steep slope** (1 mark).
- Egret - **long thin legs with a long neck and beak** (1 mark) - **wades in shallow water and catches food without wetting its body** (1 mark).
- Fish - **streamlined body with strong fins and tail** (1 mark) - **water flows past it easily, so it can hold its place in the current** (1 mark).
- **Marking note: A feature entered without an advantage earns the feature mark only.**

**(c) [2 marks]**

- **Its features (gills, fins, moist skin) work only in water, and there is no water on the hill top** (1 mark).
- **It would not find its usual food or any shelter, and would dry up in the heat** (1 mark).
- **Marking note: 'Because it is a fish' earns no mark without the reason.**

**Question 3 [Total: 11 marks]****(a) [3 marks]**

- P - **mutualism** (1 mark).
- Q - **commensalism** (1 mark).
- R - **parasitism** (1 mark).
- **Marking note: 'Symbiosis' alone is not accepted for P.**

**(b) [6 marks]**

- P - **the bee gets nectar for food** (1 mark); **the flower is pollinated, so it can form seeds** (1 mark).
- Q - **the bird gets the insects to eat** (1 mark); **the cow is neither helped nor harmed** (1 mark).
- R - **the plant on the branch gets water and food from the tree** (1 mark); **the tree is harmed and grows weak** (1 mark).
- **Marking note: Marks are for the benefit and the effect, not for naming the association again.**

**(c) [2 marks]**

- **He should not remove the bees (he should manage them instead)** (1 mark).
- **Reason: the bees pollinate the flowers, and without them the trees would set far less fruit** (1 mark).
- **Marking note: Advice with no reason earns 1 mark only.**

**Question****(a)** [4 marks]

- Any FOUR of: **long or widely spreading roots**; **small, narrow or needle-like leaves**; **thick waxy or hairy leaf surfaces**; **fleshy stems, roots or tubers that store water**; **thick fire-resistant bark**; **shedding of leaves in the dry season**; **thorns in place of leaves** (1 mark each).

- *Marking note: 'It grows well' is not a feature.\**

**(b)** [3 marks]

- Any THREE of: **stilt or prop roots that support it in the soft mud**; **breathing roots that grow up out of the mud for air**; **thick leathery leaves that cope with the salt**; **the ability to get rid of or store excess salt**; **seedlings that germinate on the parent plant** (1 mark each).

- *Marking note: Credit any clear description even if the technical name is not used.\**

**(c)** [2 marks]

- **Tall growth lets the tree reach the sunlight above the crowded forest** (1 mark).

- **Broad leaves catch as much light as possible, and the plentiful water makes the water loss affordable** (1 mark).

- *Marking note: 'To look beautiful' earns no mark.\**

**Question 5 [Total: 10 marks]****(a)** [3 marks]

- Any THREE items, each with its correct use, for example: **notebook and pencil - to record observations**; **camera or phone - to photograph what cannot be collected**; **hand lens - to examine small specimens**; **tape or rule - to measure**; **bags or envelopes - to carry leaf specimens**; **field guide or chart - to help name the organisms** (1 mark each).

- *Marking note: An item named without its use earns no mark.\**

**(b)** [3 marks]

- Any THREE of: **the name of the organism (English and local name)**; **the land form and place where it was found**; **the date it was seen**; **the features noticed and the adaptation each gives**; **the name of the recorder** (1 mark each).

- *Marking note: 'A nice drawing' is not a record.\**

**(c)** [2 marks]

- Any TWO of: **\*\*the valley is damper and the water collects there, while the plateau drains quickly\*\***; **\*\*the valley soil is deeper and richer, the plateau soil shallow and stony\*\***; **\*\*the plateau is more exposed to sun and wind\*\***; **\*\*temperatures differ between the high and the low ground\*\***; **\*\*farming and grazing differ in the two places\*\*** (1 mark each).

- \*Marking note: 'One is up and one is down' alone earns no mark.\*

**(d)** [2 marks]

- Any TWO of: **\*\*observe, draw or photograph instead of uprooting and killing\*\***; **\*\*collect only small specimens where necessary\*\***; **\*\*do not disturb nests, eggs or young animals\*\***; **\*\*keep to the paths and avoid trampling\*\***; **\*\*light no fires\*\***; **\*\*carry all rubbish away\*\*** (1 mark each).

- \*Marking note: 'Behave well' is too vague for a mark.\*

**Question 6 [Total: 8 marks]****(a)** [2 marks]

- **\*\*The natural place in which a plant or animal lives\*\*** (1 mark).  
 - **\*\*It supplies the food, water, shelter and breeding place that the organism needs\*\*** (1 mark).

- \*Marking note: 'Where animals sleep' is accepted only with the idea of food or shelter.\*

**(b)** [4 marks]

- Any FOUR of: **\*\*animals lose shelter and nesting places and move away or die\*\***; **\*\*food plants are destroyed, so food becomes scarce and competition increases\*\***; **\*\*the shade is lost, so the ground is hot and dry and shade-loving plants die\*\***; **\*\*the bare soil is washed away by rain and the streams are silted\*\***; **\*\*associations such as pollination are broken\*\***; **\*\*some kinds of plants and animals disappear from the area\*\*** (1 mark each).

- \*Marking note: Two answers that both say 'animals die' count once.\*

**(c)** [2 marks]

- Any TWO of: **\*\*replant trees and protect the seedlings from fire and animals\*\***; **\*\*set part of the slope aside as a protected area\*\***; **\*\*stop charcoal burning and use improved stoves or gas\*\***; **\*\*farm the flat land instead of the slope\*\***; **\*\*agree a closed season for hunting\*\*** (1 mark each).

- \*Marking note: 'Report them to the police' alone earns no mark.\*

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**## CURRICULUM ASSESSMENT BLUEPRINT**

**\*\*Strand:\*\* 5: Humans and the Environment | **\*\*Sub-strand:\*\* 5: Understanding****



the Environment | **Class:** Basic 7 (JHS 1)

**Content Standard:** **B7/JHS1.5.5.1** - Demonstrate understanding of diffe

# Waste Management

## ANSWER KEY

### SECTION A: OBJECTIVE TEST

Total Marks: 40

| Q  | Ans | Curriculum Area                               | Cognitive Level |
|----|-----|-----------------------------------------------|-----------------|
| 1  | C   | 16.1 Types and Sources of Waste               | Remember        |
| 2  | C   | 16.1 Types and Sources of Waste               | Remember        |
| 3  | D   | 16.1 Types and Sources of Waste               | Remember        |
| 4  | D   | 16.1 Types and Sources of Waste               | Remember        |
| 5  | C   | 16.1 Types and Sources of Waste               | Remember        |
| 6  | D   | 16.1 Types and Sources of Waste               | Remember        |
| 7  | A   | 16.1 Types and Sources of Waste               | Understand      |
| 8  | B   | 16.1 Types and Sources of Waste               | Remember        |
| 9  | C   | 16.2 Biodegradable and Non-Biodegradable W... | Remember        |
| 10 | B   | 16.2 Biodegradable and Non-Biodegradable W... | Understand      |
| 11 | B   | 16.2 Biodegradable and Non-Biodegradable W... | Remember        |
| 12 | D   | 16.2 Biodegradable and Non-Biodegradable W... | Analyse         |
| 13 | A   | 16.2 Biodegradable and Non-Biodegradable W... | Understand      |
| 14 | D   | 16.2 Biodegradable and Non-Biodegradable W... | Analyse         |
| 15 | B   | 16.2 Biodegradable and Non-Biodegradable W... | Analyse         |
| 16 | C   | 16.2 Biodegradable and Non-Biodegradable W... | Understand      |
| 17 | B   | 16.3 Good Waste Management Practices          | Remember        |
| 18 | C   | 16.3 Good Waste Management Practices          | Understand      |
| 19 | D   | 16.3 Good Waste Management Practices          | Understand      |
| 20 | A   | 16.3 Good Waste Management Practices          | Understand      |
| 21 | D   | 16.3 Good Waste Management Practices          | Remember        |
| 22 | A   | 16.3 Good Waste Management Practices          | Analyse         |
| 23 | A   | 16.3 Good Waste Management Practices          | Understand      |
| 24 | B   | 16.3 Good Waste Management Practices          | Evaluate        |
| 25 | A   | 16.3 Good Waste Management Practices          | Apply           |
| 26 | C   | 16.4 Composting and the Science Behind It     | Remember        |
| 27 | B   | 16.4 Composting and the Science Behind It     | Understand      |
| 28 | C   | 16.4 Composting and the Science Behind It     | Understand      |
| 29 | A   | 16.4 Composting and the Science Behind It     | Understand      |
| 30 | B   | 16.4 Composting and the Science Behind It     | Analyse         |
| 31 | C   | 16.4 Composting and the Science Behind It     | Understand      |
| 32 | D   | 16.4 Composting and the Science Behind It     | Understand      |
| 33 | A   | 16.5 Effects of Poor Waste Management and ... | Understand      |
| 34 | B   | 16.5 Effects of Poor Waste Management and ... | Remember        |
| 35 | A   | 16.5 Effects of Poor Waste Management and ... | Understand      |
| 36 | A   | 16.5 Effects of Poor Waste Management and ... | Apply           |
| 37 | B   | 16.5 Effects of Poor Waste Management and ... | Apply           |

| Q  | Ans | Curriculum Area                               | Cognitive Level |
|----|-----|-----------------------------------------------|-----------------|
| 38 | D   | 16.5 Effects of Poor Waste Management and ... | Apply           |
| 39 | C   | 16.5 Effects of Poor Waste Management and ... | Evaluate        |
| 40 | D   | 16.5 Effects of Poor Waste Management and ... | Evaluate        |

**Answer Distribution:**

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

**Cognitive Level Distribution:**

Analyse: 5 (12%) | Apply: 4 (10%) | Evaluate: 3 (7%) | Remember: 13 (32%) | Understand: 15 (37%)

**SECTION B: THEORY — MARKING SCHEME****Question 1 [Total: 12 marks]****(a)** [4 marks]

- A - **organic (food or kitchen) waste** (1 mark).
- B - **plastic waste** (1 mark).
- C - **glass and metal waste** (1 mark).
- D - **paper waste** (1 mark).
- **Marking note:** Accept 'biodegradable waste' for A and D only if the material is also named.\*

**(b)** [4 marks]

- A - **biodegradable** (1 mark).
- B - **non-biodegradable** (1 mark).
- C - **non-biodegradable** (1 mark).
- D - **biodegradable** (1 mark).
- **Marking note:** Mark each entry separately.\*

**(c)** [2 marks]

- **Biodegradable waste is broken down by micro-organisms into simple substances** (1 mark).
- **Non-biodegradable waste is not broken down and stays in the environment for a very long time** (1 mark).
- **Marking note:** 'One rots and the other does not' is accepted for the first mark only if decay is clearly meant.\*

**(d)** [2 marks]

- **Sort it into a separate container and send it for recycling (accept: bury it safely in a pit)** (1 mark).
- **Because glass and metal do not decay; broken glass causes injury and tins collect water in which mosquitoes breed** (1 mark).
- **Marking note:** 'Burn it' earns no mark.\*

**Question 2 [Total: 8 marks]****(a)** [3 marks]

- Any THREE of: **flies spread typhoid, cholera or diarrhoea**; **mosquitoes breeding in the tins spread malaria**; **rats and cockroaches spread disease**; **the smell and smoke affect breathing**; **sharp objects cause injury** (1 mark each).
- **Marking note:** A disease named without its link to the heap still earns the mark if the link is obvious.\*

**(b)** [3 marks]

- **The organic waste is decaying, being broken down by bacteria and fungi** (1 mark).
- **Gases such as ammonia and hydrogen sulphide are released, and these smell bad** (1 mark).
- **Mosquitoes lay eggs in the still water and breed there, spreading malaria** (1 mark).
- **Marking note: Accept 'rotting' for decay.**

**(c)** [2 marks]

- Any TWO of: **burning plastics and rubber releases poisonous smoke and gases that pollute the air and harm the lungs**; **the fire may spread and destroy property**; **useful material that could be composted or recycled is destroyed** (1 mark each).
- **Marking note: 'It is against the law' earns no mark without a scientific reason.**

**Question 3 [Total: 10 marks]****(a)** [3 marks]

- **Reduce: produce less waste by using less and avoiding unnecessary items** (1 mark).
- **Reuse: use an article again as it is, instead of throwing it away** (1 mark).
- **Recycle: process used material into new articles** (1 mark).
- **Marking note: An example alone, with no explanation, earns no mark.**

**(b)** [3 marks]

- **Reduce: one sound example, such as a refillable bottle instead of daily sachets** (1 mark).
- **Reuse: one sound example, such as tins used for seedlings** (1 mark).
- **Recycle: one sound example, such as collecting plastic bottles for a recycling centre** (1 mark).
- **Marking note: An example placed under the wrong term earns no mark.**

**(c)** [2 marks]

- **Separating waste into different containers according to its kind** (1 mark).
- **So that each kind can be composted, reused or recycled instead of being dumped** (1 mark).
- **Marking note: Accept 'sorting' for segregation.**

**(d)** [2 marks]

- **Three sensible labels, such as organic waste, plastics and paper** (1 mark).
- **A correct item named for each bin** (1 mark).
- **Marking note: Labels such as 'rubbish 1, rubbish 2' earn no mark.**

**Question 4****(a)** [4 marks]

- P - **coarse dry material: maize stalks, twigs or sticks** (1 mark).
- Q - **green plant material and kitchen waste: grass, weeds, peels** (1 mark).
- R - **topsoil or animal droppings (manure)** (1 mark).
- S - **a cover of palm fronds, dry grass or an old sack** (1 mark).
- **Marking note: Plastics or tins named in any layer earn no mark.**

**(b)** [3 marks]

- (i) **Water keeps the heap moist** (1 mark); **the micro-organisms need moisture to live and act on the waste** (1 mark).
- (ii) **The micro-organisms respire as they feed, and respiration gives out heat** (1 mark).
- **Marking note: 'The Sun heats it' earns no mark for (ii).**

**(c)** [3 marks]

- **Turning lets air (oxygen) into the heap** (1 mark).
- **The decomposing micro-organisms are aerobic and need oxygen; turning also mixes the layers for even decay** (1 mark).
- **Without turning, decay becomes slow and anaerobic and the heap smells foul** (1 mark).
- **Marking note: Accept 'the heap would rot slowly and smell' for the third mark.**

**(d)** [2 marks]

- Any TWO of: **the waste is turned into manure that feeds the garden and improves the soil**; **money is saved on fertiliser**; **less waste goes to the dump, so fewer flies, less smell and less pollution** (1 mark each).
- **Marking note: 'It is cheaper' alone, with no explanation, earns no mark.**

**Question 5 [Total: 10 marks]****(a)** [3 marks]

- Any THREE of: **research good waste management practices**; **obtain permission from the authorities and inform the traders**; **fix a date and share the members into groups with clear duties**; **collect the tools and protective items needed**; **arrange for the waste collected to be carted away** (1 mark each).
- **Marking note: 'Tell everybody' is too vague for a mark.**

**(b)** [3 marks]

- **Hand gloves - to protect the hands from cuts and germs** (1 mark).
- **Nose mask - to keep out dust, smell and germs** (1 mark).
- **Boots or closed shoes - to protect the feet from sharp objects and dirty**

water\*\* (1 mark).

- \*Marking note: No mark for an item named without a reason.\*

**(c)** [4 marks]

- \*\*Sort (segregate) the waste as it is collected\*\* (1 mark).

- \*\*Compost the organic waste\*\* (1 mark).

- \*\*Send the plastics, paper and metal for recycling or reuse\*\* (1 mark).

- \*\*Hand over only the remaining unusable waste for proper disposal, with no open burning\*\* (1 mark).

- \*Marking note: A plan that ends with everything being dumped or burnt scores no more than 1 mark.\*

## Question 6 [Total: 8 marks]

**(a)** [4 marks]

- Any FOUR of: \*\*title, date and place\*\*; \*\*aim of the project\*\*; \*\*materials and method used\*\*; \*\*findings or observations\*\*; \*\*conclusion\*\*; \*\*recommendations\*\* (1 mark each, the heading with a correct phrase of content).

- \*Marking note: A heading with nothing said about its content earns half of nothing - award the mark only when both are given.\*

**(b)** [2 marks]

- Any TWO of: \*\*provide enough covered, labelled bins and empty them regularly\*\*; \*\*fix a weekly clean-up day\*\*; \*\*assign someone to sweep daily\*\*; \*\*educate the traders with posters and announcements\*\*; \*\*enforce a by-law against dumping in gutters\*\* (1 mark each).

- \*Marking note: 'They should keep it clean' is too vague for a mark.\*

**(c)** [2 marks]

- \*\*It spreads the knowledge to the whole school, so that others change their habits\*\* (1 mark).

- \*\*It makes the findings and recommendations known so that they can be acted upon (accept: it encourages others to take part)\*\* (1 mark).

- \*Marking note: 'Because the teacher asked for it' earns no mark.\*

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## ## CURRICULUM ASSESSMENT BLUEPRINT

\*\*Strand:\*\* 5: Humans and the Environment | \*\*Sub-strand:\*\* 1: Waste Management | \*\*Class:\*\* Basic 7 (JHS 1)

\*\*Content Standard:\*\* \*\*B7/JHS1.5.1.1\*\* - Exhibit knowledge and skill of

scientific basis for management practices of types of waste in the en





# Water Cycle

## ANSWER KEY

### SECTION A: OBJECTIVE TEST

Total Marks: 40

| Q  | Ans | Curriculum Area                           | Cognitive Level |
|----|-----|-------------------------------------------|-----------------|
| 1  | D   | 3.1 Natural Sources of Water              | Remember        |
| 2  | A   | 3.1 Natural Sources of Water              | Remember        |
| 3  | C   | 3.1 Natural Sources of Water              | Remember        |
| 4  | D   | 3.1 Natural Sources of Water              | Understand      |
| 5  | A   | 3.1 Natural Sources of Water              | Understand      |
| 6  | B   | 3.2 Evaporation and Transpiration         | Remember        |
| 7  | D   | 3.2 Evaporation and Transpiration         | Remember        |
| 8  | C   | 3.2 Evaporation and Transpiration         | Understand      |
| 9  | B   | 3.2 Evaporation and Transpiration         | Remember        |
| 10 | C   | 3.2 Evaporation and Transpiration         | Understand      |
| 11 | A   | 3.2 Evaporation and Transpiration         | Analyse         |
| 12 | A   | 3.2 Evaporation and Transpiration         | Understand      |
| 13 | D   | 3.2 Evaporation and Transpiration         | Apply           |
| 14 | D   | 3.3 Condensation and Cloud Formation      | Remember        |
| 15 | C   | 3.3 Condensation and Cloud Formation      | Understand      |
| 16 | B   | 3.3 Condensation and Cloud Formation      | Remember        |
| 17 | D   | 3.3 Condensation and Cloud Formation      | Understand      |
| 18 | A   | 3.3 Condensation and Cloud Formation      | Understand      |
| 19 | B   | 3.3 Condensation and Cloud Formation      | Understand      |
| 20 | A   | 3.3 Condensation and Cloud Formation      | Understand      |
| 21 | B   | 3.4 Precipitation, Run-off and Collection | Remember        |
| 22 | A   | 3.4 Precipitation, Run-off and Collection | Understand      |
| 23 | B   | 3.4 Precipitation, Run-off and Collection | Remember        |
| 24 | C   | 3.4 Precipitation, Run-off and Collection | Remember        |
| 25 | C   | 3.4 Precipitation, Run-off and Collection | Remember        |
| 26 | D   | 3.4 Precipitation, Run-off and Collection | Apply           |
| 27 | A   | 3.4 Precipitation, Run-off and Collection | Analyse         |
| 28 | B   | 3.5 The Water Cycle as a Repeated Pattern | Understand      |
| 29 | D   | 3.5 The Water Cycle as a Repeated Pattern | Understand      |
| 30 | B   | 3.5 The Water Cycle as a Repeated Pattern | Understand      |
| 31 | A   | 3.5 The Water Cycle as a Repeated Pattern | Understand      |
| 32 | C   | 3.5 The Water Cycle as a Repeated Pattern | Analyse         |
| 33 | C   | 3.5 The Water Cycle as a Repeated Pattern | Analyse         |
| 34 | B   | 3.6 The Importance of the Water Cycle     | Understand      |
| 35 | C   | 3.6 The Importance of the Water Cycle     | Understand      |
| 36 | A   | 3.6 The Importance of the Water Cycle     | Understand      |

| Q  | Ans | Curriculum Area                       | Cognitive Level |
|----|-----|---------------------------------------|-----------------|
| 37 | D   | 3.6 The Importance of the Water Cycle | Understand      |
| 38 | B   | 3.6 The Importance of the Water Cycle | Analyse         |
| 39 | D   | 3.6 The Importance of the Water Cycle | Analyse         |
| 40 | C   | 3.6 The Importance of the Water Cycle | Evaluate        |

**Answer Distribution:**

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

**Cognitive Level Distribution:**

Analyse: 6 (15%) | Apply: 2 (5%) | Evaluate: 1 (2%) | Remember: 12 (30%) | Understand: 19 (47%)

**SECTION B: THEORY — MARKING SCHEME****Question 1 [Total: 10 marks]****(a) [5 marks]**

- A - **evaporation** (1 mark).
- B - **transpiration** (1 mark).
- C - **condensation** (1 mark).
- D - **precipitation** (accept: rainfall) (1 mark).
- E - **run-off** (accept: surface run-off, collection of water in rivers returning to the sea) (1 mark).
  - \*Marking note: 'Evaporation' written for B earns no mark; water lost from the leaves of a plant is transpiration.\*

**(b) [3 marks]**

- (i) **The sun** (1 mark).
- (ii) **It is left behind in the sea; only pure water vapour rises** (1 mark).
- (iii) **The air is cooler high up, so the vapour condenses into tiny water drops** (1 mark).
  - \*Marking note: 'It is hot' is not accepted for (i); the sun must be named.\*

**(c) [2 marks]**

- **The water returns to the sea, the place where it started, so the same water is used again** (1 mark).
- **The same stages follow one another in the same order without ending** (1 mark).
  - \*Marking note: A candidate who writes only 'because it happens every year' earns no mark.\*

**Question 2 [Total: 12 marks]****(a) [4 marks]**

- All four stages written in the correct order (2 marks; 1 mark if one stage is out of place).
- Arrows drawn between the stages to show the direction of the change (1 mark).
- An arrow from **collection back to evaporation**, closing the cycle (1 mark).
  - \*Marking note: A chart that stops at precipitation loses the last mark, however neat it is.\*

**(b)** [5 marks]

- Any FIVE of: **a source of energy - energy released during condensation warms the environment**; **a carrier of nutrients**; **it improves (raises) the water table**; **it regulates the weather pattern**; **it provides clean, fresh water**; **it supplies water for crops, animals and people** (1 mark each).
- **Marking note:** 'It gives us water' may be credited once only, as the provision of clean water.\*

**(c)** [3 marks]

- Any THREE of: **rain falling on the town, the farms or the hills**; **the river or the wells being refilled (the water table rising)**; **people using the water for drinking, washing, farming or watering animals**; **water returning to the air by evaporation from the river and transpiration from trees** (1 mark each).
- **Marking note:** Marks are for what the diagram would show, not for drawing it; a candidate who draws it correctly still earns the marks.\*

**Question 3 [Total: 10 marks]****(a)** [3 marks]

- P - **steam / water vapour rising from the water** (1 mark).
- Q - **drops of water (condensed water) on the underside of the plate** (1 mark).
- R - **the water collected in the cup** (1 mark).
- **Marking note:** 'Smoke' written for P earns no mark.\*

**(b)** [3 marks]

- P - **evaporation** (1 mark).
- Q - **condensation** (1 mark).
- R - **precipitation and collection** (accept either) (1 mark).
- **Marking note:** Accept 'rainfall' for R.\*

**(c)** [2 marks]

- Burner - **the sun** (1 mark).
- Cold metal plate - **the cool air high up in the atmosphere (where clouds form)** (1 mark).
- **Marking note:** 'The sky' alone is not accepted for the plate; the idea of cool air or cloud level is needed.\*

**(d)** [2 marks]

- Any TWO of: **in nature the sun heats the water slowly and the water does not boil**; **the model leaves out transpiration by plants**; **the model leaves out run-off and water soaking into the ground**; **in nature the wind carries the clouds far away before the rain falls**; **the model is very much smaller and lasts only a few minutes** (1 mark each).
- **Marking note:** 'It is not real water' earns no mark.\*

**Question****(a)** [3 marks]

- **Trees release water vapour into the air by transpiration** (1 mark).
- **With fewer trees, less water vapour enters the air** (1 mark).
- **Less vapour means fewer clouds form by condensation, so less rain falls** (1 mark).
- **Marking note:** The word 'transpiration' is not compulsory if the idea of water vapour from the leaves is clear.\*

**(b)** [3 marks]

- (i) **The level below the ground at which the soil or rock is filled (saturated) with water** (1 mark).
- (ii) **Rain water soaks into the soil (infiltration)** (1 mark) **and collects below the surface, raising the stored water so that wells do not dry up** (1 mark).
- **Marking note:** 'Water under the ground' alone earns the mark in (i) only if the idea of a level is given.\*

**(c)** [4 marks]

- Any TWO actions (1 mark each) with a matching reason (1 mark each): **plant trees and grasses** - **roots slow the water and help it soak into the soil**; **leave open ground, dig trenches or infiltration pits** - **the water is held until it soaks in**; **harvest rain water from roofs into tanks** - **less water runs over the ground and water is stored for the dry season**; **clear and maintain the gutters** - **run-off is carried away without flooding the streets**.
- **Marking note:** An action without a reason earns 1 mark only.\*

**Question 5 [Total: 10 marks]****(a)** [3 marks]

- Dish 1 - **48 cm<sup>3</sup>** (1 mark).
- Dish 2 - **22 cm<sup>3</sup>** (1 mark).
- Dish 3 - **6 cm<sup>3</sup>** (1 mark).
- **Marking note:** Accept answers without the unit, but a wrong unit (for example g) loses the mark for that dish.\*

**(b)** [3 marks]

- **The dish in the sun received more heat energy from the sun** (1 mark).
- **More heat makes water evaporate faster** (1 mark).
- **The cupboard was cool and the still, closed air soon held as much vapour as it could, so evaporation was slow** (1 mark).
- **Marking note:** Accept 'there was no wind in the cupboard to carry the vapour away' for the third mark.\*

**(c)** [2 marks]

- Any TWO of: **\*\*the dishes were identical, so the exposed surface was the same\*\***; **\*\*the same volume of water was used in each dish\*\***; **\*\*all the dishes were left for the same time (six hours)\*\*** (1 mark each).
- **\*Marking note: 'They used water' is not a fair-test measure and earns no mark.\***

**(d)** [2 marks]

- **\*\*The pupil is not correct\*\*** (1 mark).
- **\*\*The water has changed into water vapour and entered the air; it is not destroyed and will return as rain\*\*** (1 mark).
- **\*Marking note: No mark for the explanation if the candidate agrees with the pupil.\***

**Question 6 [Total: 8 marks]****(a)** [3 marks]

- **\*\*Evaporation leaves salts and dirt behind, so the rain that falls is fairly pure\*\*** (1 mark).
- **\*\*Falling rain picks up dust and smoke from the air\*\*** (1 mark).
- **\*\*The roof and gutters add dust, leaves, bird droppings and germs to the water\*\*** (1 mark).
- **\*Marking note: A candidate who writes only 'the tank may be dirty' earns 1 mark.\***

**(b)** [2 marks]

- Any TWO of: **\*\*boil the water before drinking\*\***; **\*\*filter it through a clean cloth or filter\*\***; **\*\*treat it with an approved water-treatment agent (for example chlorine tablets)\*\***; **\*\*keep the tank covered and clean\*\***; **\*\*clean the roof and gutters and let the first dirty water run to waste\*\*** (1 mark each).
- **\*Marking note: Accept any safe, practical method used in Ghanaian homes.\***

**(c)** [3 marks]

- Any THREE of: **\*\*the water in the tank is still part of the cycle; when it is used it evaporates or soaks into the ground\*\***; **\*\*the total amount of water on the earth does not change, it is only used again and again\*\***; **\*\*the tank holds only a very small part of the rain that falls\*\***; **\*\*collecting roof water reduces run-off and flooding\*\*** (1 mark each).
- **\*Marking note: A candidate who merely writes 'the neighbour is wrong' without a reason earns no mark.\***

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## CURRICULUM ASSESSMENT BLUEPRINT

**\*\*Strand:\*\*** 2: Cycles | **\*\*Sub-strand:\*\*** 1: Earth Science | **\*\*Class:\*\*** Basic 7 (JHS 1)

**\*\*Content**

# Integrated Science Glossary — Basic 7

This glossary contains key terms and definitions that appear throughout this book. Use it as a quick reference while studying and preparing for examinations.

## **Abiotic Factors**

Non-living components of an ecosystem such as temperature, light, water, and soil.

## **Acid**

A substance that produces hydrogen ions ( $H^+$ ) when dissolved in water. Has pH less than 7.

## **Adaptation**

A characteristic or behavior that helps an organism survive in its environment.

## **Agriculture**

The practice of farming, including cultivation of crops and rearing of animals.

## **Atom**

The smallest unit of an element that retains the properties of that element.

## **Base**

A substance that produces hydroxide ions ( $OH^-$ ) when dissolved in water. Has pH greater than 7.

## **Biotic Factors**

Living components of an ecosystem such as plants, animals, and microorganisms.

## **Carbon Dioxide**

A gas ( $CO_2$ ) produced during respiration and used by plants in photosynthesis.

## **Cell**

The basic structural and functional unit of all living organisms.

## **Chlorophyll**

The green pigment in plants that absorbs light energy for photosynthesis.

## **Climate**

The average weather conditions of a place over a long period (30+ years).

## **Compound**

A substance formed when two or more elements chemically combine in fixed proportions.

## **Conservation**

The protection and careful management of natural resources.



**Digestion**

The process of breaking down large food molecules into smaller, soluble molecules.

**Ecosystem**

A community of living organisms interacting with each other and their non-living environment.

**Electricity**

The flow of electric charge (electrons) through a conductor.

**Element**

A pure substance that cannot be broken down into simpler substances by chemical means.

**Energy**

The ability to do work or cause change. Measured in joules (J).

**Environment**

The surroundings or conditions in which an organism lives.

**Erosion**

The wearing away and removal of soil and rock by wind, water, or ice.

**Evaporation**

The process by which a liquid changes to a gas.

**Fertilizer**

A substance added to soil to increase its fertility and plant growth.

**Force**

A push or pull that can change the motion or shape of an object. Measured in newtons (N).

**Germination**

The process by which a seed begins to grow into a new plant.

**Habitat**

The natural home or environment of an organism.

**Hydroelectric**

Relating to the generation of electricity using flowing water.

**Irrigation**

The artificial application of water to land to assist crop production.

**Matter**

Anything that has mass and occupies space.

**Metamorphosis**

A complete change in form or structure during the life cycle of an organism.

**Mixture**

A combination of two or more substances that are not chemically bonded.

**Nutrient**

A substance that provides nourishment essential for growth and maintenance of life.

**Oxygen**

A gas (O<sub>2</sub>) essential for respiration in most living organisms.

**Photosynthesis**

The process by which plants make food using sunlight, water, and carbon dioxide.

**Pollution**

The introduction of harmful substances into the environment.

**Recycling**

The process of converting waste materials into new materials and objects.

**Respiration**

The process by which organisms release energy from food.

**Soil**

The upper layer of earth in which plants grow, consisting of rock particles, organic matter, water, and air.

**Solar Energy**

Energy from the sun that can be converted into electricity or heat.

**Solution**

A homogeneous mixture of two or more substances.

**Temperature**

A measure of the average kinetic energy of particles in a substance. Measured in degrees Celsius (°C) or Kelvin (K).

**Transpiration**

The loss of water vapor from plant leaves.

**Water Cycle**

The continuous movement of water between the earth's surface and the atmosphere.

**Weather**

The day-to-day conditions of the atmosphere including temperature, humidity, wind, and precipitation.

**Work**

The transfer of energy when a force moves an object through a distance. Measured in joules (J).

# Curriculum Reference

This book aligns with the following curriculum standards:

- Ghana Education Service (GES)  
Standards-Based Curriculum for Junior High Schools (2019)
- National Council for Curriculum and Assessment (NaCCA)  
Curriculum Framework for Basic Education
- West African Examinations Council (WAEC)  
BECE Syllabus for Integrated Science / Mathematics

The content has been reviewed to ensure alignment with:

- Strand and sub-strand organization
- Learning indicators and objectives
- Cognitive levels (Knowledge, Understanding, Application, Analysis)
- Core competencies (Critical Thinking, Problem Solving, Communication)

Teachers and students should use this book in conjunction with the official curriculum documents available from the Ghana Education Service.

For more information:

Ghana Education Service: [www.ges.gov.gh](http://www.ges.gov.gh)

NaCCA: [www.nacca.gov.gh](http://www.nacca.gov.gh)

WAEC: [www.waec.org](http://www.waec.org)

# About Beacon Educational Consult

Beacon Educational Consult is a Ghanaian educational publishing company dedicated to producing high-quality learning materials for Junior High School students across Ghana.

Our mission is to support students in achieving academic excellence through comprehensive, curriculum-aligned resources that prepare them for the Basic Education Certificate Examination (BECE) and beyond.

We believe that every student deserves access to excellent educational materials, and we are committed to making quality textbooks affordable and accessible to schools and families throughout Ghana.

Our books are:

- Written by experienced Ghanaian educators
- Aligned with the national curriculum
- Reviewed for accuracy and quality
- Designed to support both classroom learning and independent study

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